

Renewal Geometry and the Emergence of Lorentzian Spacetime

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Abstract

We develop a finite operational framework in which Lorentzian spacetime and vacuum Einstein dynamics are reconstructed from renewal data. Quotienting histories by equality of their conditional futures yields a canonical predictive carrier, a positive history algebra, and an intrinsic word-Gram hierarchy; together these define the *Renewal Grand Tensor*, from which relational geometry is reconstructed. Within specified finite relational classes, two complementary selection principles identify three spatial dimensions, while one additional protected signed direction yields Lorentzian signature. Calibrated renewal statistics then reconstruct the spatial metric, lapse, and shift. A variational structure is reconstructed directly from the finite dynamics. On noncollapsed continuum branches satisfying explicit stationarity, compactness, and curvature-propagation conditions, its local curvature-linear representative is of Palatini–Holst type and the limiting geometry satisfies the vacuum Einstein equation. We also construct a cutoff-uniform nonsymmetric three-dimensional class, stable across an open finite-dimensional family of renewal laws sharing the same Einstein continuum limit.

Keywords. Emergent spacetime; Lorentzian geometry; predictive dynamics; renewal processes; variational dynamics; Palatini–Holst gravity; ADM decomposition.

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1 Introduction

One of the oldest ambitions of fundamental physics is to explain why spacetime has the structure that it does, rather than to take that structure as the stage on which all other physics occurs. General relativity transformed gravity from a force propagating through spacetime into the dynamics of spacetime itself, but it still begins with a differentiable manifold carrying a Lorentzian metric. Most subsequent approaches to quantum gravity likewise replace some part of the classical geometric description while retaining another part as primitive. Causal-set theory begins with a locally finite causal order [28, 29]; in addition to stochastic sequential growth laws [2], its order-theoretic programme includes dimension estimators [3, 4] and discrete curvature/action constructions [5]. Regge calculus approximates curved manifolds by simplicial data [1], while causal dynamical triangulations exhibit a dynamically generated macroscopic four-dimensional phase and scale-dependent spectral dimension [7, 8, 31]. Quantum graphity, spin-foam and group-field approaches instead begin from graph, simplicial, representation-theoretic or field-theoretic quanta whose collective regimes are intended to recover continuum geometry [30, 9, 32, 17, 18]. Noncommutative geometry replaces the underlying space by spectral and operator-algebraic data [33].

Operational approaches have also weakened the amount of causal structure assumed in advance. Quantum causal histories place matrix algebras and completely positive evolution on a discrete causal pre-spacetime [10]; Hardy’s causaloid programme formulates operational probabilities without imposing a fixed background causal structure [11]; and process-matrix methods admit quantum correlations without a predefined global causal order [12]. Other programmes seek gravitational dynamics or geometry from thermodynamics, entropy, or quantum information [34, 35, 36, 19, 20, 21]. These approaches differ profoundly in their microscopic assumptions, but together they make clear that “emergence of spacetime” can mean several mathematically distinct things: estimating continuum dimension from discrete order, recovering geometry from quantum correlations, deriving gravitational field equations from a variational or entropic principle, or obtaining a continuum phase from combinatorial quantum data.

The question pursued here differs in its operational starting point. We ask whether the structures normally used to define spacetime can arise from the organization of *predictive histories* without taking a causal order, adjacency relation, simplicial complex, spectral triple, metric, or persistent state space as the initial geometric object. The microscopic datum is an operational renewal event: an intervention occurs, a finite outcome is registered, some distinctions remain capable of influencing later operations, and the process may subsequently renew. At this level a history is simply a sequence of operational events. It has no assigned position, distance, metric, connection, curvature, or spacetime dimension, and its ordering is not yet interpreted as motion through a pre-existing geometry. The claim below is therefore not priority for operational or pregeometric descriptions as such; it is the particular reconstruction chain by which predictive persistence, positive history structure, relational dimension, Lorentzian geometry and a certified Einstein limit are connected.

Prediction before geometry. This choice of starting point is motivated by a simple observation. Physical description is never sensitive to an arbitrary microscopic history in its full detail. What matters operationally is which distinctions in that history can still affect a future experiment. Two past histories that produce exactly the same statistics under every admissible continuation are therefore physically indistinguishable for all subsequent purposes. Identifying such histories is a predictive rather than a geometric reduction. It is closely related to causal-state constructions in computational mechanics, where pasts are quotiented by equality of their conditional futures [37, 38], to continuous-time predictive representations of stochastic processes [39, 40], and to

observable-operator and predictive-state realizations in which dynamical state is encoded through future observations and, in the controlled setting, action-conditioned future predictions [13, 14]. The predictive minimality used here is therefore not by itself the claimed novelty. The distinction is that complete intervention-conditioned future equivalence is used as the canonical pregeometric carrier on which the positive process structure and the later relational-geometric reconstruction are built.

Because future equivalence is preserved under admissible continuation, the quotient is not merely a compressed description of the past. It carries a canonical successor structure and therefore becomes the minimum persistent predictive state of the process. Persistence is thus reconstructed from the organization of future distinguishability rather than supplied as an independent state space, graph adjacency, or background coordinate system.

Renewal as a microscopic principle. This perspective leads naturally to renewal theory. Renewal processes are usually introduced as stochastic models of systems that repeatedly reset, regenerate, or lose part of their past. They underlie classical results in probability and statistical mechanics [41], while non-Markovian renewal processes include heavy-tailed and infinite-mean regimes with qualitatively different long-time behavior [42, 43, 44]. Renewal and semi-Markov ideas have also been lifted to completely positive quantum dynamics, including quantum semi-Markov and quantum renewal processes [15, 16]. In these conventional classical and quantum constructions, however, renewal occurs *inside* an already given time and state space: waiting times are measured by an external temporal parameter and the process acts on a previously specified classical or quantum state carrier.

Here we reverse that logical order. Renewal is not placed in spacetime; renewal belongs to the microscopic organization from which spacetime is reconstructed. The microscopic law may retain history through waiting-time, record, or return statistics and need not be Markovian. A local Markovian evolution can nevertheless arise after predictive compression and coarse graining, once unresolved renewal memory has collapsed onto a sufficient future state. In this sense the familiar locality of continuum evolution is naturally interpreted as a *Markovian scaling regime of a more general renewal dynamics*, rather than as the fundamental form of microscopic evolution.

There is a useful conceptual analogy with statistical mechanics and renormalization. Thermodynamic variables emerge after most microscopic information has been discarded, while renormalization retains degrees of freedom that remain relevant at longer scales. Here the reduction is instead organized by *future distinguishability*: microscopic information is discarded precisely when no admissible future can read it. What survives is therefore not an arbitrary coarse graining but a canonical predictive quotient. The geometry developed below is built on that quotient. Loosely speaking, spacetime is the stable organization of those relational distinctions that prediction cannot forget.

Main result. The central result is a finite-to-continuum reconstruction and certification theorem. Complete conditional future statistics first determine a canonical minimum predictive carrier and a positive history algebra, whose intrinsic word-Gram hierarchy together form the *Renewal Grand Tensor*. The finite relational dimension is selected in two independent ways. Exact endpoint-cycle source-rank balance among connected simple regular relations selects the tetrahedral cell K_4 without an alternating-form hypothesis; within the stronger pair-homogeneous simplex category, minimal faithful alternating completion selects the same cell. Its standard module is the three-dimensional A_3 root space. Calibrated root-race statistics then reconstruct the ten ADM variables. The positive Grand Tensor cannot itself supply a negative direction: an independently occurring protected

binary orientation is an irreducible signed datum, and adjoining one signed direction to the positive three-dimensional spatial form fixes Lorentzian inertia $(1, 3)$ before a Clifford realization is chosen.

The dynamical part is reconstructed from the complete accepted physical cylinder rather than imposed as a continuum action. A marked renewal pressure recovers the full action slope, finite curl and period tests determine whether the increments integrate to one scalar common action, and a positive Fisher–Bregman gap, with its physical test-lift and cutoff constants retained, certifies stationarity of the accepted field dynamics for that independently reconstructed action. Finite Gibbs/Kullback–Leibler identities separately compare the accepted probability rows with their reconstructed costs. After resolved memory is removed, Lorentz covariance and the finite bivector commutant classify the first-order gravitational score as Palatini–Holst plus determinant volume. A positive spatial cut margin supplies regulator-uniform Sobolev, Poincaré, counting, and compact-screen bounds. Continuous and fully discrete propagation estimates give a spacetime L^2 curvature bound from initial energy and explicit growth, transfer and differentiated-source budgets; a sharp action-current comparison can supply the source budget. Connection curvature can be identified either through a separate connection compactness certificate with a finite Hodge–temporal sufficient criterion, or, on the literal-link branch, directly from anisotropic spatial connection energy together with the measured electric and magnetic link records; the latter route requires no time derivative of the temporal connection coefficient. Interface-complete Cartan comparisons remain available for piecewise reconstructions. On the branch where these finite and cofinal certificates pass, the weak–strong Palatini limit gives

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 0} \tag{1.1}$$

distributionally on the determining test core. If a required condition fails, the construction instead returns the corresponding finite or cofinal obstruction. The result is therefore an Einstein-obstruction theorem, rather than a claim that every microscopic renewal process flows to general relativity. A failed sufficient certificate does not exclude an Einstein subsequence obtainable by a different argument.

The positive branch is nonempty and is not confined to symmetry-reduced kinematics. In addition to the homogeneous and Gowdy sectors described below, a full three-dimensional local harmonic renewal writer admits a cutoff-independent nonlinear Cauchy theorem in a fixed Sobolev ball for all ten metric components and all spatial lattice modes. More strongly, this is not an isolated engineered update: the same estimates and Einstein limit hold uniformly on a fixed open 55-dimensional family of local differential laws, with no convergence of the law parameter required along the cutoff sequence. Around finite realizations of these laws, conditional physical residual moments and a profile-transport metric define relatively open neighborhoods of history-dependent transition rows whose successful histories have the same Einstein limit in probability and, along the declared odd refinement, eventually almost surely. On the cofinally consistent initial-action branch, compatible limiting initial records can also be obtained from convergent finite preparations by solving the original reconstructed action’s complete lapse–shift constraint rows from unconstrained small free tensor records; no continuum vacuum metric is required as an input to that preparation.

These population results and the general certification theorem have different logical roles. [Theorems 6.1](#) and [6.3](#) construct nonsymmetric $3 + 1$ classes and reach the Einstein equation by cutoff-uniform evolution, consistency and finite preparation estimates, whereas [Theorem 5.4](#) is a recognition theorem for arbitrary complete gravitational renewal cylinders. Its five certificates are not the definition of the constructive population.

A separate question is whether the *unchanged finite Euler flow* of the reconstructed action itself lies on the same branch. We do not identify the constructive writer with that flow. Instead [Appendix G](#) writes the phase-compatible finite action explicitly, performs stationary connection

and Legendre reduction, and analyzes the resulting canonical system. Its regular-response theory gives exact finite preparations, while a same-event three-site certificate proves weak Poisson rank 24, a five-dimensional weak kernel, and the semisimple rapid split $(8, 8, 13)$. Under the complete graded-chart hypotheses this exact system has a boundary-selected constrained history with $O(a)$ metric and link curvature on $0 \leq t \leq ca^2$ at fixed spatial cutoff. The complementary residence and slow-compatibility results identify why this does not yet yield a forward, cutoff-uniform exact-action Cauchy theorem.

The homogeneous isotropic sector gives a complementary exact variational realization: arbitrary positive renewal variables yield the full lapse-varied vacuum Friedmann equations, so the flat and de Sitter rate profiles arise as stationary solutions rather than being prescribed in advance. A finite positive accept/reject process reconstructs the same reduced action directly from measured trial probabilities and converges quantitatively to the stationary branch. Exact conservative time blocking of this quadratic class preserves the scale-free action-law defect, so it does not generate an open basin toward the matched surface. Beyond homogeneity, a nonlinear unpolarized Gowdy T^3 regulator preserves an exact finite momentum invariant and, on its finite offset alphabet, has pathwise $O(h)$ convergence of the full Riemann tensor on smooth bounded slabs. These constructions provide complementary open, homogeneous, and nonlinear symmetry-reduced populations without asserting generic microscopic attraction to the certified branch.

The precise operational entrance is intentionally narrower than a claim that all geometric information is generated from positive probabilities alone. No predictive state space or persistent relational carrier is postulated independently: histories are identified exactly when every admissible continuation has the same conditional law, and the resulting future-equivalence quotient is the unique minimum persistent predictive carrier through which all future distinctions factor. The geometric branch also retains two kinds of information that this positive predictive quotient cannot manufacture. A protected binary orientation is an actual two-valued operational distinction, but no interpretation as past and future, positive and negative norm, or Lorentzian signature is assumed microscopically. Likewise, normalized race probabilities determine only relative rates, so absolute waiting-time and measure calibrations are required when those rates are interpreted geometrically.

A further distinction is essential. The future-equivalence class is derived as an abstract predictive state, but spacetime reconstruction must physically read that class at a cut and correlate it on the same history with the occurring root event, scheduler-separated fast path, orientation, calibration, and other declared marks. These data form a minimal faithful readable relation packet. The packet does not introduce a second predictive state: its state coordinate is the already derived future-equivalence class. Its additional content is physical occurrence and same-cut readability. The enrichment is conservative: forgetting the added marks recovers the original event cylinders exactly, while future minimization removes every unread refinement. Thus predictive persistence is derived, the physical information needed for the geometric reconstruction must actually occur together, and metric, connection, curvature, and spacetime dimension remain downstream outputs.

The Renewal Grand Tensor and Lorentzian reconstruction. The first major structural output is the object we call the *Renewal Grand Tensor*. The name does not refer to a tensor field on a pre-existing manifold. Predictively indistinguishable histories are first identified, producing the unique minimum persistent predictive carrier. Tomographically complete intervention statistics then reconstruct the finite positive process tensors, while support-minimal completely positive realization removes unobservable environmental degrees of freedom [45, 46]. The surviving mixed histories generate a finite C^* -algebra, and their intrinsic positive overlaps form a word-Gram hierarchy. Together these structures record which distinctions survive into the future, how renewal operations

compose, and how the resulting histories overlap. The Grand Tensor is therefore not geometry imposed on a renewal process; it is the canonical algebraic and positive structure retained by the process after every future-invisible distinction has been removed.

This places the construction near noncommutative and spectral geometry, but with a different starting point. In ordinary spectral geometry an algebra, Hilbert-space representation, and Dirac operator are supplied as geometric data [33]. Lorentzian variants enrich this structure with an indefinite inner product, fundamental symmetry, temporal datum, time orientation, or twist [47, 48, 49, 50, 51, 23, 52, 53]. Recent work has also examined the emergence of temporal and Lorentzian structures within twisted almost-commutative settings [54, 55]. In particular, spectral-spacetime approaches relate Lorentzian signature to a Krein product together with time-orientation data [23]. The present construction operates one level upstream. Neither spectral geometry nor a Krein structure is taken as the initial object. The positive history algebra is reconstructed from operational renewal statistics, while the protected binary orientation becomes an indefinite/Krein structure only after it acts on the predictively reduced relational carrier. Its Clifford realization then distinguishes one timelike direction.

Predictive collapse does more than compress a finite history. Because future equivalence is preserved under admissible continuation, the minimum predictive state propagates canonically from one event to the next and therefore supplies a persistent relational carrier. The local dimension is then tested by two logically distinct finite selectors. The first uses no alternating structure: for a connected simple regular relation, exact equality of the faithful endpoint-source and cycle-source ranks has the unique nontrivial solution K_4 . The second works in the more restrictive pair-homogeneous simplex category and asks for a nondegenerate alternating temporal-spatial coefficient pairing together with minimum faithful source innovation; it independently selects the same K_4 . Thus the alternating form is not the sole route to three spatial dimensions. It remains a branch axiom of the second selector and supplies only an antisymmetric coefficient pairing, not a symplectic phase space, Hamiltonian flow, metric, or Lorentzian sign. Neither selector is a universality theorem over arbitrary higher-arity microscopic systems.

The metric appears one layer later. The relational carrier supports a finite family of elementary root transitions. Their occurrence rates and waiting times provide observable first- and second-order statistics. The second-order statistics determine the spatial kinetic form, the directed first moment determines the shift, and a physical speed density fixes the remaining scale. These quantities invert uniquely to the ADM spatial metric, lapse, and shift. The metric coefficients are therefore not microscopic labels but collective statistical quantities reconstructed from the renewal process.

The geometric clock is likewise not identified with the slower renewal clock. A nondegenerate continuum bracket requires a scheduler-separated root race with total rate of order h^{-2} . Its winner and waiting-time statistics supply the absolute kinetic scale, whereas root-direction probabilities alone leave the common rate scale undetermined. The protected binary orientation then acquires its geometric meaning. The positive Grand-Tensor and root-race forms remain positive and therefore cannot produce an indefinite direction. The orientation Read supplies one independent signed line through the explicit minimal signed-extension condition; leaving the reconstructed three-dimensional spatial form positive, the minimal nondegenerate signed extension has inertia $(-, +, +, +)$, up to overall sign convention. Only after this inertia is fixed do we choose a minimal Clifford representation. The gamma matrices therefore realize a signature already determined by the one-sign extension; they do not select the signature. Thus the Lorentzian sign is neither obtained by analytic continuation of a positive diffusion tensor nor inserted through a Dirac-matrix ansatz.

From kinematics to Einstein dynamics. Producing Lorentzian kinematics is not yet a derivation of gravitational dynamics. There are important precedents in which gravitational field equations arise from thermodynamic or entanglement variational statements: Jacobson’s equation-of-state argument [34], holographic entanglement first-law derivations of linearized Einstein equations [19], and entanglement-equilibrium arguments for the semiclassical Einstein equation [20]. Causal sets provide a different discrete precedent in which order-theoretic operators reconstruct scalar curvature and an approximately local action [5]. The route here is distinct from each of these: the finite action is reconstructed from marked renewal reward–duration statistics, its exactness is tested independently, and the actually accepted dynamics is then tested for stationarity before any gravitational representative is classified.

In particular, prescribing a renewal-rate history whose reconstructed metric is de Sitter would establish only realizability of one Einstein solution. The dynamical part therefore begins instead from the complete accepted physical cylinder. Future refinement reconstructs the minimum determining-field quotient and the accepted transition kernel acting on it. On the same resolved history, a protected vector reward together with physical duration determines an analytic renewal pressure. Its first derivative restores the affine action anchor lost by a centered probability score, while its higher jets reconstruct the mixed action Gram. Finite face-curl and period tests then determine whether these increments integrate to a single scalar common action and reconstruct that action when they do.

The actually accepted transition is next compared with the update generated by this independently reconstructed action. On the interior variational branch, the Fisher–Bregman gap controls the squared represented first variation, and explicit physical test-lift bounds retain the cutoff dependence needed in the continuum passage. Exact Gibbs/Kullback–Leibler identities provide a separate comparison of finite probability rows. The field-action comparison, rather than a cost fitted after observing the transition law, supplies the stationarity test.

The gravitational handoff is likewise formulated as a reconstruction and certification problem. The protected cycle reward entering this step is initially a generic measured face reward: no Palatini, Holst, Einstein–Hilbert, or cosmological Lagrangian is named in the entrance data. After resolved cut memory is removed, binary renewal subdivision and stable additivity produce a first-order face functional without assuming a derivative at a zero face. Lorentz covariance and the residual bivector commutant then leave only the identity and internal Hodge star, so only at this stage is the local first-order gravitational score classified as Palatini–Holst plus determinant volume.

Connection stationarity yields zero torsion on the spinless branch. Global spatial compactness is not introduced as a separate geometric axiom: the physical endpoint/capacity record determines an exact cut constant, and a uniform positive margin gives Sobolev, Poincaré, low-energy-counting, and finite-rank compact-screen bounds. Curvature control is supplied by a separate positive propagation estimate. A bounded initial curvature energy together with regulator-uniform observer/transport and differentiated-source budgets yields a spacetime L^2 curvature bound, also for a fully discrete curvature record. Strong coframe convergence supplies the nonlinear product passage; strong connection convergence and interface-complete consistency separately identify its curvature. These are the two analytic inputs to the weak–strong Palatini passage. On the positive certificate branch, the limiting metric variation is

$$\chi \int_K \sqrt{-g} (G_{\mu\nu} + \Lambda g_{\mu\nu}) k^{\mu\nu}, \quad (1.2)$$

and determining stationarity therefore gives the vacuum Einstein equation distributionally.

These finite reconstructions assemble into a certification-or-obstruction alternative. For a complete gravitational renewal cylinder, the determining-field quotient, accepted kernel, renewal

pressure, common-action exactness, action gap, spatial cut constant, face linearization, memory short, connection residual, curvature-propagation budgets, and Cartan compatibility are finite or cofinally testable. If the corresponding positive margins persist, the Einstein continuum follows. If they do not, the construction returns the failed object itself: a nonzero common-action curl or period, a positive Bregman/KL row, a macroscopic cut/potential witness, a face-linearity or memory residual, a torsion/connection defect, an unbounded curvature-propagation budget, a Cartan mismatch, or a cutoff compactness defect.

The homogeneous isotropic sector addresses a narrower question not settled by the abstract Einstein handoff. Instead of prescribing a de Sitter rate profile, we begin with arbitrary positive root-symmetric variables $\kappa(t)$ and $\varrho(t)$. The reconstructed ADM map converts them bijectively to the scale factor and lapse. Rewriting the classified vacuum action in (κ, ϱ) and varying both variables before fixing the lapse yields the Hamiltonian constraint and scale equation, equivalently the full spatially flat vacuum Friedmann system. The exponential de Sitter profile therefore appears only after the stationary equations have been solved.

The same distinction persists at finite cutoff. A midpoint variational action retains independent lapse-weighted intervals, and its stationary recurrence converges at second order to the continuum flat/de Sitter branch. A finite positive accept/reject law reconstructs this action directly from its proposal distribution, accepted row, and total acceptance probability. A measured rank defect determines whether the local low-noise update and the cosmological parameter of the reconstructed action agree. On the compatible branch, finite stochastic cylinders have vanishing Euler residual and converge quantitatively to the stationary de Sitter solution.

The finite evolution mechanism is not confined to homogeneous trajectories. In the unpolarized Gowdy T^3 vacuum class both wave polarizations and a spatially varying nonlinear frame are retained. A source/characteristic/source splitting preserves an exact staggered momentum invariant, while the finite offset support and shared Hermite readout give pathwise second-jet control and hence $O(h)$ convergence of the full Riemann tensor on fixed smooth slabs. The independently varied unreduced action has the same vanishing continuum first variations. This symmetry-reduced theorem does not imply generic attraction to de Sitter or unrestricted four-dimensional regularity, but it provides a nonlinear inhomogeneous class in which full curvature compactness is derived rather than assumed.

Scope and organization. The emergence claim is relative to a precise operational entrance. Actual occurrence, complete contextual prediction, protected orientation, and absolute physical calibration are irreducible rows of the declared finite language. For gravitational dynamics one additionally needs the complete accepted field/action cylinder to occur physically. Once these records are present, however, the predictive carrier, determining field, accepted kernel, action slope, common scalar action, ADM geometry, compact spatial screen, and stationarity residual are reconstructed or certified rather than supplied as independent continuum geometry. Vanishing of the action gap and persistence of the positive noncollapse and curvature-propagation margins remain genuine physical conditions that can fail.

The dimensional claim has the same controlled scope. The K_4/A_3 result does not state that every possible microscopic theory must have three spatial dimensions. Rather, two independent finite classifications select the same cell: endpoint-cycle source-rank balance does so without an alternating-form assumption, while the alternating/minimal-source simplex selector does so in a stronger category. Their agreement removes the alternating form from the role of sole dimensional selector, while still falling short of universality over arbitrary higher-arity systems. Likewise the protected binary orientation is not a microscopic Lorentzian metric. The positive operational rows cannot reconstruct it, as shown by explicit separating models; once the A_3 spatial form is positive,

this single independent sign fixes the minimal indefinite extension to Lorentzian inertia $(1, 3)$. The subsequent Clifford matrices are a representation of that quadratic form, not the origin of its signature.

The conceptual proposal is therefore that renewal may be more fundamental than its conventional role in probability theory suggests. From the usual point of view, renewal describes memory and resetting *within* time. From the present point of view, retention and forgetting in operational history determine which distinctions survive at all, while predictive equivalence reconstructs the persistent state on which relational geometry is built. Effective Markovian locality then appears not as the microscopic starting principle but as a limiting organization of predictive memory. In this sense, the usual hierarchy

$$\boxed{\text{spacetime} \longrightarrow \text{stochastic dynamics}} \quad (1.3)$$

is replaced by

$$\boxed{\text{renewal dynamics} \longrightarrow \text{predictive structure} \longrightarrow \text{relational geometry} \longrightarrow \text{Lorentzian spacetime.}} \quad (1.4)$$

The claim is not that renewal theory in its classical form already contains general relativity. It is that when renewal is treated operationally, predictively, and with a minimal protected orientation, persistence and relational structure can be reconstructed before ordinary spacetime geometry is defined.

Sections 2 and 3 reconstruct the Renewal Grand Tensor and the calibrated Lorentzian geometry. Section 4 reconstructs the common action, compares it with the accepted dynamics, and classifies its local gravity representative. Section 5 supplies the noncollapse and curvature estimates and proves the general renewal-to-Einstein certification theorem. Section 6 then constructs a nonsymmetric cutoff-uniform Einstein class, establishes its law-space robustness, and states the compatibility with a common chronological action process. Its final subsection summarizes the separate exact-action question; the response, boundary-realization and slow-compatibility analyses are proved in Appendices G and H. Section 7 treats homogeneous and Gowdy sectors, including their operational realizations and the conservative-blocking obstruction. Section 8 discusses interpretation and open problems. The remaining appendices follow the same order as the main argument; the appendix guide identifies the proof location of each principal result.

2 Predictive reconstruction and the Renewal Grand Tensor

2.1 Operational histories and the predictive quotient

At a finite cutoff X , the operational entrance is a finite typed event protocol with a declared finite horizon, finite intervention and outcome alphabets, and normalized conditional probabilities for every reachable history. The remaining horizon is included in the cut type, so admissible continuations are compared only at compatible cuts. An unbounded-horizon protocol is also allowed when a finite-index predictive congruence is separately certified; a finite alphabet alone does not imply this property. The physical intervention letters include tomographically complete frames on each exposed finite operator port; their multitime probabilities are affine under randomization, additive under coarse graining, causal under deterministic discard, and compatible on common prefixes. A distinguished binary orientation is retained as an actual protected operational distinction. No predictive state space, persistent relational carrier, metric, scheduler, connection, or spacetime dimension is part of this entrance.

For a reachable history h ending at cut type x , let $\mathbf{F}_{X,x}$ be the set of all admissible future words within the declared horizon starting at x (or all finite words on the certified finite-index branch). Write $p_X(w \mid h)$ for the corresponding conditional cylinder probability. Define

$$h \sim_X h' \iff p_X(w \mid h) = p_X(w \mid h') \text{ for every } w \in \mathbf{F}_{X,x}. \quad (2.1)$$

Histories of different cut type are never identified. If additional actual terminal Reads or bounded readout queries are present, they are included in the future word alphabet; thus (2.1) means equality of the complete declared future signature.

Future equivalence is a typed right congruence. Indeed, if $h \sim_X h'$ and an event $a : x \rightarrow y$ is admissible, then the one-letter future gives $p_X(a \mid h) = p_X(a \mid h')$. On a positive-probability branch, the chain rule then gives

$$p_X(v \mid ha) = \frac{p_X(av \mid h)}{p_X(a \mid h)} = \frac{p_X(av \mid h')}{p_X(a \mid h')} = p_X(v \mid h'a)$$

for every continuation v . Hence $ha \sim_X h'a$.

The minimum persistent predictive carrier. Define

$$Z_{X,x}^{\min} := \mathbf{H}_{X,x} / \sim_X, \quad q_{X,x} : \mathbf{H}_{X,x} \rightarrow Z_{X,x}^{\min}. \quad (2.2)$$

For an admissible event $a : x \rightarrow y$, the right-congruence property defines

$$T_{X,a}([h]) = [ha], \quad \kappa_{X,a}([h]) = p_X(a \mid h). \quad (2.3)$$

Thus the event law itself reconstructs a deterministic predictive state machine. If $A_1, A_2, \dots$ is the realized event sequence, $H_n = A_1 \cdots A_n$, and $Z_n = q_X(H_n)$, then

$$Z_{n+1} = T_{X,A_{n+1}}(Z_n), \quad \mathbf{p}_{X,n}^{\min} = (Z_n, A_{n+1}, Z_{n+1}). \quad (2.4)$$

The persistent relation is therefore the transport of a future-equivalence class under event composition, not an independently supplied state variable.

The quotient is universal. Every reachable deterministic predictive realization $S_{X,x}$ of the same conditional event law admits a unique surjection

$$\pi_{X,x} : S_{X,x} \rightarrow Z_{X,x}^{\min} \quad (2.5)$$

intertwining branch probabilities and updates. Consequently every declared downstream readout depending only on the complete future signature factors uniquely through $Z_X^{\min}$. Generated ledgers, hidden-state labels, and other deterministic predictive presentations are therefore representations of the canonical quotient rather than additional foundational coordinates.

The physical geometric branch considered below requires more than this predictive reduction. In particular, same-cut readability of the predictive class, the protected orientation, and the absolute winner-waiting calibration of the fast root race are actual physical rows. They are not determined by the positive predictive quotient. The purpose of (2.2)–(2.4) is to remove independently postulated persistence and every future-invisible state refinement.

The quotient determines the state distinctions that prediction retains. The geometric construction also needs the corresponding state and its calibrations to be readable on the same physical history. The next completion separates this occurrence requirement from state minimization.

2.2 Readable relational completion

On a spacetime-capable branch, let $O_{X,n}$ denote the occurring root-bearing event at the n th slow cut, let $\Gamma_{X,n}$ denote the scheduler-separated fast root path between slow cuts, and let $\Xi_{X,n}$ collect the remaining same-history physical marks used by the geometric reconstruction, including the protected orientation and absolute calibrations. A raw readable relation packet is

$$\mathfrak{p}_{X,n}^{\text{rel}} = (Z_{X,n}^{\min}, \Gamma_{X,n}, O_{X,n}, \Xi_{X,n}, Z_{X,n+1}^{\min}). \quad (2.6)$$

The state coordinates in (2.6) are not new hidden variables: they are same-cut physical realizations of the predictive classes already reconstructed in (2.2). Two raw packets are identified when every admitted continuation, readout query, and terminal Read has the same conditional law. Denote the resulting finite future-minimal quotient by $\mathfrak{P}_X^{\text{rel},\min}$.

Proposition 2.1 (Conservative readable completion). *Suppose the additional readable marks in (2.6) are attached to the underlying finite operational process by normalized same-history conditional kernels. Then:*

- (i) *forgetting Γ , Ξ , and the explicit same-cut realization of the predictive class recovers every cylinder probability and branch map of the underlying operational process exactly;*
- (ii) *future-signature refinement terminates at $\mathfrak{P}_X^{\text{rel},\min}$, and every other finite readable packet sufficient for the same geometric reconstruction has a unique surjection onto it;*
- (iii) *unread refinements add neither predictive content nor source dimension; metric, coframe, connection, curvature, and Einstein data remain downstream readout outputs and are not coordinates of the relation packet.*

Thus physical readability is a conservative enrichment of a derived predictive carrier, not a second postulated state space.

The proof is given in Appendix A.3. This proposition is the interface between predictive reconstruction and the geometric branch: the first determines which state distinctions exist, while the second records which of those distinctions and physical calibrations actually occur together on one history.

Finite exact reconstruction, its horizon convention and the presentation category are treated in Appendix A. The equality tests concern the supplied probability table, not arbitrary finite samples of an unknown real-valued law.

2.3 The Renewal Grand Tensor

The predictive quotient fixes persistence; the positive multitime table fixes how the retained operations compose and overlap. These structures are assembled into one object before any metric is introduced.

The operational multitime table on the physical intervention frames reconstructs a positive causal process tensor in the quantum-comb sense [65, 66]. Its prefix-support-minimal square-root purification supplies canonical sequential memories, unique up to record-intertwining memory unitaries in the purifying category defined in Theorem A.14. The represented minimal-state projections, mixed forward maps, and their Hilbert–Schmidt reverses generate a finite history algebra $\mathcal{A}_X^{\text{hist}}$. If $[w]$ denotes the represented history word and $\hat{\tau}_{\text{reg}}$ is the normalized regular trace, define

$$\mathbb{K}_{X,r}(v, w) = \hat{\tau}_{\text{reg}}([v]^*[w]), \quad |v|, |w| \leq r. \quad (2.7)$$

The positive structural Renewal Grand Tensor is

$$\mathbb{T}_{\text{Grand},X} = \left(\mathcal{A}_X^{\text{hist}}, \{\mathbb{K}_{X,r}\}_{r \geq 0}, Z_X^{\text{min}} \right). \quad (2.8)$$

The term *Grand Tensor* denotes this algebra-valued positive moment object, not a tensor field on a manifold. Likewise, the term *history algebra* here refers to the represented C^* -algebra generated by operational history maps; it should not be confused with the history-group/decoherent- histories constructions of generalized quantum theory [27]. Each word-Gram form is a covariant sesquilinear tensor on the finite word space. Under independent composition with independently accessible operations, the predictive quotient, history algebra and word moments factor as in Proposition A.18; thus the tensor terminology also has a precise product interpretation on that category. Finite dimensionality forces the represented word hierarchy to become flat after a finite depth, at which point one further word layer reconstructs the complete represented algebra.

For the spacetime branch, (2.8) is used together with the minimal faithful readable packet $\mathfrak{P}_X^{\text{rel,min}}$ of Proposition 2.1. The distinction is deliberate: $\mathbb{T}_{\text{Grand},X}$ records the canonical positive predictive-history structure, whereas the readable packet records the actual same-history occurrence of orientation, calibration, scheduler, and root-path data required to interpret that structure geometrically. Forgetting the latter returns the positive structural tensor exactly.

Theorem 2.2 (Canonical predictive Grand-Tensor reconstruction). *Let a finite typed operational renewal process have a finite event grammar, normalized prefix-compatible conditional probabilities, tomographically complete physical intervention frames, finite operational positivity, causal trace recursion, and a protected binary orientation as described above. On an unbounded-horizon branch, assume also a common finite-dimensional sequential realization of the complete represented history operations. Then:*

- (i) *future equivalence (2.1) is a typed right congruence, and Z_X^{min} is the unique coarsest reachable deterministic predictive realization of the event law;*
- (ii) *the successor maps (2.3) are canonical and the persistent event packet (2.4) is derived from composition rather than independently postulated;*
- (iii) *the state-conditioned Hankel quotient is the unique reachable, future-separated minimum linear predictor and determines the executable forward algebra;*
- (iv) *the physical probability table reconstructs unique positive causal prefix combs whose square-root supports determine a canonical prefix-support-minimal sequential isometric class, unique up to record-intertwining memory unitaries; this is a purifying realization, not a minimization over arbitrary mixed-state memory implementations;*
- (v) *the contextual history quotient, normalized regular trace, and positive word-Gram hierarchy are canonical, and finite flatness reconstructs the complete represented monoidal $*$ -algebra;*
- (vi) *every deterministic predictive presentation and every future-operational compiler factors uniquely through Z_X^{min} ; consequently (2.8) is canonical without an independently supplied ledger, hidden state, persistent state space, preferred Kraus frame, primitive metric, or preselected history algebra;*
- (vii) *any finite same-history readable enrichment by normalized conditional kernels admits the minimal faithful quotient $\mathfrak{P}_X^{\text{rel,min}}$ of (2.6); its forgetful image is the underlying operational*

law, so same-cut readability does not alter the predictive reconstruction or introduce new geometric coordinates.

The detailed proof is given in Appendix A. Three minimizations remain logically distinct: predictive state minimization, linear Hankel minimization, and contextual C^* -history separation. Conflating them either retains future-invisible spectator structure or erases distinctions visible to an actual operational continuation.

The independence of orientation and absolute calibration, and the exact remaining calibration freedom, are established in Theorems A.19 and A.20. The next section uses these additional physical records to reconstruct geometry on the canonical carrier; it does not alter the positive history object just constructed.

3 Relational dimension and Lorentzian geometry

3.1 Endpoint–cycle selection and a secondary simplex criterion

The primary finite dimensional statement is endpoint–cycle source balance: inside connected simple regular relations it uniquely selects K_4 , without an alternating form. Its simplex representation-theoretic counterpart is the orientation-twisted equivalence of the complete cycle and endpoint sources. A secondary criterion in the pair-homogeneous simplex category combines alternating parity with an explicit minimum-source rule. That criterion selects the same cell, but its minimality requirement is a selection hypothesis, not a consequence of predictive compression.

Dimension has a different status in several neighboring emergent-geometry programmes. In causal-set theory, Myrheim–Meyer-type statistics estimate the dimension of an order assumed to be manifoldlike [3, 4, 29]; in causal dynamical triangulations the macroscopic and spectral dimensions are measured properties of the generated ensemble [7, 8]; and in entanglement-based reconstruction a best-fit spatial dimensionality can be inferred from quantum-correlation data [21]. The claims below are of a different type: they are finite classification results inside explicitly declared relational categories, not estimators of an already manifoldlike sample and not measurements of a scale-dependent effective dimension.

Proposition 3.1 (Endpoint–cycle balance selects K_4 without an alternating form). *Let G be a connected simple k -regular relation graph on $v > 1$ endpoints. If its faithful endpoint source and cycle source have equal rank, then*

$$v - 1 = \frac{kv}{2} - v + 1, \quad \text{so} \quad (4 - k)v = 4. \quad (3.1)$$

The unique nontrivial connected simple regular solution is

$$v = 4, \quad k = 3, \quad G \cong K_4. \quad (3.2)$$

Thus the associated faithful endpoint module has dimension $v - 1 = 3$. This selector contains no temporal–spatial alternating-form hypothesis.

Proof. For a connected graph the endpoint-source rank is $v - 1$, while the cycle rank is $|E| - v + 1 = kv/2 - v + 1$. Equality gives $(4 - k)v = 4$. For integers $v > 1$ and $k \geq 1$ with $k \leq v - 1$, the only possibility is $v = 4$, $k = 3$; a simple graph with these parameters is K_4 . $\square$

For the simplex selector, consider a local relation event on the minimum persistent predictive carrier of Theorem 2.2, with one future-readable source and one future-readable target. Physical

reversal turns it into a signed edge. Future minimization removes parallel labels that no future can distinguish, while pair-homogeneous recurrence completes the orbit of one nonzero relation. The relation-colour-free skeleton is therefore K_N . Writing $\mathbb{R}^N = \mathbb{R}u_0 \oplus W_N$, with $u_0 = N^{-1/2}\mathbf{1}_N$, the signed edge space decomposes as

$$C_1(K_N) = u_0 \wedge W_N \oplus \Lambda^2 W_N. \quad (3.3)$$

The first summand is the endpoint or cut source and the second the complete cycle or interference source, of dimensions $N - 1$ and $\binom{N-1}{2}$ respectively.

Proposition 3.2 (Alternating/minimal-source simplex selector). *Assume in addition that the temporal-spatial coefficient space $\mathbb{R}t \oplus W_N$ carries a nondegenerate alternating form, that the endpoint and complete signed-edge sources are faithful, that the cycle source is nonzero, and that the future-readable source innovation is minimal among conservative completions satisfying these queries. Then*

$$N = 4, \quad G_{\text{rel}} = K_4, \quad W_{\text{sp}} = W_4 \cong \mathbb{R}^3. \quad (3.4)$$

Moreover, after complexification,

$$\text{Hom}_{S_N}(\Lambda^2 W_N, W_N \otimes \text{sgn}) = \begin{cases} \mathbb{C}, & N = 4, \\ 0, & N \neq 4, \end{cases} \quad (3.5)$$

so only at $N = 4$ is the complete cycle representation one oriented copy of the spatial standard module rather than an additional inequivalent source type.

The proof is given in Appendix B. The first conclusion uses parity, nonzero interference and the stipulated source minimum. The representation-theoretic statement is a separate algebraic characterization of $N = 4$ inside the simplex family, not evidence of generic microscopic selection.

Remark 3.3 (Exact scope of the dimensional claim). The endpoint-cycle balance theorem is the primary finite classification; the alternating/minimal-source theorem is a secondary classification with different hypotheses. The first shows that the alternating form is not the sole origin of the three-dimensional spatial module. The second does not become a hidden symplectic or Hamiltonian assumption: its alternating form supplies only an antisymmetric coefficient pairing and no metric, lapse, shift, connection, curvature, or Lorentzian sign. Neither theorem is a classification of arbitrary graphs, hypergraphs, or higher-arity microscopic relation systems.

Source equivalence. Within the faithful simplex category the complete cycle source is one orientation-twisted copy of the endpoint source exactly at $N = 4$; Proposition B.8 proves uniqueness up to scale and orientation. This interprets the rank balance within that category, without asserting that an arbitrary operational law satisfies either selector.

3.2 The A_3 root race

Choose an anchored basis of W_4 in which the six unoriented A_3 roots are

$$r_1 = e_1, \quad r_2 = e_2, \quad r_3 = e_3, \quad r_4 = e_2 - e_1, \quad r_5 = e_3 - e_1, \quad r_6 = e_3 - e_2. \quad (3.6)$$

Let $k_a^\pm$ be the two directed rates associated with $\pm r_a$, and let h be the mesh scale. The predictable second and first moments are

$$\mathcal{B}_h = h^2 \sum_{a=1}^6 (k_a^+ + k_a^-) r_a r_a^\top, \quad \beta_h = h \sum_{a=1}^6 (k_a^+ - k_a^-) r_a. \quad (3.7)$$

The six dyadics $r_a r_a^\top$ form a basis of Sym_3 , so the rate sums reconstruct all six spatial inverse-metric coefficients. The rate differences map with rank three onto β_h ; their remaining three coordinates are circulating root currents. Winner probabilities alone lose the common rate scale. Observing both the winning root and its waiting time gives a faithful score frame for all directed rates.

Let $\varrho_h > 0$ be the physical speed density. The spatial metric and lapse are reconstructed by

$$\boxed{g_h = \varrho_h^{2/3} (\det \mathcal{B}_h)^{1/3} \mathcal{B}_h^{-1}, \quad N_h = \varrho_h^{1/3} (\det \mathcal{B}_h)^{1/6}.} \quad (3.8)$$

Equivalently,

$$\sqrt{\det g_h} = \varrho_h, \quad \mathcal{B}_h = N_h^2 g_h^{-1}. \quad (3.9)$$

Together with the shift β_h , these are exactly the ten ADM coefficients. The formula is an analytic bijection on the positive cone. A nondegenerate bracket requires total root rate of order h^{-2} ; identifying the geometric scheduler with a slower $O(h^{-1})$ renewal clock would make $\mathcal{B}_h \rightarrow 0$ and collapse the metric.

Linear tomography versus positive realizability. Spanning Sym_3 is a linear statement. Positive rates in a fixed root frame realize only the cone

$$\mathcal{C}_{A_3} := \left\{ \sum_{a=1}^6 s_a r_a r_a^\top : s_a > 0 \right\}, \quad s_a = h^2 (k_a^+ + k_a^-), \quad (3.10)$$

not every positive-definite matrix in that same coordinate frame. The component inequalities and the positive directed lift of a bounded shift are recorded in Appendix B. Thus the ADM map is an analytic bijection on positive bracket–density pairs, whereas realization by the declared race also requires an interior cone margin.

3.3 One signed direction and Lorentzian inertia

The Grand Tensor reconstructed in the previous section is positive: its canonical word-Gram hierarchy is a Hilbert-space Gram structure, and the root-race bracket is a positive quadratic form on the three-dimensional spatial module. Neither can contain a negative direction. The additional physical datum is a protected two-valued operational Read

$$\epsilon_X \in \{+1, -1\}, \quad (3.11)$$

with both values separated by the declared Read policy. In the support-minimal Hilbert realization it has orthogonal readout projections P_+ and P_- , $P_+ + P_- = I$, and therefore

$$J := P_+ - P_-, \quad J = J^* = J^{-1}, \quad [u, v]_J := \langle u, Jv \rangle. \quad (3.12)$$

This is the operator representation of an already occurring $\mathbb{Z}_2$ mark. The microscopic record itself carries no norm, cone, lapse, temporal coordinate, or indefinite metric.

The sign is not reconstructible from the positive packet. The separating models of Theorem A.19 keep the complete positive operational data fixed while reversing the protected orientation, and the obstruction survives common conservative refinement. The signed row is therefore irreducible relative to the declared operational language rather than a reparametrization of the positive Grand Tensor.

Definition 3.4 (Minimal signed-extension condition). A minimal signed extension of the positive spatial space (W, g) consists of one independent real line L , a negative-definite quadratic form q_L on L , and the orthogonal extension $q = q_L \oplus g$. Calibration fixes the absolute coefficient of q_L . A protected orientation selects one of the two directions on L ; it does not, by itself, determine the sign of q_L .

A binary Read has a representation $J = P_+ - P_-$ on its Hilbert carrier, but the dimensions of the eigenspaces of J need not be one and three and are not the dimension of the reconstructed tangent space. The passage from this Read to a single negatively signed geometric line is precisely [Definition 3.4](#). Neither positivity of a word Gram nor a choice of orientation alone implies it. Once this condition is imposed, Sylvester inertia determines the Lorentzian type without choosing gamma matrices. In inverse-metric conventions the calibrated line is $-N^{-2}\tau^2$; the corresponding cotangent extension with shift is [\(B.17\)](#).

Proposition 3.5 (Minimal one-sign extension has Lorentzian inertia). *Let (W, g) be the positive three-dimensional spatial quadratic space reconstructed from the K_4/A_3 root race. Adjoin one independent real line $\mathbb{R}t$ with the negative quadratic form specified by the minimal signed-extension condition, and leave $g|_W$ unchanged. Then every minimal nondegenerate orthogonal signed extension is, up to multiplication of the whole quadratic form by -1 ,*

$$\boxed{q(\tau t + w) = -N^{-2}\tau^2 + g(w, w), \quad \text{inertia}(q) = (1, 3).} \quad (3.13)$$

No choice of Clifford representation enters this conclusion.

The proof and the corresponding independence statement are given in [Appendix B](#). The point is that the inertia conclusion precedes the Clifford realization.

The signed-extension condition, rather than positivity or orientation alone, is the extra geometric hypothesis. Its role is now explicit, and the Clifford layer only represents the resulting quadratic form.

Clifford realization. Once the signed quadratic form is fixed, the explicit matrices and principal-symbol calculation in [Appendix B](#) give a faithful minimal Clifford realization. No representation-theoretic choice enters the inertia conclusion. We can now collect the reconstruction steps in a single theorem.

Theorem 3.6 (Minimal faithful $3 + 1$ ADM reconstruction from either finite selector). *Let the minimum persistent predictive carrier of [Theorem 2.2](#) admit a geometric branch with one source and one target, physical reversal, faithful endpoint data, and a local relation selected as K_4 by either [Proposition 3.1](#) or the alternating/minimal-source simplex criterion of [Proposition 3.2](#). Assume an occurring scheduler-separated root-race calibration supplies directed winners, their waiting times, and a positive physical speed density. Assume also that the protected orientation packet supplies the Read [\(3.11\)](#), independent of the positive packet, with one nontrivial signed direction in the sense of [Definition 3.4](#). Then:*

- (i) *the selected local relation cell is K_4 and its spatial module is the A_3 root space $W_4 \cong \mathbb{R}^3$;*
- (ii) *the winner–waiting statistics of the scheduler-separated root race form a faithful coordinate system for all directed rates;*
- (iii) *the moments [\(3.7\)](#) and the speed density invert uniquely to the ADM spatial metric, lapse, and shift through [\(3.8\)](#);*

- (iv) *a nondegenerate continuum bracket requires an $O(h^{-2})$ fast scheduler;*
- (v) *positivity of the reconstructed spatial form together with exactly one independent signed direction fixes Lorentzian inertia $(1, 3)$ up to overall sign, before any Clifford representation is chosen;*
- (vi) *the Clifford cell (B.15) faithfully realizes that already determined quadratic form, and the principal symbol has square (B.17);*
- (vii) *under summably correctable mixed-Gram and refinement defects, the dimension, source ranks, pair gap, ADM packet, and signed inertia persist on the canonical cofinal limit quotient.*

No persistent predictive state, metric, coframe, connection, curvature tensor, or spacetime dimension is independently supplied. Persistence is reconstructed by Theorem 2.2; the protected orientation and absolute winner–waiting calibration remain irreducible physical input rows on this branch. The alternating form is required only on the second dimensional selector and is not a prerequisite of the first.

The residual status of the last two rows is not merely a limitation of the construction: Theorem A.19 gives explicit separating models showing that neither the protected orientation nor the absolute calibration is functorially reconstructed from the remaining positive operational data, even after common cofinal refinement.

The detailed proof is given in Appendix B. The local algebraic audit is finite: on the free A_3 accumulator, all equal-endpoint word relations are generated by inverse digons, root triangles, and commutator squares of depth at most four. Global topology enters separately through a finite period bank.

4 Reconstruction of the gravitational action

The preceding construction is kinematical. We now reconstruct the variational object to which a continuum limit can be applied. The order of the argument is fixed: the complete accepted record determines the field dynamics; marked renewal statistics determine action increments; finite exactness integrates those increments; and an independent comparison tests stationarity. Only then is the local gravitational representative classified.

Three kinds of hypotheses.

Input or condition	Mathematical role
Operational records	Complete conditional futures, readable root/waiting and density data, a signed-line extension, and the same-history accepted, reward and duration records determine the finite objects being compared.
Finite structural tests	The relational selector, action curl and periods, local degree and covariance class, and nonzero calibrated Palatini coefficient specify the branch. They are not automatic consequences of positivity.
Cofinal analytic tests	Scaled physical stationarity, coframe compactness, connection compactness, curvature propagation and full Cartan consistency permit the nonlinear variational limit. Each is imposed on one common realized regulator sequence.

This distinction will also govern the limit theorem in Section 5. Its analytic hypotheses are not assumed in order to reconstruct the finite objects below.

4.1 The determining field and accepted kernel

At cutoff X , let $\tilde{\Theta}_X$ denote the finite same-history record carrying the reconstructed geometric variables used by the gravitational sector, generic protected reward and duration coordinates, and the resolved accepted-branch labels. At this stage the reward coordinates are not named as Palatini, Holst, Einstein–Hilbert, or cosmological terms; they are simply actual finite Reads (or fixed functions of actual protected record coordinates) whose variation content will be reconstructed and classified below. Two values $x, y \in \tilde{\Theta}_X$ are equivalent when every admitted future word and every declared determining coordinate has the same value from x and y , including branch-domain indicators. Put

$$\Theta_{\text{fld},X}^{\min} := \tilde{\Theta}_X / \sim_{\text{fld},X} . \quad (4.1)$$

For resolved accepted subbranches a , let $p_{a,X}(\theta)$ be their physical probabilities and $u_a^{\min}$ their descended updates. On the positive acceptance domain define

$$K_X^{\text{acc}}(\theta' \mid \theta) = \frac{1}{p_{\text{acc},X}(\theta)} \sum_{a: u_a^{\min}(\theta) = \theta'} p_{a,X}(\theta). \quad (4.2)$$

Theorem 4.1 (Determining-field and accepted-kernel reconstruction). *The quotient (4.1) is the unique coarsest finite record through which every declared determining-field future and every primitive update factors. Finite partition refinement terminates, and (4.2) is the unique table-native accepted transition reproducing all selected post-accepted field futures.*

Thus the accepted field dynamics is not selected after the fact once the complete accepted cylinder is present. What cannot be reconstructed from the marginal accepted/not-accepted bit is the complete field cylinder itself; this is a physical occurrence requirement rather than an implicit state-space assumption. The proof and the corresponding non-identifiability boundary are given in Appendix C.

4.2 Renewal pressure and common-action exactness

Let (Ω_X, p_X) be a finite faithful marked first-return law, let $\tau_X : \Omega_X \rightarrow (0, \infty)$ be the protected physical duration, and let E_X be the finite variation space. A protected cycle-reward synthesis is a linear map

$$\mathcal{G}_X : E_X \longrightarrow L^2(\Omega_X, p_X), \quad v \longmapsto g_{v,X}, \quad (4.3)$$

whose values are actual Reads or fixed functions of protected record coordinates. Define

$$\mathcal{L}_X(q, r) = \mathbb{E}_X[e^{-g_{q,X} - r\tau_X}]. \quad (4.4)$$

Since $\partial_r \mathcal{L}_X(0, 0) = -\mathbb{E}_X \tau_X < 0$, there is a unique analytic pressure $\mathcal{P}_X$ near the origin with

$$\mathcal{P}_X(0) = 0, \quad \mathcal{L}_X(q, \mathcal{P}_X(q)) = 1. \quad (4.5)$$

The normalized tilted law is $p_{X,q}(\omega) = p_X(\omega) \exp[-g_{q,X}(\omega) - \mathcal{P}_X(q)\tau_X(\omega)]$. Let $S_X v = D \log p_{X,q}|_{q=0}[v]$, let $\pi_X = D\mathcal{P}_X(0)$, and let $T_X c = c\tau_X$.

Theorem 4.2 (Renewal-pressure action reconstruction). *The marked renewal table reconstructs $\mathcal{P}_X$ and all of its finite jets. If $\bar{\tau}_X = \mathbb{E}_X \tau_X$, then*

$$\pi_X(v) = -\frac{\mathbb{E}_X g_{v,X}}{\bar{\tau}_X}, \quad \mathcal{G}_X = -S_X - T_X \pi_X. \quad (4.6)$$

Hence the centered probability score together with physical duration recovers the complete uncentered action slope, including its affine anchor. The pressure Hessian and duration cross-jets reconstruct the full marginal and mixed action Gram whenever all reward sectors occur on the same marked cycle record.

The remaining question is whether the reconstructed increments belong to one scalar action. Let $\mathcal{V}_X$ be the finite protected variation complex and let $\alpha_X \in C^1(\mathcal{V}_X; \mathbb{R})$ be the reconstructed action increment.

Theorem 4.3 (Finite common-action exactness). *There is a scalar common action $\mathcal{S}_X$ satisfying*

$$\boxed{\alpha_X = d_0 \mathcal{S}_X} \quad (4.7)$$

if and only if $d_1 \alpha_X = 0$ on every declared 2-cell and the periods of α_X vanish on one integral basis of the free part of $H_1(\mathcal{V}_X; \mathbb{Z})$. After one additive anchor is fixed on each connected component, $\mathcal{S}_X$ is unique and is reconstructed by path integration. Moreover, if P_{har} is the harmonic projection and $\sigma_{\text{coex}} > 0$ is the least singular value of d_1 on the coexact range, then

$$\text{dist}(\alpha_X, \text{Ran } d_0)^2 \leq \|P_{\text{har}} \alpha_X\|^2 + \sigma_{\text{coex}}^{-2} \|d_1 \alpha_X\|^2. \quad (4.8)$$

Thus failure of exactness supplies a finite obstruction to a common action.

Remark 4.4 (Non-circularity of the gravitational reward). The synthesis $\mathcal{G}_X$ is fixed by the protected cycle record before it is compared with the accepted kernel K_X^{acc} . Neither the transition probabilities nor a desired continuum equation are used to fit a gravitational cost after the fact. Moreover, no Palatini or Holst decomposition is assumed at this stage. That decomposition appears only after derivative-free face linearization, complete resolved-memory subtraction, and the Lorentz-equivariant commutant classification below. The subsequent Bregman/KL test is therefore an independent question: it asks whether the actually accepted dynamics is stationary for the previously reconstructed action.

Exactness identifies which scalar action the recorded increments represent. It does not yet relate that action to the actually accepted evolution. The next estimate makes this comparison at the level of field variations, with all cutoff-dependent test-lift factors retained.

4.3 Stationarity of the reconstructed action

On a compact convex chart $\Theta \subset \mathbb{R}^d$ of the reconstructed determining field, let Ψ and the reconstructed common action $\mathcal{A}$ be C^2 on an open neighbourhood of Θ . The operational potential is strictly convex, with Fisher metric

$$G(\theta) = \nabla^2 \Psi(\theta), \quad mI \preceq G(\theta) \preceq MI, \quad (4.9)$$

and, for $\lambda, L \geq 0$ and $\eta > 0$,

$$-\lambda G(\theta) \preceq D^2 \mathcal{A}(\theta) \preceq LG(\theta), \quad \eta\lambda < 1. \quad (4.10)$$

For the Bregman divergence [56]

$$D_\Psi(y, \theta) = \Psi(y) - \Psi(\theta) - \langle \nabla \Psi(\theta), y - \theta \rangle, \quad (4.11)$$

put

$$\Phi_\theta(y) = \mathcal{A}(y) + \eta^{-1} D_\Psi(y, \theta), \quad y_\theta^* = \arg \min_{y \in \Theta} \Phi_\theta(y). \quad (4.12)$$

The minimum is unique. On the branch used for unrestricted first variations, y_θ^* lies in the interior of Θ for every supported source θ . This interiority is important: a constrained minimum at a chart boundary need not have zero ordinary action gradient.

For an actually accepted transition $\theta \rightarrow y$, define

$$\Delta_\Psi(\theta, y) = \Phi_\theta(y) - \Phi_\theta(y_\theta^*) \geq 0. \quad (4.13)$$

If $K = (1 - \vartheta)I + \vartheta K^{\text{acc}}$, with state-independent $0 < \vartheta \leq 1$, has stationary law ν , set

$$\begin{aligned} \bar{\Delta}_\Psi &= \int \nu(d\theta) K^{\text{acc}}(dy \mid \theta) \Delta_\Psi(\theta, y), \\ \mathfrak{S}(\theta) &= \nabla \mathcal{A}(\theta)^* G(\theta)^{-1} \nabla \mathcal{A}(\theta). \end{aligned} \quad (4.14)$$

Theorem 4.5 (Accepted action gap controls interior stationarity). *Under the preceding hypotheses, including interiority of the supported proximal minimizers, an explicit finite constant $C_{\Psi, \mathcal{A}, \eta}$ satisfies*

$$\int \mathfrak{S}(\theta) \nu(d\theta) \leq C_{\Psi, \mathcal{A}, \eta} \bar{\Delta}_\Psi. \quad (4.15)$$

In particular, zero gap forces the represented first variation of the common action to vanish ν -almost surely.

The proof in Appendix C keeps the convexity and chart constants explicit. To pass to physical metric tests at varying cutoffs, let $v_{X, K}(k; \theta)$ be the actual represented lift of a fixed determining test k on a compact cylinder K , and put

$$B_{X, K}(k)^2 := \sup_{\theta \in \text{supp } \nu_X} \|v_{X, K}(k; \theta)\|_{G_X(\theta)}^2. \quad (4.16)$$

Duality then gives the quantitative physical-test estimate

$$\mathbb{E}_{\nu_X} |\delta \mathcal{A}_X[v_{X, K}(k)]|^2 \leq B_{X, K}(k)^2 C_X \bar{\Delta}_{\Psi, X}, \quad C_X = C_{\Psi_X, \mathcal{A}_X, \eta_X}. \quad (4.17)$$

Thus the right-hand side, rather than the unweighted gap alone, is the quantity required to vanish. On a countable determining core this gives convergence of the first variations in mean square and an almost-sure vanishing subsequence under any common coupling; the deterministic continuum handoff is applied to the resulting realized sequences. The construction and proof are given in Proposition C.7.

For a finite probability row with reference law $p_\theta(y) > 0$ and same-history action cost $c_\theta(y)$, the entropy-proximal row is

$$q_\theta^*(y) = \frac{p_\theta(y) e^{-\eta c_\theta(y)}}{\sum_z p_\theta(z) e^{-\eta c_\theta(z)}}, \quad (4.18)$$

and its row variational gap is exactly $\eta^{-1} D_{\text{KL}}(K^{\text{acc}}(\cdot \mid \theta) \| q_\theta^*)$. This is an independent likelihood test against the reconstructed cost. Equality of two finite-temperature probability rows does not by itself make every sampled field configuration a critical point of $\mathcal{A}$; the field-action estimate and its physical test lifts remain the stationarity criterion used in the Einstein passage.

The preceding tests reconstruct an action and measure its stationarity, without choosing a gravitational Lagrangian. We now restrict the local incidence and degree class and use Lorentz covariance to identify its possible bulk terms. This algebraic step is separate from the analytic limit in the next section.

4.4 Classification of the gravitational representative

Let e^I be the reconstructed coframe, ω^{IJ} a metric-compatible reconstructed Cartan connection, $B^{IJ} = e^I \wedge e^J$, and $T^I = De^I$. Before any continuum gravitational Lagrangian is named, the generic protected face reward is subjected to two finite operations described in Appendix C: binary renewal subdivision, which produces a unique first-order linear functional on its exact or summably stable branch, and subtraction of the complete resolved cut-memory log-partition. Denote the resulting generic first-order gravitational score by $A_{X,\text{sh}}^{\text{grav}}$. For later reference, write the three Lorentz-natural first-order densities (up to the overall normalization convention) as follows. The Palatini–Holst family and its boundary/topological extensions are standard first-order gravity structures [22, 68]; the point of the theorem below is the restricted invariant classification of the reconstructed score, not the introduction of these densities themselves:

$$L_{\text{Palatini}} = \frac{1}{2} \epsilon_{IJKL} e^I \wedge e^J \wedge R^{KL}, \quad (4.19)$$

$$L_{\text{Holst}} = e^I \wedge e^J \wedge R_{IJ}, \quad (4.20)$$

$$L_{\text{vol}} = \frac{1}{4!} \epsilon_{IJKL} e^I \wedge e^J \wedge e^K \wedge e^L. \quad (4.21)$$

Definition 4.6 (Local curvature-linear gravitational class). The gravitational class considered here consists of local polynomial four-form densities in an oriented coframe e and a metric-compatible Lorentz connection ω . The curvature-dependent part has bidegree $(2, 1)$ in (e, R) , and the curvature-independent part has degree four in e . Coefficients are spacetime-constant real scalars, and the only internal contractions are the Lorentz metric, its oriented volume tensor, and the represented scalar multiplicity commutant. No inverse coframe, independent torsion polynomial, derivative of curvature, additional background tensor, or matter field is admitted in this class. Densities are compared modulo exact terms whose variations vanish on the declared compactly supported bulk tests; boundary functionals are retained separately when boundary tests are allowed.

This definition specifies the incidence and degree of the reconstructed score. Here *curvature linear* is more restrictive than merely first order in derivatives of an independent connection. In particular, it is not a classification of every generally covariant first-order gravity action.

Theorem 4.7 (Lorentz-natural classification of the generic gravitational score). *Assume $A_{X,\text{sh}}^{\text{grav}}$ belongs to the polynomial local class of Definition 4.6, is real and Lorentz natural, has scalar residual typed multiplicity commutant on the bivector carrier, and represents determinant coframe volume. Then, up to coefficient normalization, its only first-order Lorentz-natural bulk terms are*

$$A_{X,\text{sh}}^{\text{grav}} \in \text{span}_{\mathbb{R}} \{L_{\text{Holst}}, L_{\text{Palatini}}, L_{\text{vol}}\}. \quad (4.22)$$

Equivalently, there are real coefficients $\alpha_X, \beta_X, \lambda_X$ such that

$$A_{X,\text{sh}}^{\text{grav}} = \alpha_X L_{\text{Holst}} + \beta_X L_{\text{Palatini}} + \lambda_X L_{\text{vol}}. \quad (4.23)$$

The hypothesis specifies the incidence and degree class, not its three coefficients or the name of a gravitational Lagrangian. The classification is not asserted for all first-order connection actions.

The classification follows because both the coframe bivector and curvature transform in $\Lambda^2 \mathbb{R}^{1,3}$ and $\text{End}_{SO^+(1,3)}(\Lambda^2 \mathbb{R}^{1,3}) = \text{span}_{\mathbb{R}} \{I, \star\}$; the scalar residual commutant excludes any additional internal tensor, and determinant volume is the only independent alternating four-coframe scalar at this order. The full proof is in Appendix C.

Compatibility at the level of physical first variations. Classification of a face score is not, by itself, convergence of the first variation of the complete physical action. Let $S_{H,X}$, $S_{P,X}$ and V_X denote the actual common-cylinder realizations of the three classified densities, with their physical cell measures and boundary conventions. On a represented determining variation v_X define

$$\mathfrak{r}_X[v_X] := \delta\mathcal{A}_X[v_X] - \alpha_X\delta S_{H,X}[v_X] - \beta_X\delta S_{P,X}[v_X] - \lambda_X\delta V_X[v_X]. \quad (4.24)$$

The gravitational reduction used below requires, on the same metric and connection determining cores,

$$\mathfrak{r}_X[v_{X,K}(k)] \longrightarrow 0. \quad (4.25)$$

At each cutoff these are comparisons of actual represented test rows; the finite banks exhaust the fixed core with the stated test-interpolation control. This condition includes the variation of any retained subdivision error, the complete resolved-memory subtraction, and any boundary remainder. The memory log-partition must be the contribution represented on that same physical record, not an unrelated convex function chosen for subtraction.

Condition (4.25) is automatic when the physical action agrees exactly, up to the specified boundary completion, with its classified representative on the chart. More generally, a uniform C^1 bound on the remainder over the physically lifted test bank is sufficient. Convergence of action values alone is not sufficient: the smooth functions $A_n(x) = n^{-1} \sin(nx)$ converge uniformly to zero, although $A'_n(0) = 1$. Thus the Einstein handoff uses variational consistency, not an unjustified interchange of differentiation and a cutoff limit.

On the spinless nondegenerate branch, the connection variation controls torsion by [Theorem C.14](#). Classification and stationarity are therefore in place. To obtain an Einstein metric, it remains to pass these variations to one common noncollapsed limit and to identify its nonlinear connection curvature.

5 The renewal-to-Einstein limit

This section proves the conditional continuum theorem. A positive spatial cut margin supplies spatial compactness estimates. Initial-energy propagation controls curvature size, while a separate connection argument identifies the weak curvature limit. The final variational passage uses these estimates on the same realized regulator sequence.

The literal-link criterion is the principal connection route presented here. Fully discrete energy propagation, the differentiated action-current estimate, a Hodge-temporal alternative and interface comparisons are retained in [Appendix D](#). They are alternative or supporting certificates, not additional conclusions inferred from a curvature norm alone.

5.1 Global noncollapse and compact spatial screens

Let the physical endpoint quotient of a global regulator carry masses $m_{v,X}$, symmetric spatial conductances $c_{e,X}$, physical cell scale h_X , and provenance-safe efficient capacities $u_{e,X}$. The Dirichlet form and its nonnegative self-adjoint Laplacian are

$$\mathcal{E}_X^{\text{sp}}(f) = \sum_{\{u,v\} \in E_X} c_{uv,X} |f(u) - f(v)|^2, \quad \mathcal{E}_X^{\text{sp}}(f) = \langle f, \Delta_X^{\text{sp}} f \rangle_{L^2(\mu_X)}. \quad (5.1)$$

The endpoint/action packet also contains a finite capacity-conductance comparison. On the positive comparison branch there are cutoff-independent constants $0 < a_- \leq a_+ < \infty$ such that

$$a_- h_X c_{e,X} \leq u_{e,X} \leq a_+ h_X c_{e,X} \quad (e \in E_X). \quad (5.2)$$

This is an explicit same-history comparison between two reconstructed physical rows; it is not an identification made after taking the continuum limit. Define the efficient and analytic cut constants

$$I_X^{\text{eff}} := \min_{0 < \mu_X(A) \leq \mu_X(V_X)/2} \frac{u_X(\partial A)}{\mu_X(A)^{2/3}}, \quad I_X^{\text{sp}} := \min_{0 < \mu_X(A) \leq \mu_X(V_X)/2} \frac{h_X c_X(\partial A)}{\mu_X(A)^{2/3}}. \quad (5.3)$$

Then

$$a_- I_X^{\text{sp}} \leq I_X^{\text{eff}} \leq a_+ I_X^{\text{sp}}. \quad (5.4)$$

Thus a positive efficient-capacity margin is exactly the operational input needed to obtain a positive spatial isoperimetric margin on this branch.

Theorem 5.1 (Cut margin and compact-screen alternative). *Assume the total physical volume is uniformly bounded, $\mu_X(V_X) \leq V_*$, the weighted degree satisfies $\sum_u c_{uv,X} \leq D_* h_X^{-2} m_{v,X}$, and (5.2) holds. If $\inf_X I_X^{\text{eff}} = I_*^{\text{eff}} > 0$, then $I_X^{\text{sp}} \geq I_*^{\text{eff}}/a_+ =: I_*^{\text{sp}} > 0$ and every mean-zero f obeys regulator-uniform Sobolev and Poincaré bounds*

$$\|f\|_{L^6(\mu_X)}^2 \leq C_S \mathcal{E}_X^{\text{sp}}(f), \quad \|f\|_{L^2(\mu_X)}^2 \leq C_P \mathcal{E}_X^{\text{sp}}(f), \quad (5.5)$$

with cutoff-independent constants C_S, C_P given explicitly in Appendix D. The low-energy counting function satisfies

$$N_X(R) \leq 1 + 4e^{3/2} V_* (C_S R)^{3/2} \quad (R > 0), \quad (5.6)$$

and for $P_{X,R} = \mathbf{1}_{[0,R]}(\Delta_X^{\text{sp}})$,

$$\|(I - P_{X,R})f\|_{L^2(\mu_X)}^2 \leq R^{-1} \mathcal{E}_X^{\text{sp}}(f). \quad (5.7)$$

Hence the spectral projections form uniformly exhausted finite-rank compact screens. If instead $I_{X_j}^{\text{eff}} > 0$ tends to zero, the minimizing cuts A_j give normalized potentials ϕ_j for the demand (D.1), defined in Appendix D, with

$$\sum_e u_{e,X_j} |\mathrm{d}\phi_j(e)| = 1, \quad \left| \sum_v b_{A_j,X_j}(v) \phi_j(v) \right| = (I_{X_j}^{\text{eff}})^{-1} \longrightarrow \infty, \quad (5.8)$$

which is an explicit macroscopic-neck obstruction. If a cut has zero capacity already at finite cutoff, that disconnected-support cut is retained directly, without dividing by its capacity.

The discrete Sobolev–Nash–heat-kernel mechanism behind this theorem is standard in spirit; see, for example, Chung [57, Chs. 10–11] and Saloff-Coste [58]. Because the regulator normalization is part of the present result, Appendix D gives the complete finite proof, including the constants, the $L^1 \rightarrow L^\infty$ heat-semigroup bound, and the derivation of the $R^{3/2}$ counting law.

Thus compact spatial control is derived on the positive cut branch and is replaced by a concrete obstruction on the collapsed branch. Uniform spatial ranks and tails alone do not establish spacetime compactness. The common-cylinder Hodge/action packet also retains convergence of transported finite screens and temporal control of the field coefficients. Appendix D gives the compact-screen upgrade from this packet to the strong convergence needed for quadratic Palatini insertions.

5.2 Curvature propagation from initial energy

The cut estimate does not control curvature. We therefore retain a separate positive energy for the actual electric and magnetic records and propagate it from the initial cut. The cancellation below is checked before taking norms, so the large skew principal operator does not enter the growth constant; compare [59, 60].

On a compact foliated cylinder $K = [0, T_*] \times \Sigma$, let $y_X(t) = (E_X(t), B_X(t))$ be the actual retained electric/magnetic curvature coefficients in the reconstructed observer frame, with positive physical Gram $M_X(t) = \text{diag}(M_{E,X}, M_{B,X}) \succ 0$. Put

$$z_X(t)^2 := y_X(t)^T M_X(t) y_X(t). \quad (5.9)$$

The finite Bianchi/transport propagation writer is required to occur in the form

$$\dot{y}_X = (K_X + L_X)y_X + f_X, \quad M_X K_X + K_X^T M_X = 0, \quad (5.10)$$

where K_X is the positive-Gram skew principal curl block, L_X contains lower-order observer, lapse, connection, and transport terms, and f_X is the complete differentiated curvature source, including any incoming boundary-energy contribution when the cylinder has a timelike spatial boundary. Define

$$a_X(t) := \frac{1}{2} \max \left\{ 0, \lambda_{\max} \left[M_X^{-1/2} (\dot{M}_X + M_X L_X + L_X^T M_X) M_X^{-1/2} \right] \right\}. \quad (5.11)$$

On the gravitational shared-link branch, f_X contains the physically transferred, volume-normalized Palatini metric row together with the differentiated Cartan and finite Bianchi defects; Appendix C keeps these terms separate.

The principal cancellation is tested on the actual record. For example, a recorded curl $C_X : V_{B,X} \rightarrow V_{E,X}$ has positive-Gram adjoint $C_X^\dagger = M_{B,X}^{-1} C_X^T M_{E,X}$, and the block $\begin{pmatrix} 0 & C_X \\ -C_X^\dagger & 0 \end{pmatrix}$ is exactly M_X -skew. Defining this adjoint does not prove that an actual geometric second curl equals $-C_X^\dagger$: their discrepancy is retained in the source or symmetric-growth record. Consequently no hidden h_X^{-1} principal norm is discarded in the estimate.

Theorem 5.2 (Initial-energy curvature propagation certificate). *Assume (5.10). Then*

$$z_X(t) \leq e^{\int_0^t a_X} \left[z_X(0) + \int_0^t e^{-\int_0^s a_X} \|f_X(s)\|_{M_X(s)} ds \right]. \quad (5.12)$$

Suppose moreover that the actual curvature field entering the Palatini test has a uniformly stable interpolation

$$R_X = \mathcal{I}_X y_X + \zeta_X, \quad \|\mathcal{I}_X y_X(t)\|_{L^2(\Sigma_t)} \leq C_I z_X(t), \quad \|\zeta_X\|_{L^2(K)} \leq C_\zeta, \quad (5.13)$$

and that the lapse is bounded by N_ . If, uniformly in X ,*

$$z_X(0) \leq Z_*, \quad \int_0^{T_*} a_X(t) dt \leq A_*, \quad \int_0^{T_*} \|f_X(t)\|_{M_X(t)} dt \leq F_*, \quad (5.14)$$

then

$$\|R_X\|_{L^2(K)} \leq C_I \sqrt{N_* T_*} e^{A_*} (Z_* + F_*) + C_\zeta. \quad (5.15)$$

Thus the curvature-size requirement needed below follows with $p = 2 > 3/2$.

The fully discrete version is Theorem D.8; Proposition D.9 gives the sharply scaled sufficient source estimate. Both are proved in Appendix D. Neither estimate alone identifies the curvature of a limiting connection.

5.3 Gauge-free anisotropic compactness from literal links

Periodic-link notation. Let $N = 2m + 1$, $h = N^{-1}$, and $\Lambda_h = (h\mathbb{Z}/\mathbb{Z})^3$. With shifts S_i set

$$D_i^+ = (S_i - I)/h, \quad D_i^- = (I - S_i^{-1})/h, \quad D_i^0 = (D_i^+ + D_i^-)/2. \quad (5.16)$$

The spatial H_h^1 norm is the sum of the discrete L^2 norm and the L^2 norms of these forward differences; coefficient norms use a fixed positive matrix inner product. The nonlinear evolution in the next section uses this same regulator.

We next identify the nonlinear curvature from the original link record. Link variables, plaquette holonomies and curvature extracted from small loops are standard ingredients of lattice gauge theory [26]. What is specific here is the anisotropic compactness statement in the original common trivialization and its use in the renewal-to-Palatini passage. Unlike a compactness assumption on the full connection, this route requires strong convergence only of the spatial coefficients; the temporal coefficient may remain weakly convergent. On the periodic branch let

$$U_{i,h} = \exp(hA_{i,h}), \quad Z_{i,h} = \frac{U_{i,h} - I}{h}, \quad (5.17)$$

and let $E_{i,h}$ and $F_{ij,h}$ denote respectively the normalized literal electric link and spatial plaquette-curvature records in the same common trivialization. All coefficient norms below use a fixed positive Euclidean matrix norm; no indefinite Lorentz norm is used in the compactness estimate.

Theorem 5.3 (Gauge-free anisotropic literal-link compactness). *On an odd periodic spatial regulator, suppose the literal links remain in one dimensionless identity chart,*

$$h|A_{i,h}| \leq \delta_*, \quad (5.18)$$

with δ_ fixed and sufficiently small, and suppose on a compact time interval*

$$\sup_h \left(\|A_h\|_{L_t^\infty L_h^2} + \|A_h\|_{L_t^2 H_h^1} + \|A_{0,h}\|_{L^2} + \|E_h\|_{L^2} + \|F_h\|_{L^2} \right) < \infty. \quad (5.19)$$

Then, after extraction in the original common trivialization,

$$\mathcal{I}_h A_h \rightarrow A \text{ strongly in } L^2, \quad \mathcal{I}_h A_{0,h} \rightharpoonup A_0 \text{ weakly in } L^2, \quad (5.20)$$

and the weak L^2 limits of the literal electric and magnetic records are exactly the components of

$$R(\omega) = d\omega + \omega \wedge \omega, \quad \omega = A_0 dt + \sum_i A_i dx^i. \quad (5.21)$$

No Coulomb condition, gauge smallness, temporal-mean bound, $D^+ A_{0,h}$ bound, or $\partial_t A_{0,h}$ bound is required. The identity-link chart and the anisotropic energy bound (5.19) remain explicit hypotheses.

The proof is given in Appendix D. Its key point is to compactify $Z_{i,h}$ rather than differentiate the logarithm in time: the literal electric identity gives a cutoff-uniform H_h^{-2} time-derivative bound for Z_h , while $Z_h - A_h \rightarrow 0$ strongly. Weak-strong passage in the exact electric equation and the second-order plaquette expansion then identify every component of (5.21).

A second route assumes compactness of the full connection and verifies the Cartan and interface residuals explicitly; see Assumption D.13 and Theorems D.14 and D.17. The limit theorem allows either route, with the respective hypotheses unchanged.

5.4 The conditional Einstein limit

The finite and analytic steps can now be assembled. The hypotheses below specify which records occur, which finite structural tests pass, and which bounds persist on a common cofinal sequence. The constructive existence question is treated separately in Section 6.

Call a cofinal regulator a *complete gravitational renewal cylinder* when the geometric packet, complete accepted determining-field cylinder, generic protected reward-duration record, and physical endpoint, propagation and Cartan records occur on compatible restrictions of one operational process. The records include the physical test lifts, action calibration, actual curvature interpolation and common-cylinder comparison maps used above. The word *gravitational* labels the sector being tested; it does not mean that an Einstein Lagrangian is stored in the entrance record.

Theorem 5.4 (General renewal-to-Einstein certification). *For a complete gravitational renewal cylinder, consider a cofinal sequence of Lorentzian relational regulators of Theorem 3.6. On every compact spacetime cylinder K , define the following five certification conditions on compatible restrictions of the same operational process.*

- (E1) *The complete accepted cylinder and generic protected reward-duration record reconstruct the determining kernel and scalar common action. Its actual first variations tend to zero on a fixed determining metric-test core. The scaled bound (4.17), with the realized subsequence of Proposition C.7, is a sufficient operational route to this condition. The physical coframe lifts of these tests have uniformly bounded, convergent coefficient tensors in the curvature and volume insertions.*
- (E2) *Complete resolved-memory subtraction and exact or summably stable binary face subdivision give the first-order score classified by Theorem 4.7, with the physical first-variation remainder condition (4.25) on the same metric and connection tests. Pointwise convergence of action values is not used in place of this condition.*
- (E3) *In a fixed physical action calibration the spacetime-constant coefficients converge, $(\alpha_X, \beta_X, \lambda_X) \rightarrow (\alpha, \beta, \lambda)$, with $\beta \neq 0$ the coefficient of L_{Palatini} . The spinless connection Euler residual tends to zero and the Cartan map is uniformly nondegenerate on the retained coframe chart.*
- (E4) *The endpoint packet satisfies the hypotheses of Theorem 5.1, with $\liminf_X I_X^{\text{eff}} > 0$. Its common-cylinder compact-screen packet, including transported-screen convergence and temporal control, gives a nondegenerate limit with $e_X \rightarrow e$ strongly in local L^2 and $\sup_X \|e_X\|_{L^6(K)} < \infty$.*
- (E5) *The actual curvature records satisfy either the continuous propagation certificate of Theorem 5.2 or the fully discrete certificate of Theorem D.8, with uniform initial, growth, transfer and source budgets and stable curvature interpolation. The scaled action-current estimate (D.48) may supply the source budget. For the connection/curvature handoff, either the connection satisfies the separate compactness certificate Assumption D.13, with Theorem D.14 as a sufficient Hodge–temporal realization and the two Cartan identification residuals in (D.69), or the periodic literal-link branch satisfies Theorem 5.3, which directly supplies strong spatial connection compactness and identifies the full nonlinear curvature from the electric and magnetic records.*

If conditions (E1)–(E5) hold on a cofinal tail, then a subsequence has zero limiting torsion and, for each determining compactly supported inverse-metric test $k^{\mu\nu}$, the metric first variation converges

to

$$\chi \int_K \sqrt{-g} (G_{\mu\nu} + \Lambda g_{\mu\nu}) k^{\mu\nu}, \quad \chi \neq 0. \quad (5.22)$$

Consequently

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 0} \quad (5.23)$$

holds distributionally on the determining test core, and on its closure in the declared test topology.

Proof outline. The calibrated connection equation and the uniform Cartan inverse give vanishing limiting torsion. The cut and transport estimates supply the strong coframe limit; the propagation estimate supplies the curvature bound; and the selected connection criterion identifies its weak limit. Strong interpolation then passes the coframe bivectors against the weak curvature, while the physical first-variation remainder tends to zero. On the torsion-free branch the Holst coframe variation vanishes and the nonzero Palatini coefficient leaves the Einstein insertion displayed above. Stationarity on the determining core proves the conclusion. The full argument, including the volume term and the common subsequence, is [Theorem D.22](#) and [Corollary D.23](#) in [Appendix D](#). $\square$

Corollary 5.5 (Einstein limit or failed certificate). *For a complete gravitational renewal cylinder, apply the finite reconstructions and conditions (E1)–(E5) on a selected cofinal sequence. If the conditions pass on a cofinal tail, the sequence has the Einstein subsequential limit of [Theorem 5.4](#). Otherwise the failing algebraic test, residual, calibrated comparison, positive margin, propagation budget or compactness tail is retained on the same physical records. The complete witness list and its cofinal quantifiers are given in [Appendix D.11](#). Failure is an obstruction to this sufficient certification procedure, not a proof that every Einstein subsequence is impossible.*

Proof. The positive conclusion is [Theorem 5.4](#). The complementary witnesses are obtained by negating the corresponding finite or cofinal tests, as detailed in [Appendix D.11](#). $\square$

Remark 5.6 (Certification is not the definition of the constructive branch). Conditions (E1)–(E5) recognize limits of general complete cylinders. They do not define the constructive law of [Theorem 6.1](#), whose estimates are proved directly. Conversely, the recognition theorem is not restricted to that law.

The nonzero Palatini coefficient is distinct from invertibility of the Palatini–Holst connection operator. A nonzero pure Holst coefficient can eliminate torsion, but its torsion-free metric variation does not supply the Einstein equation. Likewise the propagated curvature bound supplies size, whereas [\(D.69\)](#) identifies the nonlinear connection curvature. These distinctions prevent either condition from being hidden inside a generic “positive branch.”

6 A constructive nonsymmetric 3 + 1 Einstein class

6.1 The local harmonic law and uniform realization

The preceding theorem recognizes Einstein limits from operational certificates. We now address a different question: whether a nonempty nonsymmetric class admits cutoff-uniform evolution and convergence. For the declared local harmonic law, these bounds are proved directly on a fixed Sobolev neighbourhood; the five recognition conditions are not assumed as its definition. The law is not identified with the raw finite Euler flow of the reconstructed action.

The continuum analytic target in this subsection is the standard harmonic reduction of the Einstein equations. Local Sobolev well-posedness for broad classes of quasilinear hyperbolic systems,

including the reduced Einstein system, belongs to the classical theory [24]; harmonic and generalized-harmonic formulations and their stability theory are also well established [25, 74]. The new assertion here is therefore not a replacement for continuum Einstein well-posedness. It is the cutoff-uniform discrete writer, its genuinely finite renewal realization, and the quantitative convergence of that realization to the harmonic vacuum development on the stated common Sobolev neighbourhood.

We use the odd periodic regulator and the forward, backward and centred differences of (5.16). For $g = \eta + q$ in a small Lorentzian chart, write the harmonic normal row as

$$a(g)q_{tt} + 2b^i(g)\partial_i q_t - c^{ij}(g)\partial_i \partial_j q - F(g, \partial g) = 0, \quad a = -g^{00}, \quad b^i = -g^{0i}, \quad c^{ij} = g^{ij}, \quad (6.1)$$

where F is the analytic first-jet source of the harmonic-reduced vacuum Ricci equation, equivalently of its trace-reversed reduced Einstein system. This identifies the target equation of the constructive realization explicitly. Its selection is not derived again from the root race, nor is consistency of this writer used as a substitute for the independent action classification and stationarity certificates. Put

$$\mathsf{K}_b = \sum_i \{M_{bi} D_i^0 + D_i^0 M_{bi}\}, \quad (6.2)$$

and, with $Y_h = (v, D_1^+ q, D_2^+ q, D_3^+ q)$,

$$\mathsf{G}_h(q, v) = F(g, Y_h) - \sum_{i,j} D c^{ij}(g) [D_i^+ q] D_j^+ q + \sum_i D b^i(g) [D_i^+ q] v. \quad (6.3)$$

The local evolution law is

$$q_t = v, \quad a(g)v_t = \sum_{i,j} D_i^- (c^{ij}(g) D_j^+ q) - \mathsf{K}_b v + \mathsf{G}_h(q, v). \quad (6.4)$$

The divergence and skew placements are fixed by the positive quadratic principal energy. The compensator removes exactly the first-derivative terms introduced by these placements, so the smooth continuum limit is (6.1). A parity obstruction of the bare $A_3 \cong D_3$ execution graph requires one additional odd-parity execution incidence for a cutoff-uniform Cartesian Sobolev realization; this incidence is used only to execute the writer and is not added to the root race entering (3.7). The complete proof is in Appendix E. In the main theorem we use the discrete Sobolev norms

$$\|u\|_{H_h^r}^2 = \sum_{|\alpha| \leq r} \|D^\alpha u\|_h^2, \quad \|X\|_{\mathcal{X}_h^r}^2 = \|q\|_{H_h^{r+1}}^2 + \|v\|_{H_h^r}^2, \quad (6.5)$$

where D^α is formed from the commuting forward differences.

For an integer $s \geq 11$ define the compatible continuum data class

$$\begin{aligned} \mathcal{D}_s(\varepsilon) = \{(\gamma, K) : \gamma \in H^{s+1}(\mathbb{T}^3), \quad K \in H^s(\mathbb{T}^3), \\ R(\gamma) + (\text{tr}_\gamma K)^2 - |K|_\gamma^2 = 0, \quad \nabla_\gamma^i K_{ij} - \partial_j \text{tr}_\gamma K = 0, \\ \|\gamma - I\|_{H^{s+1}} + \|K\|_{H^s} < \varepsilon\}. \end{aligned} \quad (6.6)$$

This is relative openness in the vacuum constraint set; it is not an ambient open ball of fixed nonzero constraint violations.

Theorem 6.1 (Constructive nonsymmetric 3 + 1 renewal–Einstein realization). *There are $\varepsilon_s, T_s, h_s, C_s > 0$, independent of the cutoff, with the following properties.*

- (O1) For every $h < h_s$, every real ten-component initial record $X_h(0) = (q_h(0), v_h(0))$ in the fixed ball $\|q_h(0)\|_{H_h^{s+1}} + \|v_h(0)\|_{H_h^s} \leq \varepsilon_s$ has a unique solution of (6.4) on $[0, T_s]$, and

$$\sup_{t \leq T_s} (\|q_h(t)\|_{H_h^{s+1}} + \|v_h(t)\|_{H_h^s}) \leq C_s \|X_h(0)\|_{\mathcal{X}_h^s}. \quad (6.7)$$

No symmetry, Fourier-support restriction, analyticity, or cutoff-shrinking amplitude is imposed.

- (O2) Every sufficiently small $(\gamma, K) \in \mathcal{D}_s(\varepsilon_s)$ has a harmonic finite preparation for which the whole cutoff sequence converges on $[0, T_s] \times \mathbb{T}^3$ to a unique metric g with the prescribed initial data. Writing $\varepsilon_0 = \|\gamma - I\|_{H^{s+1}} + \|K\|_{H^s}$, one has

$$\boxed{G(g) = 0, \quad \sup_{t \leq T_s} (\|g_h - g\|_{H^{s-1}} + \|\partial_t g_h - \partial_t g\|_{H^{s-2}} + \|\partial_t^2 g_h - \partial_t^2 g\|_{H^{s-3}}) \leq C_s h \varepsilon_0.} \quad (6.8)$$

Here g_h denotes the canonical interpolation of $\eta + q_h$.

- (O3) For the compatible preparations in item (O2), the harmonic subsidiary field and Einstein residual obey, at the corresponding lower orders,

$$\sup_{t \leq T_s} (\|c(g_h)\|_{H^{s-2}} + \|\partial_t c(g_h)\|_{H^{s-3}} + \|G(g_h)\|_{H^{s-3}}) \leq C_s h \varepsilon_s, \quad (6.9)$$

while the positive A_3 rate lift remains in an interior cone and its spatial cut margin stays uniformly positive. In particular the reconstructed metric is uniformly noncollapsed and its metric curvature is uniformly bounded on the common slab and converges to the curvature of g at the lower Sobolev order supplied by (6.8).

- (O4) The differential writer has a genuinely finite renewal realization. With segment duration $\tau_h \asymp h^2$, fixed-degree local Taylor–Hermite steps and Sobolev-scaled finite precision produce committed successor words whose complete acceleration defect is $O(\varepsilon h^4)$ in $C_t H_h^s$ and whose differentiated defect is $O(\varepsilon h^2)$ in $C_t H_h^{s-3}$ on the common slab; [Theorem E.17](#) gives stronger rates for the explicit construction. With the compatible finite initialization specified there, every such certified history inherits the bounds above. This statement concerns finite-range information flow; it does not identify the arithmetic service time with the physical root clock.

Consequently the positive Einstein population is not confined to homogeneous or one-dimensional inhomogeneous reductions: it contains a relatively open small-data class with arbitrary three-dimensional dependence. The next result shows that the differential writer itself is also not isolated.

Remark 6.2 (Scope of the constructive law). This is a cutoff-uniform small-data realization, not a universal attraction result or an identification with exact finite Euler flow. The independent exact-action question is summarized at the end of this section and analyzed in [Appendices G and H](#).

6.2 Structural stability of renewal laws and action-prepared finite seeds

The fixed writer of [Theorem 6.1](#) is the centre of a larger finite law class. Let

$$\Lambda_h = - \sum_{i=1}^3 D_i^- D_i^+, \quad \mathcal{B} = \{B = B^T \in \mathbb{R}^{10 \times 10} : \|B\|_{\text{op}} < b_0\}, \quad (6.10)$$

where $b_0 > 0$ is fixed below the spatial coercivity margin of the common small metric chart. Thus $\mathcal{B}$ is a 55-dimensional open ball. For $B \in \mathcal{B}$ define

$$q_t = v, \quad a(g)v_t = \sum_{i,j} D_i^-(c^{ij}(g)D_j^+q) - \mathbf{K}_b v + \mathbf{G}_h(q, v) - h^2 B \Lambda_h^2 q. \quad (6.11)$$

The added fourth-order stencil has fixed execution radius. It need not be small in the top ultraviolet vector-field norm; its incidence and sign are instead controlled by the same positive shifted energy.

For a finite chronological realization of (6.11), let I_j have duration $\tau_j \asymp h^2$. A candidate physical letter e at a live prefix π carries a piecewise-Hermite segment Q_e with actual acceleration defect

$$f_e = Q_e'' - V_{B,h}(Q_e, Q_e'), \quad (6.12)$$

and, with unit cost assigned to a grammar or execution failure, define

$$c_{h,j}(\pi, e) = \varepsilon^{-2} \int_{I_j} (\|f_e\|_{H_h^s}^2 + \|\partial_t f_e\|_{H_h^{s-3}}^2) dt, \quad C_h = \sum_j c_{h,j}. \quad (6.13)$$

The cost is determined by the physical letter and its prefix, not by its probability.

For later use we also fix the canonical object coming from the *original* phase-compatible finite action. Appendix G displays that action, its unique stationary connection on the retained small chart, and its Legendre map. Write $\mathcal{L}_h^\circ(\gamma, V, N, \beta)$ for the stationary-connection-eliminated finite Lagrangian and let $V_h(\gamma, \pi, N, \beta)$ be the analytic inverse of

$$\pi = D_V \mathcal{L}_h^\circ(\gamma, V, N, \beta) \quad (6.14)$$

on the nondegenerate Legendre chart. The canonical Hamiltonian used below is therefore the displayed Legendre transform

$$\boxed{\mathcal{H}_h(\gamma, \pi, N, \beta) = \langle \pi, V_h(\gamma, \pi, N, \beta) \rangle_h - \mathcal{L}_h^\circ(\gamma, V_h(\gamma, \pi, N, \beta), N, \beta).} \quad (6.15)$$

Its original four lapse-shift rows are

$$\mathcal{C}_h(X) = (-\nabla_N \mathcal{H}_h, \nabla_\beta \mathcal{H}_h) \big|_{N=1, \beta=0}. \quad (6.16)$$

Thus $\mathcal{C}_h$ is not a residual of the local harmonic writer and (6.15) is not merely a characterization of an unnamed action. Finite zero-set preparation and transport to continuum Cauchy data are separate assertions. The latter uses comparison of the actual stationary/Legendre maps, made explicit in Assumption F.12 and Lemma F.13.

Theorem 6.3 (Structurally stable renewal laws with action-prepared finite seeds). *After decreasing the common small-data radius if necessary, the following statements hold with constants independent of the cutoff.*

- (L1) *For every $B \in \bar{\mathcal{B}}$ the law (6.11) has the same fixed-radius nonlinear lifespan as Theorem 6.1. There is an energy $\mathcal{E}_{s,h}^B$ uniformly equivalent to $\|X\|_{\mathcal{X}_h^s}^2$ such that*

$$(\mathcal{E}_{s,h}^B)' \leq C_s \mathcal{E}_{s,h}^B + C_s (\mathcal{E}_{s,h}^B)^{3/2}. \quad (6.17)$$

- (L2) Let $B_h \in \overline{\mathcal{B}}$ be arbitrary, with no convergence of B_h required. For the compatible data of (6.6), the whole sequence of solutions of (6.11) converges to the same harmonic vacuum development g as the central writer, and

$$\sum_{j=0}^3 \|q_{B_h, h}^{(j)} - q_{0, h}^{(j)}\|_{C_t H_h^{s-1-j}} \leq C_s h \|B_h\|_{\text{op}} \varepsilon. \quad (6.18)$$

Hence the metric, subsidiary, noncollapse and curvature conclusions of [Theorem 6.1](#) are uniform on the entire coefficient ball.

- (L3) Let $K_{h,j}(e \mid \pi)$ be arbitrary history-dependent finite transition rows satisfying, at every live prefix,

$$\sum_e K_{h,j}(e \mid \pi) c_{h,j}(\pi, e) < M \tau_j h^4. \quad (6.19)$$

No independence, invariant starting law, or per-letter small-defect condition is imposed. Then

$$\mathbb{E} C_h \leq M T h^4, \quad \mathbb{P}\{C_h > h^2\} \leq M T h^2. \quad (6.20)$$

On the complementary event the finite history shadows a member of the differential-law family closely enough to have the same Einstein limit. For a fixed finite grammar containing a sufficiently accurate completion at every live source, the strict conditions (6.19) form nonempty relatively open convex subsets of the product of row simplexes and may contain strictly positive rows. The construction in [Theorem E.17](#) supplies the required strict residual margin. Along the odd refinement sequence the exceptional probabilities are summable, so under any common coupling the Einstein conclusion holds eventually almost surely.

- (L4) For every fixed sufficiently small nonzero preparation amplitude t , the exact map (6.16) has a finite, nonsymmetric analytic zero-set chart generated from arbitrary small free tensor records W after removal of four fixed balancing modes, with the free-record bound taken in $\mathcal{P}_h^{s+2}$ as defined in the appendix. Its prepared records $\mathcal{I}_h(t, W)$ satisfy

$$\mathcal{C}_h(\mathcal{I}_h(t, W)) = 0, \quad \|\mathcal{I}_h(t, W)\| \leq C_s |t|. \quad (6.21)$$

The chart is relatively open in the complete finite zero set, has codimension $4N^3$, and the linearized constraint map has a right inverse obeying

$$\|\mathcal{R}_{h,t} f\| \leq C_s (\|P_{\perp} f\| + |t|^{-1} |P_0 f|). \quad (6.22)$$

No CMC, conformal-flatness, spatial symmetry, finite Fourier support, or continuum-compatible metric is imposed on the free record W . On the cofinally consistent initial-action branch of [Assumption F.12](#), every cofinal sequence of these records with $\mathcal{I}_h W_h$ convergent in $\mathcal{P}^{s+1}$ has a harmonic completion converging to the vacuum development of its limiting prepared data, for arbitrary $B_h \in \overline{\mathcal{B}}$. For merely bounded free records this conclusion is sub-sequential, not whole-sequence convergence to one fixed metric. The original four lapse-shift rows vanish identically at the initial cut independently of the subsequently chosen acceleration law.

The law-space statement is deliberately quantitative. Here structural stability means a common continuum limit in the stated Sobolev and residual topologies; it does not mean attraction under a renormalization flow. It gives a fixed open ball of differential laws and relatively open finite

probability-row neighborhoods in a physically scaled residual topology; it does not assert a fixed unscaled total-variation ball of arbitrary microscopic kernels or attraction from arbitrary leading-order forces.

Remark 6.4 (The robustness class is nontrivial but not arbitrary). The family in [Theorem 6.3](#) can differ from the central writer by a nonvanishing amount at the highest lattice frequencies. The law parameter is, however, retained during a physical history. Unrestricted rapid switching between two individually stable coefficient marks can parametrically amplify a resolved high-frequency mode. Likewise, a row may be close in ordinary total variation while accumulating order-one failure probability over $O(h^{-2})$ stages. These boundaries explain why the neighborhood is formulated by energy-compatible coefficients and complete physical residual profiles rather than by an unscaled norm on arbitrary kernels. The proofs are in [Appendix F](#).

6.3 Compatibility on one chronological process

The stability theorem controls a class of finite evolution laws. Its compatibility with the independently reconstructed action is a further question, because equality of the two finite dynamics has not been assumed. The following chronological correction preserves the specified physical marginals while retaining both the evolution-residual and stationarity scales. Let $\nu_{h,j}$ be the physical marginal at stage j and let $Q_{h,j}(y | x)$ be the independently reconstructed common-action comparator on the admitted transition support. Put

$$R_{h,j}(x, y) = \nu_{h,j}(x)Q_{h,j}(y | x), \quad \rho_{h,j+1} = \nu_{h,j}Q_{h,j}, \quad d_{h,j} = \nu_{h,j+1} - \rho_{h,j+1}. \quad (6.23)$$

For the source and target incidence maps $Ah(x) = \sum_y h(x, y)$ and $Bh(y) = \sum_x h(x, y)$, define the source-shortened operator

$$\mathcal{L}_{h,j} = \text{diag}(\rho_{h,j+1}) - Q_{h,j}^\top \text{diag}(\nu_{h,j})Q_{h,j}. \quad (6.24)$$

When $d_{h,j} \in \text{Ran } \mathcal{L}_{h,j}$ and the source-preserving Thomson correction has leverage at most $1/2$, the explicit formula ([F.39](#)) in [Appendix F.4](#) defines a nonnegative corrected coupling and its stochastic kernel. It has exactly the source and target marginals specified above. $(Q_{h,j}c)(x) = \sum_y Q_{h,j}(y | x)c(x, y)$. Its occurrence energy is

$$\mathcal{E}_{\text{occ},h,j} = \langle d_{h,j}, \mathcal{L}_{h,j}^\dagger d_{h,j} \rangle. \quad (6.25)$$

Theorem 6.5 (Same-cylinder action-occurrence Einstein compiler). *Consider the open nonsymmetric $3 + 1$ chart of [Theorems 6.1](#) and [6.3](#). Suppose at each chronological stage the preceding Thomson correction exists on the same support as the common-action comparator, and let $c_{h,j} \geq 0$ be the stable-writer residual cost with duration $\tau_{h,j}$. Assume, uniformly,*

$$\begin{aligned} m_{h,j} &:= \mathbb{E}_{R_{h,j}} c_{h,j} \leq M_0 \tau_{h,j} h^4, \\ \mathcal{E}_{\text{occ},h,j} &\leq M_1 \tau_{h,j} h^4, \quad \sum_j \tau_{h,j} \leq T. \\ \mathcal{V}_{h,j} &:= \mathbb{E}_{R_{h,j}} |c_{h,j} - Q_{h,j} c_{h,j}|^2 \leq V_0 \tau_{h,j} h^4, \end{aligned} \quad (6.26)$$

Let $\Delta_{h,j}$ be the Bregman field-action gap of the same reconstructed common action, on its supported interior proximal chart. For the compared source and target marginals retain the actual positive action drop

$$d_{\mathcal{A},h,j}^+ = [\mathbb{E}_{\nu_{h,j}} \mathcal{A}_{h,j} - \mathbb{E}_{\nu_{h,j+1}} \mathcal{A}_{h,j}]_+. \quad (6.27)$$

Assume the corrected test-lift bound ([F.52](#)), including this drop, tends to zero on the fixed determining core. Equal marginals are a sufficient special case for removal of the drop; prescribed chronological

marginals need not be equal. Finally assume the physical first-variation reduction and calibrated coefficient/torsion conditions (E2)–(E3) of [Theorem 5.4](#), including nonzero Palatini normalization, and the stated noncollapse and calibration conditions on the constructive small metric chart. Then

- (i) the corrected finite process has exactly the prescribed chronological marginals and remains on the common-action support;
- (ii) for $C_h = \sum_j c_{h,j}(X_j, X_{j+1})$,

$$\mathbb{E}C_h \leq (M_0 + \sqrt{M_1 V_0})Th^4, \quad \Pr\{C_h > h^2\} = O(h^2); \quad (6.28)$$

- (iii) on $\{C_h \leq h^2\}$ the constructive metric-jet bounds imply the literal-link budget [\(5.19\)](#); hence [Theorem 5.3](#) makes connection compactness and nonlinear curvature identification outputs of the successful history rather than independent assumptions;
- (iv) the represented first variation of the same reconstructed action tends to zero in mean square on the determining core;
- (v) every continuum cluster point on a realized cofinal subsequence where the residual and stationarity conclusions hold satisfies

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 0 \quad (6.29)$$

distributionally on that core.

Along any common coupling with $\sum_n h_n^2 < \infty$, the successful residual branch occurs eventually almost surely; no independence between cutoffs or between chronological stages is required.

This theorem is a compatibility result, not an identification of two finite dynamics. The common action is reconstructed before the chronological correction is formed, while the writer-residual moments independently test whether that action-supported process remains in the constructive Einstein basin.

6.4 Exact finite-action dynamics: a certified compatibility test

The constructive and chronological results do not identify their laws with the unchanged finite Euler flow. The stronger exact-action question is kept separate because it has its own constraint geometry and singular time scales. The action itself is no longer left implicit: [Appendix G](#) displays the finite phase-compatible action and the canonical Hamiltonian [\(6.15\)](#). The purpose of the exact-action analysis in this article is therefore precise: to test whether that concrete finite variational system has nontrivial prepared states and locally compatible constrained histories, without claiming a cutoff-uniform Euler-flow limit.

Theorem 6.6 (Exact-action preparation and fixed-cutoff compatibility). *For the phase-compatible finite action of [Theorem G.1](#), on the respective branches specified in [Appendices G](#) and [H](#):*

- (X1) *The canonical constraint system and regular finite response are well-defined on an analytic stationary-connection/Legendre chart. Transverse source elimination is lossless, the signed response has the exact positive-reproduction/scalar–trace decomposition, and nontrivial source-neutral and all-cutoff active prepared families exist.*

- (X2) *At the certified $N = 3$, $h = 1/3$ event, the weak first-Poisson block has exact rank 24 and a five-dimensional kernel. The reduced rapid spectrum has a semisimple double zero, and the full weak generator has*

$$(\dim E_+, \dim E_-, \dim E_0) = (8, 8, 13). \quad (6.30)$$

Under the complete amplitude-graded chart assumptions, the unchanged action admits an exactly constrained boundary-selected history on $0 \leq t \leq ca^2$ whose first field and multiplier tilt agree with the prepared event and whose metric and literal-link curvatures are $O(a)$.

- (X3) *These fixed-cutoff results do not give forward Cauchy stability. On the separate invariant normal-factor chart, residence through a fixed physical time requires the rapid and slow exponentially small widths of [Theorem G.27](#). Bounded-rate slow limits additionally obey the acceleration and quadratic slope compatibility conditions of [Appendix H](#). Positive-time propagation of a compatible slow branch, physical forward selection without terminal data, and uniform control under spatial refinement remain open.*

Proof. The action and Hamiltonian provenance are [Theorem G.1](#). The response and preparations are [Theorems G.3, G.5, G.9 and G.10](#). The same-event rank and spectral certificates are [Theorems G.15 and G.16](#), and the boundary history is [Theorem G.21](#). The residence estimate is [Theorem G.27](#); the slow necessary conditions are [Theorems H.5, H.16, H.19 and H.21](#). $\square$

Boundary selection fixes unstable coordinates at a terminal cut and is therefore a compatibility construction rather than a microscopic forward selection rule. None of the exact-action results is used as a hypothesis in the constructive Einstein theorem above.

7 Explicit vacuum sectors and operational diagnostics

The general limit and the nonsymmetric realization answer the principal continuum questions. The following sectors serve complementary purposes: the homogeneous model makes lapse variation and action–law matching explicit, whereas the Gowdy model retains nonlinear spatial variation and an exact discrete momentum invariant. Their special hypotheses are not substitutes for the general certification conditions.

7.1 Homogeneous renewal coordinates and the full lapse constraint

Effective homogeneous and Friedmann-type dynamics have also been extracted from microscopic quantum-geometric models, notably group-field-theory condensates [\[17, 18\]](#). The role of the present sector is different: the homogeneous geometry is first reconstructed from renewal rate and density variables, and the lapse is retained as an independent variational degree of freedom before the stationary equations are solved. The comparison is therefore one of emergent cosmological dynamics, not an identification of the underlying microscopic variables.

On the A_3 root-symmetric branch write

$$k_{\alpha,h}(t) = \frac{\kappa(t)}{8h^2}, \quad m_h(t) = \varrho(t)h^3, \quad \kappa(t) > 0, \quad \varrho(t) > 0. \quad (7.1)$$

The root identity gives $\mathcal{B} = \kappa I_3$ and reversal symmetry gives $\beta = 0$. If $g_{ij} = a^2 \delta_{ij}$, the ADM inversion [\(3.8\)](#) is therefore exactly

$$a = \varrho^{1/3}, \quad N = \varrho^{1/3} \kappa^{1/2}, \quad \varrho = a^3, \quad \kappa = \frac{N^2}{a^2}. \quad (7.2)$$

Thus κ is not the scale factor: together with the physical density it contains the lapse information that must remain independent until after variation.

On a compact interval and torus of coordinate volume V_0 , the torsion-free Palatini–volume branch of [Theorem 5.4](#), equivalently its Einstein–Hilbert representative, reduces after the standard Gibbons–Hawking–York Dirichlet boundary completion [\[61, 62\]](#) and division by the nonzero constant χV_0 , to

$$S[a, N] = \int \left[-\frac{6a\dot{a}^2}{N} - 2\Lambda Na^3 \right] dt. \quad (7.3)$$

Here the induced spatial metric is fixed at the initial and final slices, so $\delta a(t_0) = \delta a(t_1) = 0$. The lapse remains an independent bulk variation and is not gauge-fixed before the Euler equations are taken. The four-dimensional reduction and the exact cancellation of the second-derivative endpoint term by the boundary completion are written out in [Appendix I.1](#).

Theorem 7.1 (Vacuum Friedmann equations in renewal coordinates). *Let $u = \log \kappa$, $v = \log \varrho$, and $\mathcal{H} = \dot{a}/(Na)$, with $D_\tau = N^{-1}d/dt$. Under [\(7.2\)](#), the action [\(7.3\)](#) becomes*

$$S[\kappa, \varrho] = \int \left[-\frac{2\dot{\varrho}^2}{3\varrho^{4/3}\sqrt{\kappa}} - 2\Lambda\varrho^{4/3}\sqrt{\kappa} \right] dt. \quad (7.4)$$

Its Euler expressions, defined by $\delta S = \int (E_u \delta u + E_v \delta v) dt$, are

$$E_u = Na^3(3\mathcal{H}^2 - \Lambda), \quad E_v = Na^3 \left[4D_\tau \mathcal{H} + \frac{8}{3}(3\mathcal{H}^2 - \Lambda) \right]. \quad (7.5)$$

Consequently stationarity in the two independent positive renewal variables is equivalent to

$$3\mathcal{H}^2 = \Lambda, \quad D_\tau \mathcal{H} = 0, \quad (7.6)$$

which is the complete vacuum Einstein equation within the spatially flat homogeneous isotropic ansatz, including the lapse constraint.

Proof. The map $(u, v) \mapsto (\log a, \log N) = (v/3, u/2 + v/3)$ has nonzero Jacobian, so no homogeneous variation is lost. Varying [\(7.3\)](#) before fixing N gives

$$E_N = 2a^3(3\mathcal{H}^2 - \Lambda), \quad E_a = 6Na^2(2D_\tau \mathcal{H} + 3\mathcal{H}^2 - \Lambda).$$

The chain rule gives $E_u = (N/2)E_N$ and $E_v = (a/3)E_a + (N/3)E_N$, proving [\(7.5\)](#). The two equations are therefore equivalent to [\(7.6\)](#). [Appendix I.1](#) checks directly that these are exactly the 00 and spatial Einstein equations for the reconstructed metric, rather than relying on a symmetry-reduction principle. $\square$

Corollary 7.2 (Flat and de Sitter rates are stationary renewal solutions). *For $\Lambda > 0$ put $H_0 = \sqrt{\Lambda/3}$ and let $\tau(t) = \int^t N(s)ds$. Every positive connected stationary solution of [\(7.4\)](#) has*

$$a(\tau) = a_0 e^{\sigma H_0(\tau - \tau_0)}, \quad \varrho = a^3, \quad \kappa = N^2/a^2, \quad \sigma \in \{-1, 1\}. \quad (7.7)$$

After variation, choose proper-time gauge $N = 1$. On the expanding branch,

$$k_{\alpha,h}(t) = \frac{a_0^{-2} e^{-2H_0(t-t_0)}}{8h^2}, \quad m_h(t) = a_0^3 e^{3H_0(t-t_0)} h^3. \quad (7.8)$$

For $\Lambda = 0$, a is constant; for $\Lambda < 0$ there is no positive real solution in this spatially flat vacuum sector.

The exponential factor in [\(7.8\)](#) is therefore an output of the stationary renewal equations on this branch, not an input used to obtain those equations. The result does not select Λ , the expanding versus contracting sign, or the initial scale a_0 .

7.2 A finite lapse-varied renewal action

The same distinction can be kept at finite cutoff. The construction is a midpoint variational discretization in the standard sense of discrete mechanics [63], but every equation needed below is derived explicitly. Put $q = a^{3/2} = \sqrt{\varrho}$ and let $s_j = N_j \Delta t_j > 0$ be independent lapse-weighted intervals. With $\bar{q}_j = (q_j + q_{j+1})/2$, $A_0 = 8/3$, and $B_0 = 2\Lambda$, define

$$S_d(q, s) = - \sum_{j=0}^{M-1} \left[A_0 \frac{(q_{j+1} - q_j)^2}{s_j} + B_0 s_j \bar{q}_j^2 \right]. \quad (7.9)$$

This discretizes the boundary-completed reduced action, with fixed endpoint amplitudes and independently varied lapse-weighted intervals. The full endpoint convention and the conversion back to renewal variables are given in Appendix I.1.

Theorem 7.3 (Finite stationary classification and de Sitter convergence). *Let $v_j = (q_{j+1} - q_j)/s_j$. The full Euler equations of (7.9) are*

$$A_0 v_j^2 - B_0 \bar{q}_j^2 = 0, \quad (7.10)$$

$$2A_0(v_j - v_{j-1}) - B_0(s_{j-1}\bar{q}_{j-1} + s_j\bar{q}_j) = 0. \quad (7.11)$$

For $\Lambda > 0$, put $\omega = \sqrt{B_0/A_0} = \sqrt{3\Lambda/4}$. Every positive stationary solution has one common sign $\sigma \in \{-1, 1\}$ and

$$q_{j+1} = q_j \frac{1 + \sigma \omega s_j/2}{1 - \sigma \omega s_j/2}, \quad 0 < \omega s_j/2 < 1. \quad (7.12)$$

Conversely every such recurrence is stationary. If $\tau_j = \sum_{i < j} s_i$, $T = \sum_j s_j$, $h_t = \max_j s_j$, and $\omega h_t/2 \leq b < 1$, then

$$\max_j \left| \log \frac{q_j}{q_0} - \sigma \omega \tau_j \right| \leq \frac{\omega^3 T}{12(1-b^2)} h_t^2. \quad (7.13)$$

Hence $a_j = q_j^{2/3}$, $\varrho_j = q_j^2$, and in proper-time gauge $\kappa_j = q_j^{-4/3}$ converge with second-order logarithmic error to the stationary flat/de Sitter profiles of Corollary 7.2.

The proof, including the lapse variation and the exact rational recurrence, is in Appendix I.1. In particular, fixing $N = 1$ before variation would lose (7.10) and admit spurious non-Einstein homogeneous trajectories.

7.3 Measured acceptance law and action–dynamics matching

To relate the finite variational action directly to an operational probability law, we next exhibit a finite positive class in which the action is recovered from the trial probabilities themselves. For $A, B > 0$, $C \in \mathbb{R}$, and $s > 0$, consider

$$E_s(x, y) = A \frac{(y - x)^2}{s} + Bs \left(\frac{x + y}{2} \right)^2 + C(y^2 - x^2), \quad \begin{pmatrix} A & C \\ C & B \end{pmatrix} \succeq 0. \quad (7.14)$$

Within this declared finite parametric class, three measured costs identify A, B, C . Define the directly measurable rank defect

$$\mathfrak{D} = AB - C^2 \geq 0. \quad (7.15)$$

For a finite proposal alphabet F , let $p_s(y | x) > 0$ be a normalized proposal row and accept a proposal with probability

$$a_s(x, y) = e^{-E_s(x, y)/\varepsilon}. \quad (7.16)$$

Write

$$Z_s(x) = \sum_{y \in F} p_s(y | x) a_s(x, y), \quad Q_s(y | x) = \frac{p_s(y | x) a_s(x, y)}{Z_s(x)}. \quad (7.17)$$

Both Q_s and the total acceptance probability Z_s are entries of the same complete trial table.

Theorem 7.4 (Measured acceptance reconstructs the finite action). *The complete trial table reconstructs the unnormalized cost exactly:*

$$E_s(x, y) = -\varepsilon \log \frac{Z_s(x) Q_s(y | x)}{p_s(y | x)}. \quad (7.18)$$

For a path with fixed endpoints, the observed writer

$$\mathcal{A}_{\text{obs}} = \varepsilon \sum_j \left[\log \frac{Q_{s_j}(q_{j+1} | q_j)}{p_{s_j}(q_{j+1} | q_j)} + \log Z_{s_j}(q_j) \right] + C(q_M^2 - q_0^2) \quad (7.19)$$

is identically

$$\mathcal{A}_{\text{obs}} = - \sum_j \left[A \frac{(q_{j+1} - q_j)^2}{s_j} + B s_j \bar{q}_j^2 \right] \quad (7.20)$$

off shell on the parametric domain. Thus every interior-amplitude and lapse first variation agrees exactly with the reconstructed finite action.

The one-step minimizing skeleton has proper-time amplitude drift $c = -C/A$. The bulk action in (7.20) has $\Lambda_{\text{act}} = 4B/(3A)$, whereas the limiting constant-Hubble geometry of the minimizing skeleton has $\Lambda_{\text{geom}} = 4C^2/(3A^2)$. Consequently

$$G[g] + \Lambda_{\text{act}} g = \frac{4\mathfrak{D}}{3A^2} g. \quad (7.21)$$

The local operational update and the vacuum parameter of its measured bulk action therefore agree exactly if and only if $\mathfrak{D} = 0$.

The total acceptance probability in (7.18) is essential: the accepted row Q_s alone is invariant under addition of a source-dependent cost and therefore does not identify the lapse action. The complete trial record removes this ambiguity. On the rank-one branch with $A = A_0$, $B = A_0 \omega^2$, and $C = -\sigma A_0 \omega$, the cost is the square

$$E_s(x, y) = \frac{A_0}{s} \left[y - x - \sigma \omega s \frac{x + y}{2} \right]^2, \quad (7.22)$$

so its zero-cost recurrence is exactly (7.12).

Theorem 7.5 (Finite positive operational family with an Einstein limit). *Fix $T, \omega, q_0 > 0$ and a sign σ . There exists a sequence of genuinely finite accept/reject cylinders with M time stages, finite positive proposal alphabets, calibrated step $s = T/M$, rank-one costs (7.22), and a finite retry cap such that, for constants independent of M ,*

$$\mathbb{E} \mathcal{E}_M \leq CM^{-4}, \quad \mathbb{E} \max_j |q_j - q_0 e^{\sigma \omega \tau_j}| \leq CM^{-2}, \quad (7.23)$$

where $\mathcal{E}_M = A_0 \sum_j R_j^2/s$ and $R_j = q_{j+1} - q_j - \sigma \omega s \bar{q}_j$. The expected absolute first variation of (7.9) against every fixed smooth scale/lapse test is $O(M^{-2})$, including the lapse constraint. The retry-failure probability is at most M^{-6} . Under any common coupling of cutoffs, failures occur only finitely often almost surely and the successful piecewise-affine paths converge strongly in H^1 and uniformly to the stationary solution. Attaching the A_3 root race with

$$\varrho_j = q_j^2, \quad \kappa_j = q_j^{-4/3}, \quad k_{\alpha, h_x, j} = \frac{\kappa_j}{8h_x^2}, \quad m_{h_x, j} = \varrho_j h_x^3 \quad (7.24)$$

therefore gives positive renewal regulators converging to a vacuum metric with $\Lambda = 4\omega^2/3$.

Appendix I.1 gives the finite grids, temperature schedule, entropy–energy estimate, retry bound, and deterministic residual estimates used in this theorem. The construction proves nonemptiness of a matched operational class; it does not assert that an arbitrary Grand Tensor satisfies the quadratic duration law or the rank-one condition.

Flat and de Sitter regulators. The stationary profiles above give explicit geometric regulators. Theorem I.14 in Appendix I.3 collects their generator, noncollapse, curvature and summable exact-link estimates. In particular, flat and de Sitter geometries arise after solving the lapse-varied equations, rather than being inserted as prescribed microscopic rate histories.

7.4 Conservative blocking preserves the action–law defect

The preceding construction raises a distinct stability question: can ordinary time coarse graining drive a neighboring quadratic law toward the matched surface? For the conservative interval elimination of the measured quadratic cost, the answer is no. Assume $0 < s\sqrt{B/A} < 2$ and set

$$\mu = \sqrt{AB}, \quad \theta = 2 \operatorname{artanh}\left(\frac{s}{2}\sqrt{B/A}\right), \quad \rho = \frac{C}{\mu}, \quad \Delta_{\text{rel}} = 1 - \rho^2 = \frac{AB - C^2}{AB}. \quad (7.25)$$

Then the same cost can be written

$$\mathcal{E}_{\mu, \theta, C}(x, y) = \mu \coth \theta (x^2 + y^2) - 2\mu \operatorname{csch} \theta xy + C(y^2 - x^2). \quad (7.26)$$

For two successive intervals define conservative blocking by

$$(E_1 \star_0 E_2)(x, y) = \inf_{z \geq 0} \{E_1(x, z) + E_2(z, y)\}. \quad (7.27)$$

Theorem 7.6 (No transverse matching attraction under conservative blocking). *For every $\theta_1, \theta_2 > 0$,*

$$\mathcal{E}_{\mu, \theta_1, C} \star_0 \mathcal{E}_{\mu, \theta_2, C} = \mathcal{E}_{\mu, \theta_1 + \theta_2, C}. \quad (7.28)$$

Hence $AB = \mu^2$, C , $\mathfrak{D} = AB - C^2$, and the scale-free mismatch Δ_{rel} are invariant under every finite sequence of homogeneous blockings. For two identical midpoint intervals of duration s , rewritten as one interval of duration $2s$,

$$A' = A + \frac{Bs^2}{4}, \quad B' = \frac{AB}{A + Bs^2/4}, \quad C' = C. \quad (7.29)$$

Consequently $\Delta_{\text{rel}} > 0$ remains strictly positive: the matched surface $\Delta_{\text{rel}} = 0$ has no open basin of attraction for this conservative blocking operation, even after changes of time/amplitude units or common positive rescaling of the action. On finite positive amplitude alphabets with a symmetric untwisted trial kernel, the endpoint twist C is retained exactly under finite blocking, and a controlled low-temperature/grid limit recovers the same nonzero relative defect.

This is a no-go result for a specified natural coarse-graining operation, not a no-go theorem for every possible renewal renormalization. In particular, a driven or conditioned law may define a different physical flow. The theorem shows that generic Einstein matching cannot be inferred merely from conservative elimination of intermediate amplitudes.

The same blocking operation also gives an exact consistency test: local minimizers compose across intervals precisely on the matched branch. [Proposition I.8](#) in [Appendix I.1](#) proves this characterization. It tests matching; it does not generate an attracting flow.

7.5 A nonlinear inhomogeneous two-polarization vacuum sector

The unpolarized Gowdy T^3 class provides a complementary nonlinear test: a finite splitting preserves an exact momentum invariant, and the readout controls the full curvature pathwise. This is a symmetry-reduced result on a smooth slab [\[64\]](#). Write $[t_0, t_1] \times \mathbb{T}^3$, $t_0 > 0$, write

$$g = t^{-1/2} e^{\lambda/2} (-dt^2 + d\theta^2) + t [e^P (dx + Q dy)^2 + e^{-P} dy^2]. \quad (7.30)$$

Set

$$a = P_t, \quad b = P_\theta, \quad c = e^P Q_t, \quad d = e^P Q_\theta, \quad (7.31)$$

and

$$e = \frac{1}{2}(a^2 + b^2 + c^2 + d^2), \quad f = ab + cd. \quad (7.32)$$

The first-order vacuum system is

$$\begin{aligned} a_t &= b_\theta - a/t + c^2 - d^2, & b_t &= a_\theta, \\ c_t &= d_\theta - c/t - ac + bd, & d_t &= c_\theta + ad - bc, \\ P_t &= a, & Q_t &= e^{-P} c, & \lambda_t &= 2te, \end{aligned} \quad (7.33)$$

with coordinate and momentum constraints

$$P_\theta = b, \quad e^P Q_\theta = d, \quad \lambda_\theta = 2tf. \quad (7.34)$$

The identities

$$e_t - f_\theta = -\frac{a^2 + c^2}{t}, \quad f_t - e_\theta = -\frac{f}{t} \quad (7.35)$$

propagate the last constraint and preserve the zero mean of tf required for periodic λ .

For the variational test the areal gauge is imposed only after variation. With independent fields $(R, \lambda, P, Q, N, \beta)$, $R, N > 0$, and $v_u = \dot{u} - \beta u'$, use the boundary-marked ADM density

$$\begin{aligned} \mathcal{L}_G &= -\frac{v_R(v_\lambda - 4\beta')}{2N} + \frac{R}{2N}(v_P^2 + e^{2P} v_Q^2) \\ &\quad + N \left[-2R'' + \frac{R'\lambda'}{2} - \frac{R}{2} \{ (P')^2 + e^{2P} (Q')^2 \} \right]. \end{aligned} \quad (7.36)$$

Its lapse and shift equations are independent constraints. The solution of [\(7.33\)](#)–[\(7.34\)](#) lifts to those equations with $R = t$, $N = 1$, $\beta = 0$; hence the finite action test below is not obtained by first gauge-fixing away the lapse and shift variations.

Theorem 7.7 (Nonlinear two-polarization renewal regulator). *Let $X_*(t)$ be a smooth solution of [\(7.33\)](#)–[\(7.34\)](#) on a fixed slab $[t_0, t_1] \times \mathbb{T}^1$, with the spatial derivatives required below uniformly bounded and with the reconstructed metric uniformly nondegenerate. Let $h = \ell$ be the synchronized temporal and spatial mesh. Then there is a nearest-neighbour symmetric source/transport/source splitting with the following properties.*

- (G1) With $U = (u_1, u_2, u_3, u_4) = \sqrt{t}(a, b, c, d)$, $\mathcal{J}(U) = u_1 u_2 + u_3 u_4$, $D_+ = (S_+ - I)/\ell$, and $A_+ = (I + S_+)/2$, the quantity

$$\mathcal{K}_\ell = D_+ \lambda - 2A_+ \mathcal{J}(U) \quad (7.37)$$

is preserved exactly by every deterministic full step. The map is second-order consistent on the stated smooth chart.

- (G2) If $X_{j+1} = \Phi_h(t_j, X_j) + r_j$ and

$$\mathcal{E}_h = \sum_j \frac{\|r_j\|_{2,\ell}^2}{2h}, \quad \mathcal{F}_h = \ell^{-(2r+1)} \mathcal{E}_h, \quad (7.38)$$

then, under the stated smooth-chart bootstrap or an enforced offset envelope,

$$\max_j \|X_j - S_\ell X_*(t_j)\|_{r,\infty,\ell} \leq C(\epsilon_0 + h^2 + \sqrt{\mathcal{F}_h}). \quad (7.39)$$

For $r \geq 1$, the accumulated staggered-constraint error is bounded by its initial value plus $C\sqrt{\mathcal{F}_h}$.

- (G3) On the finite offset alphabet with $\|\epsilon_j\|_\infty \leq c_0 h^4$ and initial C_h^1 error $O(h^2)$, the shared tensor-product cubic Hermite readout is pathwise $O(h^2)$ in $W^{1,\infty}$ and $O(h)$ in its second derivatives. Consequently, on compact subslabs,

$$\|g_h - g_*\|_{W^{1,\infty}} + \|D^2 g_h - D^2 g_*\|_{L^\infty} + \|\text{Riem}(g_h) - \text{Riem}(g_*)\|_{L^\infty} \leq Ch, \quad (7.40)$$

where the first term is in fact $O(h^2)$. Thus $\sup_h \|\text{Riem}(g_h)\|_{L^2(K)} < \infty$ and $\|\text{Ric}(g_h)\|_{L^\infty(K)} \leq Ch$ pathwise. The Levi-Civita curvature of the limiting metric is therefore identified directly.

- (G4) An independent midpoint/centered discretization of (7.36), with N and β varied independently before restriction to areal gauge, has first-variation residual tending to zero against every fixed smooth local test. A finite positive offset/acceptance realization can be chosen with the support cap in item (G3) and obeys the exact Gibbs entropy-energy bound; its successful histories therefore inherit the pathwise curvature estimate.

The construction and proof are given in Appendix I.2. It is a nonlinear and spatially inhomogeneous population result, but its scope remains symmetry reduced and local to a smooth positive slab. Unlike the general certificate of Theorem 5.2, the curvature bound in (7.40) is obtained directly from pathwise second-jet control and requires no separately assumed all-time curvature budget on this class. It does not imply an unrestricted four-dimensional nonlinear Cauchy theorem.

The finite proposal-access estimate, including reset and retry assumptions, is Proposition I.12 in Appendix I.2. It determines when the measured process reaches the admissible histories; it is separate from the deterministic curvature estimate.

8 Interpretation and discussion

The construction developed in this paper separates several structures that are usually introduced simultaneously. Predictive persistence is obtained first by quotienting operational histories according

to their complete conditional futures. Relational geometry is then reconstructed on this future-minimal carrier from physically occurring incidence, calibration, and renewal statistics. Lorentzian signature requires one additional protected signed datum, and gravitational dynamics enters only after a common variational structure has been reconstructed and its continuum behaviour controlled.

The resulting picture is therefore not a direct identification of stochastic dynamics with spacetime geometry. It is a sequence of reconstruction and compatibility problems. Each stage has its own finite data, its own obstructions, and its own continuum requirements. This separation is important both conceptually and mathematically: the predictive quotient does not already contain a Lorentzian metric, Lorentzian kinematics does not already imply Einstein dynamics, and the existence of a reconstructed action does not imply that an arbitrary microscopic evolution is stationary for that action. In this respect the proposal should be compared componentwise with causal sets, operational causal frameworks, predictive-state methods, quantum renewal processes and information-theoretic emergence programmes, rather than understood as a claim that those neighboring literatures lack pregeometric or emergent-spacetime constructions.

8.1 Predictive persistence and Lorentzian geometry

The first structural point is that persistence need not be supplied as a microscopic state space. Histories that have identical statistics under every admissible future continuation are operationally indistinguishable and are therefore identified. Because this equivalence is preserved by continuation, the quotient carries a canonical successor structure. The minimum predictive carrier is thus reconstructed from the process itself rather than introduced as a hidden state, lattice site, or geometric point.

The Renewal Grand Tensor records the corresponding finite positive history structure: the minimum predictive carrier, represented history algebra, and positive word-Gram hierarchy. Its role is not that of a metric tensor on a pre-existing space. It records which distinctions survive predictive reduction, how represented historical operations compose, and which positive overlaps remain between them. Geometry becomes meaningful only after this predictive structure is combined with the additional same-history information required by the geometric branch.

Dimension and relational structure. The selection of three spatial dimensions is likewise conditional and structural rather than universal. Two independent finite classifications select the tetrahedral relation cell K_4 . The endpoint-cycle balance criterion does so among connected simple regular relations without invoking an alternating form, while the stronger pair-homogeneous simplex criterion selects the same cell from minimal faithful alternating completion. Both lead to the three-dimensional A_3 standard module. This differs from Myrheim-Meyer causal-set dimension estimators [3, 4, 29], from the measured spectral dimension of causal dynamical triangulations [8], and from best-fit dimensional reconstruction using entanglement data [21]: the present result is a uniqueness statement inside specified finite source categories, rather than an estimator applied to a geometry-bearing ensemble.

The endpoint-cycle balance argument is the primary result and does not depend on the parity argument. The orientation-twisted source equivalence provides its representation-theoretic interpretation; the alternating criterion is secondary and retains its explicit minimization hypothesis. At the same time, neither theorem claims that every possible microscopic relation system must belong to these categories. Higher-arity interactions, enriched incidence structures, or different primitive relational questions can lead outside the classified branch. The result should therefore be read as a pair of finite uniqueness statements within explicitly stated structural classes.

Metric reconstruction and calibration. Once the A_3 relational carrier has been selected, the metric is reconstructed from observable renewal statistics. The symmetric second moments of the directed root race determine the spatial kinetic form, the antisymmetric first moments determine the shift, and the physical speed density together with the waiting-time scale fixes the remaining ADM normalization. The spatial metric, lapse, and shift are consequently collective quantities derived from the finite renewal process rather than labels attached to microscopic events.

This reconstruction also identifies what cannot be obtained from normalized probabilities alone. Winner probabilities determine only relative rates; an absolute waiting-time calibration is required to recover their common scale. More generally, the calibration fibre described above separates physically meaningful scale information from conventional changes of units. Absolute calibration and same-cut readability are therefore genuine physical inputs of the geometric branch, even though the metric itself is not.

The Lorentzian sign. The Lorentzian signature occupies a particularly clear position in this hierarchy. Both the Grand-Tensor Gram structure and the root-race spatial form are positive. They cannot generate a negative direction by themselves. The protected binary orientation is an additional operational distinction, and the separating constructions above show that it is independent of the remaining positive data within the declared language.

Once the positive spatial module has dimension three, adjoining one independent nondegenerate signed direction yields Lorentzian inertia $(1, 3)$, up to overall convention. This conclusion precedes the Clifford representation. The gamma matrices therefore realize an already determined Lorentzian quadratic form; they are not responsible for selecting its signature. In particular, the Lorentzian direction is neither obtained by analytic continuation of a positive diffusion form nor inserted through a choice of Dirac matrices. This signature question is also logically distinct from Lorentz invariance of a discrete substrate: for example, Poisson sprinklings in causal-set theory can remain Lorentz invariant without selecting a preferred frame [6]. Here the issue is instead how one negative geometric direction enters after a positive predictive-history reconstruction.

8.2 From reconstructed action to Einstein dynamics

Lorentzian reconstruction alone is a kinematical statement. The dynamical problem begins with the complete accepted physical cylinder. The determining field is obtained from future-equivalence on that cylinder, while a marked renewal pressure reconstructs the affine action slope from protected reward and duration data. Finite curl and period tests then determine whether the resulting increments integrate to a single scalar action.

This ordering makes the variational statement noncircular. The action is reconstructed before the accepted dynamics is compared with it. The Fisher–Bregman and related finite probability identities subsequently test whether the observed evolution is stationary for that independently reconstructed action; they do not define the action by fitting a cost to the observed transition kernel.

After resolved memory has been removed, Lorentz covariance restricts the first-order gravitational score to the Palatini–Holst–volume family. Connection stationarity gives the torsion-free branch, while the positive spatial cut condition supplies regulator-uniform noncollapse and compact screens. Curvature control is obtained separately from initial energy and propagation estimates and is identified with the curvature of the limiting connection through the Cartan comparison. These ingredients allow the weak–strong variational passage to

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 0} \tag{8.1}$$

on the certified continuum branch.

The point of this construction is therefore not merely that particular renewal histories can be chosen whose reconstructed metrics happen to solve Einstein’s equation. Geometry, action provenance, stationarity, noncollapse, and curvature compactness are tested independently and are joined only at the continuum handoff.

Constructive and certification results. There are two complementary theorem levels. The general renewal-to-Einstein theorem is a recognition result: a complete gravitational renewal cylinder satisfying the stated finite and cofinal certificates has an Einstein subsequential limit. Its hypotheses deliberately separate action stationarity, first-order gravitational provenance, noncollapse, and curvature propagation.

The open $3 + 1$ population theorem addresses the converse concern that these certificates might describe an empty or highly fine-tuned class. For an explicit local writer it proves a cutoff-independent nonlinear Cauchy interval with arbitrary three-dimensional dependence in a fixed Sobolev neighbourhood. The surrounding open family of local differential laws has the same continuum Einstein limit, and the corresponding finite history-dependent transition rows admit relatively open residual-controlled neighbourhoods. The same-cylinder compiler further shows that, under explicit occurrence-energy and variance scalings, a chronological correction of the independently reconstructed action comparator preserves both the writer residual scale and the field-action stationarity scale, while the literal-link compactness theorem supplies the connection handoff on the successful branch.

The homogeneous and Gowdy constructions play a different role. The homogeneous sector makes the lapse variation and action–dynamics matching fully explicit and shows that the flat and de Sitter profiles are stationary outputs rather than prescribed rate histories. The unpolarized Gowdy sector provides a nonlinear inhomogeneous example with exact finite momentum propagation and direct full-curvature convergence. Together with the open $3 + 1$ population, these examples show that the positive branch is not restricted to a single symmetric trajectory.

Einstein matching is not an automatic attractor. The robustness results should not be confused with a universality theorem. For the conservative homogeneous blocking operation studied above, the dimensionless action–law mismatch is invariant. Coarse graining of that specific type therefore does not drive a generic microscopic law toward the Einstein-matched surface. The optimizer-composition identity instead gives an exact consistency characterization of the matched branch.

This distinction is useful more generally. Stability of an already controlled Einstein branch and dynamical selection of that branch from a larger microscopic theory are different questions. The former is established for the finite law classes treated here; the latter remains a problem of microscopic selection and renormalization.

The exact-action route. A further distinction concerns the exact Euler flow of the reconstructed finite action. The constructive local writer is action compatible but is not assumed to coincide with that raw finite flow. Appendix G therefore starts from the displayed phase-compatible action (G.11), performs stationary connection and Legendre reduction, and asks a more limited question: does the unchanged finite system possess nontrivial prepared states and locally compatible constrained histories?

The answer is positive but deliberately fixed-cutoff. The finite response admits exact transverse elimination and source-neutral preparations. At the certified $N = 3$ event the weak first-Poisson

block has rank 24, its five-dimensional kernel consists of two generalized-zero and three constant-shift directions, and the rapid generator has the semisimple $(8, 8, 13)$ stable–unstable–centre split. Under the complete graded-chart conditions a boundary-selected exact history exists for $O(a^2)$ physical time with $O(a)$ metric and literal-link curvature. The same analysis also shows why this does not yet prove a cofinal Euler-flow theorem: forward residence requires exponentially thin normal preparation, while the slow system obeys additional acceleration and slope-compatibility conditions. Thus the exact-action results serve as a nontrivial compatibility test for the reconstructed variational system; cutoff-uniform forward selection and positive-time propagation remain open.

8.3 Scope and open problems

The results establish a finite reconstruction and continuum mechanism for a specified class of operational renewal processes. They do not imply that every renewal law produces Lorentzian geometry or that every such geometry enters the Einstein branch. Actual occurrence of the required operational rows, protected orientation, absolute calibration, noncollapse, action stationarity, and curvature propagation remain genuine conditions on the microscopic realization.

Likewise, the paper does not establish a universal derivation of three spatial dimensions. The K_4/A_3 result is conditional on the finite relational categories classified above. Its strength is that two mathematically distinct selectors lead to the same spatial module, not that all conceivable pregeometric theories are forced into that category.

Microscopic selection and universality. The most immediate open problem is to understand why a broad microscopic class should enter the controlled Lorentzian–Einstein branch. The present results establish nonempty and relatively open families once the appropriate physical topology is chosen, but they do not derive an attracting renormalization flow from arbitrary renewal laws. A useful next step would be to identify natural conditioning, coarse-graining, or learning dynamics under which the action, noncollapse, and propagation margins become stable large-scale quantities.

The exact-action problem is a sharper version of the same question. One would like to determine whether the reconstructed finite action itself admits a regular cofinal branch whose exact Euler flow preserves the required constraint and slow compatibility structure on a cutoff-independent interval and enters the same Einstein continuum class as the constructive writers. The fixed-cutoff boundary construction and the two exponential residence bounds do not settle this selection and propagation problem.

Matter and internal structure. The present analysis concerns the vacuum gravitational sector. An extension to matter requires more than appending additional fields to the limiting Einstein equation. Internal symmetry, matter incidence, fermionic multiplicity, and their common action should be represented on the same finite carrier as the reconstructed spacetime structure, and their nonlinear stress must remain controlled in the continuum passage. This raises both finite operator-algebraic questions and independent compactness questions for the coupled gravity–matter system.

Residual memory. A final question concerns the information lost in the local continuum description. The microscopic dynamics considered here may be genuinely non-Markovian, whereas the controlled continuum branches are effectively local after predictive compression. It is therefore natural to ask which subleading structures retain memory of the microscopic renewal process.

Controlled departures from the Markovian scaling regime could provide a systematic language for nonlocal or history-dependent corrections to the effective geometric dynamics.

Within the scope established here, the main conclusion is more limited and more precise. A persistent state, a spatial metric, Lorentzian signature, and the vacuum Einstein equation need not all be supplied at the microscopic level. Predictive persistence can be reconstructed from future distinguishability; a calibrated relational branch can reconstruct the ADM geometry; one independent protected sign fixes the Lorentzian inertia; and, when the independently reconstructed action, noncollapse, and curvature conditions are satisfied, the corresponding continuum branch obeys vacuum Einstein dynamics. The remaining problem is to understand how broadly microscopic physics selects and stabilizes that branch.

Formalization and computational verification

The proof development is split between theorem-prover formalization and finite exact-arithmetic certificates. The structural algebraic and analytic library is implemented in **Lean 4/Mathlib**; the fixed-cutoff root-box, interval-matrix, modular Pfaffian and rational-similarity statements are verified by companion exact-arithmetic certificate checkers generated from the displayed finite action. The public source release records these two verification layers separately, so a floating-point eigensolver output is never used as a proof-bearing premise. Source code, build configuration and documentation are available at <https://github.com/AI-MathPhys/renewal-geometry>.

Appendices

The appendices retain the detailed constructions, quantitative constants and alternative sufficient criteria. Each has one mathematical purpose; all exact-action statements are separate from the general Einstein-limit proof.

Appendix	Proofs and dependencies
A	Predictive quotient, process reconstruction, history algebra and calibration.
B	Relational selectors, root statistics, positive realizability and signed geometry.
C	Determining field, common action, stationarity, local classification and torsion.
D	Noncollapse, propagation, connection alternatives, interfaces and the Einstein limit.
E	Uniform harmonic evolution and genuinely finite renewal realization.
F	Law-space robustness, finite constraint preparation and same-cylinder compatibility.
G	Explicit phase-compatible finite action, canonical Hamiltonian, exact response, fixed-cutoff certificate and boundary histories.
H	Supporting necessary slow-compatibility, acceleration and slope obstructions for the exact Euler route.
I	Homogeneous matching, conservative blocking, Gowdy and flat/de Sitter regulators.

A Operational reconstruction and positive history algebra

This appendix proves [Theorem 2.2](#) and [Proposition 2.1](#). It uses only the finite operational entrance from [Section 2](#); the final subsection isolates the additional orientation and calibration information.

A.1 Finite typed operational renewal datum

Definition A.1 (Finite typed event grammar). At cutoff X , a finite typed event grammar consists of:

- (E1) a finite set of cut types $\mathcal{O}_X$;
- (E2) finite sets $\Sigma_X(x, y)$ of primitive event letters $a : x \rightarrow y$;
- (E3) a finite-horizon prefix-closed family $\mathbf{H}_{X,x}$ of reachable typed histories ending at each cut type x , with the remaining horizon included in the type, typed concatenation and empty words; alternatively, an unbounded family with a separately certified finite-index future congruence;
- (E4) finite preparation, intervention-outcome, and terminal-Read alphabets, together with finite exposed operator ports;
- (E5) a distinguished protected binary orientation record whose two values are separated by the declared physical Read policy.

No state label, persistent relation object, metric, scheduler, connection, or spacetime semantic is assigned to a history.

For $h \in \mathbf{H}_{X,x}$ let $\Sigma_{X,h}$ be the event letters admissible after h .

Definition A.2 (Finite conditional operational renewal law). A finite conditional operational renewal law on [Definition A.1](#) assigns

$$p_X(a \mid h) \in [0, 1], \quad \sum_{a \in \Sigma_{X,h}} p_X(a \mid h) = 1, \quad (\text{A.1})$$

for every reachable history. For an admissible future word $w = a_1 \cdots a_k$ define

$$p_X(w \mid h) = \prod_{j=1}^k p_X(a_j \mid ha_1 \cdots a_{j-1}), \quad p_X(\epsilon_x \mid h) = 1. \quad (\text{A.2})$$

The physical intervention letters include, on every exposed Hermitian Choi-coordinate space, a finite physical CP frame whose real span is the full Hermitian space. The induced multitime frame probabilities are affine under randomization, additive under coarse graining, normalized on complete instruments, causal under deterministic discard, compatible on common prefixes, and satisfy the finite positivity conditions of [Theorem A.14](#). Actual bounded terminal Reads, marks, and readout queries may be included as terminal event outcomes; equivalently they may be appended to the future test bank.

The distinguished renewal semantics used in the geometric branch is carried by the admitted completion/return event family. No ordinary reset state is needed for the predictive construction below; if a stronger reset law is imposed, it is an additional property of the same event process rather than a state-space axiom.

A.2 Future equivalence and the derived persistent carrier

Definition A.3 (Complete predictive future signature). For $h \in \mathbf{H}_{X,x}$ define

$$\Sigma_X(h) = (p_X(w \mid h))_{w \in \mathbf{F}_{X,x}}, \quad (\text{A.3})$$

where $\mathbf{F}_{X,x}$ is the complete horizon-compatible future-test bank starting at x ; on an unbounded-horizon protocol it comprises all finite admissible continuations. Histories $h, h' \in \mathbf{H}_{X,x}$ are future equivalent when

$$h \sim_X h' \iff p_X(w \mid h) = p_X(w \mid h') \text{ for every } w \in \mathbf{F}_{X,x}. \quad (\text{A.4})$$

Histories of different cut type are never identified.

Theorem A.4 (History-level right congruence). *Future equivalence is a typed right congruence. If $h \sim_X h'$ and $a : x \rightarrow y$ is admissible, then*

$$p_X(a \mid h) = p_X(a \mid h'), \quad (\text{A.5})$$

and on the positive-probability branch

$$ha \sim_X h'a. \quad (\text{A.6})$$

Proof. Use the one-letter future a in [\(A.4\)](#). If the common probability is positive, then for any continuation v the chain rule gives

$$p_X(v \mid ha) = \frac{p_X(av \mid h)}{p_X(a \mid h)} = \frac{p_X(av \mid h')}{p_X(a \mid h')} = p_X(v \mid h'a).$$

Hence the complete future signatures agree after appending a . □

Definition A.5 (Minimum predictive state). For every cut type x put

$$Z_{X,x}^{\min} := \mathbf{H}_{X,x} / \sim_X, \quad q_{X,x} : \mathbf{H}_{X,x} \rightarrow Z_{X,x}^{\min}. \quad (\text{A.7})$$

For an admissible event $a : x \rightarrow y$ define

$$T_{X,a}([h]) = [ha], \quad \kappa_{X,a}([h]) = p_X(a \mid h). \quad (\text{A.8})$$

Corollary A.6 (Derived predictive state machine). *The maps in (A.8) are well defined. If $w = a_1 \cdots a_k$, $z_0 = q_X(h)$, and $z_j = T_{X,a_j}(z_{j-1})$, then*

$$p_X(w \mid h) = \prod_{j=1}^k \kappa_{X,a_j}(z_{j-1}). \quad (\text{A.9})$$

Thus the predictive state machine is reconstructed from the conditional event law.

Proof. Well-definedness is Theorem A.4; the product formula is (A.2) written on future-equivalence classes. $\square$

Definition A.7 (Deterministic predictive realization). A reachable deterministic predictive realization consists of finite sets $S_{X,x}$, history maps $s_x : \mathbf{H}_{X,x} \rightarrow S_{X,x}$, partial updates $u_a : S_{X,x} \rightarrow S_{X,y}$, and branch functions $r_a : S_{X,x} \rightarrow [0, 1]$ such that on every reachable branch

$$s_y(ha) = u_a(s_x(h)), \quad p_X(a \mid h) = r_a(s_x(h)). \quad (\text{A.10})$$

Theorem A.8 (Canonical minimum predictive realization). *The quotient $Z_X^{\min}$ is the unique coarsest reachable deterministic predictive realization. For every reachable realization S_X there is a unique surjection*

$$\pi_{X,x} : S_{X,x} \rightarrow Z_{X,x}^{\min} \quad (\text{A.11})$$

such that

$$q_x = \pi_x s_x, \quad \pi_y u_a = T_{X,a} \pi_x, \quad r_a = \kappa_{X,a} \pi_x. \quad (\text{A.12})$$

Consequently a generated ledger or hidden-state presentation, when present, is only an intermediate predictive realization and has a unique quotient onto $Z_X^{\min}$.

Proof. If $s_x(h) = s_x(h')$, induction on future-word length using (A.10) gives $p_X(w \mid h) = p_X(w \mid h')$ for every future word w . Hence $h \sim_X h'$, so $\pi_x(s_x(h)) := [h]$ is well defined. Reachability makes π_x unique and surjective. The intertwining identities follow from the definitions, and the quotient itself is a realization by Corollary A.6. $\square$

Theorem A.9 (Persistence derived from composition). *Let $A_1, A_2, \dots$ be the realized event sequence, let $H_n = A_1 \cdots A_n$, and put $Z_n = q_X(H_n)$. Then on every realized branch*

$$Z_{n+1} = T_{X,A_{n+1}}(Z_n). \quad (\text{A.13})$$

Consequently

$$\mathbf{p}_{X,n}^{\min} = (Z_n, A_{n+1}, Z_{n+1}) \quad (\text{A.14})$$

is a persistent relational event packet derived from the event process. Every other deterministic persistent predictive packet maps uniquely onto (A.14).

Proof. By definition, $Z_{n+1} = q_X(H_n A_{n+1}) = T_{X,A_{n+1}} q_X(H_n)$. For any other predictive realization, apply the universal map of Theorem A.8 at consecutive cuts. $\square$

Corollary A.10 (Future-operational transfer). *Let $\mathcal{D}$ be any downstream finite readout whose value is unchanged whenever two candidate states have the same complete declared future signature. Then there is a unique $\overline{\mathcal{D}}$ on $Z_X^{\min}$ such that*

$$\mathcal{D} = \overline{\mathcal{D}} \circ \pi_X. \quad (\text{A.15})$$

Thus deleting every future-null state coordinate preserves such downstream outputs exactly.

Proof. The readout is constant on the fibres of the universal surjection (A.11); define $\overline{\mathcal{D}}$ on a fibre by the common value. Surjectivity gives uniqueness. $\square$

A.3 Conservative readable relational completion

The predictive quotient above determines persistence but does not by itself assert that the quotient class, scheduler path, orientation, and calibration are jointly readable at one cut. We now isolate the finite conservative extension used by the geometric reconstruction.

Let $\Phi_{a,*}$ be the support-minimal Schrödinger-picture branch map for an underlying event letter a . Let $z \in Z_X^{\min}$ be its predictive source class. For a finite family of added same-history marks $m = (\gamma, \xi, z')$, where γ is a scheduler-path record, ξ contains orientation and calibration marks, and z' is the next readable predictive class, let

$$K_{X,a}(m \mid z) \geq 0, \quad \sum_m K_{X,a}(m \mid z) = 1. \quad (\text{A.16})$$

On block states $\rho = (\rho_z)_z$, define

$$[\tilde{\Phi}_{a,m,*}(\rho)]_{z'} = \sum_z K_{X,a}(m \mid z) \Phi_{a,*}(\rho_z). \quad (\text{A.17})$$

The symbol in (A.17) denotes a same-history conditional extension, not a tensor-product decomposition of the state space.

The refined branch maps are completely positive because their coefficients are nonnegative. Their normalized forgetful identity is

$$F_X \left(\sum_m \tilde{\Phi}_{a,m,*}(\rho) \right) = \Phi_{a,*}(F_X \rho); \quad (\text{A.18})$$

This also supplies the explicit realization used in [Proposition 2.1](#).

Proof of Proposition 2.1. Complete positivity follows because every target block in (A.17) is a nonnegative scalar combination of completely positive branch maps. Summing (A.16) over the added marks gives one, and therefore summing the refined branches proves (A.18). Induction on word length gives exact preservation of every underlying finite cylinder.

For future minimality, define two raw readable packets to be equivalent when all declared future signatures agree. Prefixing the same admissible event to two equal future signatures leaves them equal, so the relation is a typed right congruence. Finite partition refinement terminates because every strict round increases the number of classes. The usual fibre argument then gives its universal property: any other sufficient readable presentation identifies only packets with equal complete future signatures and therefore surjects uniquely onto the stable quotient. $\square$

A.4 Finite distinguishing depth and the presentation category

Proposition A.11 (Finite predictive reconstruction with an explicit horizon convention). *For a finite-horizon protocol, include the remaining horizon in the cut type and compare all admissible continuations. The future quotient is finite and is a right congruence on every nonterminal positive-probability branch. For an unbounded-horizon protocol admitting a reachable finite presentation with m records, the coarsest predictive partition is obtained after at most $m - 1$ strict partition refinements. A stable finite partition is sufficient for equality of all finite futures. Finite alphabets without either a horizon bound or a finite-index hypothesis do not imply a finite quotient.*

Proof. At a fixed horizon there are finitely many histories. The conditional chain rule proves right congruence; inclusion of the remaining horizon in the type ensures that the continuation av and its suffix v are both legitimate in the corresponding comparison. For a finite presentation, initialize the partition by type, immediate read values, branch domains and probabilities. At each round, additionally require equal successor blocks for each letter of positive probability. Induction shows that the partition after r rounds records equality of the immediate data and of all future tests with at most r further successor operations. Each strict refinement increases the number of blocks, which is at most m . At a fixed point, induction on word length gives equality of every finite future. Conversely, future-equivalent records cannot be separated by any round. This proves termination and the claimed quotient. Exact comparisons of the table values are part of this algebraic statement.

For the last assertion, consider a binary renewal process whose probability of the next renewal, after n consecutive nonrenewals, is $1/(n+2)$. Every such history is reachable and its immediate next-event law differs from that at every other age. The alphabet is finite but the future quotient has infinitely many states. Thus finiteness of the quotient is not a consequence of alphabet size alone. $\square$

Proposition A.12 (Terminality and naturality of predictive reduction). *Fix a typed operational law. Let a morphism $f : S \rightarrow S'$ of reachable predictive presentations satisfy $s' = f \circ s$, preserve reads and branch probabilities, and intertwine every positive-probability successor. Then $Z_X^{\min}$ is terminal in this category. Isomorphisms of typed laws induce isomorphisms of their predictive quotients, compatibly with composition.*

Proof. If two histories have the same record in S , their complete futures agree, so $f_S(s(h)) = [h]$ is well defined. It preserves the immediate data and successors by the chain rule. Reachability forces every morphism from S to the quotient to have this value, proving terminality. An isomorphism of laws carries equal future signatures to equal future signatures and therefore descends to the quotient; the inverse descends as well. The descended maps respect composition because they agree on every represented history. $\square$

A.5 The state-conditioned Hankel quotient

Fix a minimum predictive state sector (x, z) . Let $P_{x,z}$ and $F_{x,z}$ denote the finite pasts and futures meeting at that sector, and set

$$\imath_{x,z}(f, p) = \Pr(f \circ p). \quad (\text{A.19})$$

For each past p define $h_p(f) = \imath_{x,z}(f, p)$ and

$$M_{x,z} = \text{span}_{\mathbb{R}}\{h_p : p \in P_{x,z}\}. \quad (\text{A.20})$$

Theorem A.13 (Canonical typed Hankel realization). *For every primitive event $a : (x, z) \rightarrow (y, T_{X,a}z)$ there is a unique linear map*

$$A_a : M_{x,z} \longrightarrow M_{y,T_{X,a}z}, \quad A_a h_p = h_{a \circ p}. \quad (\text{A.21})$$

Every future f defines $\lambda_f(m) = m(f)$ and

$$\imath(f, p) = \lambda_f(h_p), \quad \lambda_f A_a = \lambda_{f \circ a}. \quad (\text{A.22})$$

The realization is reachable and future separated. For any other linear realization $N_{x,z}$,

$$N_{x,z}^{\text{reach}} / N_{x,z}^{\text{unobs}} \cong M_{x,z}; \quad (\text{A.23})$$

therefore every linear realization has dimension at least $\dim M_{x,z}$, with equality exactly on the reachable future-separated branch.

Proof. If $\sum_j c_j h_{p_j} = 0$, then for every target future f ,

$$\sum_j c_j h_{a \circ p_j}(f) = \sum_j c_j (f \circ a, p_j) = 0.$$

Hence appending a respects every column relation and (A.21) is well defined. Reachability is built into (A.20); every nonzero response function is detected by some future. For another realization, map its reachable state n to the function $f \mapsto \mu_f(n)$. The image is $M_{x,z}$ and the kernel is exactly the future-unobservable subspace, proving (A.23). $\square$

This is the reachable/observable Hankel construction of rational-series realization theory [67], applied separately to the typed operational sectors. The maps A_a generate the table-native executable forward algebra. A Hilbert adjoint is not introduced at this stage; it is selected only after the positive process tensor has fixed the support-minimal Hilbert realization.

A.6 Comb tomography and support-minimal sequential realization

For each exposed Hermitian Choi space select a real basis $\{F_\alpha\}$ from a spanning physical frame, and let $\{D^\alpha\}$ be its trace-dual basis, so $\text{Tr}(D^\alpha F_\beta) = \delta_\beta^\alpha$. An overcomplete frame may instead be used with a dual reconstruction frame; the Kronecker identity is required only for the selected basis. For an N -step table define

$$R^{(N)} = \sum_{\alpha_1, \dots, \alpha_N} p(\alpha_1, \dots, \alpha_N) D^{\alpha_1} \otimes \dots \otimes D^{\alpha_N}. \quad (\text{A.24})$$

Theorem A.14 (Finite process-comb tomography). *The tensor $R^{(N)}$ in (A.24) is independent of the chosen spanning physical frame and is the unique Hermitian tensor representing all multilinear intervention probabilities. It is a deterministic finite quantum process exactly when*

$$R^{(N)} \succeq 0, \quad \text{Tr}_{2k-1} R^{(k)} = I_{2k-2} \otimes R^{(k-1)}, \quad R^{(0)} = 1. \quad (\text{A.25})$$

Here support minimality means that the dilation of every prefix channel is a minimal Stinespring dilation; all purifying memory is retained until the corresponding final discard. On this branch there are memories $A_k \cong \text{supp } R^{(k)}$, $A_0 = \mathbb{C}$, and isometries

$$V_k : H_{2k-2} \otimes A_{k-1} \longrightarrow H_{2k-1} \otimes A_k \quad (\text{A.26})$$

whose sequential link followed by final memory discard realizes $R^{(N)}$. Every other prefix-support-minimal sequential isometric realization satisfies

$$V'_k = (I \otimes U_k) V_k (I \otimes U_{k-1}^*) \quad (\text{A.27})$$

for unique memory unitaries U_k on these supports. Named operators retained as part of the compared record realization are transported by these same unitaries. The minimum predictive state remains a separate classical quotient; its sharp physical readability is not inferred from purification alone.

Proof. Products of the physical frames span the full Hermitian tensor space, so the dual-frame formula is the unique representing tensor. Positivity and the partial-trace recursion are necessary by complete positivity and causal normalization. Their sufficiency follows from the sequential construction below, in the comb realization framework [65]. The explicit support-minimal construction is also given in [73, Theorem 2].

Let $|\psi_k\rangle = |(R^{(k)})^{1/2}\rangle$ with purifying space $A_k = \overline{\text{supp } R^{(k)}}$. Its ancillary Schmidt support is all of A_k , and tracing A_k gives $R^{(k)}$. Suppose the prefix through $k-1$ has been realized. Tensor $|\psi_{k-1}\rangle$ with the unnormalized maximally entangled vector on the new input and a copy of that input. This is a minimal purification of $R^{(k-1)} \otimes I_{2k-2}$, with purifying space $A_{k-1} \otimes H'_{2k-2}$. Viewed over the same base space, $|\psi_k\rangle$ purifies exactly that operator because of (A.25); its purifying space is $H_{2k-1} \otimes A_k$. Schmidt decomposition therefore supplies a unique isometry from the first purifying support into the second. Identifying the input copy with H_{2k-2} gives V_k in (A.26). Induction and final discard realize the full comb.

The prefix purifying dimension cannot be below $\text{rank } R^{(k)}$, by Schmidt rank, and the construction attains that value. For two such realizations, minimal-purification uniqueness at prefix k gives a unique U_k relating their prefix vectors. Insert that relation at consecutive prefixes. Contraction with all external prefix vectors spans $H_{2k-2} \otimes A_{k-1}$, by prefix support minimality, and therefore gives the operator identity (A.27). This proves the stated sequential, rather than merely terminal, uniqueness. It does not assert uniqueness or the same memory cost for nonisometric realizations that discard or classically compress memory between rounds. $\square$

Remark A.15 (Why support reduction is necessary). An arbitrary nonminimal positive factorization may contain spectator directions. Taking ambient Hilbert adjoints before removing them can create mixed history words absent from the operational process. The support-minimal square-root class is therefore the canonical boundary at which a history C^* -algebra becomes well defined.

A.7 History algebra, contextual separation, and flat word Grams

Choose one support-minimal realization. Let $\mathcal{A}_X^{\text{raw}}$ be the finite-dimensional $*$ -algebra generated by the represented projections onto Z_X^{min} , primitive branch maps, and their Hilbert–Schmidt adjoints. A represented element is *contextually null* when every admitted left and right history context and every terminal Read has zero matrix coefficient on it. Let $\mathcal{J}_X^{\text{ctx}}$ be the set of all such elements.

Lemma A.16 (Contextual nullity is a two-sided $*$ -ideal). *$\mathcal{J}_X^{\text{ctx}}$ is the largest two-sided $*$ -ideal invisible in every admitted history context. Hence*

$$\mathcal{A}_X^{\text{hist}} = \mathcal{A}_X^{\text{raw}} / \mathcal{J}_X^{\text{ctx}} \quad (\text{A.28})$$

is the unique minimum context-preserving separated history envelope.

Proof. If a is null, inserting any represented word on either side only enlarges the tested context, and reversing the context shows that a^* is null. Thus the null set is a two-sided $*$ -ideal. Every context-invisible ideal is contained in it by definition, which gives the universal property. $\square$

For a finite-dimensional unital C^* -algebra $\mathcal{A}$, define the normalized regular trace

$$\widehat{\tau}_{\text{reg}}^{\mathcal{A}}(a) = \frac{\text{Tr}_{\mathbb{C}}(L_a)}{\dim_{\mathbb{C}} \mathcal{A}}, \quad L_a(b) = ab. \quad (\text{A.29})$$

If $\mathcal{A} \cong \bigoplus_{\lambda} M_{n_{\lambda}}(\mathbb{C})$, then $\text{Tr}_{\mathbb{C}}(L_a) = \sum_{\lambda} n_{\lambda} \text{Tr}(a_{\lambda})$, so $\widehat{\tau}_{\text{reg}}$ is a faithful tracial state invariant under every $*$ -isomorphism.

Let $\mathcal{W}_{X, \leq r}$ have basis the composable represented history words of length at most r , and let $W_r e_w = [w]$ in $\mathcal{A}_X^{\text{hist}}$. Define

$$\mathbb{K}_{X,r}(v, w) = \widehat{\tau}_{\text{reg}}([v]^* [w]). \quad (\text{A.30})$$

Theorem A.17 (Finite flat word reconstruction). *Every $\mathbb{K}_{X,r}$ is positive. There is a least r_X such that*

$$\text{rank } \mathbb{K}_{X,r_X} = \text{rank } \mathbb{K}_{X,r_X+1}. \quad (\text{A.31})$$

At this depth the represented word span is the whole history algebra, and the panel through depth $r_X + 1$ reconstructs the object units, minimum predictive state algebra, multiplication, involution, predictive core algebras, and every represented mixed history.

Proof. Positivity follows from $\sum_{v,w} \bar{c}_v c_w \mathbb{K}(v,w) = \hat{\tau}_{\text{reg}}(a^*a) \geq 0$. Faithfulness makes the Gram kernel exactly the represented word-relation space. The nested word spans stabilize because the history algebra is finite-dimensional. Equality of two consecutive ranks therefore means equality of two consecutive spans; the stabilized span contains the units and is invariant under left multiplication by every generator, so it equals the full generated algebra. The one-step extension reconstructs the generator multiplication matrices and reversed words reconstruct the involution. $\square$

A.8 Covariance of word moments and independent tensor composition

Proposition A.18 (Tensorial covariance and independent composition). *A change of word-space coordinates $c = C\tilde{c}$ transforms the Gram matrix by $\tilde{\mathbb{K}} = C^* \mathbb{K} C$. A record-preserving unitary equivalence of the support-minimal realization induces a $*$ -isomorphism of history algebras preserving all word moments.*

Suppose two independent operational processes have product preparations, independent future tests and independently accessible local operations, including identity operations. Assume joint reachability is the product of local reachability. With the product contextual test family, their composite has

$$Z_{12}^{\min} \cong Z_1^{\min} \times Z_2^{\min}, \quad \mathcal{A}_{12}^{\text{hist}} \cong \mathcal{A}_1^{\text{hist}} \otimes \mathcal{A}_2^{\text{hist}}. \quad (\text{A.32})$$

On a rectangular bank of independent word pairs,

$$\mathbb{K}_{12}((v_1, v_2), (w_1, w_2)) = \mathbb{K}_1(v_1, w_1) \mathbb{K}_2(v_2, w_2). \quad (\text{A.33})$$

These identifications respect reassociation and interchange of the independent factors. No factorization is asserted for correlated processes or for a restricted diagonal operation grammar.

Proof. The coordinate rule follows by substituting $c = C\tilde{c}$ in the quadratic form. A realizing unitary conjugates every generator and its adjoint, carries the contextual null ideal to the corresponding null ideal, and preserves the normalized regular trace under the resulting isomorphism. For independent processes, future probabilities factor. Equality of both marginal future signatures gives equality of the joint signature; conversely, a test on one factor with discard on the other detects any unequal marginal. Product reachability therefore identifies the joint quotient with the product.

Independently accessible generators contain $a \otimes I$ and $I \otimes b$, so they generate the tensor product of the two separated finite algebras. Product contexts separate this algebra: the spans of the local context functionals are the full dual spaces after contextual separation, and their products span the joint dual. There is consequently no further contextual ideal. Finally $L_{a \otimes b} = L_a \otimes L_b$ and $\dim(\mathcal{A}_1 \otimes \mathcal{A}_2) = \dim \mathcal{A}_1 \dim \mathcal{A}_2$, so the normalized regular trace is multiplicative. This proves the Gram identity. All identifications are specified on elementary histories and therefore commute with the usual associativity and interchange maps. $\square$

A.9 Proof of Theorem 2.2

Proof of Theorem 2.2. Item (i) is Theorems A.4 and A.8. Item (ii) is Theorem A.9. Item (iii) is Theorem A.13. Item (iv) is Theorem A.14. Item (v) is Lemma A.16 and Theorem A.17. Item (vi) follows from Theorem A.8 and Corollary A.10 and support-minimal unitary uniqueness. Item (vii) is Proposition 2.1. None of these constructions introduces an independent predictive state space, metric, connection, or spacetime dimension beyond the actual operational rows explicitly retained. $\square$

A.10 Relative minimality of the operational entrance

For the purpose of the foundational audit, keep the typed potential event grammar fixed and distinguish four semantic rows. Let $\mathfrak{D}$ denote the actual-occurrence interface selecting which potential events, writers, and Reads are physically present; let $\mathbb{P}^+$ denote the complete contextual predictive law, including the finite ancillary/matrix-order positivity and causal-recursion conditions used in the comb reconstruction; let Θ denote the protected orientation section; and let Λ denote the absolute physical calibration needed when durations, rates, lengths, or measures are interpreted in physical units. This separation is an audit of information content, not an enlargement of the operational model above.

Theorem A.19 (Relative primitive-basis minimality for the paper-level entrance). *Within the fixed category of finite typed operational-predictive models used in this paper, the semantic entrance*

$$[\mathfrak{G}; \mathfrak{D}, \mathbb{P}^+, \Theta, \Lambda] \tag{A.34}$$

has the following status.

- (R1) *The rows already assumed in the paper are sufficient for the foundational reconstruction actually used below: $\mathbb{P}^+$ and composition give the minimum future quotient and persistent predictive carrier, the contextual positive table gives the support-minimal causal process and history algebra, while Θ and Λ supply precisely the signed and absolute-scale information required on the Lorentzian ADM branch.*
- (R2) *Each of $\mathfrak{D}$, $\mathbb{P}^+$, Θ , and Λ is independent of the other three relative to the fixed typed grammar: for each forgotten row there are two anchored finite models with identical reducts and inequivalent full models.*
- (R3) *These four independence obstructions survive arbitrary common conservative refinement and cofinal reindexing; hence no functorial cutoff limit can reconstruct a forgotten row from the other three.*
- (R4) *Protected orientation can be absorbed into occurrence exactly when the occurrence packet contains a named actual Read together with a physical decoder selecting one orientation section. This is a definitional bundling, not an informational reduction of the orientation choice.*

Consequently the displayed semantic entrance is a minimal known sufficient and independent basis relative to the declared operational language and target class. No claim of absolute minimality among all conceivable foundational languages is made.

Proof. Sufficiency for the predictive and positive-process part is exactly [Theorems A.4, A.8, A.14](#) and [A.17](#), [Corollary A.6](#), and [Lemma A.16](#). On the geometric branch, [Lemmas B.10](#) and [B.12](#) and [Theorem B.11](#) show how the physically calibrated root race supplies the absolute kinetic data used by the ADM inversion, while [Lemmas B.13](#) and [B.15](#) and [Corollary B.16](#) use the protected sign and do not derive it from positive data.

For row-irredundancy it is enough to give one anchored separating pair for each row. For $\mathbb{P}^+$, fix one complete binary instrument with named outcomes a, b , keep every other row fixed, and compare Bernoulli laws $\Pr(a) = p$ and $\Pr(a) = q$ with $0 < p \neq q < 1$. Forgetting the predictive row makes the models identical, whereas any full anchored isomorphism fixes the outcome names and therefore cannot identify the two probabilities.

For $\mathfrak{D}$, adjoin a potential deterministic writer $W = 0$ whose mathematical extension is unique. In one model retain only the base event as actual; in the other declare W to be an actual readable writer as well. The structural and probabilistic data agree after the occurrence incidence is forgotten, but the full models differ on whether W belongs to the physical interface. Thus structural realizability, even when unique, does not imply actual occurrence.

For Θ , take the same anchored real two-plane with named basis e_1, e_2 and positive Gram I_2 in both models, but choose the unit alternating forms $\omega(e_1, e_2) = +1$ and $\omega(e_1, e_2) = -1$. Every positive Gram datum agrees. An anchored isomorphism fixes the ordered basis and cannot reverse the alternating form, so the protected orientation is not selected by the positive packet.

For Λ , fix a projective race with winner law p_a and an internal waiting variable $S \sim \text{Exp}(1)$ independent of the winner. Two models with the same grammar, occurrence, predictive law, and orientation but physical durations $T = c_0 S$ and $T = c_1 S$, $c_0 \neq c_1$, have identical uncalibrated reducts and different absolute rates $k_a = p_a/c_i$. Hence the common physical scale is not determined by normalized/projective race data. This is the same scale direction whose loss is visible in [Lemma B.10](#) when waiting-time calibration is omitted.

To see cofinal stability, tensor either member of any separating pair with the same strictly refining finite spectator process and let every cutoff map append and later discard only spectator records. The row-forgotten theories are then naturally isomorphic at every cutoff. If a full natural isomorphism appeared on a cofinal tail, composing it with spectator discard would give an anchored isomorphism of the original finite separating pair, a contradiction. Thus common refinement or passage to a cofinal limit cannot create the missing row.

Finally, if occurrence itself contains a named actual orientation Read and a deterministic physical decoder into the two-element orientation torsor, the decoded section and a separately named Θ determine one another. If the decoder is omitted, the two opposite sections above remain compatible with the same positive occurrence statistics. The former is therefore only a change of packaging, while the latter retains genuine orientation independence. $\square$

The missing calibration scale. Winner probabilities lose the common rate scale; the following result states the complete log-affine calibration freedom and the required anchors.

Theorem A.20 (Exact calibration fibre and minimum anchor bank). *Write all positive physical scales and additive origins in log-affine coordinates $x \in \mathbb{R}^N$. Suppose the physically established incidence relations are*

$$Lx = b. \tag{A.35}$$

If the solution set is nonempty and x_0 is one solution, then:

(K1) *the complete calibration fibre is*

$$x_0 + \mathcal{K}_L, \quad \mathcal{K}_L := \text{Ker } L, \tag{A.36}$$

of dimension $N - \text{rank } L$;

(K2) *exactly $\dim \mathcal{K}_L$ independent scalar anchors are necessary and sufficient to determine one calibration in this affine fibre;*

(K3) *if $U : \mathbb{R}^d \rightarrow \mathbb{R}^N$ encodes conventional changes of base units with $LU = 0$, the intrinsic physical residual is*

$$\mathcal{K}_L^{\text{phys}} = \mathcal{K}_L / \text{Im } U, \quad (\text{A.37})$$

and one anchor per independent quotient direction is necessary and sufficient;

(K4) *winner probabilities for a root race instantiate a one-dimensional common-rate direction of this type: multiplying all directed rates by the same positive constant leaves the winner law unchanged, whereas the waiting-time scale separates that direction.*

Proof. If $Lx = Lx_0 = b$, then $x - x_0 \in \text{Ker } L$, and conversely every $x_0 + k$, $k \in \text{Ker } L$, solves (A.35); this gives (A.36) and its dimension by rank-nullity. A basis of $\mathcal{K}_L$ and the corresponding dual linear functionals give a sufficient scalar anchor bank. Any smaller bank has nontrivial common kernel on $\mathcal{K}_L$ and is therefore insufficient. Quotienting the subgroup of pure unit conventions gives (A.37) and the same argument on the quotient gives the intrinsic anchor count. For the root race, the winner law is $p_b = k_b / \sum_c k_c$ and is invariant under $k_b \mapsto ck_b$; the exponential waiting time has total rate $\sum_c k_c$ and therefore fixes the missing common scale. $\square$

B Minimal relational dimension and ADM reconstruction

This appendix supplies the algebraic and statistical details of [Theorem 3.6](#). It assumes the carrier of Section 2, keeps the two relational selectors separate, and then treats calibration, positive realizability, signed extension and Clifford representation.

B.1 Endpoint-cycle balance and the simplex comparison

The primary regular-graph classification is proved directly in [Proposition 3.1](#). The stronger simplex category is developed next; its extra assumptions are not used in that proof.

Assumption B.1 (Minimal faithful predictive relation branch). A local relation event has one future-readable source and one future-readable target. Physical reversal retains both orientations. Future-minimality identifies parallel relation labels exactly when every continuation identifies them. A nonempty relation set is invariant under a group acting transitively on unordered endpoint pairs.

Lemma B.2 (Pair-homogeneous completion). *Under [Assumption B.1](#), the relation-colour-free local skeleton is the complete graph K_N on the endpoint set.*

Proof. The selected branch of the derived persistent predictive carrier contains exactly one future-readable source and one future-readable target, so one occurring relation gives one unoriented pair after physical reversal. The orbit of that pair under a group transitive on $\binom{X}{2}$ is the complete set of unordered pairs. Future-minimality removes only unread parallel labels, not distinct endpoint pairs. Hence the skeleton is K_N . $\square$

Let $X = \{1, \dots, N\}$, let $u_0 = N^{-1/2} \mathbf{1}_N$, and write $\mathbb{R}^N = \mathbb{R}u_0 \oplus W_N$. Identify the signed edge $[i \rightarrow j]$ with $e_i \wedge e_j \in \Lambda^2 \mathbb{R}^N$. The boundary map is $\partial(e_i \wedge e_j) = e_j - e_i$.

Theorem B.3 (Cut-cycle decomposition). *There is an orthogonal decomposition*

$$C_1(K_N) = u_0 \wedge W_N \oplus \Lambda^2 W_N, \quad (\text{B.1})$$

with

$$\partial(u_0 \wedge w) = \sqrt{N} w, \quad \text{Ker } \partial = \Lambda^2 W_N. \quad (\text{B.2})$$

Consequently

$$\dim \text{Ran } \partial = N - 1, \quad \dim \text{Ker } \partial = \binom{N-1}{2}. \quad (\text{B.3})$$

Proof. The exterior square of $\mathbb{R}^N = \mathbb{R}u_0 \oplus W_N$ is $u_0 \wedge W_N \oplus \Lambda^2 W_N$. For $x, y \in \mathbb{R}^N$,

$$\partial(x \wedge y) = \left(\sum_i x_i \right) y - \left(\sum_i y_i \right) x.$$

Taking $x = u_0, y = w \in W_N$ gives the first identity. If both vectors lie in W_N , both coordinate sums vanish, so the second summand is in the kernel. Dimension counting shows that it is the whole kernel and gives (B.3). $\square$

Remark B.4 (Metric tomography does not select N). The root dyadics of a complete simplex define an isomorphism

$$\mathbb{R}^{\binom{N}{2}} \longrightarrow \text{Sym}(W_N), \quad (a_{ij}) \longmapsto \sum_{i < j} a_{ij} (e_j - e_i)(e_j - e_i)^\top$$

for every N . Indeed, a vanishing weighted sum is a graph Laplacian whose off-diagonal entries are $-a_{ij}$. Thus the equality $\binom{N}{2} = \dim \text{Sym}(W_N)$ is universal and cannot by itself select three spatial dimensions. The paper uses two different non-tomographic selectors: exact endpoint-cycle source-rank balance, and the stronger alternating/minimal-source simplex criterion.

Remark B.5 (Conditional character of the selector). The alternating form in the next theorem is part of the definition of the selected relational category. It is not reconstructed from the positive word Grams and does not, by itself, specify a metric, a Hamiltonian, a symplectic phase space of continuum fields, or a Lorentzian sign. Accordingly the theorem asserts uniqueness only among faithful pair-homogeneous simplex cells satisfying this alternating/minimal-source query. It is the second dimensional selector, not the sole derivation of K_4 ; the regular-graph source-balance result below uses no alternating form.

Theorem B.6 (Minimal faithful K_4 selector on the alternating simplex branch). *Assume, in addition to Assumption B.1, that:*

- (D1) *the temporal-spatial coefficient space is $\mathbb{R}t \oplus W_N$ and carries a nondegenerate alternating form;*
- (D2) *the endpoint and complete signed-edge source Grams are faithful;*
- (D3) *the cycle source is nonzero;*
- (D4) *among conservative completions satisfying these queries, the future-readable source innovation is minimal.*

Then

$$N = 4, \quad G_{\text{rel}} = K_4, \quad W_{\text{sp}} = W_4 \cong \mathbb{R}^3. \quad (\text{B.4})$$

The faithful endpoint and edge source ranks are $r_0(N) = N - 1$ and $r_1(N) = \binom{N}{2}$.

Proof. The alternating coefficient space has dimension $1 + \dim W_N = N$. A nondegenerate alternating form exists only in even dimension, so N is even. A nonzero cycle source requires $\Lambda^2 W_N \neq 0$, hence $N \geq 3$; together with evenness this gives $N \geq 4$. On the even integers $N \geq 4$, both $N - 1$ and $\binom{N}{2}$ are strictly increasing. The unique minimum of the pair of faithful source ranks is therefore attained at $N = 4$. The graph conclusion follows from [Lemma B.2](#), and $\dim W_4 = 3$. $\square$

Remark B.7 (The minimization category). The minimum in [Theorem B.6](#) is taken over the declared admissible simplex cells, with source cost ordered by the pair $(N - 1, \binom{N}{2})$; equivalently one may use any cost strictly increasing in both faithful ranks. It is not a theorem that a fixed law with more future-distinguishable endpoints has a conservative four-endpoint compression. Such a compression would contradict faithfulness unless the additional distinctions were genuinely unread. The independent endpoint–cycle balance condition is not inferred from this minimum rule.

A representation-theoretic strengthening explains why the cycle packet is particularly economical at $N = 4$. After complexification, $W_N \cong S^{(N-1,1)}$, $\Lambda^2 W_N \cong S^{(N-2,1,1)}$, and $W_N \otimes \text{sgn} \cong S^{(2,1^{N-2})}$. These Young diagrams coincide exactly for $N = 4$. Hence

$$\text{Hom}_{S_N}(\Lambda^2 W_N, W_N \otimes \text{sgn}) = \begin{cases} \mathbb{C}, & N = 4, \\ 0, & N \neq 4, \end{cases} \quad (\text{B.5})$$

on the complexified nontrivial simplex branch; the corresponding real intertwiner space is one-dimensional over $\mathbb{R}$ at $N = 4$. Thus only K_4 identifies the entire cycle representation with one oriented spatial copy rather than an additional inequivalent source type.

The source-equivalence condition is

$$\Lambda^2 W_N \cong W_N \otimes \text{sgn}, \quad N = 4. \quad (\text{B.6})$$

Proposition B.8 (Faithful orientation-twisted source equivalence). *For $N \geq 3$, an S_N -equivariant isomorphism $\Lambda^2 W_N \rightarrow W_N \otimes \text{sgn}$ exists if and only if $N = 4$. At $N = 4$ its space is one dimensional. With the standard invariant Euclidean forms, an isometric choice is unique up to sign.*

Proof. An isomorphism requires $(N - 1)(N - 2)/2 = N - 1$, which gives $N = 4$. On the three-dimensional space W_4 , the Hodge identification $\Lambda^2 W_4 \rightarrow W_4$ satisfies $*(gu \wedge gv) = \det(g)g*(u \wedge v)$. The determinant of the standard permutation representation restricted to W_4 is $\text{sgn}(g)$, since its complementary constant line is fixed. Thus Hodge identification is the required twisted intertwiner. The complexified standard representation is irreducible and its commuting endomorphisms are scalar; composing any two intertwiners proves uniqueness up to a real scalar. Normalization of the invariant forms restricts that scalar to ± 1 . $\square$

B.2 The A_3 source frame

In the anchored basis of [\(3.6\)](#), define $Q_a = r_a r_a^\top$.

Lemma B.9 (Root dyadics span Sym_3). *The six matrices $Q_1, \dots, Q_6$ form a basis of Sym_3 .*

Proof. Q_1, Q_2, Q_3 are the three diagonal matrix units. For $i < j$,

$$(e_j - e_i)(e_j - e_i)^\top - e_i e_i^\top - e_j e_j^\top = -(e_i e_j^\top + e_j e_i^\top).$$

The three difference roots therefore recover the three symmetric off-diagonal matrix units. The six matrices are linearly independent and span the six-dimensional space Sym_3 . $\square$

Let the twelve oriented roots race with positive rates $k_a^\pm$. Put $K = \sum_a (k_a^+ + k_a^-)$, let A be the winning oriented root, and let T be the first waiting time.

Lemma B.10 (Winner–waiting score is faithful). *For each oriented branch b , the log-rate score is*

$$s_b = \mathbf{1}_{\{A=b\}} - k_b T. \quad (\text{B.7})$$

Its Gram matrix is diagonal and positive:

$$\mathbb{E}[s_b s_c] = \delta_{bc} \frac{k_b}{K}. \quad (\text{B.8})$$

Winner probabilities alone have covariance $\text{diag}(p) - pp^\top$ and therefore lose the common rate scale.

Proof. For independent exponential clocks, $T \sim \text{Exp}(K)$, the winner has probability $p_b = k_b/K$, and A and T are independent. Therefore

$$\begin{aligned} \mathbb{E}[s_b s_c] &= \delta_{bc} p_b - p_b k_c \mathbb{E}T - k_b p_c \mathbb{E}T + k_b k_c \mathbb{E}T^2 \\ &= \delta_{bc} \frac{k_b}{K} - 2 \frac{k_b k_c}{K^2} + 2 \frac{k_b k_c}{K^2}, \end{aligned}$$

which proves (B.8). The covariance of the winner indicator is singular in the all-ones rate direction because probabilities are unchanged by common rescaling. $\square$

B.3 Positive-rate realizability

The dyadic span does not make every positive matrix a positive rate sum. For the cone (3.10), the explicit inversion is

$$\begin{aligned} s_4 &= -\mathcal{B}_{12}, & s_5 &= -\mathcal{B}_{13}, & s_6 &= -\mathcal{B}_{23}, \\ s_1 &= \mathcal{B}_{11} + \mathcal{B}_{12} + \mathcal{B}_{13}, & s_2 &= \mathcal{B}_{22} + \mathcal{B}_{12} + \mathcal{B}_{23}, & s_3 &= \mathcal{B}_{33} + \mathcal{B}_{13} + \mathcal{B}_{23}. \end{aligned} \quad (\text{B.9})$$

These six strict inequalities are a finite realizability test. The ADM formula is an analytic bijection on abstract positive bracket–density pairs; its realization by the declared positive race is restricted to (3.10). In the orthonormal D_3 coordinates used for the constructive small-data class, the isotropic bracket is an interior point of the corresponding cone. A bounded shift also has a positive directed lift at sufficiently fine cutoff: for a fixed right inverse J_R of $R = (r_1 \cdots r_6)$, choose $d = h^{-1} J_R \beta$ and $k_a^\pm = (s_a/h^2 \pm d_a)/2$. Positivity holds whenever $h|(J_R \beta)_a| < s_a$. Thus local positive realizability follows from an interior margin, not from the dimension of the dyadic span alone.

B.4 ADM inversion and scheduler scale

Define $\mathcal{B}_h$ and β_h by (3.7).

Theorem B.11 (Complete finite ADM frame). *The six rate sums map isomorphically to $\mathcal{B}_h \in \text{Sym}_3$. The six rate differences map with rank three onto $\beta_h \in \mathbb{R}^3$ and have a three-dimensional kernel, the tetrahedral circulation space. For every $\mathcal{B}_h \succ 0$ and $\varrho_h > 0$, the formula (3.8) is the unique solution of*

$$\mathcal{B}_h = N_h^2 g_h^{-1}, \quad \varrho_h = \sqrt{\det g_h}. \quad (\text{B.10})$$

Hence the root rates, waiting-time scale, and speed density reconstruct the six spatial metric components, three shift components, and lapse.

Proof. The first assertion is Lemma B.9. The difference map has image containing r_1, r_2, r_3 , hence has rank three; rank-nullity gives a three-dimensional kernel. For the inversion, take determinants in $\mathcal{B} = N^2 g^{-1}$: $\det \mathcal{B} = N^6 / (\det g) = N^6 / \varrho^2$. Thus $N = \varrho^{1/3} (\det \mathcal{B})^{1/6}$. Substituting back gives $g = N^2 \mathcal{B}^{-1}$, which is exactly (3.8). Positivity makes the solution unique. $\square$

Lemma B.12 (Fast-clock lower bound). *Suppose $\|r_a\| \leq R$ and $\mathcal{B}_h \succeq cI_3$. Then*

$$\sum_a (k_a^+ + k_a^-) \geq \frac{3c}{R^2 h^2}. \quad (\text{B.11})$$

In particular, a total rate $O(h^{-1})$ produces $\mathcal{B}_h = O(h)$ and a collapsed continuum bracket.

Proof. Taking traces in (3.7),

$$3c \leq \text{tr } \mathcal{B}_h \leq R^2 h^2 \sum_a (k_a^+ + k_a^-).$$

This gives (B.11). The final assertion follows by substitution. $\square$

B.5 Finite Krein orientation from the protected sign

Let $\epsilon_X \in \{+1, -1\}$ be the actual protected orientation Read of (3.11) on the finite support-minimal relational carrier $\mathcal{H}_J$. Functional calculus of this classical two-point Read gives orthogonal projections $P_\pm$ with $P_+ + P_- = I$; define $J = P_+ - P_-$. Thus

$$\mathcal{H}_J = \mathcal{H}_J^+ \oplus \mathcal{H}_J^-, \quad \mathcal{H}_J^\pm = \text{Ran } P_\pm, \quad J|_{\mathcal{H}_J^\pm} = \pm I. \quad (\text{B.12})$$

Assume both summands are nonzero. Define

$$[u, v]_J = \langle u, Jv \rangle. \quad (\text{B.13})$$

Then $[\cdot, \cdot]_J$ is a nondegenerate Hermitian form with one positive and one negative sector in the orientation factor, hence a finite Krein structure. This statement uses only the protected sign involution and the support-minimal Hilbert realization; no continuum metric is involved.

Lemma B.13 (Positive geometry and Krein signature are distinct layers). *The positive word-Gram hierarchy of Theorem A.17 remains positive after adjoining the protected orientation record, while the indefinite sign is carried entirely by J . Consequently no positive combination of the spatial root dyadics can replace the Krein involution, and the Lorentzian signature of the principal symbol requires the signed layer in addition to the A_3 bracket.*

Proof. The word-Gram hierarchy is a Gram matrix for the faithful positive regular trace and is therefore positive semidefinite. Likewise every spatial bracket is a positive sum $\sum_a c_a r_a r_a^\top$ with $c_a > 0$. Such a form has no negative direction. By contrast, J has both $+1$ and -1 spectral subspaces by (B.12); the form (B.13) is therefore indefinite. The two structures are independent in exactly the required sense: the spatial geometry remains positive while J supplies the sign subsequently represented by the temporal Clifford generator. $\square$

Theorem B.14 (Irreducible sign and Lorentzian inertia before Clifford realization). *Let W_4 carry the positive spatial form reconstructed by Theorem B.11. The complete positive Grand-Tensor packet cannot supply a negative direction, while the protected orientation row is independent of that packet by Theorem A.19. If the geometric extension retains the positive restriction on W_4 and satisfies Definition 3.4 on exactly one independent line $\mathbb{R}t$, then, up to overall sign,*

$$q(\tau t + w) = -a^2 \tau^2 + g(w, w), \quad a > 0, \quad \text{inertia}(q) = (1, 3). \quad (\text{B.14})$$

The conclusion depends only on positivity of the spatial form, independence of the signed row, and one-line minimality; it does not depend on a choice of gamma matrices.

Proof. The positive word-Gram and root-race forms have no negative eigenvalues by Lemma B.13. The separating pair in the proof of Theorem A.19 shows that reversing the protected orientation leaves all positive operational data unchanged, so its orientation is not a derived eigenvalue of the positive packet. Assignment of a negative quadratic form to the added line is the separate signed-extension condition. A minimal extension introduces one real line and no additional signed directions. Nondegeneracy makes its coefficient nonzero; choosing the relative sign opposite to the positive spatial restriction gives one negative eigenvalue and the three positive eigenvalues of g . Sylvester's law of inertia makes this count invariant under every real change of basis. Multiplying the entire form by -1 reverses the sign convention but not the Lorentzian type. No Clifford representation has been used. $\square$

B.6 Clifford realization of the signed quadratic form

The inertia has already been fixed by Theorem B.14. We now give the explicit representation used in the geometric reconstruction.

The matrix representation is now introduced only after Proposition 3.5. A minimal depth-two realization uses binary factors A, B to represent the four Clifford axes. The physical sign content remains the single Read (3.11); the second matrix factor is representation-theoretic multiplicity and is not a second operational time-orientation choice. Set

$$\gamma^0 = -iX_A \otimes I_B, \quad \gamma^1 = Z_A \otimes Z_B, \quad \gamma^2 = Z_A \otimes Y_B, \quad \gamma^3 = Y_A \otimes I_B. \quad (\text{B.15})$$

They satisfy $(\gamma^0)^2 = -I$, $(\gamma^a)^2 = I$, and $\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 0$ for $\mu \neq \nu$; together they generate $M_4(\mathbb{C}) \cong \text{Cl}_{1,3}(\mathbb{C})$. If $E_h^\top E_h = g_h^{-1}$, the principal symbol

$$\sigma_h(\xi) = \gamma^0 N_h^{-1}(\xi_0 - \beta_h \cdot \vec{\xi}) + \gamma^a (E_h)_a^i \xi_i \quad (\text{B.16})$$

has square

$$\sigma_h(\xi)^2 = \left[-N_h^{-2}(\xi_0 - \beta_h \cdot \vec{\xi})^2 + g_h^{ij} \xi_i \xi_j \right] I. \quad (\text{B.17})$$

The Clifford cell therefore represents the Lorentzian form fixed in (3.13); it does not create its one-minus-three-plus signature.

Lemma B.15 (Depth-two Lorentzian Clifford cell). *The matrices γ^μ satisfy*

$$(\gamma^0)^2 = -I, \quad (\gamma^a)^2 = I, \quad \{\gamma^\mu, \gamma^\nu\} = 0 \quad (\mu \neq \nu), \quad (\text{B.18})$$

and generate $M_4(\mathbb{C}) \cong \text{Cl}_{1,3}(\mathbb{C})$.

Proof. The square relations follow from $X^2 = Y^2 = Z^2 = I$. Every mixed product contains exactly one local Pauli anticommutation, so the mixed anticommutators vanish. The generators contain $X_A \otimes I$ and $Y_A \otimes I$, hence also $Z_A \otimes I$. Multiplication by γ^1 and γ^2 then gives $I \otimes Z_B$ and $I \otimes Y_B$, hence $I \otimes X_B$. Both local matrix factors are generated, so the total algebra is $M_2(\mathbb{C}) \otimes M_2(\mathbb{C}) = M_4(\mathbb{C})$. $\square$

Corollary B.16 (Lorentzian principal square). *Let $E_h^\top E_h = g_h^{-1}$ and define σ_h by (B.16). Then (B.17) holds. If $g_h \succ 0$ and $N_h > 0$, the characteristic quadratic form realizes the already determined inertia $(1, 3)$ in the convention $(-, +, +, +)$.*

Proof. The temporal-spatial cross terms vanish by the Clifford anticommutation relations. The temporal square contributes $-N_h^{-2}(\xi_0 - \beta_h \cdot \vec{\xi})^2$, while the spatial square is $\delta^{ab}(E_h)_a^i (E_h)_b^j \xi_i \xi_j = g_h^{ij} \xi_i \xi_j$. $\square$

B.7 Cofinal transport and proof of Theorem 3.6

For adjacent cutoffs, let the complete old/detail positive form have block matrix

$$M_Y = \begin{pmatrix} A_{Y/X} & B_{Y/X} \\ B_{Y/X}^* & D_{Y/X} \end{pmatrix}, \quad (\text{B.19})$$

and Schur complement $S_{Y/X} = A_{Y/X} - B_{Y/X} D_{Y/X}^\dagger B_{Y/X}^*$. Define

$$\sigma_{Y/X} = \|S_{Y/X} - M_X\|_{\text{op}}, \quad \theta_{Y/X} = \|B_{Y/X} D_{Y/X}^\dagger\|_{\text{op}}, \quad \mu_{Y/X} = \lambda_{\min}^+(D_{Y/X}). \quad (\text{B.20})$$

The harmonic lift

$$\mathcal{H}_{Y/X} u = J_{Y/X} u - K_{Y/X} D_{Y/X}^\dagger B_{Y/X}^* u \quad (\text{B.21})$$

minimizes the fine energy over detail completions and has energy $\langle u, S_{Y/X} u \rangle$. This is ordinary completion of the square.

If

$$\sum_X (\sigma_X + \theta_X + \mu_X^{-1}) < \infty, \quad (\text{B.22})$$

then the finite forms and their regular isolated spectral projections are Cauchy after harmonic transport. In particular, a separated threshold rank, a positive pair gap, and a finite-dimensional Clifford source persist. This is the finite-dimensional core of the common-screen limit used here.

Proof of Theorem 3.6. The predictive carrier is supplied by Theorem A.9. For the regular-graph route, the local dimension follows directly from Proposition 3.1. For the pair-homogeneous simplex route, the local graph is complete by Lemma B.2 and the same K_4/A_3 module follows from Theorems B.3 and B.6. Thus the alternating form is needed only on the second route. The root sums and differences, faithful rate coordinates, ADM inversion, and clock lower bound are Lemmas B.9, B.10 and B.12 and Theorem B.11. The positive/signed separation is Lemma B.13; Theorem B.14 fixes Lorentzian inertia before the Clifford matrices are introduced, and Lemma B.15

and [Corollary B.16](#) realizes that form. Finally, [\(B.22\)](#) makes every corrected finite matrix packet Cauchy in operator norm on each protected finite screen; spectral separation therefore preserves the ranks and positive gaps, while the Schur mismatch and leakage tails vanish. The canonical limit quotient consequently retains the K_4/A_3 module, the physically occurring temporal calibration, the ADM packet, and the Lorentzian Clifford cell. $\square$

Remark B.17 (Exact dimension scope). The result applies to the derived future-minimal, one-source/one-target, reciprocal, pair-homogeneous, faithful, and minimal faithful branch. A deliberately higher-arity boundary, a relation-coloured theory, a nonfaithful cycle quotient, or an enriched cross-product packet changes the primitive query and is not a counterexample. In particular, a seven-dimensional vector cross product $\Lambda^2 W \rightarrow W$ has a fourteen-dimensional kernel and requires an additional stable G_2 three-form; it is an enriched theory rather than an equal-cost realization of the faithful K_4 branch.

C Action reconstruction, stationarity and gravitational classification

This appendix proves the finite action and stationarity statements of [Section 4](#). The determining-field quotient, renewal pressure and common action are reconstructed before comparison with the accepted law; the final steps classify the restricted local representative and control torsion. Continuum estimates are in [Appendix D](#).

C.1 Accepted determining-field quotient and kernel

Let $\tilde{\Theta}_X$ be the finite accepted ledger of [Equation \(4.1\)](#). Two states x, y are future equivalent when every declared determining coordinate and every admitted future word agrees from x and y , including all branch-domain indicators. Let $\Theta_{\text{fld}, X}^{\min}$ be the quotient.

Proof of [Theorem 4.1](#). Future equality is a typed right congruence. If $x \sim_{\text{fld}, X} y$, then for every primitive branch a and continuation w , every determining read satisfies

$$d(u_w u_a(x)) = d(u_{wa}(x)) = d(u_{wa}(y)) = d(u_w u_a(y)),$$

so every primitive update descends to the quotient. At each refinement round, split two states whenever an immediate coordinate differs or one primitive update reaches different blocks. The k th partition identifies precisely the states not separated by a future of length at most k . Every strict round increases the number of finite blocks, hence the refinement terminates. Its stable partition has the universal property by the usual fibre argument.

Accepted branch probabilities and target quotient classes are constant on the stable fibres. Summing the resolved accepted branches whose target lies in a specified quotient class gives [\(4.2\)](#), and these sums are exactly the probabilities of the corresponding post-accepted Reads. Uniqueness follows because those Reads separate the quotient classes. $\square$

Remark C.1 (Occurrence boundary). The theorem reconstructs the dynamical state and kernel from the *complete* accepted cylinder. It does not reconstruct that cylinder from an accepted binary clock alone. Indeed, an identity accepted kernel and a full-support reset kernel can be coupled to the same state-independent acceptance bit while producing different terminal field laws. Thus occurrence of the complete accepted field record is a genuine physical input row.

C.2 Renewal pressure and the common scalar action

Let (Ω_X, p_X) be a faithful marked first-return law, let $\tau_X > 0$ be the protected physical duration, and let $\mathcal{G}_X : E_X \rightarrow L^2(\Omega_X, p_X)$ be the reward synthesis of (4.3). Put

$$\mathcal{L}_X(q, r) = \mathbb{E}_X[e^{-g_{q,X} - r\tau_X}], \quad \bar{\tau}_X = \mathbb{E}_X \tau_X, \quad T_X c = c\tau_X.$$

For the score, differentiating the tilted law also gives

$$S_X v = -\mathcal{G}_X v - \pi_X(v)\tau_X. \quad (\text{C.1})$$

Proof of Theorem 4.2. The transform is a finite sum of known probabilities times known protected writer values. At the origin, $\mathcal{L}_X(0, 0) = 1$ and $\partial_r \mathcal{L}_X(0, 0) = -\bar{\tau}_X < 0$, so the analytic implicit-function theorem gives the pressure root. Differentiating $\mathcal{L}_X(q, \mathcal{P}_X(q)) = 1$ in direction v gives

$$0 = -\mathbb{E}_X g_{v,X} - \pi_X(v)\bar{\tau}_X,$$

which proves the first identity. Differentiating the normalized tilted law gives the score formula and hence the synthesis reconstruction. $\square$

Corollary C.2 (Pressure-jet reconstruction of mixed action blocks). *With $\bar{\tau}_X(q) = \mathbb{E}_{X,q} \tau_X$, $\delta_X = D\bar{\tau}_X(0)$, and $m_{2,X} = \mathbb{E}_X \tau_X^2$,*

$$D^2 \mathcal{P}_X(0) = \bar{\tau}_X^{-1} S_X^* S_X, \quad \delta_X = T_X^* S_X, \quad (\text{C.2})$$

$$\mathcal{G}_X^* \mathcal{G}_X = \bar{\tau}_X D^2 \mathcal{P}_X(0) + \delta_X^* \pi_X + \pi_X^* \delta_X + m_{2,X} \pi_X^* \pi_X. \quad (\text{C.3})$$

Consequently one pressure surface reconstructs every marginal and mixed action block represented on the same marked cycle record.

Proof. Differentiate the pressure normalization twice. The first factors are the centered scores $-S_X u$ and $-S_X v$, which gives the first identity. Differentiating the tilted mean duration gives $\delta_X = T_X^* S_X$; expanding $\mathcal{G}_X = -S_X - T_X \pi_X$ gives the second formula. $\square$

Let $\mathcal{V}_X$ be the finite protected variation complex, with coboundaries $d_0 : C^0 \rightarrow C^1$ and $d_1 : C^1 \rightarrow C^2$, $d_1 d_0 = 0$, and let $\alpha_X \in C^1(\mathcal{V}_X; \mathbb{R})$ be the reconstructed physical action increment.

Proof of Theorem 4.3. If $\alpha_X = d_0 \mathcal{S}_X$, telescoping gives zero face sums and zero periods. Conversely, fix a base configuration in each connected component and define $\mathcal{S}_X(q)$ by summing α_X along a path from the base to q . Face closure makes the sum invariant under elementary 2-cell moves and the period conditions remove the remaining homology class. One anchor on each connected component removes its additive constant. For the estimate use the finite Hodge decomposition $C^1 = \text{Ran } d_0 \oplus \mathcal{H}^1 \oplus \text{Ran } d_1^*$ and the singular-value bound $\|d_1 \beta\| \geq \sigma_{\text{coex}} \|\beta\|$ on the coexact range. $\square$

Remark C.3 (Provenance is not stationarity). Reconstructing a common action does not prove that the observed accepted transition is generated by it. The accepted/action comparison below is an independent same-history test. Conversely, fitting a cost to the observed kernel after the fact would be tautological and has no stationarity content.

C.3 Fisher–Bregman stationarity and physical test lifts

Let Θ , Ψ , G , $\mathcal{A}$, and η satisfy the convex-chart and interiority hypotheses of [Theorem 4.5](#). Define D_Ψ , Φ_θ , y_θ^* , Δ_Ψ , and $\bar{\Delta}_\Psi$ by (4.11)–(4.14).

Theorem C.4 (Quantitative accepted-action stationarity). *Put*

$$\mu_\Phi = (\eta^{-1} - \lambda)m > 0, \quad \Lambda_\Phi = (\eta^{-1} + L)M, \quad \ell = \max\{\lambda, L\}. \quad (\text{C.4})$$

Then

$$\int D_\Psi(y, \theta) \nu(d\theta) K^{\text{acc}}(dy \mid \theta) \leq \eta \bar{\Delta}_\Psi, \quad (\text{C.5})$$

$$\int \|y - \theta\|^2 \nu(d\theta) K^{\text{acc}}(dy \mid \theta) \leq \frac{2\eta}{m} \bar{\Delta}_\Psi, \quad (\text{C.6})$$

and

$$\int \mathfrak{S}(\theta) \nu(d\theta) \leq C_{\Psi, \mathcal{A}, \eta} \bar{\Delta}_\Psi, \quad (\text{C.7})$$

where one admissible constant is

$$C_{\Psi, \mathcal{A}, \eta} = \frac{4\Lambda_\Phi^2}{m\mu_\Phi} + \frac{4M^2(1 + \eta\ell)^2}{m^2\eta}. \quad (\text{C.8})$$

Proof. State independence of $\vartheta > 0$ and stationarity of ν imply $\nu K^{\text{acc}} = \nu$. Since y_θ^* minimizes Φ_θ and $\Phi_\theta(\theta) = \mathcal{A}(\theta)$,

$$\mathcal{A}(y) - \mathcal{A}(\theta) + \eta^{-1} D_\Psi(y, \theta) \leq \Delta_\Psi(\theta, y).$$

The action difference cancels on averaging, proving (C.5). The lower Hessian bound gives $D_\Psi(y, \theta) \geq m\|y - \theta\|^2/2$, hence (C.6).

The Hessian of Φ_θ lies between $\mu_\Phi I$ and $\Lambda_\Phi I$ on the convex chart. Interiority of y_θ^* gives $\nabla \Phi_\theta(y_\theta^*) = 0$. Strong convexity and the upper Hessian bound therefore imply

$$\|y - y_\theta^*\|^2 \leq \frac{2\Delta_\Psi(\theta, y)}{\mu_\Phi}, \quad \|\nabla \Phi_\theta(y)\|^2 \leq \frac{2\Lambda_\Phi^2}{\mu_\Phi} \Delta_\Psi(\theta, y).$$

For

$$r_\Psi(\theta, y) = \eta \nabla \Phi_\theta(y) = \nabla \Psi(y) - \nabla \Psi(\theta) + \eta \nabla \mathcal{A}(y)$$

we have

$$\eta \nabla \mathcal{A}(\theta) = r_\Psi - [\nabla \Psi(y) - \nabla \Psi(\theta)] + \eta [\nabla \mathcal{A}(\theta) - \nabla \mathcal{A}(y)].$$

The last two terms together have norm at most $M(1 + \eta\ell)\|y - \theta\|$. Squaring with $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$, using $G^{-1} \preceq m^{-1}I$, and averaging yields (C.7) with (C.8). $\square$

Remark C.5 (Boundary minima). The first two inequalities do not require interiority, but the ordinary gradient conclusion does. For example, on $\Theta = [0, 1]$ with $\Psi(x) = x^2/2$ and $\mathcal{A}(x) = x$, the source $\theta = 0$ has proximal minimum $y_\theta^* = 0$. The stationary identity transition supported at zero has zero gap although $\mathcal{A}'(0) = 1$. A boundary formulation must retain the normal-cone or projected variational residual. It cannot be substituted for unrestricted determining metric variations without checking their admissibility.

Proposition C.6 (Action-gap control for nonstationary marginals). *Let $\pi(d\theta, dy)$ be any coupling with source α and target β in the convex chart of [Theorem C.4](#), and retain the interiority hypothesis. Set*

$$\bar{\Delta}_\pi = \mathbb{E}_\pi \Delta_\Psi, \quad d_{\mathcal{A}} = \mathbb{E}_\alpha \mathcal{A} - \mathbb{E}_\beta \mathcal{A}, \quad C_1 = \frac{4\Lambda_\Phi^2}{m\mu_\Phi}, \quad C_2 = \frac{4M^2(1+\eta\ell)^2}{m^2\eta}.$$

Then

$$\mathbb{E}_\pi D_\Psi(y, \theta) \leq \eta(\bar{\Delta}_\pi + d_{\mathcal{A}}), \quad \mathbb{E}_\alpha \mathfrak{S} \leq (C_1 + C_2)\bar{\Delta}_\pi + C_2 d_{\mathcal{A}}. \quad (\text{C.9})$$

In particular the latter right side may be enlarged by replacing $d_{\mathcal{A}}$ with its positive part. For a represented physical test with lift bound $B(k)$,

$$\mathbb{E}_\alpha |D\mathcal{A}[v(k)]|^2 \leq B(k)^2 [(C_1 + C_2)\bar{\Delta}_\pi + C_2(d_{\mathcal{A}})_+]. \quad (\text{C.10})$$

No invariant law is required. Equality of source and target marginals recovers the stationary estimate.

Proof. The pointwise inequality in the stationary proof is $\eta^{-1}D_\Psi(y, \theta) \leq \Delta_\Psi(\theta, y) + \mathcal{A}(\theta) - \mathcal{A}(y)$. Averaging gives the first bound, and hence $\mathbb{E}_\pi \|y - \theta\|^2 \leq 2\eta(\bar{\Delta}_\pi + d_{\mathcal{A}})/m$. The pointwise gradient argument in [Theorem C.4](#) does not use stationarity. Its proximal-gradient term contributes $C_1\bar{\Delta}_\pi$, and its displacement term contributes $C_2(\bar{\Delta}_\pi + d_{\mathcal{A}})$. This proves the second bound. Positive-metric duality gives the test estimate. A zero proximal gap at one chronological step need not force zero gradient: it can instead describe genuine action descent, represented by $d_{\mathcal{A}} > 0$. $\square$

Proposition C.7 (Cutoff-aware stationarity on the determining test core). *Let k be a fixed physical test and let its represented lift satisfy [\(4.16\)](#). Then [\(4.17\)](#) holds. If*

$$B_{X,K}(k)^2 C_X \bar{\Delta}_{\Psi,X} \rightarrow 0 \quad (\text{C.11})$$

for each test in a countable determining core, the corresponding first variations converge to zero in mean square. Under any common coupling there is a deterministic subsequence of cutoffs on which all these first variations converge to zero almost surely.

Proof. At each represented field value, duality for the positive G_X pairing gives

$$|D\mathcal{A}_X[v]|^2 \leq \|D\mathcal{A}_X\|_{G_X^{-1}}^2 \|v\|_{G_X}^2.$$

Take the supported supremum of the second factor and apply [Theorem C.4](#). This proves the mean-square estimate. Enumerate the determining core as $k_1, k_2, \dots$. Select increasing cutoffs X_j so that the right-hand side of [\(4.17\)](#) is at most 2^{-j} for each of $k_1, \dots, k_j$. For a fixed k_m , the sum over $j \geq m$ of the expected squared residuals is finite. Tonelli's theorem implies that the corresponding sum of squared residuals is finite almost surely, so the residual tends to zero. A countable intersection gives the assertion for the whole core. No independence between cutoffs is used. Uniform continuity of the limiting variational functional on the declared test topology then extends zero from the core to all determining tests. $\square$

Corollary C.8 (Finite Gibbs/KL certificate). *For a finite reference row $p_\theta(y) > 0$ and reconstructed same-history cost $c_\theta(y)$, the row q_θ^* of [\(4.18\)](#) is the unique entropy-proximal row and*

$$\Delta_{\text{Gibbs}, \theta} = \eta^{-1} D_{\text{KL}}(K^{\text{acc}}(\cdot \mid \theta) \| q_\theta^*) \quad (\text{C.12})$$

is exactly its row variational gap.

Proof. For any probability row q ,

$$\sum_y q(y) c_\theta(y) + \eta^{-1} D_{\text{KL}}(q \| p_\theta) = \eta^{-1} D_{\text{KL}}(q \| q_\theta^*) - \eta^{-1} \log \sum_z p_\theta(z) e^{-\eta c_\theta(z)}.$$

Relative entropy is nonnegative and vanishes only at equality. This proves a statement about probability rows; it does not identify their sampled points with interior critical points of the field action. $\square$

C.4 Derivative-free face linearization and complete memory shorting

The first-order gravitational score is not assumed as a derivative at a zero face. It is obtained from renewal subdivision after all resolved cut-memory coordinates have been retained.

Theorem C.9 (Derivative-free renewal face linearization). *Let E be a finite-dimensional real normed space, $U \subset E$ a balanced convex neighborhood of zero, and $A : U \rightarrow \mathbb{R}$ with $A(0) = 0$. If*

$$A(z) = 2A(z/2) \tag{C.13}$$

for every $z \in U$, A is bounded near zero, and

$$|A(x+y) - A(x) - A(y)| \leq H \|x\| \|y\|$$

near zero, then A is the restriction of a unique real linear functional. More generally, if the renormalized scores $L_m(z) = 2^m A_m(2^{-m}z)$ have summable adjacent-cutoff defects and their additivity defect tends to zero, then L_m converges locally uniformly to one continuous linear functional.

Proof. For the exact branch iterate (C.13):

$$A(x+y) - A(x) - A(y) = 2^m [A(2^{-m}(x+y)) - A(2^{-m}x) - A(2^{-m}y)].$$

The right-hand side is bounded by $H 2^{-m} \|x\| \|y\|$ and therefore vanishes as $m \rightarrow \infty$. Thus A is additive; boundedness near zero gives continuity and hence real homogeneity. On the stable branch the adjacent-cutoff estimate makes L_m locally uniformly Cauchy; passing to the limit in the additivity defect gives a continuous linear limit. $\square$

Theorem C.10 (Complete resolved-memory subtraction). *Let the resolved cut-memory state be $\rho_{\text{cq}} = \bigoplus_\omega p_\omega \rho_\omega$ with faithful blocks and linear perturbation $K_{\text{cq}}(z) = \bigoplus_\omega K_\omega(z)$. Put*

$$M_{\text{cq}}(z) = \log \text{Tr} \exp(\log \rho_{\text{cq}} + K_{\text{cq}}(z)) - \bar{m}(z), \tag{C.14}$$

where $\bar{m}$ is the average linear slope. Then $M_{\text{cq}}(0) = DM_{\text{cq}}(0) = 0$, its Hessian is the sum of a classical branch covariance and a positive Kubo–Mori form, and

$$A_{\text{sh}} = A_{\text{coarse}} - M_{\text{cq}} \tag{C.15}$$

is the canonical fully memory-shortened face score. Thus unresolved memory is removed at the complete log-partition level rather than only to second order.

Proof. The block exponential and trace split over the classical label. Duhamel’s formula gives the first derivative in each faithful block and the Kubo–Mori quadratic form as the second logarithmic derivative. Differentiating the outer finite log-sum-exp adds the covariance of the branch means. Both terms are positive. Subtracting the complete log-partition therefore removes exactly the resolved cut-memory contribution and leaves the displayed shortened score. $\square$

C.5 Local gravitational classification and variational consistency

We use the densities (4.21), with the same normalization and boundary convention.

Proposition C.11 (Classification in the prescribed incidence and degree class). *Within Definition 4.6, the real Lorentz-natural bulk space is generated by L_{Holst} , L_{Palatini} and L_{vol} . The assertion holds before torsion is set to zero.*

Proof. Let $V = \mathbb{R}^{1,3}$. Antisymmetry of the exterior products identifies the curvature-dependent coefficient with an invariant bilinear pairing between two copies of $\Lambda^2 V$. The Lorentz metric identifies one copy with its dual. On real bivectors the internal Hodge operator satisfies $\star^2 = -I$; after complexification its $+i$ and $-i$ eigenspaces are the two nonisomorphic irreducible bivector representations of the connected Lorentz group. Schur's lemma makes a commuting complex operator scalar on each eigenspace. Reality makes the two scalars complex conjugate, leaving precisely $uI + v\star$, $u, v \in \mathbb{R}$. These two pairings give $e^I \wedge e^J \wedge R_{IJ}$ and $\frac{1}{2}\epsilon_{IJKL}e^I \wedge e^J \wedge R^{KL}$. Scalar multiplicity excludes another coefficient operator on an internal factor. The alternating part of a four-coframe coefficient is an invariant in $\Lambda^4 V^*$, a one-dimensional space generated by the oriented volume tensor. Symmetric metric contractions do not create an additional alternating four-coframe invariant. This proves the asserted span. Exact terms are dealt with by the boundary convention in the definition, not by deleting physical boundary variations. $\square$

Remark C.12 (Topological terms and torsion dependence). Euler and Pontryagin densities are quadratic in curvature and do not belong to this bidegree class. Their local bulk variations are exact, but their boundary and global information need not vanish. The Nieh–Yan identity is

$$d(e^I \wedge T_I) = T^I \wedge T_I - e^I \wedge e^J \wedge R_{IJ}. \quad (\text{C.16})$$

It follows from $De^I = T^I$ and $DT^I = R^I{}_J \wedge e^J$, with contraction turning the exterior covariant derivative into d . Thus enlarging the class to include $T^I \wedge T_I$ introduces a relation to the Holst term modulo an explicitly retained boundary density; it is not justified by the bivector commutant alone. More general inverse-coframe or torsion invariants require a different classification. The distinction between bulk equations, variational principles and boundary charges is essential in first-order gravity [68]. Neither the Palatini coefficient nor its ratio to the volume coefficient is fixed by Lorentz covariance.

Lemma C.13 (Transfer of a classified variational representative). *Suppose the classified curvature and volume variations converge on each physical determining test, and (4.25) holds on those tests. Then the complete action variation has the same limit. If that complete variation tends to zero, the limiting classified variation vanishes. The assertion also holds on one common realized subsequence of stochastic records satisfying these hypotheses.*

Proof. Use (4.24), the convergence of the three calibrated coefficients, and the vanishing remainder. All operations take place on the same test and the same sequence. In the stochastic case the argument is applied after the common almost-sure subsequence has been selected; no independent subsequences are multiplied together. $\square$

C.6 Connection stationarity and torsion

Let $B^{IJ} = e^I \wedge e^J$, $T^I = De^I$, and $P_{\alpha,\beta} = \alpha I + \beta\star$, with the same spacetime-constant coefficients as in (4.23).

Theorem C.14 (Connection stationarity and the torsion alternative). *If $(\alpha, \beta) \neq (0, 0)$, then*

$$P_{\alpha, \beta}^{-1} = \frac{\alpha I - \beta \star}{\alpha^2 + \beta^2}. \quad (\text{C.17})$$

For the Palatini–Holst action the spinless connection Euler equation is $D(P_{\alpha, \beta} B) = 0$. For a nondegenerate coframe, the Cartan map

$$\mathcal{K}_e(T)^{IJ} = T^I \wedge e^J - e^I \wedge T^J$$

is injective. Consequently connection stationarity implies $T = 0$; at finite cutoff, a uniform lower singular-value bound on $\mathcal{K}_{e, X}$ controls the torsion norm by the connection Euler residual.

Proof. The inverse formula follows from $\star^2 = -I$ on Lorentzian bivectors. Varying the connection and integrating by parts gives the Euler source. To prove injectivity, expand $T^I = \frac{1}{2} T^I_{KL} e^K \wedge e^L$ in a coframe basis. Choosing a fourth index in $T^I \wedge e^J - e^I \wedge T^J = 0$ first eliminates all components with $K, L \neq I$; the remaining coefficients satisfy pairwise sign relations that, in four dimensions, force them all to vanish. Inverting the two maps gives the finite estimate. $\square$

D Noncollapse, curvature compactness and the Einstein limit

This appendix proves the estimates and limiting argument of Section 5. It uses the finite geometry and action from Sections 3 and 4; temporal compactness, curvature propagation and variational consistency remain separate hypotheses on one common physical sequence. The alternative connection routes are kept in distinct subsections.

D.1 Cut estimates and spatial compactness

For a nonempty half-volume cut A , write $A^c = V_X \setminus A$ and define the mean-zero demand

$$b_{A, X}(v) = \mu_X(A)^{2/3} \left[\frac{m_{v, X} \mathbf{1}_A(v)}{\mu_X(A)} - \frac{m_{v, X} \mathbf{1}_{A^c}(v)}{\mu_X(A^c)} \right]. \quad (\text{D.1})$$

This fixes the normalization of the dual cut witness below.

$$C_S = \frac{128 D_*}{(I_*^{\text{sp}})^2}, \quad C_P = V_*^{2/3} C_S. \quad (\text{D.2})$$

Let a finite physical endpoint graph have vertex masses $m_{v, X}$, measure μ_X , symmetric conductances $c_{uv, X}$, and efficient edge capacities $u_{e, X}$. We use the two cut constants I_X^{eff} and I_X^{sp} of (5.3); on the positive spatial branch they are related by the finite capacity–conductance comparison (5.2).

Definition D.1 (Spatial Dirichlet normalization). On the finite weighted graph $G_X = (V_X, E_X)$ set

$$\mathcal{E}_X^{\text{sp}}(f) = \sum_{\{u, v\} \in E_X} c_{uv, X} |f(u) - f(v)|^2, \quad \langle f, \Delta_X^{\text{sp}} f \rangle_{L^2(\mu_X)} = \mathcal{E}_X^{\text{sp}}(f). \quad (\text{D.3})$$

All scalar Sobolev estimates below are first stated for real-valued functions; the real weighted Laplacian has the same spectrum after complexification. Let $P_{0, X}$ be the orthogonal projection onto the mean-zero subspace and put $T_X(t) = e^{-t \Delta_X^{\text{sp}}} P_{0, X}$. Since the conductances are symmetric and nonnegative, $e^{-t \Delta_X^{\text{sp}}}$ is a finite Markov semigroup and is contractive on every $L^q(\mu_X)$, $1 \leq q \leq \infty$.

Lemma D.2 (Cut inequality implies the half-volume L^1 Sobolev bound). *Assume*

$$h_X c_X(\partial A) \geq I_*^{\text{sp}} \mu_X(A)^{2/3} \quad (\text{D.4})$$

whenever $0 < \mu_X(A) \leq \mu_X(V_X)/2$. If $g \geq 0$ and $\mu_X(\text{supp } g) \leq \mu_X(V_X)/2$, then

$$\|g\|_{L^{3/2}(\mu_X)} \leq \frac{h_X}{I_*^{\text{sp}}} \sum_{\{u,v\} \in E_X} c_{uv,X} |g(u) - g(v)|. \quad (\text{D.5})$$

Proof. For $t \geq 0$ put $A_t = \{v : g(v) > t\}$. Since $A_t \subseteq \text{supp } g$, every nonempty A_t lies on the half-volume side of (D.4). The layer-cake identity gives $g = \int_0^\infty \mathbf{1}_{A_t} dt$. Minkowski's integral inequality in $L^{3/2}(\mu_X)$ therefore yields

$$\|g\|_{3/2} \leq \int_0^\infty \|\mathbf{1}_{A_t}\|_{3/2} dt = \int_0^\infty \mu_X(A_t)^{2/3} dt. \quad (\text{D.6})$$

Applying the cut inequality to every level set and then using the finite coarea identity gives

$$\|g\|_{3/2} \leq \frac{h_X}{I_*^{\text{sp}}} \int_0^\infty c_X(\partial A_t) dt \quad (\text{D.7})$$

$$= \frac{h_X}{I_*^{\text{sp}}} \sum_{\{u,v\} \in E_X} c_{uv,X} |g(u) - g(v)|, \quad (\text{D.8})$$

which is (D.5). $\square$

Lemma D.3 (Discrete L^6 Sobolev and Poincaré inequalities). *Assume (D.4), $\mu_X(V_X) \leq V_*$, and*

$$\sum_u c_{uv,X} \leq D_* h_X^{-2} m_{v,X} \quad (v \in V_X). \quad (\text{D.9})$$

Then every mean-zero f satisfies

$$\|f\|_6^2 \leq C_S \mathcal{E}_X^{\text{sp}}(f), \quad C_S = \frac{128 D_*}{(I_*^{\text{sp}})^2}, \quad (\text{D.10})$$

and

$$\|f\|_2^2 \leq C_P \mathcal{E}_X^{\text{sp}}(f), \quad C_P = V_*^{2/3} C_S. \quad (\text{D.11})$$

In particular $\lambda_1(\Delta_X^{\text{sp}}|_{\mathbf{1}^\perp}) \geq C_P^{-1}$.

Proof. First suppose $f \geq 0$ and $\mu_X(\text{supp } f) \leq \mu_X(V_X)/2$. Apply Lemma D.2 to $g = f^4$. Since $\|f^4\|_{3/2} = \|f\|_6^4$ and $|a^4 - b^4| \leq 4(a^3 + b^3)|a - b|$ for $a, b \geq 0$,

$$\|f\|_6^4 \leq \frac{4h_X}{I_*^{\text{sp}}} \sum_{\{u,v\}} c_{uv,X} (f(u)^3 + f(v)^3) |f(u) - f(v)| \quad (\text{D.12})$$

$$\leq \frac{4h_X}{I_*^{\text{sp}}} \left[\sum_{\{u,v\}} c_{uv,X} (f(u)^3 + f(v)^3)^2 \right]^{1/2} \mathcal{E}_X^{\text{sp}}(f)^{1/2}. \quad (\text{D.13})$$

Using $(a^3 + b^3)^2 \leq 2(a^6 + b^6)$ and (D.9),

$$\sum_{\{u,v\}} c_{uv,X} (f(u)^3 + f(v)^3)^2 \leq 2 \sum_v f(v)^6 \sum_u c_{uv,X} \quad (\text{D.14})$$

$$\leq 2D_* h_X^{-2} \|f\|_6^6. \quad (\text{D.15})$$

Substitution into (D.13) and division by $\|f\|_6^3$ gives

$$\|f\|_6^2 \leq \frac{32D_*}{(I_*^{\text{sp}})^2} \mathcal{E}_X^{\text{sp}}(f) \quad (\text{D.16})$$

for every half-volume-supported nonnegative f .

For a general real f , choose a weighted median m so that $\mu_X(f > m)$ and $\mu_X(f < m)$ are each at most half the total volume. Put $f_+ = (f - m)_+$ and $f_- = (m - f)_+$. Truncation is 1-Lipschitz and, edgewise,

$$|f_+(u) - f_+(v)|^2 + |f_-(u) - f_-(v)|^2 \leq |f(u) - f(v)|^2, \quad (\text{D.17})$$

so $\mathcal{E}(f_+) + \mathcal{E}(f_-) \leq \mathcal{E}(f)$. Since the two functions have half-volume support, (D.16) gives

$$\|f - m\|_6^2 = (\|f_+\|_6^6 + \|f_-\|_6^6)^{1/3} \quad (\text{D.18})$$

$$\leq \|f_+\|_6^2 + \|f_-\|_6^2 \leq \frac{32D_*}{(I_*^{\text{sp}})^2} \mathcal{E}(f). \quad (\text{D.19})$$

If f has mean zero, then

$$|m| \mu_X(V_X)^{1/6} \leq \|f - m\|_6, \quad (\text{D.20})$$

because $|m| = |\mu_X(V_X)^{-1} \int (f - m) d\mu_X|$ and Hölder gives $\|f - m\|_1 \leq \mu_X(V_X)^{5/6} \|f - m\|_6$. Therefore $\|f\|_6 \leq 2\|f - m\|_6$, which yields (D.10) with the stated factor four. Finally,

$$\|f\|_2^2 \leq \mu_X(V_X)^{2/3} \|f\|_6^2 \leq V_*^{2/3} C_S \mathcal{E}(f), \quad (\text{D.21})$$

which is (D.11). Rayleigh–Ritz gives the spectral-gap bound. $\square$

Lemma D.4 (Nash inequality, ultracontractivity, and spectral counting). *Under the hypotheses of Lemma D.3, every mean-zero f obeys*

$$\|f\|_2^{10/3} \leq C_S \mathcal{E}_X^{\text{sp}}(f) \|f\|_1^{4/3}. \quad (\text{D.22})$$

Moreover, for every $t > 0$,

$$\|T_X(t)\|_{L^1 \rightarrow L^\infty} \leq 4 \left(\frac{3C_S}{2t} \right)^{3/2}. \quad (\text{D.23})$$

Consequently, if $N_X(R) = \text{rank } \mathbf{1}_{[0,R]}(\Delta_X^{\text{sp}})$, then for $R > 0$

$$N_X(R) \leq 1 + 4e^{3/2} V_* (C_S R)^{3/2}. \quad (\text{D.24})$$

Proof. Interpolation between L^1 and L^6 gives $\|f\|_2 \leq \|f\|_1^{2/5} \|f\|_6^{3/5}$. Raising this inequality to the power 10/3 and using (D.10) yields (D.22).

Let $F_t = T_X(t)f$ and $u(t) = \|F_t\|_2^2$. Since F_t has mean zero and the finite Markov semigroup is L^1 -contractive,

$$\|F_t\|_1 \leq \|P_{0,X} f\|_1 \leq 2\|f\|_1. \quad (\text{D.25})$$

Self-adjointness and finite dimensionality permit differentiation under the spectral decomposition:

$$u'(t) = -2\mathcal{E}_X^{\text{sp}}(F_t). \quad (\text{D.26})$$

Combining (D.22), (D.25), and (D.26) gives

$$u'(t) \leq -\frac{2}{C_S 2^{4/3} \|f\|_1^{4/3}} u(t)^{5/3}. \quad (\text{D.27})$$

Integrating the differential inequality for $u^{-2/3}$ yields

$$\|T_X(t)f\|_2^2 \leq \frac{3^{3/2}}{2} C_S^{3/2} t^{-3/2} \|f\|_1^2. \quad (\text{D.28})$$

Because $T_X(t/2)$ is self-adjoint and $T_X(t) = T_X(t/2)T_X(t/2)$ on the mean-zero subspace, duality and the semigroup property give

$$\|T_X(t)\|_{1 \rightarrow \infty} \leq \|T_X(t/2)\|_{1 \rightarrow 2} \|T_X(t/2)\|_{2 \rightarrow \infty} \quad (\text{D.29})$$

$$= \|T_X(t/2)\|_{1 \rightarrow 2}^2 \leq 4 \left(\frac{3C_S}{2t} \right)^{3/2}, \quad (\text{D.30})$$

which is (D.23). This is the standard Sobolev–Nash ultracontractivity route in finite weighted form; compare [57, Chs. 10–11] and [58].

Let $K_t^0(v, w)$ be the integral kernel of $T_X(t)$ with respect to μ_X . The spectral expansion gives $K_t^0(v, v) = \sum_{j \geq 1} e^{-t\lambda_j} |\varphi_j(v)|^2 \geq 0$, and therefore

$$\text{Tr}(T_X(t)) = \sum_v m_{v,X} K_t^0(v, v) \quad (\text{D.31})$$

$$\leq \mu_X(V_X) \|T_X(t)\|_{1 \rightarrow \infty} \quad (\text{D.32})$$

$$\leq 4V_* \left(\frac{3C_S}{2t} \right)^{3/2}. \quad (\text{D.33})$$

Since the positive cut margin implies connectedness, the zero eigenvalue is simple. Hence

$$(N_X(R) - 1)e^{-tR} \leq \sum_{0 < \lambda_j \leq R} e^{-t\lambda_j} \leq \text{Tr}(T_X(t)). \quad (\text{D.34})$$

Choosing $t = 3/(2R)$ gives (D.24). $\square$

Theorem D.5 (Operational Sobolev, Poincaré, and compact-screen bounds). *Assume uniformly that $\mu_X(V_X) \leq V_*$, (D.9) holds, the capacity–conductance comparison (5.2) holds, and $I_X^{\text{eff}} \geq I_*^{\text{eff}} > 0$. Then $I_X^{\text{sp}} \geq I_*^{\text{eff}}/a_+ = I_*^{\text{sp}} > 0$ and the estimates (D.10), (D.11), and (D.24) hold. In addition,*

$$\|(I - P_{X,R})f\|_2^2 \leq R^{-1} \mathcal{E}_X^{\text{sp}}(f), \quad P_{X,R} = \mathbf{1}_{[0,R]}(\Delta_X^{\text{sp}}). \quad (\text{D.35})$$

Thus $P_{X,R}$ is a uniformly exhausted finite-rank compact-screen family. A sequence of positive cut constants tending to zero gives the normalized cut/potential obstruction (5.8); a zero-capacity cut is retained directly as disconnected support.

Proof. The comparison (5.4) converts the efficient-capacity margin into (D.4). The Sobolev and Poincaré estimates are Lemma D.3; the counting law is Lemma D.4. By the spectral theorem, $\Delta_X^{\text{sp}} \succeq R(I - P_{X,R})$, so (D.35) follows immediately. For a positive minimizing cut A , set $\phi = \mathbf{1}_A/u_X(\partial A)$. Then $\sum_e u_{e,X} |d\phi(e)| = 1$, and (D.1) gives $|\sum_v b_{A,X}(v)\phi(v)| = \mu_X(A)^{2/3}/u_X(\partial A) = (I_X^{\text{eff}})^{-1}$. This is an explicit feasible dual witness. A zero-capacity cut is recorded directly as disconnected support. $\square$

Proposition D.6 (Compact positive-screen upgrade). *Let $Y_X \rightarrow Y$ in a Hilbert space and let positive operators $\mathbb{A}_X$ satisfy $mI \preceq \mathbb{A}_X \preceq MI$ and $\mathbb{A}_X^{-1/2} \rightarrow \mathbb{A}^{-1/2}$ strongly. Put $Z_X = \mathbb{A}_X^{1/2} Y_X$*

and suppose $Z_X \rightarrow Z$. If for every R there are finite-rank screens $S_{X,R} \rightarrow S_R$ in operator norm, with S_R compact, and

$$\lim_{R \rightarrow \infty} \sup_X \|(I - S_{X,R})Z_X\| = 0, \quad \lim_{R \rightarrow \infty} \|(I - S_R)Z\| = 0,$$

then $Y_X \rightarrow Y$ strongly. Consequently every bounded quadratic insertion in Y_X converges in L^1 on the corresponding compact spacetime cylinder.

Proof. For fixed R , compactness of S_R converts weak convergence into $S_R Z_X \rightarrow S_R Z$ strongly, and operator-norm convergence gives $S_{X,R} Z_X - S_R Z_X \rightarrow 0$. Decompose

$$Z_X - Z = (I - S_{X,R})Z_X + (S_{X,R}Z_X - S_R Z) + (S_R - I)Z.$$

Taking the limsup and then $R \rightarrow \infty$ gives $Z_X \rightarrow Z$ strongly. Uniform coercivity and strong convergence of the inverse square roots then give $Y_X \rightarrow Y$. The quadratic statement follows from $\|B(Y_X, Y_X) - B(Y, Y)\|_1 \leq C\|Y_X - Y\|_2(\|Y_X\|_2 + \|Y\|_2)$. $\square$

D.2 Strong coframe interpolation

Lemma D.7 (Strong interpolation for the reconstructed coframes). *Let K be a compact spacetime cylinder with finite reconstructed volume measure. Suppose the common-cylinder interpolants of the finite coframes satisfy*

$$e_X \rightarrow e \text{ strongly in } L^2(K), \quad \sup_X \|e_X\|_{L^6(K)} \leq M_6 < \infty. \quad (\text{D.36})$$

Then $e \in L^6(K)$ and, for every $2 \leq q < 6$,

$$e_X \rightarrow e \text{ strongly in } L^q(K). \quad (\text{D.37})$$

More precisely, if $\theta_q = 3/q - 1/2 \in (0, 1]$, then

$$\|e_X - e\|_{L^q} \leq \|e_X - e\|_{L^2}^{\theta_q} \|e_X - e\|_{L^6}^{1-\theta_q}. \quad (\text{D.38})$$

If $p > 3/2$ and $p' = p/(p-1)$, then $2p' < 6$ and

$$e_X \wedge e_X \longrightarrow e \wedge e \text{ strongly in } L^{p'}(K). \quad (\text{D.39})$$

Proof. The uniform L^6 bound gives a weakly convergent subsequence in $L^6(K)$. Because K has finite measure, weak L^6 convergence implies distributional convergence, while the strong L^2 convergence has the unique distributional limit e . Hence every weak L^6 cluster point is e , so $e \in L^6(K)$ and $\|e\|_6 \leq M_6$. The difference $e_X - e$ is therefore uniformly bounded in L^6 . The Riesz–Thorin/Hölder interpolation identity $1/q = \theta_q/2 + (1 - \theta_q)/6$ gives (D.38); its right-hand side tends to zero for $q < 6$, proving (D.37).

For $p > 3/2$, $p' < 3$ and hence $q = 2p' < 6$. The exterior product is a bounded bilinear map on the finite-dimensional coframe fibres, so pointwise $|a \wedge b| \leq C|a||b|$. Therefore Hölder's inequality gives

$$\|e_X \wedge e_X - e \wedge e\|_{L^{p'}} \leq C\|(e_X - e) \wedge e_X + e \wedge (e_X - e)\|_{L^{p'}} \quad (\text{D.40})$$

$$\leq C(\|e_X\|_{L^{2p'}} + \|e\|_{L^{2p'}})\|e_X - e\|_{L^{2p'}} \longrightarrow 0, \quad (\text{D.41})$$

which proves (D.39). $\square$

D.3 Curvature-energy propagation

The differential estimate underlying [Theorem 5.2](#) is

$$\frac{1}{2} \frac{d}{dt} z_X^2 \leq a_X z_X^2 + z_X \|f_X\|_{M_X}, \quad (\text{D.42})$$

Proof of [Theorem 5.2](#). Differentiate $z_X^2 = y_X^T M_X y_X$ and substitute (5.10). The principal term cancels exactly because $M_X K_X + K_X^T M_X = 0$. By (5.11), the remaining quadratic term is at most $2a_X z_X^2$, while Cauchy–Schwarz in the positive M_X -Gram bounds the source term by $2z_X \|f_X\|_{M_X}$. Applying the integrating-factor argument to $(z_X^2 + \varepsilon^2)^{1/2}$ and sending $\varepsilon \downarrow 0$ gives (5.12). The interpolation estimate in (5.13), integration against the physical lapse and slice measure, and the uniform budgets give (5.15). $\square$

D.4 Fully discrete propagation with physical transfers

The preceding estimate has a fully discrete counterpart, so a differentiable curvature path is not needed merely to propagate its size. At one spatial cutoff, take a partition of the selected time interval with lengths $s_j > 0$. Let $Y_j = M_j^{1/2} y_j$ be the normalized actual curvature records and let T_j be their recorded physical transfer in these coordinates. Suppose

$$Y_{j+1} - T_j Y_j = s_j (K_j + L_j) \frac{Y_{j+1} + T_j Y_j}{2} + \varepsilon_j, \quad (\text{D.43})$$

where $K_j^T = -K_j$, $(L_j + L_j^T)/2 \preceq a_j I$, $\|T_j\|_{\text{op}} \leq e^{d_j}$, and $a_j, d_j \geq 0$. The residual ε_j includes the complete source, solver and coordinate-transfer errors of the actual step. In particular, nonlinear changes of field coordinates are not assumed to commute with midpoint discretization.

Theorem D.8 (Fully discrete initial-energy curvature certificate). *Assume $s_j a_j/2 \leq b < 1$, and set*

$$D_X = \sum_j d_j, \quad A_X^d = \sum_j s_j a_j, \quad \mathcal{D}_X = \sum_j \frac{\|\varepsilon_j\|^2}{s_j}, \quad T_X = \sum_j s_j. \quad (\text{D.44})$$

Each recorded linear step is uniquely solvable, and

$$\max_j \|Y_j\| \leq e^{D_X + A_X^d/(1-b)} \left(\|Y_0\| + \frac{\sqrt{T_X \mathcal{D}_X}}{1-b} \right). \quad (\text{D.45})$$

There is no restriction involving $s_j \|K_j\|_{\text{op}}$. With a uniformly stable interval interpolation of these actual curvature records, a uniform lapse bound, and a bounded reconstruction error as in (5.13), uniform bounds on the quantities in (D.45) imply $\sup_X \|R_X\|_{L^2(K)} < \infty$.

The statement concerns the represented linear step with its coefficients fixed. For an implicit nonlinear evolution, existence of that evolution is not inferred from this matrix inverse; its actual coefficient and residual records must satisfy (D.43). The proof in this appendix uses the skew cancellation before taking norms and retains the physical transfer between successive Grams.

Proof. Write $A_j = K_j + L_j$. For any real vector v ,

$$\langle v, (I - s_j A_j/2)v \rangle \geq (1 - s_j a_j/2) \|v\|^2.$$

Cauchy–Schwarz gives $\|(I - s_j A_j/2)v\| \geq (1 - s_j a_j/2) \|v\|$. Thus the finite matrix is invertible and its inverse norm is at most $(1 - s_j a_j/2)^{-1}$.

Let $v = T_j Y_j$ and first let w solve the step with zero residual. Pair $w - v = s_j A_j(w + v)/2$ with $(w + v)/2$ to obtain

$$\|w\|^2 - \|v\|^2 \leq \frac{s_j a_j}{2} (\|w\| + \|v\|)^2.$$

Dividing by $\|w\| + \|v\|$ when this number is nonzero, and treating the zero case directly, gives

$$\|w\| \leq \frac{1 + s_j a_j/2}{1 - s_j a_j/2} \|v\|.$$

The inhomogeneous solution adds $(I - s_j A_j/2)^{-1} \varepsilon_j$. Consequently

$$\|Y_{j+1}\| \leq e^{d_j} \frac{1 + s_j a_j/2}{1 - s_j a_j/2} \|Y_j\| + \frac{\|\varepsilon_j\|}{1 - b}.$$

For $0 \leq x \leq b$, $\log((1+x)/(1-x)) \leq 2x/(1-b)$. Every product of step factors is therefore bounded by $\exp(D_X + A_X^d/(1-b))$. Iteration and weighted Cauchy–Schwarz,

$$\sum_j \|\varepsilon_j\| \leq \sqrt{T_X \sum_j \|\varepsilon_j\|^2 / s_j},$$

prove (D.45). The size of the skew matrix K_j has entered neither estimate.

For raw curvature records y_j with Gram M_j , the transfer is $T_j = M_{j+1}^{1/2} \mathcal{T}_j M_j^{-1/2}$, where $\mathcal{T}_j$ is the actual recorded frame/transport map. Its norm excess is retained in d_j . Stable interval interpolation bounds the reconstructed slice norm by a fixed multiple of the largest endpoint norm. Integrating with the physical lapse and adding the bounded reconstruction error yields, for an appropriate interpolation constant C_I ,

$$\|R_X\|_{L^2(K)} \leq C_I \sqrt{N_* T_X} e^{D_X + A_X^d/(1-b)} \left(\|Y_0\| + \frac{\sqrt{T_X \mathcal{D}_X}}{1-b} \right) + C_\zeta.$$

All interface and incoming boundary contributions must already be included in the actual transfer, residual or reconstruction comparison. This proof does not assert that arbitrary nonlinear coordinate changes preserve the midpoint recurrence: any such discrepancy is part of ε_j . $\square$

D.5 Differentiated action-current control

The forcing budget can itself be checked against the reconstructed action, provided the actual differentiated current has the required incidence. Write $\mathbf{a}_X = D\mathcal{A}_X$ for its action-gradient covector and suppose

$$f_X = \mathsf{T}_X \mathbf{a}_X + r_X^{\text{src}}. \quad (\text{D.46})$$

The transfer T_X includes the physical metric-test lift, the nonzero Palatini normalization, trace reversal, the differentiated Bianchi current, and its frame comparison. Torsion, Cartan, boundary and discretization contributions not in this map remain in r_X^{src} .

Proposition D.9 (Sharp differentiated action-current certificate). *Let $H_X \succ 0$ be the actual source Gram, transferred to the curvature-energy space. The least squared amplification from the action dual norm is*

$$\Gamma_X = \|H_X^{1/2} \mathsf{T}_X G_X^{1/2}\|_{\text{op}}^2, \quad \|\mathsf{T}_X \mathbf{a}\|_{H_X}^2 \leq \Gamma_X \|\mathbf{a}\|_{G_X^{-1}}^2. \quad (\text{D.47})$$

Consequently a pointwise certificate $\|\mathbf{a}_X(t)\|_{G_X^{-1}}^2 \leq C_X(t)\Delta_X(t)$ gives

$$\int_0^{T_*} \|f_X\|_{H_X} dt \leq \sqrt{2T_* \int_0^{T_*} (\Gamma_X C_X \Delta_X + \|r_X^{\text{src}}\|_{H_X}^2) dt}. \quad (\text{D.48})$$

For the normalized discrete source $\varepsilon_j/s_j = \mathsf{T}_j \mathbf{a}_j + r_j^{\text{src}}$, the corresponding squared-energy estimate is $\mathcal{D}_X \leq 2 \sum_j s_j (\Gamma_j C_j \Delta_j + \|r_j^{\text{src}}\|_{H_j}^2)$.

This replaces a separately supplied source-size bound by an explicitly scaled action-side bound on the verified transfer branch. It does not turn a small undifferentiated Euler residual into a small Bianchi current: a single centered grid derivative can have $\Gamma_X = h_X^{-2}$. The sharpness example and the expected-energy version of the certificate are proved in this appendix. An estimate only in expectation yields a bound in probability; almost-sure uniform curvature control additionally uses the pathwise or summable source conditions stated there.

Differentiated gravitational source. The source norm is not the norm of an undifferentiated Einstein residual. On the actual shared-link branch the forcing contains the differentiated, volume-normalized Palatini metric row, together with the differentiated Cartan/torsion defect and the differential Bianchi defect. Lower-order transport terms are retained in a_X . To state a quantitative criterion, use the physical incidence (D.46), with all untransferred contributions collected in r_X^{src} . Its verification is an assertion about the actual action, connection and observation records; it is not obtained simply by introducing an auxiliary source variable.

The pointwise source estimate used in Proposition D.9 is

$$\|f_X\|_{H_X}^2 \leq 2\Gamma_X C_X \Delta_X + 2\|r_X^{\text{src}}\|_{H_X}^2. \quad (\text{D.49})$$

With deterministic supported upper bounds on Γ_X , it also holds in expectation when the action-gradient estimate is only in expectation. The proof below retains the expected product when no such upper bound is available.

Proof of Proposition D.9. Write $\mathbf{a} = G_X^{1/2} v$. Then $\|\mathbf{a}\|_{G_X^{-1}} = \|v\|$ and

$$\|\mathsf{T}_X \mathbf{a}\|_{H_X} = \|H_X^{1/2} \mathsf{T}_X G_X^{1/2} v\|.$$

The maximum squared ratio is the largest squared singular value. A unit maximizing right singular vector attains it, so the constant is sharp and also supplies a finite witness to large source amplification. Apply $\|u + v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$ to (D.46). Time Cauchy–Schwarz proves (D.48); summing the same pointwise inequality with weights s_j proves its discrete counterpart. For varying field values the argument is pointwise before averaging. Without a deterministic supported bound, the expected estimate must retain $\mathbb{E}[\Gamma_X \|\mathbf{a}_X\|_{G_X^{-1}}^2]$, not a product of separately averaged factors. $\square$

Proposition D.10 (Sharp grid-scale source amplification). *On a periodic grid of m sites, with m divisible by four, spacing h and inner product $h \sum_i u_i v_i$, let $(D_h u)_i = (u_{i+1} - u_{i-1})/(2h)$. With the same positive source and target norms,*

$$\|D_h\|_{2 \rightarrow 2}^2 = h^{-2}. \quad (\text{D.50})$$

In particular, a vanishing unscaled action-gradient norm does not imply a bounded differentiated current.

Proof. The Fourier multiplier is $i \sin(2\pi k/m)/h$, whose maximum modulus is h^{-1} and is attained at $k = m/4$. The same singular value occurs on the real sine/cosine subspace. Choose a unit vector v_h in that subspace and set $\mathbf{a}_h = h^{1/2}v_h$. Then $\|\mathbf{a}_h\|_2^2 = h \rightarrow 0$, whereas $\|D_h \mathbf{a}_h\|_2^2 = h^{-1} \rightarrow \infty$. $\square$

Thus, for this derivative transfer and uniformly controlled remaining factors, $C_X \Delta_X = O(h_X^2)$ suffices for bounded source and $o(h_X^2)$ suffices for vanishing source. These rates are not inferred from $\Delta_X \rightarrow 0$. Cancellations in the actual Bianchi transfer may improve Γ_X , but they must be retained in that map and its physical Gram.

Corollary D.11 (Expected curvature energy and a sufficient pathwise upgrade). *Suppose the finite propagation equation holds pathwise on $[0, T_*]$, with $\int a_X \leq A_*$ deterministically. Suppose its forcing satisfies (D.46), with the target norm already transferred to the curvature-energy norm, and $\mathbb{E}\|\mathbf{a}_X(t)\|_{G_X^{-1}}^2 \leq C_X(t)\Delta_X(t)$. Using deterministic supported upper bounds on Γ_X , one obtains*

$$\begin{aligned} \mathbb{E} \sup_{t \leq T_*} z_X(t)^2 &\leq 2e^{2A_*} \left[\mathbb{E} z_X(0)^2 + 2T_* \int_0^{T_*} \Gamma_X C_X \Delta_X \, dt \right. \\ &\quad \left. + 2T_* \mathbb{E} \int_0^{T_*} \|r_X^{\text{src}}\|_{H_X}^2 \, dt \right]. \end{aligned} \quad (\text{D.51})$$

Together with the stable interpolation and lapse bounds, uniformity of these budgets gives a uniform expected squared spacetime curvature norm. For a countable cofinal sequence, a sufficient pathwise strengthening is

$$\sup_X z_X(0) < \infty \text{ almost surely, } \quad \sum_X \mathbb{E} \int_0^{T_*} \|f_X(t)\|_{H_X}^2 \, dt < \infty, \quad (\text{D.52})$$

with pathwise uniform growth, interpolation, lapse and boundary budgets. Under these conditions $\sup_X \|R_X\|_{L^2(K)} < \infty$ almost surely.

Proof. The pathwise estimate gives $\sup_t z_X(t) \leq e^{A_*}(z_X(0) + \int \|f_X\|)$. Square and apply Cauchy–Schwarz in time to obtain

$$\mathbb{E} \sup_t z_X(t)^2 \leq 2e^{2A_*} \left(\mathbb{E} z_X(0)^2 + T_* \mathbb{E} \int_0^{T_*} \|f_X\|_{H_X}^2 \, dt \right).$$

Apply Proposition D.9 and the interpolation estimate. For the last assertion, Tonelli makes the sum of the nonnegative integrated source energies finite almost surely. Their supremum is then finite, and the deterministic propagation estimate applies to the whole realized sequence. A uniform expected norm alone gives boundedness in probability by Markov’s inequality; it is not substituted for this pathwise argument. $\square$

D.6 Literal-link anisotropic compactness

For the periodic route of Theorem 5.3, let S_i denote the positive spatial shift and write $U_i = \exp(hA_i) = I + hZ_i$. Define the literal electric coefficient by the exact link identity

$$E_i U_i = \partial_t U_i / h - A_0 U_i + U_i S_i A_0, \quad (\text{D.53})$$

with the same sign convention as the Cartan writer. The spatial magnetic record F_{ij} is the h^{-2} logarithmic plaquette coefficient on the common identity chart. These are actual finite link observables rather than curvatures assigned after interpolation.

Lemma D.12 (Exact link evolution and negative-norm time control). *The coordinate $Z_i = (U_i - I)/h$ satisfies*

$$\partial_t Z_i = E_i U_i + D_i^+ A_0 + Z_i S_i A_0 - A_0 Z_i. \quad (\text{D.54})$$

On the chart $h|A_i| \leq \delta_*$,

$$|Z_i| \leq C|A_i|, \quad |Z_i - A_i| \leq Ch|A_i|^2, \quad |D_j^+ Z_i| \leq C|D_j^+ A_i|, \quad (\text{D.55})$$

$$\|\partial_t Z\|_{H_h^{-2}} \leq C(\|E\|_{2,h} + (1 + \|Z\|_{2,h})\|A_0\|_{2,h}). \quad (\text{D.56})$$

The constants are independent of the cutoff.

Proof. Multiply (D.53) on the right by U_i^{-1} and use $U_i = I + hZ_i$; rearrangement gives (D.54). The first two pointwise estimates are the exponential integral formula and its quadratic remainder. The differential of $Q_h(A) = (e^{hA} - I)/h$ has norm at most $e^{h|A|}$ on the fixed chart, so integration along the segment between neighboring coefficients gives the difference estimate.

Pair (D.54) with a matrix test ϕ of H_h^2 norm at most one. The uniform discrete embedding bounds $\|\phi\|_{\infty,h}$ and $\|D^-\phi\|_{2,h}$. Summation by parts moves D_i^+ from A_0 to ϕ , while the two product terms have L_h^1 norm at most $C\|Z\|_{2,h}\|A_0\|_{2,h}$. The electric term is bounded by $C\|E\|_{2,h}$ because U_i is uniformly bounded. Taking the supremum over tests proves (D.56). $\square$

Proof of Theorem 5.3. The hypotheses and Lemma D.12 give a uniform $L_t^\infty L_h^2 \cap L_t^2 H_h^1$ bound for Z_h and an $L_t^2 H_h^{-2}$ bound for $\partial_t Z_h$. For a fixed Fourier radius R , every coefficient of the projection onto $|k| \leq R$ is bounded in $H^1(0, T)$; hence the projected family is precompact in L^2 . The spatial H^1 estimate gives the uniform tail bound

$$\|(I - P_R)\mathcal{I}_h Z_h\|_{L_{t,x}^2} \leq CR^{-1}\|Z_h\|_{L_t^2 H_h^1}. \quad (\text{D.57})$$

A diagonal extraction followed by $R \rightarrow \infty$ therefore yields strong L^2 compactness of $\mathcal{I}_h Z_h$.

The spatial Sobolev inequality and interpolation give

$$\int_0^T \|A_h\|_{10/3,h}^{10/3} dt \leq \|A_h\|_{L_t^\infty L_h^2}^{4/3} \|A_h\|_{L_t^2 L_h^6}^2 \leq C. \quad (\text{D.58})$$

Using $h|A_i| \leq \delta_*$ and (D.55),

$$\|Z_h - A_h\|_{L^2}^2 \leq Ch^2 \int |A_h|^4 \leq C\delta_*^{2/3} h^{4/3} \int |A_h|^{10/3} \rightarrow 0. \quad (\text{D.59})$$

Thus A_h has the same strong spatial limit, while the assumed bound gives $A_{0,h} \rightharpoonup A_0$ after extraction.

Pass to the exact temporal identity. Since $U_i - I = hZ_i$, $\|E_i(U_i - I)\|_1 \leq h\|E_i\|_2\|Z_i\|_2 \rightarrow 0$. The shifted temporal arrays $S_i A_{0,h}$ have the same weak limit as $A_{0,h}$, and the remaining products pass by weak-strong duality. Summation by parts gives

$$\partial_t A_i = E_i + \partial_i A_0 + A_i A_0 - A_0 A_i \quad (\text{D.60})$$

in distributions.

For the magnetic record, the four-link plaquette expansion on the common identity chart has linear part equal to the discrete curl and quadratic part $[A_i, A_j]$. Translation errors are bounded in L^1 by $Ch\|D^+ A\|_2\|A\|_2$, while the cubic remainder is bounded by $Ch\|A\|_3^3$ and tends to zero by (D.58). Hence

$$F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j] \quad (\text{D.61})$$

in the weak limit. Equations (D.60)–(D.61) are exactly the components of $R(\omega) = d\omega + \omega \wedge \omega$. No strong compactness of A_0 has been used. $\square$

D.7 Full-connection compactness: a Hodge–temporal criterion

A separate connection compactness certificate.

Assumption D.13 (Connection compactness on a common physical cylinder). After the declared physical frame identifications, the actual interpolated connections form a relatively compact family in $L^2(K)$ on each compact cylinder K . The identifications and positive norms are fixed or uniformly controlled on overlaps. No rapidly varying unrecorded gauge transformation is used to assert convergence. A common subsequence is selected on a countable compact exhaustion.

The assumption is distinct from a curvature-size bound and from smallness of the connection Euler residual. The following gives a sufficient realization in finite quantities. It also identifies the additional records needed to replace the assumption by estimates.

On the periodic odd Cartesian regulator let $A_h = (A_{1,h}, A_{2,h}, A_{3,h})$ be the spatial connection coefficients and $A_{0,h}$ the temporal coefficient, all in one common trivialization. With the forward and backward differences of (5.16), set

$$(d_h A_h)_{ij} = D_i^+ A_{j,h} - D_j^+ A_{i,h}, \quad \delta_h A_h = - \sum_i D_i^- A_{i,h}. \quad (\text{D.62})$$

The internal coefficient norm is positive; it is not the indefinite Lorentz contraction. All norms in the following theorem are over the common finite time interval and spatial torus, with uniformly comparable physical measures. Let $\mathcal{I}_h$ be the trigonometric spatial interpolation.

Theorem D.14 (Finite Hodge–temporal sufficient criterion). *Suppose the actual spatial curvature writer obeys*

$$d_h A_h = R_h^{\text{sp}} - \mathcal{Q}_h(A_h) + \epsilon_h, \quad \|\mathcal{Q}_h(A_h)\|_{L_{t,h}^2} \leq C \|A_h\|_{L_{t,h}^4}^2, \quad (\text{D.63})$$

and that the following budget is uniform in h :

$$\begin{aligned} & \|(A_{0,h}, A_h)\|_{L_{t,h}^2} + \|R_h^{\text{sp}}\|_{L_{t,h}^2} + \|A_h\|_{L_{t,h}^4}^2 + \|\epsilon_h\|_{L_{t,h}^2} + \|\delta_h A_h\|_{L_{t,h}^2} \\ & + \|D^+ A_{0,h}\|_{L_{t,h}^2} + \|\partial_t \mathcal{I}_h(A_{0,h}, A_h)\|_{L_t^2 H_x^{-1}} \leq C_K. \end{aligned} \quad (\text{D.64})$$

Then the interpolated full connections are relatively compact in $L^2(K)$ and satisfy [Assumption D.13](#).

The proof in this appendix uses an exact discrete Hodge identity and a finite-mode temporal compactness argument. The codifferential, L^4 product and temporal budgets are additional physical conditions; they are not inferred from $\|R_h\|_2$ alone. For piecewise reconstructions, jumps and frame transfers must be included in the actual temporal derivative and incidence residual. The theorem does not establish existence of a gauge satisfying these estimates for an arbitrary Lorentz connection.

The compactness result here is a quantitative sufficient criterion in a common gauge, not a gauge-selection theorem. It is an instance of the spatial compactness and temporal regularity mechanism underlying [69]; the particular finite normalization is proved directly.

Lemma D.15 (Exact periodic discrete Hodge identity). *For a vector-valued periodic spatial one-cochain $A = (A_1, A_2, A_3)$,*

$$\sum_{i,j} \|D_i^+ A_j\|_h^2 = \sum_{i < j} \|D_i^+ A_j - D_j^+ A_i\|_h^2 + \left\| \sum_i D_i^- A_i \right\|_h^2. \quad (\text{D.65})$$

All coefficient norms use the same positive inner product.

Proof. For a Fourier frequency k let $d_i = (e^{2\pi i h k_i} - 1)/h$. The symbol of D_i^- is $-\bar{d}_i$. On the coefficient vector $z = (\hat{A}_1, \hat{A}_2, \hat{A}_3)$ the identity is

$$\sum_{i < j} |d_i z_j - d_j z_i|^2 + \left| \sum_i \bar{d}_i z_i \right|^2 = \left(\sum_i |d_i|^2 \right) \left(\sum_j |z_j|^2 \right).$$

Expand the two squares; their cross terms cancel. Apply this identity to each internal coordinate and sum using Parseval. No continuum product rule or curvature identity is used in this step. $\square$

Proof of Theorem D.14. The incidence (D.63) and the stated product bound give a uniform $L^2_{t,h}$ bound for $d_h A_h$. Equation (D.65) then bounds every spatial forward derivative of A_h in that norm. The temporal coefficient has its own spatial derivative bound in (D.64). Uniform equivalence of the forward-difference Sobolev norm and the trigonometric H^1 norm on the odd frequency cube therefore gives

$$\|\Omega_h\|_{L^2_t H^1_x} + \|\partial_t \Omega_h\|_{L^2_t H^{-1}_x} \leq C_K, \quad \Omega_h = \mathcal{I}_h(A_{0,h}, A_h).$$

For completeness fix a Fourier radius R and let P_R be the common orthogonal projection onto $|k| \leq R$. The spatial estimate gives

$$\|(I - P_R)\Omega_h\|_{L^2_{t,x}}^2 \leq C_K(1 + R^2)^{-1}.$$

Each of the finitely many coefficients of $P_R \Omega_h$ is bounded in $L^2(0, T)$, and its distributional time derivative is bounded in $L^2(0, T)$: pair the H^{-1} derivative with that fixed smooth Fourier vector. Thus these coefficients are bounded in $H^1(0, T)$ and relatively compact in $L^2(0, T)$. Choose a subsequence successively for $R = 1, 2, \dots$. The uniform tail estimate makes the diagonal subsequence Cauchy in $L^2_{t,x}$. Uniformly comparable physical norms give the asserted strong compactness of the actual connections. Restriction and a second diagonal argument give a common subsequence on a countable cylinder exhaustion. $\square$

D.8 Transported screens and general curvature identification

Proposition D.16 (Transported-screen version of connection compactness). *Let $\Omega_X \in L^2(0, T; \mathcal{H})$ be uniformly bounded after common physical interpolation into a Hilbert space $\mathcal{H}$. For every R suppose there are time-independent finite-rank operators $P_{X,R}$ and P_R such that $\|P_{X,R} - P_R\|_{\text{op}} \rightarrow 0$, and*

$$\lim_{R \rightarrow \infty} \sup_X \|(I - P_{X,R})\Omega_X\|_{L^2_t \mathcal{H}} = 0. \quad (\text{D.66})$$

Assume the coefficients of $P_R \Omega_X$ in a fixed finite basis of its range form a relatively compact family in $L^2(0, T)$ for each R . Then Ω_X is relatively compact in $L^2(0, T; \mathcal{H})$.

Proof. For fixed R , operator-norm convergence and the uniform L^2 bound give $(P_{X,R} - P_R)\Omega_X \rightarrow 0$ in $L^2_t \mathcal{H}$. Extract successive subsequences on which the finitely many coefficients of $P_R \Omega_X$ converge. For the diagonal subsequence, split a difference into its two screen tails and its finite-rank part. First take the cutoff limit at fixed R and then let $R \rightarrow \infty$, using (D.66). This gives a Cauchy sequence in the complete space $L^2_t \mathcal{H}$. $\square$

A bounded H^1 time norm for each transported coefficient is one sufficient condition in the proposition. If screens or frames depend on time, their actual derivatives contribute to those coefficient bounds. Spatial rank counts by themselves do not provide this temporal compactness. This formulation allows non-Cartesian regulators without identifying arbitrary finite graph connections with smooth-gauge connections in advance.

Identification with the actual connection curvature. Curvature size and curvature identification are separate questions. Let $\mathcal{T}_{X,K}$ be the finite determining curvature-test bank, and let $\mathcal{R}_X(\omega_X)$ denote the finite Cartan writer on the common-cylinder interpolation. Define

$$\mathfrak{C}_X(K) := \sup_{\substack{\Phi \in \mathcal{T}_{X,K} \\ \|\Phi\|_{L^2(K)} \leq 1}} \left| \int_K \langle R_X - \mathcal{R}_X(\omega_X), \Phi \rangle \right|. \quad (\text{D.67})$$

The banks exhaust a fixed countable determining smooth core, with compatible test restrictions and vanishing test-interpolation errors. Their geometric consistency is recorded separately by

$$\mathfrak{I}_X(\Phi) := \int_K \langle \mathcal{R}_X(\omega_X), \Phi \rangle - \langle d_{\text{dist}}\omega_X + \omega_X \wedge \omega_X, \Phi \rangle. \quad (\text{D.68})$$

The distributional derivative includes every interface contribution of a piecewise connection. Thus matching two finite writers alone is not used to identify the nonlinear curvature of a weakly convergent connection.

For the propagated $p = 2$ route the operational hypotheses are

$$\omega_X \rightarrow \omega \text{ strongly in } L^2_{\text{loc}}, \quad \mathfrak{C}_X(K) \rightarrow 0, \quad \mathfrak{I}_X(\Phi) \rightarrow 0 \quad (\text{D.69})$$

The theorem below gives the more general L^p statement.

Theorem D.17 (Weak curvature compactness with an identified connection limit). *Let K be a compact cylinder and $1 < p < \infty$. A bound*

$$\sup_X \|R_X\|_{L^p(K)} \leq C_{K,p} < \infty \quad (\text{D.70})$$

gives a weakly convergent subsequence

$$R_X \rightharpoonup R \quad \text{in } L^p(K). \quad (\text{D.71})$$

If, in addition, $\omega_X \rightarrow \omega$ strongly in L^2_{loc} and

$$R_X - [d_{\text{dist}}\omega_X + \omega_X \wedge \omega_X] \rightarrow 0 \quad \text{in distributions,} \quad (\text{D.72})$$

then $R = d\omega + \omega \wedge \omega$ distributionally. The finite-writer conditions (D.67)–(D.68), with exhaustive determining tests and their interpolation control, give (D.72). In particular, the propagated $p = 2$ bound gives the stated connection-curvature conclusion.

Proof. For $1 < p < \infty$, reflexivity gives weak sequential compactness of bounded balls of $L^p(K)$. Strong L^2 convergence of the connection gives convergence of its distributional derivative, and

$$\|\omega_X \wedge \omega_X - \omega \wedge \omega\|_{L^1(K)} \leq C(\|\omega_X\|_2 + \|\omega\|_2)\|\omega_X - \omega\|_2 \rightarrow 0.$$

Thus (D.72) identifies the distributional limit of R_X . Weak L^p convergence has the same distributional limit, proving the assertion. For the finite-writer version, split the tested residual into $R_X - \mathcal{R}_X(\omega_X)$ and the discrepancy (D.68). The first tends to zero on the exhaustive core by (D.67), and the second by the stated consistency. The declared test-interpolation control extends the identity to the determining test topology. A diagonal subsequence works on a countable compact exhaustion.

If the curvature norms are instead unbounded, choose a subsequence on which they diverge and retain the unit fields $R_X/\|R_X\|_{L^p(K)}$ together with their diverging scales. This witnesses failure of the specified curvature-size certificate, not nonexistence of every possible weaker limit. $\square$

For a piecewise smooth connection, its full distributional derivative includes face measures. A bound on cell interiors alone therefore does not control the curvature entering the continuum variation. The following construction specifies a sufficient volume realization without discarding these measures.

Work on a periodic rectangular four-dimensional domain with a uniform cubical mesh of side h , or on a compact interior subdomain together with its neighbouring cells. Polynomial degrees are fixed, positive Grams and physical measures are uniformly comparable to the reference ones, and all face coefficients are compared in a common local trivialization. For an oriented interior face f , set $J_f = n_f^b \wedge [\omega_h]_f$, with the sign chosen so that

D.9 Interfaces and distributional curvature

For discontinuous connection reconstructions, the identification has a concrete sufficient realization. On a uniformly regular cubical chart put $J_f = n_f^b \wedge [\omega_h]_f$ at each interior face and

$$\mathcal{J}_h^2 = \sum_f h^{-1} \|J_f\|_{L^2(f)}^2. \quad (\text{D.73})$$

A bounded-overlap volume lifting $\mathcal{L}_h J$ of the face measures gives

$$\begin{aligned} R_h^{\text{lift}} &= \text{d}_b \omega_h + \omega_h \wedge \omega_h + \mathcal{L}_h J, \\ \|R_h^{\text{lift}}\|_{L^2(K)} &\leq \|\text{d}_b \omega_h\|_2 + C \|\omega_h\|_4^2 + C \mathcal{J}_h. \end{aligned} \quad (\text{D.74})$$

Its discrepancy from the full distributional curvature is at most $C_K h \mathcal{J}_h \|\nabla \Phi\|_\infty$ on a fixed smooth test. The construction and proof are given in [Theorem D.19](#). These are explicit additional interface budgets, not consequences of a small connection Euler residual. The lifting is used only when the action's actual curvature writer agrees with it on the determining tests, up to a vanishing recorded discrepancy.

$$\text{d}_{\text{dist}} \omega_h = \text{d}_b \omega_h + \sum_f J_f \delta_f. \quad (\text{D.75})$$

Let $\eta \geq 0$ be a fixed smooth function supported in $(-1/4, 1/4)$ with integral one. On the normal prism of f , define

$$(\mathcal{L}_f J_f)(y + r n_f) = h^{-1} \eta(r/h) J_f(y), \quad \mathcal{L}_h J = \sum_f \mathcal{L}_f J_f, \quad (\text{D.76})$$

and extend each term by zero outside its prism.

Lemma D.18 (Interface lifting and its distributional error). *With $\mathcal{J}_h$ defined in [\(D.73\)](#),*

$$\|\mathcal{L}_h J\|_2 \leq C \mathcal{J}_h, \quad \left| \left\langle \mathcal{L}_h J - \sum_f J_f \delta_f, \Phi \right\rangle \right| \leq C_K h \mathcal{J}_h \|\nabla \Phi\|_\infty \quad (\text{D.77})$$

for each fixed smooth compactly supported or periodic test Φ .

Proof. Integration in the normal coordinate gives

$$\|\mathcal{L}_f J_f\|_2^2 = h^{-1} \|\eta\|_{L^2(\mathbb{R})}^2 \|J_f\|_{L^2(f)}^2.$$

The prisms have bounded overlap depending only on dimension. Pointwise Cauchy–Schwarz over the overlapping prisms proves the first bound. For the second, subtract $\Phi(y)$ from $\Phi(y + rn_f)$ and use $|r| \leq h/4$. The total error is at most $Ch\|\nabla\Phi\|_\infty \sum_f \|J_f\|_{L^1(f)}$. Moreover

$$\sum_f h\|J_f\|_{L^1(f)} \leq \left(\sum_f h^{-1}\|J_f\|_{L^2(f)}^2 \right)^{1/2} \left(\sum_f h^3|f| \right)^{1/2} \leq C_K h \mathcal{J}_h.$$

There are $O_K(h^{-4})$ faces of area h^3 , which proves the last inequality and the result. $\square$

Theorem D.19 (Constructive connection-and-interface curvature criterion). *Suppose*

$$\|\mathrm{d}_b\omega_h\|_{L^2(K)} + \|\omega_h\|_{L^4(K)}^2 + \mathcal{J}_h \leq C_K. \quad (\text{D.78})$$

Then the volume curvature R_h^{lift} in (D.74) is uniformly bounded in $L^2(K)$ and has the displayed distributional error. If also $\omega_h \rightarrow \omega$ strongly in L^2_{loc} , every weak L^2 cluster point equals $\mathrm{d}\omega + \omega \wedge \omega$ distributionally.

Proof. The matrix exterior product satisfies $|\omega_h \wedge \omega_h| \leq C|\omega_h|^2$. Hölder’s inequality and Lemma D.18 give (D.74). The discrepancy from $\mathrm{d}_{\text{dist}}\omega_h + \omega_h \wedge \omega_h$ is exactly the lifting error in (D.77), which tends to zero on each fixed test. Apply Theorem D.17 with $p = 2$. $\square$

The theorem is a sufficient identification and size criterion with its own explicit connection and jump budgets. A stationary connection estimate in L^2 does not imply (D.78). Nor is R_h^{lift} silently substituted for the action’s curvature writer: that writer must agree with the lift on the determining tests up to a vanishing recorded discrepancy. On a conforming connection reconstruction $J_f = 0$, and the interface part is absent.

D.9.1 The weak–strong Palatini passage

D.10 Weak–strong variational passage

Lemma D.20 (Weak–strong curvature pairing and volume convergence). *Assume (D.36), let $p > 3/2$, and suppose*

$$R_X \rightharpoonup R \quad \text{weakly in } L^p(K). \quad (\text{D.79})$$

Then, for every bounded compactly supported coefficient tensor Φ ,

$$\int_K \langle \Phi(e_X \wedge e_X), R_X \rangle \longrightarrow \int_K \langle \Phi(e \wedge e), R \rangle. \quad (\text{D.80})$$

The same conclusion holds with an internal Hodge star applied to either bivector factor. Moreover

$$e_X \wedge e_X \wedge e_X \wedge e_X \longrightarrow e \wedge e \wedge e \wedge e \quad \text{strongly in } L^1(K). \quad (\text{D.81})$$

Proof. Weak convergence in L^p implies $\sup_X \|R_X\|_{L^p} < \infty$. Put $B_X = \Phi(e_X \wedge e_X)$ and $B = \Phi(e \wedge e)$. By Lemma D.7, $B_X \rightarrow B$ strongly in $L^{p'}$. Split the difference as

$$\int_K \langle B_X, R_X \rangle - \int_K \langle B, R \rangle = \int_K \langle B_X - B, R_X \rangle + \int_K \langle B, R_X - R \rangle. \quad (\text{D.82})$$

The first term tends to zero by Hölder and the uniform L^p bound on R_X ; the second tends to zero by the definition of weak L^p convergence, because $B \in L^{p'}$. The internal Hodge star is a fixed bounded linear map on the bivector fibre and changes none of these estimates.

For the volume term, [Lemma D.7](#) with $q = 4$ gives $e_X \rightarrow e$ strongly in $L^4(K)$ and uniform L^4 boundedness. Expanding the fourfold difference one factor at a time and applying Hölder with four L^4 factors gives

$$\|e_X^{\wedge 4} - e^{\wedge 4}\|_{L^1} \tag{D.83}$$

$$\leq C\|e_X - e\|_{L^4}(\|e_X\|_{L^4} + \|e\|_{L^4})^3 \rightarrow 0, \tag{D.84}$$

which is [\(D.81\)](#). $\square$

Remark D.21 (Weak–strong passage without compensated compactness). The product passage above does not use a div–curl or compensated-compactness lemma. Such a device would be needed if both the coframe bivector and curvature were only weakly convergent. Here the compact-screen hypothesis gives strong L^2 convergence of the coframe together with a uniform L^6 bound; the explicit interpolation in [Lemma D.7](#) upgrades this to strong $L^{2p'}$, so ordinary weak–strong duality suffices. At this stage all fields are already represented by their canonical common-cylinder interpolants; equivalently, the same estimates may be read cellwise with the physical cell measures because the displayed L^q hypotheses are precisely the cutoff-uniform interpolation bounds used in the handoff.

Theorem D.22 (Localized renewal–Palatini handoff). *Assume the face-linearization and complete memory-short branches, the polynomial local class of [Definition 4.6](#), real Lorentz naturality, scalar residual typed multiplicity commutant, and represented determinant volume. Assume in addition the physical first-variation remainder condition [\(4.25\)](#). Let the constant calibrated coefficients converge to (α, β, λ) with $\beta \neq 0$, and let the spinless connection Euler residual vanish with the uniform Cartan inverse of [Theorem C.14](#). On every compact K , assume [\(D.36\)](#), nondegeneracy of the limiting coframe, and either the continuous or the discrete initial-energy curvature certificate. Assume also that the connection curvature is identified either by [Theorem D.17](#) or by the literal-link route of [Theorem 5.3](#), and assume the bounded, convergent physical test lifts of (E1) in [Theorem 5.4](#). Then the limiting score has the classified form*

$$\alpha L_{\text{Holst}} + \beta L_{\text{Palatini}} + \lambda L_{\text{vol}}, \tag{D.85}$$

limiting torsion vanishes, and the metric first variations converge to

$$\chi \int_K \sqrt{-g} (G_{\mu\nu} + \Lambda g_{\mu\nu}) k^{\mu\nu}, \quad \chi \neq 0. \tag{D.86}$$

Here χ is the nonzero Palatini normalization and Λ is the corresponding calibrated volume-to-Palatini coefficient ratio, in the density conventions of [\(4.21\)](#).

Proof. Both coframe bivectors and Lorentz curvature lie in $\Lambda^2 \mathbb{R}^{1,3}$. After complexification the bivector module splits into nonisomorphic self-dual and anti-self-dual irreducibles. The real commutant is therefore

$$\text{End}_{SO^+(1,3)}(\Lambda^2 \mathbb{R}^{1,3}) = \text{span}_{\mathbb{R}}\{I, \star\}. \tag{D.87}$$

The residual typed commutant supplies no additional internal tensor, and the only alternating four-coframe scalar at this order is determinant volume. Thus [\(D.85\)](#) follows in the stated first-order class. The calibrated coefficient limits ensure that these are the coefficients of the limiting variational functional, rather than unrelated values chosen after taking the limit. The first-variation remainder condition removes the unclassified part on the actual physical tests; no interchange of differentiation and merely pointwise convergence is being assumed.

By [Theorem C.14](#), the vanishing connection Euler residual and the uniform inverse bounds give vanishing torsion. On the full-connection route, strong local L^2 convergence of ω_X passes the Cartan product directly. On the literal-link route the spatial coefficients converge strongly in L^2 while the temporal coefficient converges weakly in L^2 ; strong convergence of the coframe is sufficient to pass every torsion product by weak–strong duality. The distributional derivative of e_X passes against every fixed smooth test, so the limit is a metric-compatible torsion-free represented connection. Its curvature is identified by the selected connection route, not merely by matching finite curvature labels.

The continuous or discrete propagation theorem supplies $\sup_X \|R_X\|_{L^2(K)} < \infty$. After passage to the identified weak subsequence, [Lemma D.7](#) gives $e_X \wedge e_X \rightarrow e \wedge e$ strongly in L^2 . The curvature terms therefore converge by [Lemma D.20](#) with $p = p' = 2$. The same argument applies to their physical coframe variations: their coefficient tensors are uniformly bounded and converge, so multiplying the strong L^2 bivectors by these tensors preserves strong convergence. The four-coframe volume term and its test insertions converge in L^1 by the same lemma. No product of two uncontrolled weak limits is used.

For the torsion-free connection, the first Bianchi identity removes the Holst coframe variation. The Palatini and volume variations give the Einstein and cosmological insertions in [\(D.86\)](#). The coefficient of the first is nonzero because $\beta \neq 0$. A nonzero coefficient pair alone would only justify the torsion step, and would not justify division by this Einstein coefficient. $\square$

Corollary D.23 (Vacuum equation from certified accepted-action variations). *Under the preceding hypotheses, vanishing realized common-action variations on the determining metric-test core imply*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$$

distributionally on every compact cylinder. The scaled hypothesis [\(C.11\)](#) is a sufficient stochastic route on the almost-sure subsequence of [Proposition C.7](#), provided the remaining compactness and identification budgets hold on the same realized sequence.

Proof. The handoff identifies the limit of each certified first variation with [\(D.86\)](#). Its value is zero on the determining core, and $\chi \neq 0$ permits division. Continuity in the declared test topology extends the identity to the closure of that core. In the stochastic case, first take the realized vanishing subsequence supplied by [Proposition C.7](#); the assumed pathwise compactness budgets then permit the curvature subsequence used in the handoff. A mean-square bound is not interpreted as stationarity of every finite sample. $\square$

D.11 The failed-certificate witnesses

For reference, the complete failure bank consists of a common-action curl or period; a nonvanishing realized metric variation or failure of its scaled action-gap bound; a vanishing Palatini normalization or failed coupling calibration; a cut/potential witness; a face-linearity, memory or torsion defect; an unbounded initial, growth, transfer or differentiated-source budget; an unstable curvature interpolation; a Cartan or interface mismatch; failure of the literal-link chart/energy criterion on that route; or failure of the common-cylinder compactness conditions.

Proof of Corollary 5.5. For a complete cylinder, [Theorem 4.1](#) reconstructs the finite determining field and accepted kernel. The reward-pressure and common-action theorems either reconstruct the common action or expose its curl/period failure. The accepted-action theorem and the physical test-lift estimate give a quantitative stationarity certificate with every cutoff constant retained.

The spatial cut theorem gives the stated spatial screen or a cut witness, while the common-cylinder comparison packet separately records the temporal and transported-screen conditions needed for strong coframe convergence.

Face linearization and memory subtraction give the classified first-order score on their positive branches. The calibrated coupling and Cartan tests supply the nonzero Einstein coefficient and torsion control. Continuous or fully discrete propagation bounds the actual curvature from the finite initial energy and growth/transfer/source records; the sharp current transfer may supply the last budget from scaled action errors. Finally the connection and interface comparisons identify that curvature. If all certificates pass, [Theorem D.22](#) and [Corollary D.23](#) prove the positive branch, using a diagonal subsequence on the compact exhaustion.

For completeness, the cofinal witnesses in the complementary branch are obtained from the quantifiers of the failed certificates. A nonnegative residual which does not tend to zero has a subsequence bounded below by some $\epsilon > 0$. An unbounded budget has a subsequence tending to infinity. A failed strictly positive lower margin has a subsequence tending to zero; for the cut margin the finite graph supplies the corresponding potential witness. Failure of a uniform compactness tail supplies an $\epsilon > 0$ and normalized components outside arbitrarily large declared screens. Failure of a required calibrated Cauchy comparison retains the separated pairs of finite records themselves. A finite algebraic failure retains its actual curl, period or operator residual. Discarding a finite prefix is allowed when checking the cofinal tail. These records identify which sufficient certificate failed; they do not establish that every alternative compactness argument or every other subsequence must fail. $\square$

Remark D.24 (What is discharged and what remains dynamical). Neither a continuum curvature bound nor an unidentified nonlinear connection product is inserted into the conclusion. The former follows from the continuous or discrete propagation certificate; the latter is handled by strong connection convergence and the full distributional Cartan comparison, including interfaces. The differentiated source budget can follow from (D.48), whereas bounded symmetric growth, physical occurrence and noncollapse remain genuine conditions of the microscopic evolution. A bounded mean curvature norm does not by itself supply the pathwise version. The fixed-amplitude nonsymmetric Cauchy theorem is supplied independently by [Appendix E](#) for its declared local writer; the Gowdy sector below remains a complementary nonlinear symmetry-reduced class with pathwise full-curvature control.

E Constructive nonsymmetric evolution and finite renewal realization

This section proves [Theorem 6.1](#) directly from the declared local writer. The purpose is to supply a fixed-radius, genuinely three-dimensional nonlinear population theorem on the same reconstructed Lorentzian coefficient chart used by the general certification theorem, without invoking (E1)–(E5) as independent assumptions. The evolution law is declared explicitly and is kept logically distinct from the exact finite Euler flow studied in [Appendix G](#). No symmetry reduction or analytic Fourier tail is used.

E.1 Execution graph, exact difference calculus, and the local writer

Let $N = 2m + 1$, $h = N^{-1}$ and $\Lambda_h = (h\mathbb{Z}/\mathbb{Z})^3$. Let $\mathcal{I}_h$ denote the real trigonometric interpolation on the signed frequency cube $\{-m, \dots, m\}^3$, and let $\mathcal{S}_h$ denote grid sampling. The tetrahedral

$A_3 \cong D_3$ root execution directions may be represented as

$$\mathcal{R}_F = \{\pm(e_i + e_j), \pm(e_i - e_j) : 1 \leq i < j \leq 3\}. \quad (\text{E.1})$$

All these integer vectors have even coordinate sum. This harmless parity at fixed smooth frequency becomes relevant for a cutoff-uniform Cartesian Sobolev realization.

Lemma E.1 (Root-parity obstruction and a minimal execution connector). *On the odd periodic grid, the root Dirichlet form*

$$\mathcal{D}_F(u) = \sum_{r \in \mathcal{R}_F / \{\pm 1\}} \|(S_r - I)u/h\|_h^2 \quad (\text{E.2})$$

is not uniformly coercive for $\|\nabla \mathcal{I}_h u\|_{L^2}^2$. Moreover the root-graph distance from 0 to e_1 is $N - 1$. If the oriented execution edges $\pm e_1$ are adjoined, then with

$$\mathcal{D}_+(u) = \mathcal{D}_F(u) + \|D_{e_1}^+ u\|_h^2, \quad \mathcal{D}_C(u) = \sum_i \|D_i^+ u\|_h^2, \quad (\text{E.3})$$

one has

$$\frac{1}{5} \mathcal{D}_C(u) \leq \mathcal{D}_+(u) \leq 9 \mathcal{D}_C(u). \quad (\text{E.4})$$

At least one added fixed direction of odd coordinate sum is necessary within this fixed-direction class. The connector is an execution incidence only; it does not enter the A_3 moment reconstruction.

Proof. Take $u_h(x) = \exp(2\pi i m(x_1 + x_2 + x_3))$. Every difference in a minus root vanishes and every plus-root multiplier remains bounded, while $\|\nabla \mathcal{I}_h u_h\|_2 \asymp m$ diverges. The same argument applies to every fixed-radius word in even-parity directions. A root path ending at e_1 has integer displacement $e_1 + Nz$ with even coordinate sum; since N is odd, $\sum z_i$ is odd and one coordinate has modulus at least $N - 1$. The explicit alternating pair $-e_1 + e_2, -e_1 - e_2$, repeated m times, attains that distance.

For the connector, the identity

$$S_{e_2} - I = S_{e_1}^{-1} [(S_{e_1+e_2} - I) - (S_{e_1} - I)]$$

and its e_3 analogue give $\mathcal{D}_C \leq 5\mathcal{D}_+$. Conversely each root difference is bounded by the sum of the two participating coordinate differences, so $\mathcal{D}_F \leq 8\mathcal{D}_C$. Adding the connector gives the upper bound. If every added direction had even coordinate sum, the high-frequency test above would remain unchanged. $\square$

Define $S_i, D_i^\pm, D_i^0$ by (5.16), use $\langle u, v \rangle_h = h^3 \sum_x u(x) \cdot v(x)$, and for a multi-index put $D^\alpha = \prod_i (D_i^+)^{\alpha_i}$ and $S^\alpha = \prod_i S_i^{\alpha_i}$. Set

$$\|u\|_{r,h}^2 = \sum_{|\alpha| \leq r} \|D^\alpha u\|_h^2, \quad \|X\|_{\mathcal{X}_h^r}^2 = \|q\|_{r+1,h}^2 + \|v\|_{r,h}^2. \quad (\text{E.5})$$

Lemma E.2 (Exact shifted product rule and uniform Sobolev calculus). *On the periodic array space,*

$$(D_i^+)^* = -D_i^-, \quad (D_i^0)^* = -D_i^0, \quad (\text{E.6})$$

and

$$D^\alpha(fw) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (S^{\alpha-\beta} D^\beta f) D^{\alpha-\beta} w. \quad (\text{E.7})$$

For $r \geq 3$, the norms (E.5) are uniformly equivalent to the trigonometric H^r norms, form a uniform product algebra, and the shifted commutator

$$\mathcal{C}_\alpha(f, w) = D^\alpha(fw) - (S^\alpha f)D^\alpha w \quad (\text{E.8})$$

satisfies, for a constant coefficient f_* ,

$$\|\mathcal{C}_\alpha(f, w)\|_h \leq C_r \|f - f_*\|_{r,h} \|w\|_{r-1,h}, \quad (\text{E.9})$$

$$\|D_i^- \mathcal{C}_\alpha(f, w)\|_h + \|D_i^0 \mathcal{C}_\alpha(f, w)\|_h \leq C_r \|f - f_*\|_{r+1,h} \|w\|_{r,h}. \quad (\text{E.10})$$

The constants are independent of h .

Proof. Translation of the finite sum gives the adjoints, and $D_i^+(fw) = (S_i f)D_i^+ w + (D_i^+ f)w$ gives (E.7) by iteration. The forward symbol $d_i(k) = (e^{2\pi i h k_i} - 1)/h$ satisfies $4|k_i| \leq |d_i(k)| \leq 2\pi|k_i|$ on $|k_i| \leq m$. Parseval therefore gives uniform norm equivalence. Multiplication is convolution on the finite frequency group and the exact identity $d_i([k+p]) = d_i(k) + e^{2\pi i h k_i} d_i(p)$ gives the usual uniform weighted convolution estimate. In every commutator term at least one difference falls on f ; one additional output difference gives the second line. Since $r > 3/2$, the required Fourier ℓ^1 weights are summable uniformly in the finite cube. Analytic coefficient compositions follow from their convergent power series on a smaller ball. $\square$

For the harmonic reduced vacuum equation use the coefficient functions in (6.1). The skew transport is (6.2) and the first-jet compensator is (6.3). Pointwise multiplication operators are denoted M_f .

Proposition E.3 (Locality and continuum consistency of the writer). *The acceleration in (6.4) at a site depends only on the site and the twelve coordinate-stencil neighbours $\{\pm e_i, \pm(e_i - e_j) : i < j\}$. These lie within two hops of the execution graph obtained in Lemma E.1. On every smooth metric jet, the continuum limit of the second row is exactly (6.1).*

Proof. The flux at x is

$$h^{-2} \sum_{i,j} \{c^{ij}(x)[q(x + he_j) - q(x)] - c^{ij}(x - he_i)[q(x - he_i + he_j) - q(x - he_i)]\}.$$

The skew transport uses only $x \pm he_i$, and the compensator only first forward differences. In the smooth limit the divergence contributes $(\partial_i c^{ij})\partial_j q$ and the skew transport contributes $(\partial_i b^i)v$ in addition to the desired second- and mixed-derivative terms. The two coefficient-differential terms in (6.3) cancel exactly these additions by the ordinary chain rule. No discrete Leibniz rule is invoked. $\square$

E.2 Shifted nonlinear energy and a cutoff-independent lifespan

On a sufficiently small fixed $\mathcal{X}_h^s$ ball, $a \geq a_* > 0$ and $c \succeq c_* I$. Put $a_\alpha = S^\alpha a$, $c_\alpha = S^\alpha c$, $q_\alpha = D^\alpha q$, and $v_\alpha = D^\alpha v$, and define

$$\mathcal{E}_{s,h}(X) = \frac{1}{2} \sum_{|\alpha| \leq s} \left[\langle v_\alpha, a_\alpha v_\alpha \rangle_h + \sum_{i,j} \langle D_i^+ q_\alpha, c_\alpha^{ij} D_j^+ q_\alpha \rangle_h + \|q_\alpha\|_h^2 \right]. \quad (\text{E.11})$$

It is uniformly equivalent to $\|X\|_{\mathcal{X}_h^s}^2$.

Theorem E.4 (All-component shifted nonlinear energy). *For $s \geq 11$, every solution of (6.4) remaining in the fixed chart satisfies*

$$\frac{d}{dt} \mathcal{E}_{s,h} \leq C_s \mathcal{E}_{s,h} + C_s \mathcal{E}_{s,h}^{3/2}, \quad (\text{E.12})$$

with C_s independent of h .

Proof. Apply D^α to the mass row. The exact shifted product rule gives

$$a_\alpha \dot{v}_\alpha = \sum_{i,j} D_i^- (c_\alpha^{ij} D_j^+ q_\alpha) - \mathbf{K}_{b_\alpha} v_\alpha + \mathbf{R}_\alpha, \quad (\text{E.13})$$

where every term in $\mathbf{R}_\alpha$ contains either a shifted commutator or the quadratic first-jet source. In particular the outer-divergence commutator is retained, rather than treated as an undifferentiated error. Lemma E.2 and the analytic coefficient bounds imply

$$\left(\sum_{|\alpha| \leq s} \|\mathbf{R}_\alpha\|_h^2 \right)^{1/2} \leq C_s \|X\|_{\mathcal{X}_h^s}^2. \quad (\text{E.14})$$

Pair (E.13) with v_α . Summation by parts cancels the divergence against the derivative of the spatial energy, while $\langle v_\alpha, \mathbf{K}_{b_\alpha} v_\alpha \rangle_h = 0$. Thus

$$\begin{aligned} \dot{\mathcal{E}}_{s,h} = \sum_{|\alpha| \leq s} & \left[\frac{1}{2} \langle v_\alpha, \dot{a}_\alpha v_\alpha \rangle_h + \frac{1}{2} \sum_{i,j} \langle D_i^+ q_\alpha, \dot{c}_\alpha^{ij} D_j^+ q_\alpha \rangle_h \right. \\ & \left. + \langle q_\alpha, v_\alpha \rangle_h + \langle v_\alpha, \mathbf{R}_\alpha \rangle_h \right]. \end{aligned} \quad (\text{E.15})$$

The coefficient time derivatives are analytic multiples of v and have ℓ^∞ size $O(\|X\|_{\mathcal{X}_h^s})$. Energy equivalence, Cauchy–Schwarz and (E.14) prove the result. The translated coefficient in each differentiated row is essential: replacing it by an unshifted continuum Leibniz rule would leave uncanceled top-order terms. $\square$

Corollary E.5 (Common nonlinear time and forced stability). *There are $\varepsilon_s, T_s, C_s > 0$, independent of h , such that every $\|X_h(0)\|_{\mathcal{X}_h^s} \leq \varepsilon_s$ has a unique solution through T_s satisfying (6.7). If an acceleration defect f_h is added, then on the same smaller chart*

$$(\mathcal{E}_{s,h}^{1/2})' \leq C_s \mathcal{E}_{s,h}^{1/2} + C_s \mathcal{E}_{s,h} + C_s \|f_h\|_{s,h}. \quad (\text{E.16})$$

Two top-bounded records satisfy the corresponding lower-order difference estimate

$$\sup_{t \leq T_s} \|X_h - Y_h\|_{\mathcal{X}_h^r} \leq C_s \left(\|X_h(0) - Y_h(0)\|_{\mathcal{X}_h^r} + \int_0^{T_s} \|f_h - \tilde{f}_h\|_{r,h} dt \right) \quad (\text{E.17})$$

for $3 \leq r \leq s - 1$.

Proof. The finite ODE is analytic on the coefficient chart. A scalar comparison for (E.12) keeps its energy strictly inside a smaller fixed ball on a time chosen before the cutoff; finite-dimensional continuation then gives the common lifespan. Adding the differentiated force to the remainder

proves (E.16). For the difference $X - Y = (q - p, v - w) = (\delta q, \delta v)$, write the principal row with the coefficients of q . Apart from $a(q)(f - \tilde{f})$, the additional source is exactly

$$\begin{aligned} \mathcal{R}_{X,Y} = & \sum_{i,j} D_i^- ((c^{ij}(q) - c^{ij}(p)) D_j^+ p) - \mathbf{K}_{b(q)-b(p)} w \\ & + \mathbf{G}_h(q, v) - \mathbf{G}_h(p, w) - (a(q) - a(p)) V_{0,h}(p, w), \end{aligned} \quad (\text{E.18})$$

where $V_{0,h}$ is the unforced acceleration. Writing the mass difference this way cancels the contribution of $\tilde{f}$ inside w_t ; it does not require a pointwise bound on that force. For $3 \leq r \leq s - 1$,

$$\|\mathcal{R}_{X,Y}\|_{r,h} \leq C_s \|X - Y\|_{\mathcal{X}_h^r}.$$

Indeed the flux term uses $p \in H_h^{r+2}$, the transport term uses $w \in H_h^{r+1}$, and the mass term uses $V_{0,h}(p, w) \in H_h^r$. These are all supplied by the top ball and $r \leq s - 1$. Analytic coefficient differences are bounded by $\|\delta q\|_{r+1,h}$; the first-jet source uses this norm and $\|\delta v\|_{r,h}$.

Use the quadratic energy (E.11) for $(\delta q, \delta v)$ with the shifted coefficients of q . The same principal cancellations and commutator bounds give

$$(\mathcal{E}_{\text{diff},r}^{1/2})' \leq C_s \mathcal{E}_{\text{diff},r}^{1/2} + C_s \|f - \tilde{f}\|_{r,h}.$$

Gronwall proves (E.17). No top-order force has been estimated by a bare inverse-mesh operator norm, so the constant has no $e^{CT/h}$ factor. $\square$

E.3 Compatible initial data, consistency, and subsidiary propagation

All Sobolev norms in this subsection are spatial norms on the unit torus; $C_t H^r$ denotes the supremum over the common time interval. Constants may depend on s and the fixed coefficient chart, but not on h . In particular, interpolation is not assumed to commute with multiplication or with the nonlinear coefficient maps.

Lemma E.6 (Quantitative sampling and nonlinear interpolation). *Let $r \geq 2$ be an integer, let $\mathcal{P}_h = \mathcal{I}_h \mathcal{S}_h$, and let u_h, w_h be grid arrays. On the odd frequency cube,*

$$\|\mathcal{P}_h u - u\|_{H^r} \leq C_r h \|u\|_{H^{r+1}}, \quad \|\mathcal{P}_h u\|_{H^r} \leq C_r \|u\|_{H^r}, \quad (\text{E.19})$$

$$\|\mathcal{I}_h D_i^\pm u_h - \partial_i \mathcal{I}_h u_h\|_{H^r} \leq C_r h \|u_h\|_{r+2,h}, \quad (\text{E.20})$$

$$\|\mathcal{I}_h(u_h w_h) - (\mathcal{I}_h u_h)(\mathcal{I}_h w_h)\|_{H^r} \leq C_r h \|u_h\|_{r+1,h} \|w_h\|_{r+1,h}. \quad (\text{E.21})$$

The first estimate also holds with h replaced by h^j and H^{r+1} by H^{r+j} for every fixed positive integer j . If A is an analytic coefficient map on the common small chart, then

$$\|\mathcal{I}_h A(u_h) - A(\mathcal{I}_h u_h)\|_{H^r} \leq C_{r,A} h \|u_h - u_*\|_{r+1,h} \quad (\text{E.22})$$

for a constant base point u_ . These estimates admit time-differentiated versions by the ordinary chain rule, with one spatial order reserved for each controlled time derivative.*

Proof. For u with Fourier coefficients $\hat{u}(k)$, the coefficient of its interpolant at $k \in \{-m, \dots, m\}^3$ is $\sum_{\ell \in \mathbb{Z}^3} \hat{u}(k + N\ell)$. The omitted high-frequency tail has H^r norm at most $CN^{-j} \|u\|_{H^{r+j}}$. For the folded part, apply Cauchy-Schwarz in $\ell \neq 0$ with weights $\langle k + N\ell \rangle^{r+j}$. Uniformly in the retained k ,

$$\sum_{\ell \neq 0} \langle k + N\ell \rangle^{-2(r+j)} \leq C_r N^{-2(r+j)}.$$

Multiplication by $\langle k \rangle^{2r} \leq C_r N^{2r}$ and summation in k prove the error estimate. The same argument with $j = 0$, together with the retained coefficients, proves stability since $r > 3/2$. The symbol estimate $|(e^{\pm i h \xi} - 1)/(\pm h) - i \xi| \leq C h |\xi|^2$ proves (E.20); averaging gives the corresponding estimate for D_i^0 .

Since $\mathcal{I}_h(u_h w_h) = \mathcal{P}_h[(\mathcal{I}_h u_h)(\mathcal{I}_h w_h)]$, the product estimate follows from sampling consistency and the H^{r+1} product inequality. For composition use $\mathcal{I}_h A(u_h) = \mathcal{P}_h[A(\mathcal{I}_h u_h)]$ and the analytic composition bound on the smaller common chart. Constant terms are interpolated exactly. Differentiating these identities, rather than postulating a discrete chain rule, gives the last assertion. $\square$

Lemma E.7 (Uniform physical time jets). *An unforced solution on the common top ball satisfies*

$$\|q_h\|_{C_t H_h^{s+1}} + \|q_{h,t}\|_{C_t H_h^s} + \|q_{h,tt}\|_{C_t H_h^{s-1}} + \|q_{h,ttt}\|_{C_t H_h^{s-2}} \leq C_s \varepsilon. \quad (\text{E.23})$$

The same bound holds uniformly for the coefficient family (6.11) with B constant during the history.

Proof. The first two terms follow from the nonlinear energy. The divergence row maps H_h^{s+1} to H_h^{s-1} , the skew transport maps H_h^s to H_h^{s-1} , and the first-jet source is controlled by the product calculus. Multiplication by $a(g_h)^{-1}$ preserves that space. Differentiating this same equation once replaces q_h by $q_{h,t}$ in the principal spatial term and v_h by $q_{h,tt}$ in the transport term. Every coefficient time derivative is an analytic multiple of v_h . The resulting terms are bounded in H_h^{s-2} . For the enlarged family, $h^2 \Lambda_h^2 : H_h^{r+2} \rightarrow H_h^r$ is uniformly bounded, by $h^2 \ell_h \leq 12$ in Fourier variables; apply this first to q_h and then to v_h . No time derivative of B occurs. $\square$

For $(\gamma, K) \in \mathcal{D}_s(\varepsilon)$ choose

$$\begin{aligned} g_{00}^0 &= -1, & g_{0i}^0 &= 0, & g_{ij}^0 &= \gamma_{ij}, \\ v_{ij}^0 &= -2K_{ij}, & v_{00}^0 &= 2 \operatorname{tr}_\gamma K, & v_{0i}^0 &= \gamma^{jk} (\partial_j \gamma_{ki} - \tfrac{1}{2} \partial_i \gamma_{jk}). \end{aligned} \quad (\text{E.24})$$

Initialize $q_h(0) = \mathcal{S}_h(g^0 - \eta)$ and $v_h(0) = \mathcal{S}_h v^0$. These are samples of initial data only, not of a future Einstein solution. Finite rounding of this preparation is addressed in Theorem E.17.

Use the lower-index gauge covector $c_b(g) = g_{b\mu} g^{\alpha\beta} \Gamma_{\alpha\beta}^\mu$ and set

$$\mathfrak{H}_{ab}(g, c) = \nabla_{(a} c_{b)} - \tfrac{1}{2} g_{ab} \nabla^d c_d, \quad r_{h,ab} = G_{ab}(g_h) - \mathfrak{H}_{ab}(g_h, c(g_h)), \quad g_h = \eta + \mathcal{I}_h q_h. \quad (\text{E.25})$$

Thus r_h is the actual continuum reduced residual of the interpolant. The conversion from the normal row (6.1) to r_h is a bounded analytic pointwise trace-reversal map on the common chart. This is the harmonic reduction convention used here; compare [74, Sec. 2.1].

Lemma E.8 (Consistency of the complete reduced residual). *For the unforced central writer,*

$$\|r_h\|_{C_t H^{s-2}} + \|\partial_t r_h\|_{C_t H^{s-3}} \leq C_s h \varepsilon. \quad (\text{E.26})$$

For the B -writer the same estimate holds with a uniform constant on $\overline{\mathcal{B}}$. In particular the residual includes all interpolation, mass, flux, transport and source errors.

Proof. Write $\tilde{q} = \mathcal{I}_h q_h$, $\tilde{v} = \mathcal{I}_h v_h$. For a representative spatial flux insert the two intermediate quantities $\partial_i \mathcal{I}_h(c_h^{ij} D_j^+ q_h)$ and $\partial_i[(\mathcal{I}_h c_h^{ij})(\mathcal{I}_h D_j^+ q_h)]$. Their differences from the discrete flux and the continuum flux are bounded in H^{s-2} , respectively, by

$$Ch \|c_h^{ij} D_j^+ q_h\|_{H_h^s}, \quad Ch \|c_h^{ij}\|_{H_h^s} \|D_j^+ q_h\|_{H_h^s}, \quad Ch \|q_h\|_{H_h^{s+1}},$$

with constant coefficient factors separated when necessary. The last term uses composition consistency and $\|\mathcal{I}_h D_j^+ q_h - \partial_j \tilde{q}\|_{H^{s-1}} \leq Ch \|q_h\|_{H_h^{s+1}}$. The continuum expansion of the flux contains $(\partial_i c^{ij}) \partial_j \tilde{q}$. The expansion of the skew transport contains $(\partial_i b^i) \tilde{v}$. The two compensator terms cancel precisely these contributions, with the signs in (6.3). Their remaining discrepancies are estimated by Lemma E.6.

For the mass term one must also retain

$$\mathcal{I}_h(a_h v_{h,t}) - a(g_h) \mathcal{I}_h v_{h,t}.$$

Its H^{s-2} norm is $O(h\varepsilon)$ because $v_{h,t}$ is bounded in H_h^{s-1} . The two skew-transport terms use $v_h \in H_h^s$; the analytic first-jet source is controlled in the same way. This proves the undifferentiated normal residual bound.

Differentiate these decompositions once in physical time. The highest spatially differentiated unknown is now $v_h \in H_h^s$, and the highest mass jet is $v_{h,tt} \in H_h^{s-2}$. The identical estimates therefore apply in H^{s-3} . Trace reversal and its time derivative are uniformly bounded multipliers, proving (E.26). Finally, for the enlarged writer, the extra residual and its time derivative satisfy

$$h^2 \|B \Lambda_h^2 q_h\|_{H_h^{s-2}} + h^2 \|B \Lambda_h^2 v_h\|_{H_h^{s-3}} \leq C_s h \|B\| \varepsilon,$$

by the multiplier bounds. This proves uniformity for the family. $\square$

Lemma E.9 (Initial harmonic and normal-constraint errors). *For (E.24) and its sampled preparation,*

$$\|c(g_h)(0)\|_{H^{s-2}} + \|\partial_t c(g_h)(0)\|_{H^{s-3}} \leq C_s h \varepsilon. \quad (\text{E.27})$$

The estimate uses the actual finite acceleration, not an assigned continuum acceleration.

Proof. At a unit-lapse, zero-shift initial slice, direct contraction gives

$$c^0 = \frac{1}{2}(v_{00} + \gamma^{ij} v_{ij}), \quad c^i = -\gamma^{ij} v_{0j} + \gamma^{jk} \Gamma_{jk}^i(\gamma).$$

The unrounded preparation therefore has $c = 0$ exactly. Sampling stability, product consistency and the one-derivative definition of c give $\|c(g_h)(0)\|_{H^{s-2}} \leq Ch\varepsilon$.

The Gauss–Codazzi normal components G_{0b} at this slice depend only on the spatial metric, its first two spatial derivatives and its extrinsic curvature; they contain no metric acceleration. The vacuum constraint identities in (6.6) and sampling consistency give $\|G_{0b}(g_h)(0)\|_{H^{s-3}} \leq Ch\varepsilon$. In (E.25), the coefficient of $\partial_t c_b$ in $\mathfrak{H}_{0b}$ is $1/2$ on this slice. All its other terms involve c , tangential derivatives of c , and bounded first-jet coefficients. Consequently

$$\|\partial_t c(0)\|_{H^{s-3}} \leq C(\|G_0(0)\|_{H^{s-3}} + \|r_0(0)\|_{H^{s-3}} + \|c(0)\|_{H^{s-2}}).$$

The residual lemma proves the assertion. This calculation explains exactly where the four initial constraints, in addition to $c(0) = 0$, are used. $\square$

Lemma E.10 (Forced subsidiary energy on the common interval). *For the harmonic initialization,*

$$\sup_{t \leq T_s} (\|c(g_h)\|_{H^{s-2}} + \|\partial_t c(g_h)\|_{H^{s-3}} + \|G(g_h)\|_{H^{s-3}}) \leq C_s h \varepsilon. \quad (\text{E.28})$$

Proof. The contracted Bianchi identity applied to the actual smooth interpolant and (E.25) gives the exact equation

$$\square_{g_h} c_b + \text{Ric}_b^d(g_h) c_d = -2\nabla^a r_{h,ab}. \quad (\text{E.29})$$

There is no additional unspecified discretization term: all such terms are already in r_h . The identity follows by commuting the two covariant derivatives in $\nabla^a \mathfrak{H}_{ab}$; it is the forced form of the usual harmonic subsidiary equation [74, Sec. 2.1].

For $k = s - 3$ use the positive coordinate wave energy

$$\mathcal{E}_{c,k} = \frac{1}{2} \sum_{|\alpha| \leq k} \int_{\mathbb{T}^3} (a |\partial_t \partial^\alpha c|^2 + c^{ij} \partial_i \partial^\alpha c \cdot \partial_j \partial^\alpha c + |\partial^\alpha c|^2) dx.$$

Here the index contractions in the energy are Euclidean positive contractions on the four components, and a, c^{ij} are the hyperbolic principal coefficients. The top metric and time-jet bounds give uniform coercivity and uniform coefficient, commutator and Ricci-product bounds. Pairing each commuted equation with $\partial_t \partial^\alpha c$, integrating the spatial principal term by parts, and estimating the mixed transport and lower-order terms gives

$$(\mathcal{E}_{c,k}^{1/2})' \leq C_s \mathcal{E}_{c,k}^{1/2} + C_s (\|r_h\|_{H^{k+1}} + \|\partial_t r_h\|_{H^k}). \quad (\text{E.30})$$

In particular the differentiated source is not bounded by the undifferentiated Euler residual alone. The initial-gauge lemma and (E.26) give the first two terms of (E.28) by Gronwall. Its last term follows from $G = \mathfrak{H} + r$. The common nonlinear existence interval was established before this subsidiary estimate and is not a constraint-preserving bootstrap assumption. $\square$

Lemma E.11 (Lower-order Cauchy comparison and acceleration recovery). *Let g be a solution of the continuum harmonic normal row on the common top ball and let X_h solve the central writer. Set $Y_h = \mathcal{S}_h(g - \eta, \partial_t g)$ and $e_{0,h} = \|X_h(0) - Y_h(0)\|_{\mathcal{X}_h^{s-2}}$. Then*

$$\begin{aligned} & \|q_h - \mathcal{S}_h(g - \eta)\|_{C_t H_h^{s-1}} + \|v_h - \mathcal{S}_h \partial_t g\|_{C_t H_h^{s-2}} \\ & + \|q_{h,tt} - \mathcal{S}_h \partial_t^2 g\|_{C_t H_h^{s-3}} \leq C_s (e_{0,h} + h\varepsilon). \end{aligned} \quad (\text{E.31})$$

Proof. Sampling the continuum normal row gives the finite row with an acceleration force of $C_t H_h^{s-2}$ norm at most $Ch\varepsilon$. This follows from the same flux, transport and composition decompositions as in Lemma E.8, now applied to a sampled continuum field. The three derivatives used by the second-order spatial consistency error are available from $g - \eta \in H^{s+1}$. Apply (E.17) with $r = s - 2$. This gives the first two terms without an inverse-mesh exponential. For the last term subtract the mass rows and divide by $a(g_h)$: two spatial derivatives of the H_h^{s-1} position difference and one of the H_h^{s-2} velocity difference lie in H_h^{s-3} ; coefficient differences multiply uniformly bounded continuum jets. The sampled residual is smaller than the same right-hand side. This proves the acceleration estimate independently of any differentiation of a convergence statement. $\square$

Theorem E.12 (Whole-sequence nonsymmetric vacuum limit). *For sufficiently small $(\gamma, K) \in \mathcal{D}_s(\varepsilon)$, the complete sequence of interpolated solutions converges on $[0, T_s] \times \mathbb{T}^3$ to the unique harmonic vacuum development with those initial data, and*

$$G(g) = 0. \quad (\text{E.32})$$

Moreover,

$$\sup_{t \leq T_s} (\|g_h - g\|_{H^{s-1}} + \|\partial_t g_h - \partial_t g\|_{H^{s-2}} + \|\partial_t^2 g_h - \partial_t^2 g\|_{H^{s-3}}) \leq C_s h\varepsilon, \quad (\text{E.33})$$

and

$$\| \text{Riem}(g_h) - \text{Riem}(g) \|_{C_t H^{s-3}} \leq C_s h \varepsilon. \quad (\text{E.34})$$

Proof. The top energy and [Lemma E.7](#) give compactness in $C_t H^{s+1-\delta} \times C_t H^{s-\delta}$ for $0 < \delta < 1$. Indeed, a fixed finite spatial Fourier screen has uniformly Lipschitz coefficients, while its complementary tails are bounded by the top norms; interpolation between adjacent Sobolev orders gives uniform time continuity at the displayed orders. Arzelà–Ascoli on each screen and a diagonal extraction produce a cluster point, retaining the top norms by weak lower semicontinuity. The time-jet bounds identify its velocity and acceleration distributionally.

The complete residual lemma passes the finite row to the continuum harmonic reduced equation. The subsidiary lemma gives $c(g) = 0$, hence $G(g) = 0$. Sampling consistency identifies its initial data. To see uniqueness, subtract two continuum reduced equations with these initial data. The coordinate wave energy at order $s-2$ controls the position in H^{s-1} and velocity in H^{s-2} ; coefficient differences multiply the bounded top jets of the other solution. The product estimate and integration by parts give $E_{\text{diff}}^{1/2}(t) \leq C \int_0^t E_{\text{diff}}^{1/2}$, so the difference vanishes. Therefore all cluster points coincide and the whole cutoff sequence converges.

Only after constructing this limit do we sample it for an error estimate. Its initial samples agree with the initialization, so $e_{0,h} = 0$ in [Lemma E.11](#). Add (E.19) for g , $\partial_t g$ and $\partial_t^2 g$ to obtain (E.33). On a nondegenerate chart, curvature is a smooth expression in the inverse metric, its first derivatives and its full second jet. The three slots in (E.33) control all second space–time derivatives in H^{s-3} . The product algebra at this index proves (E.34). No future continuum metric enters the finite construction; it is used here only for the error comparison. $\square$

Corollary E.13 (Direct constructive Einstein limit). *The Einstein conclusion of [Theorem E.12](#) follows from the cutoff-uniform nonlinear energy, consistency of the declared local writer, harmonic subsidiary estimate, and uniqueness of the limiting reduced Cauchy problem. No item (E1)–(E5) of [Theorem 5.4](#) is invoked as an independent hypothesis. Accordingly those five items function as sufficient certification conditions for the broader class of complete renewal cylinders, while [Theorem 6.1](#) supplies an independently constructed non-symmetry-reduced positive population.*

E.4 Positive rate lift and finite renewal words

The small metric chart has an explicit positive A_3 rate lift. In the D_3 realization

$$r_a \in \{(1, 1, 0), (1, -1, 0), (1, 0, 1), (1, 0, -1), (0, 1, 1), (0, 1, -1)\}, \quad (\text{E.35})$$

put $B = N_{\text{ADM}}^2 \gamma^{-1}$, $\varrho = \sqrt{\det \gamma}$ and, for a coordinate pair ij with remaining index k ,

$$c_{ij,\pm} = \frac{B_{ii} + B_{jj} - B_{kk}}{4} \pm \frac{B_{ij}}{2}, \quad d_a = \frac{1}{4} r_a \cdot \beta, \quad k_a^\pm = \frac{c_a}{2h^2} \pm \frac{d_a}{2h}. \quad (\text{E.36})$$

At the flat point all $c_a = 1/4$, so a fixed small metric chart makes all rates positive and comparable to h^{-2} . Direct calculation gives

$$h^2 \sum_a (k_a^+ + k_a^-) r_a r_a^T = B, \quad h \sum_a (k_a^+ - k_a^-) r_a = \beta, \quad (\text{E.37})$$

so the ADM reconstruction returns the same metric. The resulting weighted periodic graph has a cutoff-uniform cut margin by perturbation of the flat D_3 regulator and [Theorem 5.1](#).

To obtain finite histories, use a fixed-degree local Taylor–Hermite construction. The construction below makes both the spatial derivative loss and the precision convention explicit. It applies to the central writer and to every fixed $B \in \bar{B}$; when the coefficient is selected operationally it is retained as a finite initial mark.

Lemma E.14 (Inverse-mesh time jets for a local finite writer). *Let $\mathcal{F}_{B,h}(q, v) = (v, V_{B,h}(q, v))$ denote the differential writer on the common $\mathcal{X}_h^s$ ball. For each fixed $1 \leq j \leq 9$,*

$$\|D^j \mathcal{F}_{B,h}(X)\|_{(\mathcal{X}_h^s)^j \rightarrow \mathcal{X}_h^s} \leq C_{s,j} h^{-1}, \quad \|\mathcal{F}_{B,h}(X)\|_{\mathcal{X}_h^s} \leq C_s h^{-1} \|X\|_{\mathcal{X}_h^s}. \quad (\text{E.38})$$

Consequently an exact local solution from a source of size at most ε has

$$\|\partial_t^j q\|_{H_h^s} \leq C_{s,j} \varepsilon h^{1-j} \quad (1 \leq j \leq 8). \quad (\text{E.39})$$

Each initial jet through order seven is a function of a fixed-radius neighbourhood of the source, independent of h .

Proof. The inverse inequality $\|u\|_{r+1,h} \leq C_r h^{-1} \|u\|_{r,h}$ follows from the bounded frequency cube. The acceleration has two spatial derivatives of q and one of v , whereas the phase norm contains one extra derivative of q . Thus it loses at most one order in that norm. The product calculus bounds all analytic coefficient derivatives with the same single loss. The added term satisfies $h^2 \|\Lambda_h^2 q\|_{H_h^s} \leq C h^{-1} \|q\|_{H_h^{s+1}}$, so the bounds are uniform in B . Define successive vector jets by $J_1 = \mathcal{F}_{B,h}$ and $J_{j+1} = D J_j \mathcal{F}_{B,h}$. Repeated differentiation gives $\|J_j(X)\|_{\mathcal{X}_h^s} \leq C_{s,j} \varepsilon h^{-j}$. For $j = 1$ the assertion is $q_t = v$; for $j \geq 2$ the q -jet is the v -component of J_{j-1} . This proves (E.39). Each differentiation composes a fixed local stencil a finite number of times; it does not enlarge its radius with the cutoff. These estimates deliberately retain inverse-mesh powers and do not assume additional continuum regularity. $\square$

At a source $X = (q, v)$, let T_X be the degree-seven Taylor polynomial of the position, using the jets of the actual differential writer at that source. For $0 \leq t \leq \tau$, set

$$T_X(t) = q + \sum_{j=1}^7 \frac{t^j}{j!} q^{(j)}(0), \quad X_1 = (T_X(\tau), T_X'(\tau)), \quad J_{B,h}(X) = D V_{B,h}(X) [\mathcal{F}_{B,h}(X)]. \quad (\text{E.40})$$

The endpoint-corrected polynomial Q_X is the unique polynomial of degree at most seven whose value, velocity, acceleration and jerk at zero are $(q, v, V_{B,h}(X), J_{B,h}(X))$, and whose corresponding jets at τ are $(X_1, V_{B,h}(X_1), J_{B,h}(X_1))$.

Lemma E.15 (Complete Taylor–Hermite residual). *For $c_- h^2 \leq \tau \leq c_+ h^2$ and sufficiently small h , the polynomial Q_X remains in the common chart and its actual defect $f_X = Q_X'' - V_{B,h}(Q_X, Q_X')$ satisfies*

$$\|f_X\|_{C(I; H_h^s)} \leq C_s \varepsilon h^5, \quad \|\partial_t f_X\|_{C(I; H_h^{s-3})} \leq C_s \varepsilon h^3. \quad (\text{E.41})$$

These bounds concern the complete interval, not only collocation points.

Proof. Compare T_X with the exact local solution from X , used only in this proof. Taylor’s integral remainder and (E.39) give

$$\|T_X^{(j)} - q^{(j)}\|_{C(I; H_h^s)} \leq C_s \varepsilon h^{-7} \tau^{8-j} \quad (0 \leq j \leq 3). \quad (\text{E.42})$$

The position error in H_h^{s+1} costs one inverse power of h . Together with the velocity error, this gives a phase error $O(\varepsilon h^7)$. The acceleration Lipschitz bound in (E.38) then yields $\|T_X'' - V_{B,h}(T_X, T_X')\|_{H_h^s} \leq C \varepsilon h^5$. Differentiating this defect once gives a bound $C \varepsilon h^3$ in H_h^s : the third Taylor remainder is $O(\varepsilon h^3)$, $D J_{B,h}$ has norm at most $C h^{-2}$ from $\mathcal{X}_h^s$ to H_h^s , and the remaining term is $D_v V_{B,h}$ applied to the acceleration defect.

Put $\delta_2 = V_{B,h}(X_1) - T_X''(\tau)$ and $\delta_3 = J_{B,h}(X_1) - T_X'''(\tau)$. The preceding bounds give $\|\delta_2\|_{H_h^s} \leq C\varepsilon h^5$ and $\|\delta_3\|_{H_h^s} \leq C\varepsilon h^3$. The correction is explicitly

$$Q_X(t) - T_X(t) = \tau^2 \delta_2 H_2(t/\tau) + \tau^3 \delta_3 H_3(t/\tau), \quad \begin{cases} H_2(z) = -\frac{1}{2}z^4(z-1)^2(4z-5), \\ H_3(z) = \frac{1}{6}z^4(z-1)^3. \end{cases} \quad (\text{E.43})$$

Their jets through order three vanish at zero, and their endpoint jets are the unit acceleration and unit jerk jets, respectively. Existence and uniqueness also follow because a degree-seven polynomial with zero jets through order three at both endpoints is divisible by $t^4(t-\tau)^4$ and must vanish. The correction has second derivative $O(\varepsilon h^5)$, third derivative $O(\varepsilon h^3)$ in H_h^s , and phase norm $O(\varepsilon h^7)$. The same local Lipschitz estimates therefore prove (E.41), even at the stronger spatial order for its differentiated local term. Only H_h^{s-3} is retained in the asserted physical residual topology. $\square$

Lemma E.16 (Sobolev-scaled finite precision). *Let Q_X be the local polynomial of Lemma E.15, and let $\tilde{Q}_X$ be a degree-seven endpoint Hermite perturbation. Set $\delta Q = \tilde{Q}_X - Q_X$ and let*

$$b_h = \max_{\substack{e \in \{0,1\} \\ 0 \leq j \leq 3}} \tau^j \|\delta Q^{(j)}(e\tau)\|_{H_h^s}. \quad (\text{E.44})$$

Then $\|\delta Q^{(j)}\|_{C(I; H_h^s)} \leq C\tau^{-j}b_h$ for $0 \leq j \leq 3$. If $b_h \leq c\varepsilon\tau^2h^5$ and the source lies in a smaller common chart, the perturbation remains in the local phase tube. The changes of the complete residual and its time derivative are bounded by $C\tau^{-2}b_h$ and $C\tau^{-3}b_h$, respectively. Hence the bounds (E.41) are preserved.

Proof. The Hermite map on $[0, 1]$ is a fixed finite-dimensional linear inverse. Rescaling its input jets and its derivatives proves the first estimate. For the nonlinear terms use $\|\delta(Q, Q')\|_{\mathcal{X}_h^s} \leq C(h^{-1} + \tau^{-1})b_h$ and (E.38). Since $\tau/h = O(h)$, these terms are bounded by $C\tau^{-2}b_h$ in the residual. Differentiating once, write $\partial_t f = Q^{(3)} - J_{B,h}(Q, Q') - D_v V_{B,h}(Q, Q')f$. The local bounds for $DJ_{B,h}$, $D_v V_{B,h}$ and its differential, together with $\|f_X\|_{H_h^s} \leq C\varepsilon h^5$, give $C\tau^{-3}b_h$. The quadratic perturbation term is absorbed using the stated smallness of b_h . These estimates are applied to a perturbation of the controlled local polynomial, not to arbitrary polynomials having only bounded position and velocity. $\square$

Theorem E.17 (Finite local successor words). *Fix $T < T_s$. There is a finite chronological Taylor–Hermite realization with $\tau_j \asymp h^2$, fixed execution radius and shared endpoint jets through order three, such that*

$$\sup_{t \leq T} \|(Q_h, Q'_h)\|_{\mathcal{X}_h^s} \leq C_s \varepsilon, \quad \|f_h^{\text{num}}\|_{C_t H_h^s} \leq C_s \varepsilon h^5, \quad \|\partial_t f_h^{\text{num}}\|_{C_t H_h^{s-3}} \leq C_s \varepsilon h^3. \quad (\text{E.45})$$

In particular the weaker $O(\varepsilon h^4)$ and $O(\varepsilon h^2)$ residual bounds in the main text hold. A sufficient precision convention is to approximate each scaled endpoint jet $\tau_j^\ell Q^{(\ell)}$ pointwise to error at most $c_s \varepsilon h^{s+10}$ relative to the local construction, using finer intermediate accuracy as needed. The starting finite record may differ from the compatible preparation by $O(\varepsilon h^4)$ in $\mathcal{X}_h^s$. Every resulting history has the same Einstein limit.

Proof. At each source compute the finite jet bank (E.40), round its successor value and velocity, and attach finite approximations to the acceleration and jerk of that successor. Reuse these latter caches at the next source. Interpolate the two endpoint banks by the Hermite polynomial. The field and all the jet maps are finite local analytic maps on bounded source sets, so their finite arithmetic

tolerances can be chosen to meet the stated endpoint error. At fixed h all registers may be chosen from bounded rational grids; with a finite horizon this gives a genuinely finite alphabet.

A pointwise error η on the unit-volume grid obeys $\|\delta u\|_{H_h^s} \leq C_s h^{-s} \eta$. The stated scaled precision therefore makes $b_h \leq C_s \varepsilon h^{10}$, which is smaller than $\varepsilon \tau_j^2 h^5 \asymp \varepsilon h^9$ on all sufficiently fine grids. Source cache and successor rounding errors are both included in this endpoint comparison. The two preceding lemmas give the residual bounds on every interval. Shared jets make Q_h globally C^3 in time; hence f_h^{num} is C^1 , with no uncounted delta function at a cut. The radius is bounded by the fixed number of local jet compositions and endpoint evaluations, not by the number of sites or time steps.

Until a putative first guard exit, the forced energy bounds the cumulative forcing by $C\varepsilon T h^5$. After decreasing the initial radius, this cannot close the strict guard margin, so the bound holds on the whole interval. The lower-order difference estimate gives position and velocity shadowing, and the actual acceleration equation gives the second time-jet estimate. Combining with [Theorems E.12](#) and [F.3](#) proves the claimed limit. Moreover the physical stage cost is $c_{h,j} \leq C\tau_j(h^{10} + h^6)$, strictly below $M\tau_j h^4$ for any fixed $M > 0$ on a sufficiently fine tail. This supplies the strict margin used for nonempty probability-row neighborhoods. The construction concerns finite information range and precision; it makes no hardware-latency claim. $\square$

The results above prove the population statement required in the main text. They do not identify the local writer with the exact finite Euler equations of the reconstructed action. That stronger question has a different finite constraint and response structure, to which we now turn.

F Law-space robustness, finite preparations and chronological compatibility

This appendix proves [Theorem 6.3](#) and [Theorem 6.5](#). The cutoff-uniform estimates of [Appendix E](#) are the input: first the law family is enlarged, then finite initial constraints are prepared, and finally the chronological coupling is compared with the independently reconstructed action without identifying the two finite dynamics.

F.1 A fixed open family of local differential writers

Put

$$\Lambda_h = - \sum_{i=1}^3 D_i^- D_i^+, \quad \ell_h(k) = 4h^{-2} \sum_i \sin^2(\pi h k_i), \quad 0 \leq h^2 \ell_h(k) \leq 12. \quad (\text{F.1})$$

On the metric chart of [Appendix E](#), let $c \succeq c_* I$. Choose $0 < b_0 < c_*/24$ and set

$$\mathcal{B} = \{B = B^T \in \mathbb{R}^{10 \times 10} : \|B\|_{\text{op}} < b_0\}. \quad (\text{F.2})$$

This is an open ball of dimension 55. For $B \in \mathcal{B}$ define

$$q_t = v, \quad a(g)v_t = \sum_{i,j} D_i^-(c^{ij}(g)D_j^+ q) - \mathsf{K}_b v + \mathsf{G}_h(q, v) - h^2 B \Lambda_h^2 q. \quad (\text{F.3})$$

The coefficient B is constant during one physical history. It may vary with the cutoff or be sampled once as a finite initial mark.

Lemma F.1 (Uniform fourth-order multiplier bounds). *For every array w ,*

$$h^2 \|\Lambda_h w\|_h^2 \leq 12 \sum_i \|D_i^+ w\|_h^2. \quad (\text{F.4})$$

At every fixed index used below,

$$h^2 \|\Lambda_h^2 q\|_{s-1,h} \leq C_s \|q\|_{s+1,h}, \quad h^2 \|\Lambda_h^2 q\|_{s-2,h} \leq C_s h \|q\|_{s+1,h}, \quad (\text{F.5})$$

$$h^2 \|\Lambda_h^2 q\|_{s-3,h} \leq C_s h^2 \|q\|_{s+1,h}, \quad h^2 \|\Lambda_h^2 v\|_{s-3,h} \leq C_s h \|v\|_{s,h}. \quad (\text{F.6})$$

Proof. Parseval reduces the first claim to $h^2 \ell_h(k)^2 \leq 12 \ell_h(k)$. Uniform equivalence of the forward-difference Sobolev norm with the Fourier weight $(1 + \ell_h)^{r/2}$ reduces the four remaining claims to the bounded ratios

$$\frac{h^2 \ell_h^2}{1 + \ell_h}, \quad \frac{h^2 \ell_h^2}{(1 + \ell_h)^{3/2}}, \quad \frac{h^2 \ell_h^2}{(1 + \ell_h)^2},$$

which are bounded by 12, $\sqrt{12}h$, and h^2 , respectively. $\square$

The stencil Λ_h^2 has fixed coordinate radius two. Its contribution is the negative q -gradient of $h^2 \langle \Lambda_h q, B \Lambda_h q \rangle_h / 2$; this identity concerns the added term only and does not identify the complete local writer with the raw finite Euler flow of the reconstructed signed action.

For $q_\alpha = D^\alpha q$ define

$$\mathcal{E}_{s,h}^B = \mathcal{E}_{s,h} + \frac{h^2}{2} \sum_{|\alpha| \leq s} \langle \Lambda_h q_\alpha, B \Lambda_h q_\alpha \rangle_h, \quad (\text{F.7})$$

where $\mathcal{E}_{s,h}$ is (E.11).

Theorem F.2 (Fixed-radius propagation on an open writer ball). *For $s \geq 11$, uniformly for $B \in \overline{\mathcal{B}}$ on a sufficiently small common Cauchy chart,*

$$c_s \|X\|_{\mathcal{X}_h^s}^2 \leq \mathcal{E}_{s,h}^B \leq C_s \|X\|_{\mathcal{X}_h^s}^2, \quad (\mathcal{E}_{s,h}^B)' \leq C_s \mathcal{E}_{s,h}^B + C_s (\mathcal{E}_{s,h}^B)^{3/2}. \quad (\text{F.8})$$

Consequently there are $\varepsilon_s, T_s > 0$, independent of h and B , such that every initial record with $\|X_h(0)\|_{\mathcal{X}_h^s} \leq \varepsilon_s$ has a unique solution through T_s and

$$\sup_{t \leq T_s} \|X_h(t)\|_{\mathcal{X}_h^s} \leq C_s \|X_h(0)\|_{\mathcal{X}_h^s}. \quad (\text{F.9})$$

With an additional acceleration defect f ,

$$(\sqrt{\mathcal{E}_{s,h}^B})' \leq C_s \sqrt{\mathcal{E}_{s,h}^B} + C_s \mathcal{E}_{s,h}^B + C_s \|f\|_{s,h}. \quad (\text{F.10})$$

Proof. By Lemma F.1, the absolute value of the added quadratic energy at each commuted level is at most $6b_0 \sum_i \|D_i^+ q_\alpha\|_h^2$. The pre-existing spatial energy is at least $c_* \sum_i \|D_i^+ q_\alpha\|_h^2 / 2$, so $b_0 < c_*/24$ leaves a cutoff-independent positive margin.

Commute (F.3) with D^α . Since B and Λ_h commute with every lattice shift, the new principal term is $-h^2 B \Lambda_h^2 q_\alpha$. Pairing with v_α gives exactly

$$-h^2 \langle v_\alpha, B \Lambda_h^2 q_\alpha \rangle_h = -\frac{d}{dt} \frac{h^2}{2} \langle \Lambda_h q_\alpha, B \Lambda_h q_\alpha \rangle_h.$$

Thus the additional high-frequency term cancels at principal order rather than being estimated as a small top-order force. Every remaining commutator is one of those controlled in Theorem E.4;

the only change is that the actual acceleration appears in analytic mass-coefficient terms, and [Lemma F.1](#) bounds it uniformly in H_h^{s-1} . This gives (F.8). The finite-dimensional vector field is analytic on the common chart, and the same first-exit argument used for (E.16) proves the lifespan and the forced estimate. $\square$

Theorem F.3 (Uniform Einstein limit for the whole coefficient family). *Let $X_{B,h}$ and $X_{0,h}$ have the same sufficiently small compatible initial data. On their common interval,*

$$\sum_{j=0}^3 \|q_{B,h}^{(j)} - q_{0,h}^{(j)}\|_{C_t H_h^{s-1-j}} \leq C_s h \|B\|_{\text{op}} \varepsilon. \quad (\text{F.11})$$

For any sequence $B_h \in \overline{\mathcal{B}}$, with no convergence of B_h required, the whole interpolated sequence converges to the same harmonic vacuum development as the central writer. In particular the metric and its first two time derivatives satisfy the $O(h\varepsilon)$ bounds of [Theorem E.12](#), and the subsidiary field, rate positivity, cut margin and noncollapse estimates remain uniform.

Proof. For comparison only, regard the added term on the B -trajectory as an external force

$$f_{B,h} = -h^2 a(g_{B,h})^{-1} B \Lambda_h^2 q_{B,h}.$$

Both trajectories already exist on the common top chart by [Theorem F.2](#). The lower-order bounds in [Lemma F.1](#) give

$$\|f_{B,h}\|_{C_t H_h^{s-2}} + \|\partial_t f_{B,h}\|_{C_t H_h^{s-3}} \leq C_s h \|B\| \varepsilon,$$

while the top H_h^{s-1} norm need not be small. Apply the forced difference estimate (E.17) at the reserved lower orders to get the first two jet slots. Subtracting the acceleration rows gives the second time derivative; differentiating once and using the velocity differential of the local vector field gives the third. This proves (F.11). Add the central-writer convergence of [Theorem E.12](#). All constants are uniform on $\overline{\mathcal{B}}$, so no convergence of B_h is needed. $\square$

Proposition F.4 (Rapid switching is not included). *For two distinct scalar marks $B_i = \sigma_i I$ in the admissible interval, one resolved corner mode has frequency*

$$\Omega_{\sigma,h}(k)^2 = \ell_h(k)(1 + \sigma h^2 \ell_h(k)). \quad (\text{F.12})$$

Holding each law for a quarter period gives transfer matrices whose two-step product has eigenvalues $-\Omega_{\sigma_1,h}/\Omega_{\sigma_2,h}$ and its reciprocal. Their modulus ratio stays separated from one at the ultraviolet corner when $\sigma_1 \neq \sigma_2$. Repeated $O(h)$ -time switching can therefore amplify that mode like $\exp(cT/h)$ on a fixed slab.

Proof. The flat scalar mode is an oscillator. A quarter-period transfer is $\begin{pmatrix} 0 & \Omega^{-1} \\ -\Omega & 0 \end{pmatrix}$. Multiplying the two transfers gives the stated diagonal matrix. At the corner $h^2 \ell_h(k) \rightarrow 12$, so the frequency ratio approaches $\sqrt{(1+12\sigma_1)/(1+12\sigma_2)} \neq 1$. $\square$

F.2 Relatively open neighborhoods of chronological transition rows

Fix $B \in \overline{\mathcal{B}}$ and $T < T_s$. Partition $[0, T]$ into cells I_j with $c_- h^2 \leq \tau_j \leq c_+ h^2$. A chronological letter is a finite physical polynomial metric segment Q_e with shared endpoint value, velocity, acceleration and jerk records. Before a possible guard exit define

$$f_e(t) = Q_e''(t) - V_{B,h}(Q_e(t), Q_e'(t)) \quad (\text{F.13})$$

and the cost (6.13); a grammar or execution failure has cost one and terminates the word. The common-energy margin is chosen so that any word with total cost below one is evaluated entirely inside the coefficient chart.

Lemma F.5 (A stopped physical cost controls the whole word). *If $C_h \leq d_h^2 < 1$ and d_h is sufficiently small, then no failure or guard exit occurs and*

$$\|f_h\|_{L_t^2 H_h^s} + \|\partial_t f_h\|_{L_t^2 H_h^{s-3}} \leq 2\varepsilon d_h. \quad (\text{F.14})$$

Relative to the exact unforced B-history with the same initial record,

$$\|Q_h - q_{B,h}\|_{C_t H_h^s} + \|Q'_h - q'_{B,h}\|_{C_t H_h^{s-1}} + \|Q''_h - q''_{B,h}\|_{C_t H_h^{s-3}} \leq C_s \varepsilon d_h. \quad (\text{F.15})$$

Proof. Until a first guard hit, Cauchy–Schwarz gives $\int_0^{t^*} \|f_h\|_{s,h} dt \leq \sqrt{T} \varepsilon d_h$. For small d_h the forced energy (F.10) contradicts a first exit through the strict guard margin; a failure is excluded because it would already contribute unit cost. Hence the integrals are over the whole interval and (F.14) follows. Shared endpoint value, velocity, acceleration and jerk make f_h globally H^1 in time with values in H_h^{s-3} , rather than merely piecewise H^1 . The Hilbert-valued one-dimensional Sobolev estimate therefore gives

$$\|f_h\|_{C_t H_h^{s-3}} \leq C_T (\|f_h\|_{L_t^2 H_h^{s-3}} + \|\partial_t f_h\|_{L_t^2 H_h^{s-3}}) \leq C_T \varepsilon d_h.$$

Subtract the exact and forced mass equations at the reserved lower order, use (E.17) and Gronwall, and then use this pointwise force bound in the actual acceleration difference. This proves the final C_t slot without omitting a temporal jump contribution. $\square$

Theorem F.6 (A chronological law neighborhood with the same Einstein limit). *Let $K_{h,j}(e \mid \pi)$ be an arbitrary finite transition row at each live prefix, possibly depending on the entire past. If*

$$\sum_e K_{h,j}(e \mid \pi) c_{h,j}(\pi, e) < M \tau_j h^4 \quad (\text{F.16})$$

for every live prefix, then

$$\mathbb{E} C_h \leq M T h^4, \quad \mathbb{P}\{C_h > h^2\} \leq M T h^2. \quad (\text{F.17})$$

On $\{C_h \leq h^2\}$ the history obeys Lemma F.5 with $d_h = h$ and therefore has the same Einstein limit as Theorem F.3. At fixed cutoff and finite grammar the strict inequalities (F.16) define nonempty relatively open convex subsets of the product of row simplexes when a uniformly accurate completion is available at every live source. They contain strictly positive rows whenever all candidate letters have finite cost. Along the odd sequence $h = (2m+1)^{-1}$, the exceptional probabilities are summable, so the conclusion holds eventually almost surely under any common coupling.

Proof. Assign zero future cost to stopped prefixes. Conditional on any live prefix, (F.16) bounds the expected stage cost by $M \tau_j h^4$, independently of the probability of reaching that prefix. Summation gives the expectation bound; Markov at threshold h^2 gives the probability bound. The deterministic cost lemma applies on the complement. The series $\sum_m (2m+1)^{-2}$ converges, so the first Borel–Cantelli lemma gives the eventual almost-sure assertion without independence between cutoffs. At fixed finite grammar each strict conditional mean bound is an open affine half-space in one row simplex; finite products preserve relative openness and convexity. Small positive mixing preserves a strict margin. $\square$

For an observable neighborhood description, attach to each ordinary letter the profile

$$\Phi_e = (f_e/\varepsilon, \partial_t f_e/\varepsilon, 0), \quad (\text{F.18})$$

and to a failure the vector $(0, 0, 1)$ in $L^2(I_j; H_h^s) \oplus L^2(I_j; H_h^{s-3}) \oplus \mathbb{R}$. Let $d(e, e') = \|\Phi_e - \Phi_{e'}\|$ and let $W_{2,d}$ be the finite optimal transport distance induced by d .

Proposition F.7 (Physical profile transport certifies a row neighborhood). *For finite rows K, P ,*

$$\left(\sum_e K(e) c_e \right)^{1/2} \leq \left(\sum_e P(e) c_e \right)^{1/2} + W_{2,d}(K, P). \quad (\text{F.19})$$

If $\sum_e P(e) c_e \leq M_0 \tau_j h^4$, then every row satisfying

$$W_{2,d}(K, P) < \gamma \sqrt{\tau_j} h^2 \quad (\text{F.20})$$

obeys (F.16) for any $M > (\sqrt{M_0} + \gamma)^2$. These strict balls are relatively open in the finite probability simplex.

Proof. For any coupling of K and P , Minkowski in the profile Hilbert space gives $\|\Phi_e\|_{L^2} \leq \|\Phi_{e'}\|_{L^2} + \|\Phi_e - \Phi_{e'}\|_{L^2}$. Average under the coupling and minimize. The finite coupling polytope is compact, so a minimizer exists. Continuity in the row probabilities follows from finite-dimensional optimal transport. $\square$

The scale in (F.20) is intentional. A fixed unscaled total-variation ball is not stable under refinement: if there are $J_h \asymp h^{-2}$ stages and an absorbing failure of probability p_h is added independently at every live stage, then the failed-word probability is $1 - (1 - p_h)^{J_h}$, which tends to zero exactly when $J_h p_h \rightarrow 0$. Thus even $p_h \rightarrow 0$ can be too large. The theorem controls the accumulated physical defect rather than hiding this refined-word effect.

F.3 Exact finite initial constraints from free records

We now remove the need to start the constructive population from a continuum constraint solution. This subsection uses the original reconstructed finite action only at the initial cut. Let the odd phase derivative be

$$\widehat{\delta_i u}(k) = i \kappa_i(k) \widehat{u}(k), \quad \kappa_i(k) = 2h^{-1} \sin(\pi h k_i), \quad \Omega_h^2 = - \sum_i \delta_i^2. \quad (\text{F.21})$$

For $r \geq 3$ put

$$\mathcal{P}_h^r = H_h^{r+2}(\text{Sym}_3) \times H_h^{r+1}(\text{Sym}_3), \quad \mathcal{Y}_h^r = H_h^r(\mathbb{R} \oplus \mathbb{R}^3), \quad (\text{F.22})$$

and write $X = (u, \pi)$, $\gamma = I + u$. Let $\mathcal{C}_h : \mathcal{P}_h^r \rightarrow \mathcal{Y}_h^r$ be (6.16).

Lemma F.8 (Uniform original-action constraint calculus). *On a common ball $\mathcal{C}_h$ is analytic, $\mathcal{C}_h(0) = 0$, and*

$$D\mathcal{C}_h(0)(u, \pi) = \left(\delta_i \delta_j u_{ij} - \sum_i \delta_i^2 \text{tr } u, -2\delta_j \pi^{ij} \right) =: L_h(u, \pi). \quad (\text{F.23})$$

Writing $\mathcal{C}_h = L_h + \mathcal{N}_h$,

$$\|\mathcal{N}_h(X)\|_{\mathcal{Y}_h^r} \leq C_r \|X\|^2, \quad \|D\mathcal{N}_h(X)\| \leq C_r \|X\|, \quad (\text{F.24})$$

with cutoff-independent constants. No lapse or shift time derivative and no metric acceleration appears in $\mathcal{C}_h$.

Proof. Stationary connection and Legendre elimination are analytic on the retained small chart. In a lapse or shift derivative, every spatial load has one phase derivative; exact skew-adjointness transfers it to the stationary connection, so the output loses at most the declared Sobolev orders. At flat data the quadratic phase-compatible action has the Fierz–Pauli rows displayed in (F.23); the full link remainder starts at cubic order in the flat connection and contributes no linear initial row. Taylor’s theorem gives (F.24). $\square$

Let P_0 be spatial averaging in $\mathcal{Y}_h^r$ and $P_\perp = I - P_0$. On mean-zero targets define

$$R_h(f, m) = \left(\frac{1}{2}(\Omega_h^2)^{-1} f I, \frac{1}{2}\mathcal{L}_h^{\text{vec}}\mathcal{D}_h^{-1}m \right), \quad (\text{F.25})$$

where $(\mathcal{L}_h^{\text{vec}}Y)_{ij} = \delta_i Y_j + \delta_j Y_i - \frac{2}{3}\delta_{ij}\delta_k Y_k$ and $\mathcal{D}_h Y = -\delta_j(\mathcal{L}_h^{\text{vec}}Y)_{ij}$.

Lemma F.9 (Range inverse and four missing means). *One has $L_h R_h = I$ on the mean-zero target and $\|R_h\| \leq C_r$ uniformly. For $d = N^3$,*

$$\text{rank } L_h = 4d - 4, \quad \dim \ker L_h = 8d + 4. \quad (\text{F.26})$$

For each sufficiently small $z \in \ker L_h$ there is a unique mean-zero $y_h(z) = O(\|z\|^2)$ such that

$$P_\perp \mathcal{C}_h(z + R_h y_h(z)) = 0. \quad (\text{F.27})$$

The remaining mean map $\Theta_h(z) = P_0 \mathcal{C}_h(z + R_h y_h(z))$ is analytic and has zero constant and linear jets.

Proof. For $u = \phi I$ the scalar row equals $2\Omega_h^2 \phi$. For $\pi = \mathcal{L}_h^{\text{vec}}Y$ the vector row equals $2\mathcal{D}_h Y$, whose nonzero-mode symbol is $|\kappa|^2 I + \kappa \kappa^T / 3$ and is uniformly invertible on the unit odd torus. This proves the right inverse and rank. The nonlinear range equation is $y = -P_\perp \mathcal{N}_h(z + R_h y)$; the bounds of Lemma F.8 make it a uniform contraction on a ball of radius $C\|z\|^2$. The expansion of Θ_h follows because $P_0 L_h = 0$. $\square$

To resolve the four means, let $T_i \in \text{Sym}_3$ have its two transverse off-diagonal entries equal to $1/\sqrt{2}$ and all other entries zero, and define

$$u_* = \sum_{i=1}^3 T_i \cos(2\pi x_i), \quad p(\vartheta) = -\frac{2}{3}\kappa I + \sum_{i=1}^3 b_i T_i \sin(2\pi x_i), \quad \vartheta = (\kappa, b_1, b_2, b_3). \quad (\text{F.28})$$

These are balancing directions only; arbitrary complementary modes remain free.

Lemma F.10 (Exact four-mode mean calibration). *With $\omega_h = 2h^{-1} \sin(\pi h)$, the quadratic mean jet of the full retained action on the seed (F.28) is*

$$Q_h(\vartheta) = \begin{pmatrix} \frac{2}{3}\kappa^2 - \frac{1}{2} \sum_i b_i^2 - \frac{3}{8}\omega_h^2 \\ -\frac{1}{2}\omega_h b_1 \\ -\frac{1}{2}\omega_h b_2 \\ -\frac{1}{2}\omega_h b_3 \end{pmatrix}. \quad (\text{F.29})$$

It has the exact root

$$\vartheta_h^* = (3\omega_h/4, 0, 0, 0), \quad DQ_h(\vartheta_h^*) = \text{diag}(\omega_h, -\omega_h/2, -\omega_h/2, -\omega_h/2), \quad (\text{F.30})$$

whose inverse is uniformly bounded. The opposite sign of the first component gives the second branch.

Proof. Keep spatially constant lapse and shift independent and expand the canonical Hamiltonian to quadratic order about flat data. For the transverse trace-free configuration seed and momentum equal to a homogeneous trace plus transverse trace-free modes, the coefficient is

$$\mathcal{H}_h^{[2]}(u, \pi; N, \beta) = N \left(\|\pi\|_h^2 - \frac{1}{2} \|\text{tr } \pi\|_h^2 + \frac{1}{4} \sum_i \|\delta_i u\|_h^2 \right) + \sum_i \beta^i \langle \pi, \delta_i u \rangle_h.$$

The complete link remainder starts at cubic order in the stationary connection at flat data, so it does not modify this quadratic mean coefficient. Taking minus the lapse derivative and the three shift derivatives, then using orthogonality of the constant and unit Fourier modes on every odd grid, gives (F.29). Direct differentiation yields (F.30); ω_h is uniformly separated from zero on the declared grids. $\square$

Let $\mathcal{Z}_h = \ker L_h$ and remove the four balancing seed directions from an arbitrary small free record, leaving W in a fixed complementary subspace. Write $Z_h(\vartheta)$ for the seed and, for $t \neq 0$, define

$$\mathcal{Q}_h(t, \vartheta, W) = t^{-2} \Theta_h(t[Z_h(\vartheta) + W]). \quad (\text{F.31})$$

Since Θ_h has zero constant and linear jets, the integral Taylor formula extends this map analytically through $t = 0$.

Theorem F.11 (Exact finite preparation from free initial records). *There are $t_0, \rho, d_0, h_0 > 0$, independent of h , such that for $0 < |t| < t_0$, $h < h_0$, and every complementary free record W with $\|W\| < \rho$, there is a unique $\vartheta_h(t, W)$ near ϑ_h^* for which (F.31) vanishes. The resulting record*

$$\mathfrak{J}_h(t, W) = t[Z_h(\vartheta_h(t, W)) + W] + R_h y_h(t[Z_h(\vartheta_h(t, W)) + W]) \quad (\text{F.32})$$

satisfies

$$\mathcal{E}_h(\mathfrak{J}_h(t, W)) = 0, \quad \|\mathfrak{J}_h(t, W)\|_{\mathcal{P}_h^r} \leq C_r |t|. \quad (\text{F.33})$$

At every such record $D\mathcal{E}_h$ is onto and admits a right inverse with

$$\|\mathcal{R}_{h,t} f\|_{\mathcal{P}_h^r} \leq C_r (\|P_\perp f\|_{\mathcal{Y}_h^r} + |t|^{-1} |P_0 f|). \quad (\text{F.34})$$

For fixed $t \neq 0$, the image contains a relatively open neighborhood in the complete finite zero set; that zero set has codimension $4N^3$ in the $12N^3$ -dimensional canonical phase. No symmetry, CMC, conformal-flatness, frequency-support restriction, or continuum constraint solution is imposed on W .

Proof. The analytic extension of (F.31) converges in C^1 uniformly to the quadratic map of Lemma F.10 as $|t| + \|W\| \rightarrow 0$. Its root has a uniform inverse margin. Therefore $\vartheta \mapsto \vartheta - [DQ_h(\vartheta_h^*)]^{-1} \mathcal{Q}_h(t, \vartheta, W)$ is a contraction on one cutoff-independent ball after choosing its radius first and then t_0, ρ . The range equation was already solved in Lemma F.9, giving the complete zero.

For the derivative, eliminate the mean-zero block with the uniformly bounded range inverse. Along a balancing direction the remaining mean derivative is $t^2 D_\vartheta \mathcal{Q}_h$, while a range direction contributes only $O(t)$ to the mean. Solving for the balancing coefficient and multiplying by the physical factor t gives exactly the $|t|^{-1}$ term in (F.34). The inverse function theorem on the zero set then gives relative openness and the codimension count. The same estimate shows why uniformity through the symmetric tip $t = 0$ is neither claimed nor true in general. $\square$

The harmonic completion of a prepared record sets $g_{00} = -1$, $g_{0i} = 0$, $g_{ij} = \gamma_{ij}$ and uses its actual canonical metric velocity w for the spatial slot, with

$$v_{ij} = w_{ij}, \quad v_{00} = -\gamma^{ij}w_{ij}, \quad v_{0i} = \gamma^{jk}(\delta_j\gamma_{ki} - \frac{1}{2}\delta_i\gamma_{jk}). \quad (\text{F.35})$$

These lapse-shift velocities do not enter the four initial rows (6.16).

Assumption F.12 (Cofinal consistency of initial-action elimination). Let $\mathscr{W}_h$ denote the actual canonical spatial metric velocity used in (F.35). The continuum maps $\mathcal{C}$ and $\mathscr{W}$ use the same action normalization and momentum convention; $\mathcal{C} = 0$ is equivalent to the vacuum constraints for (γ, K) with $K = -\mathscr{W}/2$ at unit lapse and zero shift. On the bounded preparation chart in $\mathcal{P}_h^{s+2}$, $s \geq 11$, the actual eliminated maps satisfy

$$\begin{aligned} & \|\mathcal{I}_h\mathcal{C}_h(X_h) - \mathcal{C}(\mathcal{I}_hX_h)\|_{H^s} + \|\mathcal{I}_h\mathscr{W}_h(X_h) - \mathscr{W}(\mathcal{I}_hX_h)\|_{H^s} \\ & \leq \chi_h\|X_h\|_{\mathcal{P}_h^{s+2}}, \quad \chi_h \longrightarrow 0. \end{aligned} \quad (\text{F.36})$$

The same comparisons hold for the first derivatives used in the normalized range and four-mean equations, with their declared Sobolev buffer. The phase derivative, right inverse, complementary free spaces and balancing seeds use the common Fourier identifications. This condition concerns the original initial-action map, not the subsequently chosen harmonic writer.

Lemma F.13 (Consistency survives a uniformly regular elimination). *Suppose an eliminated variable z_h is defined by an actual equation $F_h(x_h, z_h) = 0$. On a common comparison ball assume a preconditioner A_h has uniformly bounded norm and $\|I - A_h D_z F_h\| \leq \kappa < 1$. If a comparison value z_h^{cmp} satisfies $\|F_h(x_h, z_h^{\text{cmp}})\| \leq \epsilon_h$, then*

$$\|z_h - z_h^{\text{cmp}}\| \leq \frac{\|A_h\|}{1 - \kappa} \epsilon_h. \quad (\text{F.37})$$

With corresponding C^1 consistency of F_h and its derivatives, the eliminated map is C^1 consistent. Thus finite compositions of the phase differences, pointwise analytic coefficient maps and uniformly regular stationary/Legendre solves satisfy [Assumption F.12](#) when their raw equations have the stated Sobolev consistency. Bounds on the inverse blocks must be retained; fixed cutoff analyticity alone is not sufficient.

Proof. The map $z \mapsto z - A_h F_h(x_h, z)$ is a contraction on the comparison ball. Subtract its fixed-point equation at z_h from its value at z_h^{cmp} and absorb $\kappa\|z_h - z_h^{\text{cmp}}\|$. This gives (F.37). For derivatives differentiate $F_h(x_h, z_h(x_h)) = 0$ and use $D_x z_h = -(D_z F_h)^{-1} D_x F_h$. The inverse difference identity $U^{-1} - V^{-1} = U^{-1}(V - U)V^{-1}$ and the common inverse bounds transfer C^1 consistency. The symbol $2h^{-1}\sin(\pi h k_i)$ differs from $2\pi k_i$ by a controlled higher-order symbol; the sampling and product estimates of [Lemma E.6](#) therefore verify the raw comparisons for fixed compositions with the stated derivative buffer. Iterating the elimination argument proves the sufficient criterion. $\square$

Proposition F.14 (Convergent preparation charts and subsequential preparation). *Fix $0 < |t| < t_0$ and assume [Assumption F.12](#). Let W_h lie in the fixed complementary free ball of [Theorem F.11](#), with its bound taken in $\mathcal{P}_h^{s+2}$. If $\mathcal{I}_h W_h$ converges in $\mathcal{P}^{s+1}$ under the declared comparisons, then the prepared records and their harmonic completions converge in the Cauchy topology $H^{s+1} \times H^s$ to compatible vacuum initial data. For an arbitrary bounded sequence W_h , the same conclusion holds after subsequence extraction, but the limiting data need not be unique.*

Proof. The prepared phase records have a uniform $\mathcal{P}_h^{s+2}$ bound. Trigonometric interpolation and Rellich compactness give subsequential convergence in $\mathcal{P}^{s+1}$. Equation (F.36) and the exact finite equality $\mathcal{C}_h = 0$ make every such limit solve $\mathcal{C} = 0$. The velocity comparison and the phase-difference consistency transfer the harmonic completion to its continuum counterpart.

For a convergent free record, retain the normalized range fixed point and the four-mean contraction in the proof of [Theorem F.11](#). Their comparison maps converge in C^1 by [Assumption F.12](#); the explicit Fourier right inverse and the four balancing modes have the same limits. Their contraction constants remain strictly below one on the chosen ball. Every cluster point is consequently a fixed point of the same continuum range/mean maps with the same free record. Fixed-point uniqueness identifies all cluster points. Quantitatively, for either preconditioned contraction $\mathcal{T}_h$, a sampled limiting fixed point z_h^{cmp} obeys

$$\|z_h - z_h^{\text{cmp}}\| \leq (1 - \kappa)^{-1} \|\mathcal{T}_h(z_h^{\text{cmp}}) - z_h^{\text{cmp}}\|.$$

The right side contains the free-record comparison and the actual map consistency defect. Its constants may depend on the fixed nonzero t ; no uniform estimate through the singular tip $t = 0$ is inferred. For merely bounded free records, first extract a convergent free subsequence and use the same argument. Alternating between two different limiting free records is not a claim of whole-sequence convergence. $\square$

Corollary F.15 (Action-prepared inputs populate the open law family). *Fix sufficiently small nonzero t and the cofinally consistent branch of [Assumption F.12](#). For a convergent sequence of free records as in [Proposition F.14](#), and any sequence $B_h \in \overline{\mathcal{B}}$, the harmonically completed histories exist on a common interval and converge to the vacuum Einstein development of the limiting prepared data. No convergence of B_h is required. With only the bounded free-record hypothesis, this is a subsequential statement. The four original lapse-shift rows vanish exactly at the initial cut, independently of B_h .*

Proof. The preceding proposition supplies convergent compatible Cauchy data and keeps them in the fixed top ball. Apply the lower-order stability estimate to their initial discrepancy and the $O(h)$ consistency force. The proof of [Theorem E.12](#) then gives the continuum development, with an error bounded by the initial comparison error plus $C_s h \varepsilon$; no $O(h)$ rate is assigned to an unquantified convergent input sequence. The uniform comparison in [Theorem F.3](#) removes any need for convergence of B_h . The chronological row theorem applies with these same initial records and its stated residual conditions. Exact vanishing of the initial constraints is a fixed-cutoff fact already established before the acceleration law is selected. $\square$

The three results of this section give an open population in a precise sense, but not universal attraction. The coefficient ball is fixed and genuinely open; the transition-row neighborhoods are open in the scaled complete-profile topology appropriate to $O(h^{-2})$ stages; and the initial records are relatively open in the exact finite constraint surface. The raw exact-action evolution itself remains the separate question addressed next.

F.4 Same-cylinder action occurrence and geometric compactness

This section proves [Theorem 6.5](#). The finite correction below changes only the chronological occurrence weights on the already reconstructed action support; it does not fit a new action to the corrected transition law.

Let X, Y be finite stage spaces, let α be a faithful source law and let $Q(y | x)$ be a stochastic comparator on an admitted support $\mathcal{E}$. Set $R(x, y) = \alpha(x)Q(y | x)$, $\rho(y) = \sum_x R(x, y)$ and $d = \beta - \rho$ for a prescribed target law β . On edge arrays use

$$\|h\|_{R^{-1}}^2 = \sum_{(x,y) \in \mathcal{E}} \frac{h(x,y)^2}{R(x,y)}, \quad (Ah)(x) = \sum_y h(x,y), \quad (Bh)(y) = \sum_x h(x,y). \quad (\text{F.38})$$

For the main-text notation, the source-preserving correction is When $d_{h,j} \in \text{Ran } \mathcal{L}_{h,j}$, put $u_{h,j} = \mathcal{L}_{h,j}^\dagger d_{h,j}$ and

$$h_{h,j}^*(x, y) = R_{h,j}(x, y)\{u_{h,j}(y) - (Q_{h,j}u_{h,j})(x)\}. \quad (\text{F.39})$$

If the leverage is at most $1/2$, then $\pi_{h,j}^\text{T} = R_{h,j} + h_{h,j}^*$ is nonnegative and $K_{h,j}^\text{T}(y | x) = \pi_{h,j}^\text{T}(x, y)/\nu_{h,j}(x)$ is a stochastic row with source marginal $\nu_{h,j}$ and target marginal $\nu_{h,j+1}$. For an edge observable c , write

Theorem F.16 (Chronological source-shortened Thomson correction). *Put*

$$P_s = I - A^* \text{diag}(\alpha)^{-1} A, \quad \mathcal{L}_{\alpha,Q} = B P_s B^*. \quad (\text{F.40})$$

Then

$$\mathcal{L}_{\alpha,Q} = \text{diag}(\rho) - Q^\text{T} \text{diag}(\alpha) Q \succeq 0, \quad (\text{F.41})$$

and

$$\langle u, \mathcal{L}_{\alpha,Q} u \rangle = \sum_x \alpha(x) \text{Var}_{Q(x,\cdot)}(u). \quad (\text{F.42})$$

The constraints $Ah = 0$, $Bh = d$ are solvable if and only if $d \in \text{Ran } \mathcal{L}_{\alpha,Q}$. On that branch the unique minimum- R^{-1} -norm solution is

$$h_*(x, y) = R(x, y)\{u(y) - (Qu)(x)\}, \quad u = \mathcal{L}_{\alpha,Q}^\dagger d, \quad (\text{F.43})$$

with

$$\|h_*\|_{R^{-1}}^2 = \langle d, \mathcal{L}_{\alpha,Q}^\dagger d \rangle. \quad (\text{F.44})$$

If $\max_{(x,y) \in \mathcal{E}} |u(y) - (Qu)(x)| \leq 1/2$, then $R + h_$ is nonnegative and has source marginal α and target marginal β .*

Proof. The weighted adjoints are $(A^*f)(x, y) = R(x, y)f(x)$ and $(B^*g)(x, y) = R(x, y)g(y)$. Since $AA^* = \text{diag}(\alpha)$, P_s is the orthogonal projection onto $\ker A$. Direct multiplication gives (F.41), and expanding the quadratic form gives (F.42). The Moore–Penrose normal equations for $B|_{\ker A}$ give (F.43) and (F.44). Finally $R + h_* = R\{1 + u(y) - (Qu)(x)\}$, so the leverage bound gives positivity while the two linear constraints give the required marginals. $\square$

For a real edge observable c , write $\bar{c}(x) = (Qc)(x)$ and

$$\mathcal{V}_R(c) = \sum_{x,y} R(x, y) |c(x, y) - \bar{c}(x)|^2. \quad (\text{F.45})$$

Lemma F.17 (Variance-sensitive source-preserving transfer). *If $Ah = 0$, then*

$$\left| \sum_{x,y} h(x, y) c(x, y) \right| \leq \|h\|_{R^{-1}} \mathcal{V}_R(c)^{1/2}. \quad (\text{F.46})$$

Hence the Thomson coupling $\pi^\text{T} = R + h_$ satisfies*

$$|\mathbb{E}_{\pi^\text{T}} c - \mathbb{E}_R c| \leq \sqrt{\mathcal{E}_{\text{occ}} \mathcal{V}_R(c)}. \quad (\text{F.47})$$

If $0 \leq c \leq B$ and $m = \mathbb{E}_R c$, then $\mathcal{V}_R(c) \leq Bm$.

Proof. Because $Ah = 0$, the source-only function $\bar{c}(x)$ is orthogonal to h . Thus $\sum hc = \sum h(c - \bar{c})$, and weighted Cauchy–Schwarz proves the first inequality. Insert $h = h_*$ for the second. The final statement follows rowwise from $\text{Var}(c) \leq \mathbb{E}(c^2) \leq B\mathbb{E}c$. $\square$

Consider now chronological stages $j = 0, \dots, J - 1$ with faithful marginals ν_j and corrected couplings π_j^T . Starting with $X_0 \sim \nu_0$ and using the corrected transition rows gives $X_j \sim \nu_j$ for every j by induction. Therefore, for nonnegative costs c_j ,

$$\mathbb{E} \sum_j c_j(X_j, X_{j+1}) \leq \sum_j \left(m_j + \sqrt{\mathcal{E}_{\text{occ},j} \mathcal{V}_j} \right), \quad (\text{F.48})$$

where $m_j = \mathbb{E}_{R_j} c_j$ and $\mathcal{V}_j = \mathcal{V}_{R_j}(c_j)$. Under (6.26),

$$\mathbb{E}C_h \leq (M_0 + \sqrt{M_1 V_0})Th^4, \quad \Pr\{C_h > h^2\} \leq (M_0 + \sqrt{M_1 V_0})Th^2. \quad (\text{F.49})$$

Markov’s inequality proves the probability estimate, and the first Borel–Cantelli lemma gives an almost-sure successful tail along any refinement with $\sum_n h_n^2 < \infty$; no independence is used.

The same centered estimate applies to the field-action gap. If $\Delta_j \geq 0$ is bounded by B_j on the finite support and $a_j = \mathbb{E}_{R_j} \Delta_j$, then

$$\mathbb{E}_{\pi_j^T} \Delta_j \leq a_j + \sqrt{\mathcal{E}_{\text{occ},j} \mathcal{V}_{R_j}(\Delta_j)} \leq a_j + \sqrt{B_j \mathcal{E}_{\text{occ},j} a_j}. \quad (\text{F.50})$$

For moving chronological marginals, this gap transfer alone is not a stationarity theorem. Apply [Proposition C.6](#) to the corrected coupling, using the same common-action chart at both ends. With $C_{1,h,j}, C_{2,h,j}$ as in that proposition and $\bar{\Delta}_{h,j}^T = \mathbb{E}_{\pi_{h,j}^T} \Delta_{h,j}$, put

$$U_{h,j}(k) = B_{h,j}(k) \left[(C_{1,h,j} + C_{2,h,j}) \bar{\Delta}_{h,j}^T + C_{2,h,j} d_{\mathcal{A},h,j}^+ \right]^{1/2}. \quad (\text{F.51})$$

If the actual determining first variation is represented as a finite sum of stage insertions with coefficients $w_{h,j}(k)$, Minkowski gives

$$(\mathbb{E}|\delta\mathcal{A}_h[k]|^2)^{1/2} \leq \sum_j |w_{h,j}(k)| U_{h,j}(k) =: \mathfrak{U}_h(k). \quad (\text{F.52})$$

The required scaling is $\mathfrak{U}_h(k) \rightarrow 0$ for each fixed test. For a single represented field insertion the sum has one term. The weights, chart pullbacks and lift bounds are those of the physical action; they are not normalized after the estimate. An independent direct mean-square first-variation certificate may be used instead. Small gap alone cannot suppress a nonzero chronological action descent.

Proposition F.18 (Constructive metric jets imply the literal-link budget). *On the successful event $C_h \leq h^2$ of the nonsymmetric constructive writer, the canonical metric interpolation remains in the fixed small chart and obeys the cutoff-uniform metric-jet bounds of [Theorems 6.1](#) and [6.3](#). On the spinless regular Cartan branch these bounds imply*

$$\sup_h \left(\|A_h\|_{L_t^\infty H_h^s} + \|\partial_t A_h\|_{L_t^\infty H_h^{s-1}} + \|A_{0,h}\|_{L^2} + \|E_h\|_{L^2} + \|F_h\|_{L^2} \right) < \infty \quad (\text{F.53})$$

for fixed $s \geq 3$. In particular (5.18)–(5.19) hold on every sufficiently fine successful cutoff.

Proof. The canonical ADM coframe is an analytic finite-dimensional function of the metric on the fixed nondegenerate chart. Uniform discrete Moser estimates therefore transfer the metric bounds and their first two time derivatives to e_h . The stationary Cartan equation has a uniformly invertible algebraic principal block and an h -weighted quadratic connection remainder; the same Neumann estimate used in the spinless Cartan branch gives a uniform H_h^s bound for A_h . Differentiating that algebraic equation once in time gives the H_h^{s-1} bound for $\partial_t A_h$. The normalized plaquette is an analytic four-link function whose apparent h^{-2} singularity is removable, and the electric link is the exact temporal link identity; these give the final two L^2 bounds. Sobolev embedding gives $h\|A_h\|_\infty \rightarrow 0$, hence one common identity chart on a cofinal tail. $\square$

Proof of Theorem 6.5. The Thomson theorem gives the corrected chronological marginals and support. Equation (F.48) with the assumed scaling gives (6.28). On its successful event, Proposition F.18 and Theorem 5.3 identify the connection curvature. Equation (F.50) and the nonstationary estimate (F.52), with its explicitly assumed scaling, give mean-square vanishing of the represented first variations. The diagonal/Tonelli argument of Proposition C.7 supplies a deterministic subsequence on which the countable determining core vanishes almost surely. The classified Palatini–Holst–volume representative, nonzero Palatini normalization, noncollapse and strong coframe compactness then permit the weak–strong Palatini passage. On the literal-link route the curvature is already identified by Theorem 5.3; hence the limiting variation is $\chi \int \sqrt{-g}(G_{\mu\nu} + \Lambda g_{\mu\nu})k^{\mu\nu}$ and vanishes on the determining core. Borel–Cantelli intersects the successful residual tail with the full-measure stationarity subsequence without any independence assumption. $\square$

G Exact-action response, preparation and boundary realization

This appendix makes the exact-action branch mathematically explicit. It first writes the phase-compatible finite Palatini/link action, its stationary connection solve and the resulting canonical Hamiltonian. It then proves the response and fixed-cutoff realization assertions summarized in Theorem 6.6. These statements concern the original finite action, not the harmonic evolution writer. Static all-cutoff results, the three-site certified event, the graded boundary chart and the separate normal-factor chart retain their individual hypotheses and constants; Appendix H records the supporting necessary slow-compatibility conditions.

G.1 Explicit phase-compatible finite action and canonical Hamiltonian

The exact-action statements below refer to one concrete finite variational object. We record it here so that the canonical system is reconstructible from the article itself rather than from an external coefficient table.

On the odd periodic cubic regulator let e be the canonical ADM coframe, let A_0 be an independent temporal Lorentz connection, and let

$$U_i = \exp(hA_i), \quad A_\mu \in \mathfrak{so}(1, 3). \quad (\text{G.1})$$

On the retained logarithm branch define the literal electric and plaquette curvatures by

$$\begin{aligned} E_i &= h^{-1}(\dot{U}_i U_i^{-1} + A_0 - \text{Ad}_{U_i} A_{0,+i}), \\ B_{ij} &= h^{-2} \log(U_i U_{j,+i} U_{i,+j}^{-1} U_j^{-1}). \end{aligned} \quad (\text{G.2})$$

Let b be the fixed nondegenerate invariant pairing on $\mathfrak{so}(1,3)$. The coframe coefficient arrays $\Pi_i(e), \Sigma_{ij}(e)$ are defined by the Palatini contraction

$$\chi \det(e) r(e, F) = \sum_i b(\Pi_i, E_i) + \sum_{i < j} b(\Sigma_{ij}, B_{ij}), \quad \chi \neq 0, \quad (\text{G.3})$$

and

$$\mathcal{P}_h^{\text{full}} = \int \langle \chi \det(e) (r(e, F) - 2\Lambda), 1 \rangle_h dt. \quad (\text{G.4})$$

The link logarithms share edges, so the exact stationary-connection action contains compensating terms. Put

$$\mathcal{J}_Z V = \sum_{k \geq 0} \frac{(\text{ad}_Z)^k V}{(k+1)!}. \quad (\text{G.5})$$

For an oriented ij plaquette set $(X_1, X_2, X_3, X_4) = (A_i, A_{j,+i}, -A_{i,+j}, -A_j)$ and define

$$\begin{aligned} \mathcal{K}_h &= \sum_i b(\Pi_i, (\mathcal{J}_{hA_i} - I)\dot{A}_i), \\ q_h &= - \sum_i b(\Pi_i, [A_i, A_{0,+i}]) + \frac{1}{2} \sum_{i < j} \sum_{r < s} b(\Sigma_{ij}, [X_r, X_s]), \\ q &= \sum_i b(\Pi_i, [A_0, A_i]) + \sum_{i < j} b(\Sigma_{ij}, [A_i, A_j]). \end{aligned} \quad (\text{G.6})$$

The completed signed action is

$$\boxed{\mathcal{P}_{h,D}^c = \mathcal{P}_h^{\text{full}} + \int \langle q - q_h - \mathcal{K}_h, 1 \rangle_h dt - \left[\sum_i \langle b(\Pi_i, A_i), 1 \rangle_h \right]_0^T}. \quad (\text{G.7})$$

The kinetic subtraction is exact; it removes the time derivative of the connection from the stationary connection equation rather than truncating a link expansion.

For the metric-compatible branch use the real commuting skew-adjoint phase derivatives

$$\widehat{\delta_i u}(k) = i\kappa_i(k)\widehat{u}(k), \quad \kappa_i(k) = \frac{2}{h} \sin(\pi h k_i), \quad -m \leq k_i \leq m, \quad (\text{G.8})$$

on $N = 2m + 1$ sites. Exact summation by parts puts (G.7) in the connection normal form

$$\mathcal{P}_{h,D}^c = \int \left\langle -2\chi\Lambda \det e + f_h(e) \cdot A + \frac{1}{2} A^\top C(e) A + r_h(e, A), 1 \right\rangle_h dt, \quad (\text{G.9})$$

where

$$f_{h,0} = H \sum_i D_i^- \Pi_i, \quad f_{h,i} = H \left(-\partial_t \Pi_i - \sum_{j \neq i} D_j^- \Sigma_{ji} \right). \quad (\text{G.10})$$

Replace only the spatial D_i^- in this load by δ_i , obtaining f_h° , and define the phase-compatible action

$$\boxed{\mathcal{P}_{h,D}^\circ = \mathcal{P}_{h,D}^c + \int \langle (f_h^\circ - f_h) \cdot A, 1 \rangle_h dt.} \quad (\text{G.11})$$

The quadratic connection matrix has inertia (12, 12) and, on the retained compact coframe chart, a uniform least singular value $c_* > 0$. Moreover, with $N_h = \partial_A r_h$,

$$\|N_h(e, A) - N_h(e, \tilde{A})\|_{p,h} \leq ChR\|A - \tilde{A}\|_{p,h} \quad (\text{G.12})$$

on $\|A\|_\infty, \|\tilde{A}\|_\infty \leq R$. Hence for hR sufficiently small the stationary row

$$\gamma_h^\circ(e, A) := f_h^\circ(e) + \mathbb{C}(e)A + N_h(e, A) = 0 \quad (\text{G.13})$$

has a unique analytic solution $A = \mathcal{A}_h(e, \partial_t e)$ on the declared chart, with no Fourier mode removed.

Substitute the canonical ADM coframe $e = e(\gamma, N, \beta)$ and the stationary connection $\mathcal{A}_h$ in (G.11). The resulting time-local reduced Lagrangian is denoted $\mathcal{L}_h^\circ(\gamma, V, N, \beta)$, $V = \dot{\gamma}$. On the Legendre chart used in this article the velocity Hessian is nonsingular. Thus

$$\pi = D_V \mathcal{L}_h^\circ(\gamma, V, N, \beta) \quad (\text{G.14})$$

has the unique analytic inverse $V_h(\gamma, \pi, N, \beta)$ and

$$\boxed{\mathcal{H}_h(\gamma, \pi, N, \beta) = \langle \pi, V_h \rangle_h - \mathcal{L}_h^\circ(\gamma, V_h, N, \beta)}. \quad (\text{G.15})$$

This is the Hamiltonian denoted H_h below; its lapse and shift are multipliers, not evolved canonical coordinates.

At the flat unit-lapse, zero-shift reference with $\Lambda = 0$, write $X = (u, p)$, $t_u = \text{tr } u$, $v_j = \delta_i u_{ij}$ and $\Delta_{\delta,h} = \sum_i \delta_i^2$. The unit-multiplier quadratic Hamiltonian is

$$\begin{aligned} H_2(u, p) &= \|p\|_h^2 - \frac{1}{2} \|\text{tr } p\|_h^2 + \frac{1}{4} \sum_i \|\delta_i u\|_h^2 - \frac{1}{2} \|v\|_h^2 \\ &\quad + \frac{1}{2} \langle v, \delta t_u \rangle_h - \frac{1}{4} \|\delta t_u\|_h^2. \end{aligned} \quad (\text{G.16})$$

The exact quadratic Ward identities follow because the phase derivatives commute; in particular the linear lapse–shift rows are precisely the Fierz–Pauli constraint rows used in Lemma F.8.

Theorem G.1 (Finite-action and canonical provenance). *On the compact nondegenerate coframe chart and for sufficiently small hR :*

- (i) (G.11) is an explicit analytic finite action in the coframe, temporal connection and spatial link variables;
- (ii) its stationary connection is the unique analytic solution of (G.13);
- (iii) stationary elimination followed by the nondegenerate Legendre transform gives the canonical Hamiltonian (G.15), equivalently (6.15);
- (iv) at flat data its quadratic canonical part is (G.16), so the original initial constraint map has the linearization stated in Lemma F.8.

Consequently every exact-action matrix, source, Poisson bracket and finite-dimensional certificate used below is a derivative or basis restriction of one displayed finite action.

Proof. Differentiation of the ordered link products gives (G.9). The compensators (G.6) remove the connection-velocity term exactly. The Cartan quadratic block is a coframe congruence of the invertible flat Cartan block, hence has the stated inertia and uniform singular-value margin. The analytic remainder estimate (G.12) makes (G.13) a contraction after multiplication by $C(e)^{-1}$. Analytic dependence follows from the finite-dimensional implicit-function theorem. The Legendre statement is the same argument for the reduced velocity Hessian. Expanding the phase-compatible action to second order and substituting the linear stationary connection gives (G.16); commutation and skew-adjointness of the δ_i give the exact quadratic Ward identities and the displayed constraint linearization. $\square$

G.2 Canonical constraints and the finite characteristic quotient

Let $X = (u, \pi) \in \mathbb{R}^{12V}$ and let $\lambda = (N, \beta) \in \mathbb{R}^{4V}$. For the canonical Hamiltonian $H_h(X, \lambda)$ put

$$P_h = \nabla_\lambda H_h, \quad F_h^{\text{can}} = \mathbb{J} \nabla_X H_h, \quad B_h = D_X P_h F_h^{\text{can}}, \quad M_h = D_\lambda P_h. \quad (\text{G.17})$$

The exact Euler system is

$$X_t = F_h^{\text{can}}(X, \lambda), \quad P_h(X, \lambda) = 0, \quad (\text{G.18})$$

and differentiating the constraint gives

$$B_h + M_h \dot{\lambda} = 0. \quad (\text{G.19})$$

Phase-compatible multiplier homogeneity gives

$$H_h(X, c\lambda) = cH_h(X, \lambda), \quad M_h \lambda = 0, \quad \langle \lambda, B_h \rangle_h = 0. \quad (\text{G.20})$$

These are global scaling identities, not local discrete diffeomorphism identities.

Introduce multiplier momenta and the extended constraint bracket

$$\mathcal{A}_h = D_X P_h \mathbb{J} (D_X P_h)^*, \quad \mathfrak{D}_h = \begin{pmatrix} 0 & -M_h \\ M_h & \mathcal{A}_h \end{pmatrix}. \quad (\text{G.21})$$

Theorem G.2 (Finite characteristic quotient on a regular one-null branch). *Suppose $D_X P_h$ is onto, $\ker M_h = \text{span}\{\lambda\}$ and (G.19) is solvable. Then*

$$\ker \mathfrak{D}_h = \text{span} \left\{ \begin{pmatrix} \lambda \\ 0 \end{pmatrix}, \begin{pmatrix} -M_h^\dagger B_h \\ \lambda \end{pmatrix} \right\}, \quad \text{rank } \mathfrak{D}_h = 8V - 2, \quad (\text{G.22})$$

and the local characteristic quotient has dimension $12V - 2$.

Proof. If $(r, s) \in \ker \mathfrak{D}_h$, the first row gives $s = c\lambda$. Differentiating the homogeneity identity in X and pairing with the Hamiltonian vector field gives $\mathcal{A}_h \lambda = B_h$. The second row therefore gives $r = -cM_h^\dagger B_h + d\lambda$. The extended phase has dimension $20V$ and the $8V$ independent primary and secondary constraints leave dimension $12V$; quotienting the two characteristic directions gives $12V - 2$. $\square$

For the flat Cartan tangent, let P_{tr} be the orthogonal projection onto scalar spatial matrices. Its signed operator is

$$K_0 = 2I - 6P_{\text{tr}}, \quad K_0^{-1} = \frac{1}{2}I - \frac{3}{4}P_{\text{tr}}. \quad (\text{G.23})$$

G.3 An exact regular transverse block and lossless source elimination

Use the odd-grid phase derivatives

$$\widehat{\delta_i u}(k) = i\kappa_i(k)\widehat{u}(k), \quad \kappa_i(k) = 2h^{-1} \sin(\pi h k_i), \quad \Omega_h^2 = -\sum_i \delta_i^2. \quad (\text{G.24})$$

Let

$$\mathcal{T}_h = \{\beta : \widehat{\beta}(0) = 0, \kappa(k)^T \widehat{\beta}(k) = 0 \ (k \neq 0)\} \quad (\text{G.25})$$

be the transverse-shift space, of dimension $2(V-1)$. On the scalar-momentum family the first source restricted to $\mathcal{T}_h$ has the form

$$A_h \beta = -\frac{h\kappa_h}{2} \mathcal{L}_h \left(-\frac{1}{2} \text{curl}_\delta \beta\right), \quad (\text{G.26})$$

where κ_h is uniformly bounded above and below and $\mathcal{L}_h w$ is the symmetric matrix with zero diagonal and $(\mathcal{L}_h w)_{ij} = S_i w_j + S_j w_i$.

Theorem G.3 (Exact all-cutoff transverse rank and natural-order bounds). *At every odd cutoff,*

$$\text{rank } A_h = 2(V-1), \quad \text{P}_{\text{tr}} A_h = 0, \quad (\text{G.27})$$

and

$$ch^2 \|\Omega_h \beta\|_h^2 \leq \|A_h \beta\|_h^2 \leq Ch^2 \|\Omega_h \beta\|_h^2, \quad \beta \in \mathcal{T}_h. \quad (\text{G.28})$$

The same estimates hold after multiplication by a scalar Fourier Sobolev weight, and

$$A_h^* K_0^{-1} A_h = \frac{1}{2} A_h^* A_h \succ 0. \quad (\text{G.29})$$

Proof. At one Fourier mode the off-diagonal map is unitarily equivalent to $w \mapsto (w_1 + w_2, w_1 + w_3, w_2 + w_3)$, whose squared singular values are 4, 1, 1. For transverse β , $\|\text{curl}_\delta \beta\|_h = \|\Omega_h \beta\|_h$. Substitution in (G.26) proves the two-sided estimate and injectivity. The output diagonal vanishes, so the trace projection vanishes and K_0^{-1} acts by 1/2 on this range. $\square$

Write the complete leading graded source as $F_h = (A_h, G_h)$ with target $(0, d_h)$, and put $\overline{G}_h = (I - \Pi_{A,h})G_h$.

Corollary G.4 (Lossless elimination of the transverse source columns). *One has*

$$\text{rank } F_h = (2V-2) + \text{rank } \overline{G}_h, \quad (\text{G.30})$$

and

$$F_h^* K_0^{-1} F_h \sim \text{diag} \left(\frac{1}{2} A_h^* A_h, \overline{G}_h^* K_0^{-1} \overline{G}_h \right). \quad (\text{G.31})$$

The triangular column transformation producing this congruence leaves the actual target unchanged and preserves every remaining source equation.

Proof. Use the coefficient $C_h = (A_h^* A_h)^{-1} A_h^* G_h$ and right-multiply by $\begin{pmatrix} I & -C_h \\ 0 & I \end{pmatrix}$. The transformed source is $(A_h, \overline{G}_h)$ with orthogonal column ranges. Since $K_0^{-1} A_h = A_h/2$, the signed cross term vanishes as well. $\square$

G.4 Exact scalar–trace reconstruction of the signed response

Work after any regular positive block elimination and let F denote the remaining full-rank source with actual target b . Define

$$\Pi = F(F^*F)^{-1}F^*, \quad w = F(F^*F)^{-1}b, \quad E\tau = \tau I/\sqrt{3}, \quad (\text{G.32})$$

then

$$C = E^*\Pi E, \quad H = 2I - 3C, \quad \zeta = E^*w. \quad (\text{G.33})$$

Theorem G.5 (Exact scalar–trace response reconstruction). *The signed Gram $F^*K_0^{-1}F$ is invertible if and only if H is invertible. If $\chi = H^{-1}\zeta$, then the original physical response is*

$$v = w - 3(I - \Pi)E\chi, \quad E^*v = -\chi, \quad (\text{G.34})$$

and

$$\|v\|^2 = \|w\|^2 + 9\langle \chi, (I - C)\chi \rangle. \quad (\text{G.35})$$

Consequently

$$v_h \rightarrow 0 \iff w_h \rightarrow 0 \text{ and } H_h^{-1}E^*w_h \rightarrow 0 \quad (\text{G.36})$$

on any regular cofinal family.

Proof. Write $F = QR$ with $Q^*Q = I$. The signed Gram is congruent to $(2I - 3Q^*EE^*Q)/4$. The nonzero eigenvalues of Q^*EE^*Q and $E^*QQ^*E = C$ agree, proving the invertibility equivalence. The response is characterized by $F^*v = b$ and $K_0v \in \text{Ran } F$. The first identity gives $\Pi v = w$ and the second gives $(I - \Pi)v = 3(I - \Pi)EE^*v$. Applying E^* yields $H(E^*v) = -\zeta$, hence $E^*v = -\chi$ and (G.34). Orthogonality of the two displayed terms gives (G.35). $\square$

Diagonalizing the compression C gives the trace-angle form. Equivalently, with the positive polar decomposition $F = QR$ and $T = Q^*\text{P}_{\text{tr}}Q$, $\eta = R^{-1}b$, one obtains

$$v = (2I - 3\text{P}_{\text{tr}})Q(2I - 3T)^{-1}\eta. \quad (\text{G.37})$$

If $Tu_j = \mu_j u_j$ and $\eta_j = \langle u_j, \eta \rangle$, then

$$\|\mathcal{E}^0\|^2 = \sum_j \frac{4 - 3\mu_j}{(2 - 3\mu_j)^2} |\eta_j|^2, \quad (\text{G.38})$$

so $\mu = 2/3$ is the unique signed trace resonance. This is the spectral form of Theorem G.5, not a different response law.

For a cofinal regular family define $\nu_h = \sum_j |\eta_{h,j}|^2 \delta_{\mu_{h,j}}$. Then $\|\mathcal{E}_h^0\| \rightarrow 0$ if and only if $\nu_h([0, 1]) \rightarrow 0$ and

$$\lim_{\delta \downarrow 0} \limsup_{h \rightarrow 0} \int_{|2-3\mu| < \delta} \frac{d\nu_h(\mu)}{|2-3\mu|^2} = 0. \quad (\text{G.39})$$

The proof is the direct split of (G.38) into a region separated from $2/3$ and a shrinking neighborhood of the resonance.

G.5 Time-dependent response and the necessary clock test

Static response control is insufficient for a spacetime comparison. Let $F(t), b(t)$ be C^r with full column rank and regular signed Gram in a fixed physical tangent space, and let $\Pi, w, C, H, \zeta, \chi, v$ be the preceding objects.

Theorem G.6 (Derivative hierarchy in scalar–trace coordinates). *Put $\chi_0 = \chi$ and, for $1 \leq j \leq r$,*

$$\chi_j = H^{-1} \left[\zeta^{(j)} + 3 \sum_{\ell=1}^j \binom{j}{\ell} C^{(\ell)} \chi_{j-\ell} \right]. \quad (\text{G.40})$$

Then $\chi_j = \chi^{(j)} = -E^ v^{(j)}$. In particular*

$$\chi' = H^{-1}(\zeta' + 3C'\chi), \quad v' = w' + 3\Pi'E\chi - 3(I - \Pi)E\chi'. \quad (\text{G.41})$$

Thus a small static response does not control the differentiated response without bounds on the transported source geometry.

Proof. Differentiate $H\chi = \zeta$ repeatedly and use $H^{(\ell)} = -3C^{(\ell)}$. The identity $E^*v = -\chi$ holds at every time; differentiating (G.34) gives the second formula. $\square$

The normalized exact continuation has an active/weak amplitude grading $S_a = \text{diag}(aI_A, a^2I_W)$. If u_a is the normalized source coefficient entering the multiplier solve, then

$$\|\lambda_t\|_h^2 = a^{-2} \|(u_a)_A\|_h^2 + a^{-4} \|(u_a)_W\|_h^2. \quad (\text{G.42})$$

Proposition G.7 (Necessary lapse–shift clock condition for harmonic shadowing). *Suppose $a(h) \rightarrow 0$ and an exact-action metric is to shadow a small harmonic metric in a topology controlling the initial first time derivative, with the same lapse, shift and spatial metric at the initial cut. Then necessarily*

$$a(h)^{-1} \|(u_{a(h)})_A\|_h \rightarrow 0, \quad a(h)^{-2} \|(u_{a(h)})_W\|_h \rightarrow 0. \quad (\text{G.43})$$

This requirement is independent of smallness of the instantaneous electric coefficient.

Proof. At $N = 1$, $\beta = 0$ in ADM coordinates, $\partial_t g_{00} = -2\partial_t N$ and $\partial_t g_{0i} = \gamma_{ij}\partial_t \beta^j$. First-jet shadowing therefore forces $\|\lambda_t\| \rightarrow 0$. Equation (G.42) gives the two conditions. $\square$

G.6 Prepared exact-action families

The generic probe-source calculation remains useful for comparison. For the smooth first jet used in the original response audit, the actual weak source obeys

$$\|\mathcal{I}_h d_{h,\rho} - h^2 d_\rho^{(2)}\|_{H^r} \leq C_r h^4, \quad c_r h^2 \leq \|d_{h,\rho}\|_{H_h^r} \leq C_r h^2. \quad (\text{G.44})$$

Thus the generic weak Ward defect is genuinely second order, but [Theorem G.5](#) shows why its norm alone does not control the physical response.

We next record two exact preparations with different purposes.

A rationally detuned all-cutoff active family. Let $N = 2m + 1$, $h = N^{-1}$ and define the one-dimensional product-defect operator

$$D_h u = \delta(fu) - f\delta u - (\delta f)u, \quad f(x) = \cos(2\pi x). \quad (\text{G.45})$$

It is self-adjoint, has kernel the constants, and its nonconstant part is an irreducible zero-diagonal Jacobi matrix. For positive amplitudes $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ put

$$U_\alpha = \sum_{i=1}^3 \alpha_i T_i \cos(2\pi x_i), \quad p_{h,\alpha} = -s_{h,\alpha} I, \quad s_{h,\alpha} = \omega_h \sqrt{\frac{\alpha_1^2 + \alpha_2^2 + \alpha_3^2}{6}}, \quad (\text{G.46})$$

where T_i is the transverse off-diagonal tensor used in (F.28). The diagonal lapse part of the complete first source is

$$\mathcal{D}_{h,\alpha} n = \begin{pmatrix} (\alpha_2 D_{2,h} - \alpha_3 D_{3,h})n \\ (\alpha_3 D_{3,h} - \alpha_1 D_{1,h})n \\ (\alpha_1 D_{1,h} - \alpha_2 D_{2,h})n \end{pmatrix}. \quad (\text{G.47})$$

Lemma G.8 (Finite rational detuning). *For every sufficiently fine odd cutoff there is a rational θ_h with $1 < \theta_h < 1 + h$ such that, for $\alpha_h = (1, \theta_h, 1)$,*

$$\|\mathcal{D}_{h,\alpha_h} n\|_h \geq ch^6 \|n\|_h \quad (\bar{n} = 0). \quad (\text{G.48})$$

The selection can be made by a finite algebraic search at each cutoff.

Proof. The Jacobi representation of D_h has simple algebraic spectrum Σ_h . Its reciprocal-eigenvalue sum is $O(h^{-3})$. The union of ratios θ for which $|\lambda_a - \theta\lambda_b| < 4h^6$ has measure $O(h^2)$, smaller than a fixed fraction of the interval $(1, 1 + h)$ for small h . Choose a dyadic rational outside that union and sufficiently close to the selected point. In the tensor product eigenbasis, the three squared components of (G.47) vanish simultaneously only for the constant mode, and the spectral separation gives the stated margin. $\square$

Theorem G.9 (All-cutoff regular active source). *The exact original-action initial constraints have analytic preparations*

$$\mathcal{C}_h(X_{h,\alpha_h}(a)) = 0, \quad X_{h,\alpha_h}(a) = a(U_{\alpha_h}, p_{h,\alpha_h}) + O(a^2), \quad (\text{G.49})$$

uniformly for all sufficiently fine odd cutoffs. On their first jets the complete leading active source has

$$\ker \mathcal{A}_h^{\text{act}} = \mathbb{R}(1, 0) \oplus \{(0, \beta) : \text{curl}_\delta \beta = 0\}, \quad \text{rank } \mathcal{A}_h^{\text{act}} = 3V - 3, \quad (\text{G.50})$$

its range is trace-free, and

$$(\mathcal{A}_h^{\text{act}})^* K_0^{-1} \mathcal{A}_h^{\text{act}} = \frac{1}{2} (\mathcal{A}_h^{\text{act}})^* \mathcal{A}_h^{\text{act}} \succeq ch^{16} I. \quad (\text{G.51})$$

Hence the complete remaining signed rank problem is confined to the weak columns.

Proof. The exact preparation follows from the finite initial-constraint chart and the nondegenerate four-mean calibration; the $O(h)$ detuning changes the prepared data by $O(|a|h)$ and not their continuum first jet. The diagonal lapse margin is Lemma G.8. The transverse shift block is controlled by Theorem G.3. Together they leave only constant lapse and curl-free shifts in the first-source kernel. The entire first source is trace-free on this three-amplitude family, so K_0^{-1} acts by $1/2$ on its range. The lapse estimate costs h^{-6} and the transverse reconstruction an additional two powers, giving a sufficient smallest singular value ch^8 and therefore the displayed Gram bound. $\square$

A source-neutral nontrivial family. For each axis let T_i be the transverse off-diagonal tensor and let P_i be the transverse diagonal trace-free tensor with entries $+1, -1$ on the two coordinates orthogonal to axis i . Put

$$U = \sum_{i=1}^3 r_i [T_i \cos(2\pi x_i + \phi_i) + \sigma_i P_i \sin(2\pi x_i + \phi_i)], \quad r_i > 0, \quad \sigma_i \in \{-1, 1\}. \quad (\text{G.52})$$

The two polarizations on each axis are orthogonal and have equal Frobenius norm.

Theorem G.10 (Exact source-neutral finite preparation). *For the family (G.52), the complete weak quadratic target vanishes at every odd cutoff. On compact nonzero parameter sets the original finite constraint map admits exact preparations*

$$\mathcal{C}_h(X_h(a)) = 0, \quad X_h(a) = a(U, p_h) + O(a^2) \quad (\text{G.53})$$

uniformly for sufficiently fine odd cutoffs. At a concrete three-site seed the complete leading signed Gram is nonsingular; hence a relatively open balanced neighborhood has an exact local original-action continuation for small nonzero a . On that branch the literal initial temporal and spatial link curvatures obey

$$\|E_h(a, 0)\|_h + \|F_h^{\text{sp}}(a, 0)\|_h \leq C_h |a|. \quad (\text{G.54})$$

The constant in this last estimate is fixed-cutoff; no common physical time is inferred.

Proof. The weak quadratic drift on one axis is a sum of two polarization invariants: the norm difference and the Frobenius cross product. Equal norms and orthogonality make both vanish, independently of h . The linear constraints hold because the configurations and oscillatory momenta are transverse and trace-free. The nonconstant rows are solved by the uniform range inverse of the initial-constraint chart. The four means are solved by a homogeneous trace momentum and three oscillatory balancing parameters; the derivative of the quadratic mean map is uniformly invertible on compact nonzero polarization sets. This gives (G.53). At the explicit three-site seed, finite exact elimination gives a nonzero determinant for the complete signed Gram, and analyticity preserves it on a neighborhood. Because the normalized source target starts one amplitude order later on the balanced branch, its true finite-amplitude response is $O_h(a)$. Substitution in the literal temporal-link derivative and spatial plaquette formulas gives (G.54). $\square$

G.7 Direct transverse response and the positive scalar flux equation

A further weak-neutral preparation can be chosen so that the leading active range is trace-free at every sufficiently fine odd cutoff. On this family the transverse source admits an exact least-squares factorization. Let T_h be the transverse source, $B_h = T_h^* T_h$, and let the normalized positive normal matrix H_h be the finite-band operator obtained by factoring the phase curl and coefficient multiplication. It satisfies

$$\frac{9}{16} I \preceq H_h \preceq 324 I. \quad (\text{G.55})$$

The actual pseudoinverse has the factorization

$$T_h^\dagger = \frac{2\sqrt{2}}{h\omega_h} \Omega_h^{-1} H_h^{-1} R_h \hat{\mathcal{L}}_h^* \iota_o^*, \quad (\text{G.56})$$

with the operators defined by the physical Hodge and coefficient factorization of the source; no inverse-then-project substitution is made.

Theorem G.11 (Actual direct transverse geometric suppression). *For the actual prepared transverse target $b_{\beta,h}$ and direct solution $\beta_{0,h} = -2B_h^{-1}b_{\beta,h}$,*

$$\|\beta_{0,h}\|_{H_h^{s+3}} \leq C_s h^{-1}, \quad \|\mathfrak{E}_h(\beta_{0,h})\|_{H_h^s} \leq C_s h, \quad (\text{G.57})$$

while the folded-face tail is exponentially small. These are estimates of the actual direct source response, not of a prescribed low-frequency projection.

Proof. The factorization (G.56) and (G.55) give two derivatives of Sobolev gain for B_h^{-1} at cost h^{-2} . The prepared transverse target is $O(h)$ on a fixed Fourier bank. Applying the geometric observation operator, whose symbol contributes the compensating h^2 factor at the required order, gives the second estimate. Exponential off-diagonal decay of H_h^{-1} in cyclic Fourier distance gives the folded-face tail. $\square$

After eliminating the transverse variable, the actual leading lapse solves

$$\mathcal{S}_h n_h = g_h. \quad (\text{G.58})$$

Let Q_h project onto the physical off-diagonal complement and let $K_{Q,h}$ be the induced positive coefficient metric.

Theorem G.12 (Exact positive scalar energy). *At every odd cutoff,*

$$\frac{1}{324}I \preceq K_{Q,h} \preceq \frac{16}{9}I, \quad (\text{G.59})$$

and the unchanged scalar operator is

$$\mathcal{S}_h = \frac{1}{2}\mathcal{D}_h^* \mathcal{D}_h + \frac{h^2}{2}\sigma_h^* K_{Q,h}^{-1} \sigma_h. \quad (\text{G.60})$$

Hence

$$2\langle n, \mathcal{S}_h n \rangle_h = \|\mathcal{D}_h n\|_h^2 + h^2 \langle \sigma_h(n), K_{Q,h}^{-1} \sigma_h(n) \rangle_h. \quad (\text{G.61})$$

On the bare diagonal soft subspace the normalized full energy diverges under refinement, so a pole predicted by the diagonal compression alone is not a pole of the full physical scalar energy. This compression result does not by itself prove a uniform inverse for (G.58).

Proof. The coefficient map giving the off-diagonal source is uniformly invertible, so compression of its positive inverse metric gives (G.59). Orthogonal projection of the physical source yields the exact residual factor $hM_h^{-*}Q_h K_{Q,h}^{-1}\sigma_h(n)$; squaring and adding the diagonal source gives (G.60). On the finite-dimensional soft subspace the limiting flux is nonzero on every nonzero vector, while the bare diagonal contribution is higher order; compactness therefore forces the normalized full energy to diverge. $\square$

G.8 Dynamical tangency: an exact obstruction and an upstream escape direction

Exact preparation is not the same as invariance of the prepared set under the original action vector field. Let $\mathcal{W}$ be the weak curl-free multiplier space and let $\mathbb{K}_1(Y)$ be the first Poisson matrix at a canonical first jet Y . For a leading multiplier tilt ℓ define the weak null space

$$\mathcal{N}_W(Y) = \{\nu \in \mathcal{W} : \mathbb{K}_1(Y)[\nu, \mu] = 0 \text{ for every multiplier } \mu\}, \quad (\text{G.62})$$

and the first tangency functional

$$\Delta_\nu(Y, \ell) = Dd_{2,W}(Y)[F_1Y + G\ell, \nu] + \mathbb{K}_1(F_1Y + G\ell)[\nu, \ell]. \quad (\text{G.63})$$

It is determined entirely by the canonical first jet and leading multiplier tilt; higher preparation coefficients do not enter this leading null test.

Theorem G.13 (Initial-layer obstruction on an exactly prepared branch). *There is a nontrivial exactly prepared three-site original-action family X_a, λ_a with*

$$\mathbf{B}_W(X_a, \lambda_a) = 0, \quad (\lambda_t)_W(a, 0) = 0, \quad \|\text{Riem}(g_h)(a, 0)\| = O_h(a), \quad (\text{G.64})$$

but with a weak null test ν_* satisfying $\Delta_{\nu_*}(Y, \ell) \neq 0$. Along the exact continuation,

$$\left. \frac{d}{dt} \mathbf{B}_W[\nu_*] \right|_{t=0} = a^2 \Delta_{\nu_*}(Y, \ell) + O_h(a^3), \quad (\text{G.65})$$

which forces a weak multiplier acceleration of order a^{-2} in the normalized grading. For every fixed S the exact histories exist on $0 \leq t \leq a^3 S$ for sufficiently small a , and the unscaled rates and curvature remain $O_h(a)$ there. Nevertheless no constants $T, C, a_0 > 0$ can make

$$\sup_{0 \leq t \leq T} \|R_{0i0j}(g_h)(a, t)\| \leq Ca \quad (\text{G.66})$$

hold for every $0 < a < a_0$ on this family.

Proof. Taylor expansion of the exact weak drift gives $D_X \mathbf{B}_W = a\mathcal{C}_\ell + O_h(a^2)$ and $D_\lambda \mathbf{B}_W = a\mathcal{K}_W + O_h(a^2)$. Pairing with a Poisson-null ν_* removes the unknown active multiplier acceleration at leading order and yields (G.65). Rescale time by $t = a^3 s$; the finite-dimensional analytic vector field converges on every fixed s -interval to a linear displacement system, giving the shrinking layer and the nonzero limiting curvature slope. If a uniform Ca bound held on a fixed physical interval, choose a fixed s beyond the corresponding normalized bound and then take a sufficiently small so that $a^3 s < T$, contradicting that limiting slope. $\square$

The leading multiplier tilt has a finite affine freedom after the lower quadratic, normalization and cubic-mean conditions are imposed. On the preceding canonical first jet the tangency functional is constant and nonzero on that entire admissible tilt family; hence changing only the leading multiplier tilt cannot repair the obstruction.

The canonical first jet itself has additional admissible directions. Write $\mathcal{L}(Y, \ell)$ for the complete lower incidence consisting of the quadratic drift, multiplier normalizations and cubic mean rows. At the selected base point its multiplier derivative has constant rank, and one canonical direction Z_2 admits an exact multiplier tangent m_2 with

$$D\mathcal{L}(Y, \ell)[Z_2, m_2] = 0. \quad (\text{G.67})$$

Theorem G.14 (The fixed-first-jet tangency obstruction is deformable). *For the exact lower-incidence tangent (Z_2, m_2) above, one weak null test $\nu_\diamond$ remains Poisson-null to first order and*

$$D\Delta_{\nu_\diamond}(Y, \ell)[Z_2, m_2] \neq 0. \quad (\text{G.68})$$

Thus the nonzero tangency obstruction on the preceding prepared first jet is not invariant under all admissible canonical first-jet deformations of the unchanged finite action.

Proof. Differentiate every row of the finite lower-incidence map in the selected canonical direction and solve the resulting characteristic-zero linear system for m_2 . The same weak test has zero first variation of its Poisson-null condition in each coordinate direction of the local first-jet chart. The tangency functional is quadratic in the simultaneous first-jet/multiplier increment, so its derivative is obtained exactly by polarization. Direct finite algebra gives a nonzero rational coefficient; hence the derivative is nonzero over characteristic zero. The calculation establishes a tangent, not an exact nonlinear root. $\square$

G.9 Certified same-event weak Poisson and rapid spectral packet

The spectral input for the conditional fixed-cutoff boundary theorem is recorded at one prepared event of the unchanged $N = 3$, $h = 1/3$ action. We specify that event rather than treating it as unnamed numerical data.

Let $\Lambda_3 = (\mathbb{Z}/3\mathbb{Z})^3$, $\delta_i = 3(S_i - S_i^{-1})$ and $\omega = 3\sqrt{3}$. For $i \in \{1, 2, 3\}$ let T_i be the symmetric matrix with unit entries in the two off-diagonal positions transverse to axis i , let

$$P_1 = \text{diag}(0, 1, -1), \quad P_2 = \text{diag}(1, 0, -1), \quad P_3 = \text{diag}(1, -1, 0), \quad (\text{G.69})$$

and put $f_i(n) = \cos(2\pi n_i/3)$. For the four mixed wave vectors

$$k_1 = (1, 1, 0), \quad k_2 = (1, 0, 1), \quad k_3 = (0, 1, 1), \quad k_4 = (1, -1, 0), \quad (\text{G.70})$$

if $i < j$ are the nonzero coordinate indices of k and m is the remaining index, set

$$v_k = k_j e_i - k_i e_j, \quad w_k = e_m, \quad T_k = v_k w_k^\top + w_k v_k^\top, \quad Q_k = v_k v_k^\top - 2w_k w_k^\top, \quad (\text{G.71})$$

and

$$\phi_k(n) = \cos(2\pi k \cdot n/3) + \frac{1}{\sqrt{3}} \sin(2\pi k \cdot n/3). \quad (\text{G.72})$$

Thus f_i takes values $(1, -1/2, -1/2)$ along its axis and ϕ_k takes the exact values $(1, 0, -1)$ along its wave coordinate. The ten-profile first-field synthesis is

$$U = \sum_{i=1}^3 \alpha_i T_i f_i + \sum_{r=1}^4 \xi_r T_{k_r} \phi_{k_r},$$

$$p_R = 3(\text{diag}(d_1, d_2, d_3) - (d_1 + d_2 + d_3)I) + \frac{3}{2} \sum_i \alpha_i P_i f_i + \frac{3}{2} \sum_r \xi_r Q_{k_r} \phi_{k_r}, \quad p = \sqrt{3} p_R. \quad (\text{G.73})$$

Write $Y = (U, p_R)$. Fix $\alpha_1 = 1$ and

$$d_2 = -\frac{43544048601}{5000000000}, \quad d_1(q) = \frac{\frac{1}{4}(1 + \alpha_2^2 + \alpha_3^2) + \frac{4}{3} \sum_{r=1}^4 \xi_r^2 - d_2 d_3}{d_2 + d_3}, \quad (\text{G.74})$$

with $q = (\xi_1, \xi_2, \xi_3, \xi_4, \alpha_2, \alpha_3, d_3)$. The rational centre is

coordinate	q_c
ξ_1	0.01400959084176856057974567007441966391629696585723
ξ_2	-0.06670489748267782012274481444791177741880525540001
ξ_3	-0.14112980566976204815602958227436097020231483266058
ξ_4	0.17215742625921464595401418943881609717177416130840
α_2	1.18861735067592515913699417669956560893857219610030
α_3	1.13194679076910720016041060163499890274825493906714
d_3	-7.81058367048738968959916622423363949380428649561582

where each terminating decimal denotes the corresponding exact rational. Put

$$\mathcal{B}_* = \{q : \|q - q_c\|_\infty \leq 10^{-40}\}. \quad (\text{G.75})$$

The seven preparation/Fredholm equations have a unique exact root $q_* \in \mathcal{B}_*$ with

$$\|q_* - q_c\|_\infty < 3.28 \times 10^{-49}. \quad (\text{G.76})$$

We set $Y_* = Y(q_*)$. The leading multiplier tilt ℓ_* is the mean-normalized solution of the exact quadratic preparation equation

$$\mathbb{K}_1(Y_*)\ell_* + \widehat{B}_2(Y_*) = 0, \quad (\text{G.77})$$

where $\widehat{B}_2$ is the complete quadratic consistency-drift coefficient obtained from the Hamiltonian (G.15). Thus (Y_*, ℓ_*) and the root box are determined by the displayed finite action and exact rational data. The interval rank and spectral certificates below do not by themselves verify the complete graded-chart assumption used by the boundary theorem.

The weak multiplier space splits as

$$\mathbb{R}^{29} = \mathcal{H} \oplus \mathcal{G}, \quad \dim \mathcal{H} = 3, \quad \dim \mathcal{G} = 26, \quad (\text{G.78})$$

where $\mathcal{H}$ is the exact constant-shift space. Translation invariance gives

$$K_{WW} = \begin{pmatrix} 0 & 0 \\ 0 & K_g \end{pmatrix}, \quad K_g^\top = -K_g. \quad (\text{G.79})$$

After eliminating the regular active block, let D_W be the symmetric weak mass and let D_{26} be its Schur complement on $\mathcal{G}$. The finite mass certificate gives

$$\text{inertia}(D_W) = (18, 11, 0), \quad \text{inertia}(D_{26}) = (15, 11, 0). \quad (\text{G.80})$$

Certificate convention. The matrices in this subsection are not additional dynamical assumptions. They are obtained by differentiating (G.15) on the carrier (G.73), restricting to the displayed physical multiplier bases, and performing the stated Schur eliminations. The proof-bearing numerical data consist of directed interval endpoints or exact rationals: the root-box preconditioner, the principal minor certificate, the four integral coefficient matrices in (G.84), and the rational similarity data used below. High-precision floating eigenvectors are used only to choose rational candidate transforms; every inequality entering a theorem is rechecked by outward interval or exact integer/rational arithmetic.

Theorem G.15 (Exact rank of the weak first-Poisson block). *On $\mathcal{B}_*$,*

$$\text{rank } K_g = 24, \quad \text{rank } K_{WW} = 24. \quad (\text{G.81})$$

Hence

$$\ker K_{WW} = \mathcal{Z}_g \oplus \mathcal{H}, \quad \dim \mathcal{Z}_g = 2, \quad \dim \mathcal{H} = 3. \quad (\text{G.82})$$

Proof. For the lower bound, delete gradient indices 2 and 24 in the fixed zero-based ordering. The resulting principal 24×24 interval block K_{24} admits a verified preconditioner R_{24} with

$$\|I - R_{24}K_{24}\|_\infty < 8.80 \times 10^{-14}, \quad \|K_{24}^{-1}\|_\infty < 2.56 \times 10^{-2}. \quad (\text{G.83})$$

The Neumann series therefore proves $\text{rank } K_g \geq 24$ throughout the box.

For the upper bound, on the ten-profile carrier the exact gradient block has the form

$$K_g(Y) = 1458 \sum_{r=1}^4 \xi_r M_r, \quad M_r^\top = -M_r, \quad \max_{i,j,r} |(M_r)_{ij}| \leq 6, \quad (\text{G.84})$$

with integral 26×26 matrices M_r . Let $P(\xi) = \text{Pf}(\sum_r \xi_r M_r)$, a homogeneous degree-thirteen integer polynomial. After setting $\xi_4 = 1$, its degree in each of the remaining three variables is at most thirteen. Exact evaluation on the full grid $\{0, \dots, 13\}^3$ gives zero for each of the primes

$$1000000007, \quad 1000000009, \quad 1000000033, \quad 1000000087. \quad (\text{G.85})$$

Repeated univariate interpolation therefore makes every coefficient divisible by their product. On the other hand each coefficient obeys

$$|c| \leq 25!! 24^{13} = 6929388469032406070392258560000, \quad (\text{G.86})$$

and the prime product is $1000000136000004854000053496000180873 > 2|c|$. Thus every coefficient is zero in characteristic zero. The Pfaffian vanishes identically, so every K_g has rank at most twenty-four. Combining both bounds proves (G.81); the three exact constant shifts give (G.82). $\square$

The constant shifts factor from the generalized weak pencil:

$$\det(K_{WW} + \lambda D_W) = \lambda^3 \det(D_{HH}) \det(K_g + \lambda D_{26}). \quad (\text{G.87})$$

Thus the nontrivial rapid spectrum is carried by the 26×26 generator $A_g = -D_{26}^{-1} K_g$.

Theorem G.16 (Certified weak spectral trichotomy). *On $\mathcal{B}_*$ the reduced generator A_g has six simple positive real eigenvalues and their six negative partners, one simple complex quartet with two eigenvalues in each open half-plane, four simple purely imaginary conjugate pairs, and a semisimple double zero. The nonzero hyperbolic spectrum obeys*

$$|\text{Re } \lambda| > 0.0015069. \quad (\text{G.88})$$

Consequently the full 29×29 weak generator B_W has an invariant real splitting

$$\mathbb{R}^{29} = E_- \oplus E_0 \oplus E_+, \quad (\dim E_+, \dim E_-, \dim E_0) = (8, 8, 13), \quad (\text{G.89})$$

with constants $M \geq 1$, $\kappa > 0$ such that

$$\|e^{-sB_W}|_{E_+}\| \leq M e^{-\kappa s}, \quad \|e^{sB_W}|_{E_-}\| \leq M e^{-\kappa s}, \quad \sup_{s \in \mathbb{R}} \|e^{sB_W}|_{E_0}\| \leq M. \quad (\text{G.90})$$

Proof. A verified rational similarity transform for A_g puts the interval family inside disjoint Gershgorin components. The similarity error on the complete box is less than 1.558×10^{-25} . The separated components consist of six positive and six negative real singletons, one quartet separated from the imaginary axis, four imaginary conjugate singleton pairs, and one two-eigenvalue component at the origin. Reality and the skew/symmetric pencil symmetry $\lambda \mapsto -\lambda$ imply that the imaginary singleton pairs are exactly imaginary. The only possible final two-eigenvalue cluster is a real pair, an imaginary pair, or a double zero. By Theorem G.15, $\ker A_g = \ker K_g$ has dimension two; therefore the final cluster is exactly a semisimple double zero. Counting modes gives eight stable, eight unstable and ten reduced centre directions; restoring the three exact constant shifts gives (G.89). Finite-dimensional spectral calculus gives (G.90). $\square$

The preceding theorems prove Theorem G.16. Their role is local and fixed-cutoff: no spatial-refinement statement is extracted from the numerical constants.

G.10 A complete amplitude-graded finite chart

We now fix $N = 3$, $h = 1/3$, $V = 27$ and the mean lapse equal to one. The parameter $a > 0$ is amplitude, not mesh size. We give the exact hypotheses needed for the boundary and slow-time results, rather than suppressing the finite coefficient problem in the phrase “regular branch.” All sources, normalizations, constraints and readouts below are restrictions of the same original Hamiltonian H_h .

Assumption G.17 (Amplitude-graded exact-action chart). The original finite system has an analytic prepared record $Q_a^0 = (X_a^0, \lambda_a^0)$ with $P_h(Q_a^0) = 0$, $X_a^0 = aY_* + O(a^2)$ and $\lambda_a^0 = e + a\ell_* + O(a^2)$. The normalized multiplier rate r solves $B_h + M_h r = 0$ with zero mean-lapse derivative. The following data and inequalities refer to a common neighborhood and fixed-cutoff constants.

(C1) All 108 constraint rows are retained, in 104 nonconstant and four mean rows. Their divided canonical normal derivative is an onto map $\mathcal{C}_0 : \mathbb{R}^{324} \rightarrow \mathbb{R}^{108}$. Choose $J_c : \mathbb{R}^{108} \rightarrow \mathbb{R}^{324}$ with $\mathcal{C}_0 J_c$ invertible, and $E : \mathbb{R}^{216} \rightarrow \ker \mathcal{C}_0$, with free projection $\Lambda E = I$, $\Lambda J_c = 0$. The normalized multipliers split into 78 active and 29 weak coordinates, with a fixed shear H .

(C2) For $w = (q, p_A, p_W)$ the substitution

$$\begin{aligned} X &= X_a^0 + a^2(Eq + J_c n), \\ \lambda_A &= \lambda_{a,A}^0 + a^2(p_A - H p_W), \quad \lambda_W = \lambda_{a,W}^0 + a p_W \end{aligned} \tag{G.91}$$

into P_h , followed by division of the nonconstant and mean rows by a^2 and a^3 , defines an analytic map $\mathcal{P}(a, n, w)$. It satisfies $\mathcal{P}(a, 0, 0) = 0$, $\mathcal{P}(0, n, w) = \mathcal{C}_0 J_c n$, and $D_q \mathcal{P} = O(a)$, $D_p \mathcal{P} = O(a^2)$, with these factors preserved by the fixed-order derivatives used below.

(C3) In unbalanced free coordinates $v = (v_s, v_f) \in \mathbb{R}^{294} \oplus \mathbb{R}^{29}$, and time $s = t/a^3$, the original constrained field is

$$\begin{aligned} v'_s &= a^3 b_s + R_s(a, v), \\ v'_f &= C_{\text{red}} v_s + B v_f + a^3 b_w + R_f(a, v), \end{aligned} \tag{G.92}$$

where R_s, R_f are analytic and

$$\begin{aligned} |R_s| &\leq C(a^4 + a^2|v| + a|v|^2), \quad |D_v R_s| \leq C(a^2 + a|v|), \\ |R_f| &\leq C(a^4 + a|v| + |v|^2), \quad |D_v R_f| \leq C(a + |v|). \end{aligned} \tag{G.93}$$

The transport identity is $C_{\text{red}} b_s + B b_w = 0$. For $D = \text{diag}(0_{294}, B)$ there are commuting projections $\Pi_+ + \Pi_- = I$ and constants $M_D, \kappa > 0$ such that

$$\|e^{sD} \Pi_-\| \leq M_D, \quad \|e^{-sD} \Pi_+\| \leq M_D e^{-\kappa s} \quad (s \geq 0). \tag{G.94}$$

No rank is prescribed in the boundary argument. At the certified event of [Theorem G.16](#), the weak block B is the generator B_W of [\(G.89\)](#), Π_+ is the eight-dimensional unstable projector and Π_- contains the stable and bounded-centre subspaces.

(C4) The original canonical vector field $F = \mathbb{J} \nabla_X H_h$ has bounded derivatives on the physical chart. The metric and ordered-link readouts are their actual smooth fixed-cutoff jet functions. In particular, metric curvature uses $(X, \lambda, X_t, \lambda_t, X_{tt})$, with no second time derivative of lapse or shift. The stationary spatial connection z has derivative orders $z_X = O(1)$, $z_{\lambda_A} = O(a)$ and $z_{\lambda_W} = O(a^2)$ on the relevant tubes, with their differentiated bounds. The temporal connection has bounded derivatives. At zero amplitude the reference curvature vanishes.

These are analytic divisibilities and actual source/readout comparisons, not merely formal powers along one curve. For example, the divided mean rows in item (C2) retain the four means rather than remove them. The certified event of [Theorems G.15](#) and [G.16](#) supplies the weak rank and spectral part of this chart at one concrete prepared shape. The analytic divisibility, source and readout bounds in the present assumption are the remaining finite chart conditions used by the boundary argument. No constant in this fixed- $N = 3$ certificate is asserted to be uniform at other lattice sizes.

Lemma G.18 (Exact nonlinear constraint coordinates). *Under [Assumption G.17](#), there is an analytic $n = n(a, w)$ such that $P_h(\Theta_a(w)) = 0$, where Θ_a is [\(G.91\)](#) with this value of n . Moreover,*

$$\begin{aligned} n(a, 0) &= 0, & n &= O(a|q| + a|q|^2 + a^2|p|), \\ D_q n &= O(a), & D_p n &= O(a^2), \\ D_q \mathcal{X}_a &= O(a^2), & D_p \mathcal{X}_a &= O(a^4), & D_q^2 \mathcal{X}_a &= O(a^3), & D_q D_p \mathcal{X}_a, D_p^2 \mathcal{X}_a &= O(a^4), \end{aligned} \quad (\text{G.95})$$

where $\mathcal{X}_a$ is the canonical component of Θ_a . For positive small a this is a coordinate chart on the full 323-dimensional normalized constraint leaf.

Proof. The derivative $D_n \mathcal{P}(0, 0, 0) = \mathcal{C}_0 J_c$ is invertible. The analytic implicit-function theorem therefore solves $\mathcal{P}(a, n, w) = 0$ locally. At $a = 0$ the unique solution is $n = 0$, and $\mathcal{P}(a, 0, 0) = 0$ gives $n(a, 0) = 0$. Differentiating the implicit equation gives $D_q n = -(D_n \mathcal{P})^{-1} D_q \mathcal{P}$ and the analogous formula for p . The assumed analytic factors give the stated derivative orders, including their further derivatives. Integration from $w = 0$ yields the bound on n . Differentiating [\(G.91\)](#) yields the remaining bounds. The free projection recovers q , and the affine multiplier formulas recover p_A, p_W ; invertibility of the normal derivative completes the coordinate inverse. All original constraint rows vanish because their divided equations vanish, with $a > 0$. $\square$

Evaluation of Θ_a is a coordinate solve, not a force projecting an unconstrained physical history back onto the constraint set.

G.11 Nonlinear two-point construction on an amplitude-squared interval

In the balanced coordinate $v = (ax, y)$, $w = (x, y)$, $x = (q, p_A)$, the exact reduced vector field $\mathcal{H}_a$ is

$$(\mathcal{H}_a)_q = a\Lambda F(\Theta_a(w)), \quad (\mathcal{H}_a)_A = a(r_A + aHr_W)(\Theta_a(w)), \quad (\mathcal{H}_a)_W = a^2 r_W(\Theta_a(w)). \quad (\text{G.96})$$

The inverse multiplier solve in r is the actual signed solve, not a positive surrogate. Set

$$w_{\text{ref}}(s) = (a^2 s b_s, a^3 s b_w), \quad f_a = \mathcal{H}_a(w_{\text{ref}}) - w'_{\text{ref}}. \quad (\text{G.97})$$

Lemma G.19 (Balanced derivative and reference residual estimates). *On $|w| \leq M_0 a$ and for $1 \leq S \leq c/a$,*

$$\begin{aligned} \|D_w \mathcal{H}_a - D\| &\leq C_{M_0} a, & \|D_w^j \mathcal{H}_a\| &\leq C_j \quad (2 \leq j \leq 4), \\ \|D_w^j (\mathcal{H}_a)_q\| &\leq C_j a^2 \quad (1 \leq j \leq 3), \\ \|f_a\|_\infty &\leq C a^3, & \|f'_a\|_\infty &\leq C a^4, & \|w_{\text{ref}}\|_\infty &\leq C c a, & |w'_{\text{ref}}| &\leq C a^2. \end{aligned} \quad (\text{G.98})$$

Proof. Conjugate [\(G.92\)](#) by $\text{diag}(aI_{294}, I_{29})$. The perturbation of D has blocks $D_s R_s$, $a^{-1} D_f R_s$, $a C_{\text{red}} + a D_s R_f$, and $D_f R_f$. The bounds [\(G.93\)](#) make each $O(a)$ on this tube. Analyticity and the factor a in the entire slow remainder remove the apparent singular division and bound higher

balanced derivatives. The canonical row in (G.96), together with (G.95) and bounded derivatives of F , gives its extra a^2 factor. The unbalanced reference is $v_{\text{ref}} = a^3 s(b_s, b_w)$. Its linear terms cancel by $C_{\text{red}} b_s + B b_w = 0$. The balanced residual is consequently bounded by sums of a^3 , $a^4 s$ and $a^6 s^2$ in the slow row and a^4 , $a^4 s$, $a^6 s^2$ in the weak row. These are $O(a^3)$ for $s \leq c/a$. Differentiating along $v'_{\text{ref}} = a^3(b_s, b_w)$ and using the derivative bounds gives $O(a^4)$. The other two estimates follow directly from the reference formula. $\square$

Define the forward–backward Green operator

$$(\mathcal{G}_S f)(s) = \int_0^s e^{(s-r)D} \Pi_- f(r) \, dr - \int_s^S e^{(s-r)D} \Pi_+ f(r) \, dr. \quad (\text{G.99})$$

It satisfies $\|\mathcal{G}_S\| \leq C_G(1+S)$. The factor S comes from the bounded center, not the exponentially decaying kernels. Put $N_a(s, e) = \mathcal{H}_a(w_{\text{ref}} + e) - \mathcal{H}_a(w_{\text{ref}}) - De$ and iterate

$$e^{[0]} = 0, \quad e^{[m+1]} = \mathcal{G}_S\{f_a + N_a(\cdot, e^{[m]})\}, \quad w^{[m]} = w_{\text{ref}} + e^{[m]}. \quad (\text{G.100})$$

The forward–backward placement is the standard decoupling mechanism for unstable boundary problems [70]; the amplitude estimates below pertain to this particular finite field.

Theorem G.20 (Exact constrained boundary history). *For sufficiently small fixed $c, a_0 > 0$, $0 < a < a_0$ and $1 \leq S \leq c/a$, (G.100) is a contraction on $\|e\|_\infty \leq Ka^2$. Its unique fixed point gives*

$$\begin{aligned} w'_* &= \mathcal{H}_a(w_*), & \Pi_-(w_*(0) - w_{\text{ref}}(0)) &= 0, & \Pi_+(w_*(S) - w_{\text{ref}}(S)) &= 0, \\ \|w_* - w_{\text{ref}}\|_\infty &\leq Ca^2, & \|(w_* - w_{\text{ref}})'\|_\infty &\leq Ca^3. \end{aligned} \quad (\text{G.101})$$

The reconstructed curve $\Theta_a(w_*(t/a^3))$ solves the original full normalized action equations. Every ideal iterate also satisfies all constraints, without in general satisfying the evolution equation.

Proof. On the indicated ball, Lemma G.19 gives $\text{Lip}_e N_a \leq L_0 a$. Choose c and then a_0 so that $q = C_G(1+S)L_0 a \leq 1/4$. Also $\|\mathcal{G}_S f_a\| \leq C(1+S)a^3 \leq C_0 a^2$. Taking $K = 2C_0$ makes the ball invariant, and the contraction theorem gives the unique fixed point. The integral equation is equivalent to the specified differential and boundary conditions.

For the derivative, write $F_e = f_a + N_a(s, e_*)$. Integration by parts in (G.99) gives

$$e'_*(s) = e^{sD} \Pi_- F_e(0) + e^{(s-S)D} \Pi_+ F_e(S) + \mathcal{G}_S F'_e(s).$$

Here $\|F_e\| \leq Ca^3$ and $\|F'_e\| \leq Ca^4 + L_0 a \|e'_*\|$, because the derivative at fixed e is a difference of $D\mathcal{H}_a$ applied to w'_{ref} . The endpoint factors are bounded in their prescribed time directions. Absorption using the same $q < 1$ proves $\|e'_*\| \leq Ca^3$.

The reconstructed velocity and the original vector field agree in every free coordinate by (G.96). Both are tangent to $P_h = 0$. Their difference is therefore a canonical normal vector annihilated by the invertible normal constraint derivative, and is zero. This recovers the complete canonical evolution, not just its free projection. Constraint satisfaction of all iterates follows separately from the definition of Θ_a . $\square$

Theorem G.21 (Exact-action boundary realization on a graded chart). *Fix $N = 3$ and $h = 1/3$, and assume the complete amplitude-graded exact-action chart of Assumption G.17, including its analytic divisibilities, source estimates, transport identity and physical readout bounds on one common neighborhood. The certified weak spectrum supplies the spectral part of that assumption but not its other rows. Then there are $c, a_0, C > 0$ such that, for every $0 < a < a_0$, the unchanged*

original finite action admits an exact boundary-selected history $Q_{*,a} = (X_{*,a}, \lambda_{*,a})$ of (G.18), with all 108 original constraint rows identically zero, on $0 \leq t \leq ca^2$. Its initial record preserves the certified first field and first multiplier tilt,

$$X_{*,a}(0) = aY_* + O(a^2), \quad \lambda_{*,a}(0) = e + a\ell_* + O(a^2). \quad (\text{G.102})$$

Relative to the exact prepared record through the same first jet, the boundary selector spends only higher preparation coefficients:

$$\Delta X(0) = O(a^4), \quad \Delta \lambda_A(0) = O(a^4), \quad \Delta \lambda_W(0) = O(a^3). \quad (\text{G.103})$$

In the actual metric and ordered-link readouts,

$$\sup_{0 \leq t \leq ca^2} (\| \text{Riem}(g_{*,a}) \| + \| E_{*,a}^{\text{link}} \| + \| F_{*,a}^{\text{sp,link}} \|) \leq Ca. \quad (\text{G.104})$$

The boundary construction is rank-free: the nonexpanding coordinates are fixed at the initial cut and the E_+ coordinates of (G.89) are fixed at the terminal cut. Thus no prescribed value of the unstable dimension is used in the Green argument.

The ideal nonlinear iterates remain exactly constrained and converge to this history. After $m \geq 2$ iterations their readout errors are bounded by $C2^{-m}$, Ca^22^{-m} and Ca^42^{-m} for the metric, temporal-link and spatial-link curvatures, respectively. Hence $m = 3 + \lceil p \log_2(1/a) \rceil$ gives errors $O(a^p)$, $O(a^{p+2})$, and $O(a^{p+4})$ on the entire interval.

The same history also satisfies $X_t, \lambda_t, X_{tt} = O(a)$ and $Q_*(t) - Q_a^0 = O(a^3)$ on its full interval.

Proof of the anchoring and curvature assertions. At $s = 0$ the forward-backward correction is $O(a^2)$ in the balanced coordinates. Inserting this estimate in (G.91), together with $D_q n = O(a)$ and $D_p n = O(a^2)$, gives the three orders in (G.103); in particular the first field and first multiplier tilt are unchanged. Throughout the interval the slow coordinates of w_* are $O(a)$, its weak coordinates are $O(a^2)$, and their scaled rates are $O(a^2)$ and $O(a^3)$, respectively. The physical coordinate formulas and $t = a^3 s$ give displacement $O(a^3)$ and $X_t, \lambda_t = O(a)$. Differentiating the unchanged canonical equation gives $X_{tt} = D_X F X_t + D_\lambda F \lambda_t = O(a)$. Spatial differentiation is bounded at this fixed cutoff. The smooth metric readout therefore has curvature $O(a)$; equivalently its Gauss–Codazzi expressions require these jets and first, but not second, multiplier derivatives. The stationary connection orders in the certificate give the same bound for its actual ordered-link curvatures. No equality of metric and link curvature at a fixed cutoff is used. $\square$

G.12 Approximate curves, time jets, and readout accuracy

Proposition G.22 (Accuracy of the constrained nonlinear iterates). *For $m \geq 2$ write $\rho_m = 2^{-m}$ and $\delta_m = w^{[m]} - w_*$. On the entire boundary interval,*

$$\max_{0 \leq j \leq 2} \|\partial_s^j \delta_m\|_\infty \leq Ca^2 \rho_m, \quad \|\partial_s^j (\delta_m)_q\|_\infty \leq Ca^4 \rho_m \quad (j = 1, 2). \quad (\text{G.105})$$

After nonlinear reconstruction the physical errors are

quantity	X	λ_A	λ_W	X_t	$(\lambda_A)_t$	$(\lambda_W)_t$	X_{tt}
error/($C\rho_m$)	a^4	a^4	a^3	a^3	a	1	1

(G.106)

The actual readout errors satisfy

$$\begin{aligned} \|\text{Riem}(g_m) - \text{Riem}(g_*)\|_\infty &\leq C\rho_m, \\ \|E_m^{\text{link}} - E_*^{\text{link}}\|_\infty &\leq Ca^2 \rho_m, \\ \|F_m^{\text{sp,link}} - F_*^{\text{sp,link}}\|_\infty &\leq Ca^4 \rho_m. \end{aligned} \quad (\text{G.107})$$

Proof. The contraction gives $\|\delta_m\| \leq Ca^24^{-m}$. Differentiate the Green equation: $\delta'_m = D\delta_m + N_a(s, e^{[m-1]}) - N_a(s, e_*)$. This bounds its first derivative with a fixed index loss. A second differentiation uses bounded second derivatives of $\mathcal{H}_a$, $w'_{\text{ref}} = O(a^2)$ and $e'_* = O(a^3)$, with another fixed index loss. These losses are absorbed by $C2^{-m}$. The free canonical row of D is zero, and $(w^{[m]})'_q = (\mathcal{H}_a)_q(w^{[m-1]})$; its derivative bounds gain the extra a^2 in (G.105).

A physical derivative costs a^{-3} . The multiplier coordinates are affine, so their errors follow immediately. For X_t and X_{tt} use the chain rule with (G.95). The leading scaled canonical derivative errors are $a^2(a^4\rho_m)$ and $a^4(a^2\rho_m)$, both $O(a^6\rho_m)$. The second scaled derivative has the same leading order; products involving second chart derivatives are smaller. Division by a^3 and a^6 yields the table. Smoothness of the metric jet readout gives its stated error. For the stationary spatial connection, the three source derivative orders give $\Delta z = O(a^4\rho_m)$ and $\Delta z_t = O(a^2\rho_m)$: the direct contributions to the latter are $a^3\rho_m$, $a \cdot a\rho_m$ and $a^2 \cdot \rho_m$. Changes of coefficients multiplied by the exact $O(a)$ rates are no larger. The temporal connection has error $O(a^3\rho_m)$. The actual temporal-link formula uses z_t and the shifted temporal connection; the spatial-link formula uses only spatial connection values. Their smooth fixed-cutoff expressions yield the last two estimates. $\square$

Taking $m = 3 + \lceil p \log_2(1/a) \rceil$ proves the accuracy assertion in Theorem G.21. The ideal sweeps contain exact integrals and exact constraint-coordinate solves. A numerical implementation needs a separate residual certificate.

Proposition G.23 (A posteriori boundary residual estimate). *Let $\tilde{w}$ remain in the same contraction tube and set $r = \tilde{w}' - \mathcal{H}_a(\tilde{w})$, $d_- = \Pi_-(\tilde{w}(0) - w_{\text{ref}}(0))$, $d_+ = \Pi_+(\tilde{w}(S) - w_{\text{ref}}(S))$. Then*

$$\|\tilde{w} - w_*\|_\infty \leq \frac{M_D(|d_-| + |d_+|) + C_G(1 + S)\|r\|_\infty}{1 - q}, \quad q \leq \frac{1}{4}. \quad (\text{G.108})$$

To obtain curvature accuracy one additionally controls the actual first two scaled derivatives of this error and the differentiated normal solve; a position residual alone is not a curvature certificate.

Proof. Variation of constants gives the same Green fixed-point equation with additional terms $e^{sD}d_- + e^{(s-S)D}d_+ + \mathcal{G}_{sr}$. Subtract the exact equation and absorb its q -Lipschitz term. The final statement follows from the physical derivative losses in (G.106). For an inexact normal solve, constraint residuals are its actual divided residuals multiplied back by a^2 and a^3 , not exact zeros. $\square$

G.13 Invariant graphs and exact normal expansion

The boundary construction selects an exact history but does not make its initial-value problem stable. The following formulation separates the nonlinear invariance required for a residence estimate from a frozen spectral computation. We use the invariant-graph method in finite dimension; see [71] for the broader invariant-foliation setting.

Lemma G.24 (A quantitative forward invariant-graph criterion). *Consider a localized C^3 field*

$$\dot{u} = A_u u + b_u + N_u(u, n), \quad \dot{n} = A_n n + b_n + N_n(u, n) \quad (\text{G.109})$$

on finite-dimensional spaces, with bounded nonlinear derivatives through order three. Suppose for $t \geq 0$ $\|e^{-tA_u}\| \leq Me^{-\beta t}$, $\|e^{tA_n}\| \leq Me^{\alpha t}$, and choose $\max\{0, \alpha\} < \nu$ with $3\nu < \beta$. If the global nonlinear Lipschitz constant L satisfies

$$\max_{j=1,2,3} ML((j\nu - \alpha)^{-1} + (\beta - j\nu)^{-1}) < 1, \quad (\text{G.110})$$

then the forward weighted trajectories define a C^3 invariant graph $u = h(n)$. On any open set where localization agrees with the original field, the graph is locally invariant for that original field.

Proof. For fixed $n(0) = n_0$ solve

$$\begin{aligned} n(t) &= e^{tA_n} n_0 + \int_0^t e^{(t-r)A_n} [b_n + N_n(u(r), n(r))] dr, \\ u(t) &= - \int_t^\infty e^{(t-r)A_u} [b_u + N_u(u(r), n(r))] dr. \end{aligned}$$

In the norm $\sup_{t \geq 0} e^{-\nu t} |(u(t), n(t))|$, the two kernels have bounds $M/(\nu - \alpha)$ and $M/(\beta - \nu)$. The map is a contraction and defines a unique weighted trajectory. Set $h(n_0) = u(0)$. Time translation preserves the equations and the weighted class, so uniqueness proves graph invariance. Differentiating in n_0 gives a linear integral equation with the same contractive principal term. Second and third derivatives have sources composed of products of lower derivatives; these belong to the weights 2ν and 3ν . The corresponding three inequalities in (G.110) solve these equations and the strict spectral margins justify their continuous dependence. This proves C^3 regularity. Restricting a graph trajectory to an interval on which the localization is inactive yields the original equation. $\square$

Affine forcing can displace this graph from the coordinate origin. Consequently neither passage through a selected reference point nor equality of its tangent with the frozen spectral subspace is a consequence of this lemma alone. Also, for a nonstationary invariant manifold its tangent need not be invariant under the Jacobian at each point. These distinctions matter when extracting the certificate below from an actual action chart.

Assumption G.25 (Two-scale invariant-graph and normal-factor certificate). On the original normalized 323-dimensional constraint leaf there is a nonempty local rapid-selected graph $\mathcal{S}_a$ of dimension 312. Inside it there is a nonempty local graph $\mathcal{D}_a$ of dimension 311. Both are locally invariant under the original field. They may be obtained from Lemma G.24 in successive rapid and slow coordinates, provided its quantitative extension and spectral-gap hypotheses hold on the respective charts. The following additional physical comparisons hold throughout those charts.

(N1) In rapid time $s = t/a^3$, the eleven-component normal $v = u - h_r(n)$ satisfies $v' = N_a(u, n)v$, with $\|N_a - J_u\|$ sufficiently small uniformly in the chart, where J_u has spectrum in the open right half-plane. Its physical normal displacement is uniformly comparable to $a|v|$.

(N2) Within $\mathcal{S}_a$, use slow time $\tau = t/a$. The scalar normal $\omega = w - h_s(n)$ to $\mathcal{D}_a$ satisfies

$$\omega_\tau = [\lambda_+(a) + r_a(w, n)]\omega, \quad \lambda_+(a) = \gamma + O(a), \quad |r_a| \leq \gamma/4, \quad \gamma > 0. \quad (\text{G.111})$$

Its physical normal displacement is uniformly comparable to $a|\omega|$. The certified normal radii are uniformly bounded above; no uniform lower bound for the slow chart radius is assumed.

The normal factorizations are exact consequences of graph invariance and the corresponding derivative estimates, when those estimates are verified. Indeed, subtract the graph equation from $u' - Dh_r(n)n'$ at $u = h_r(n) + v$ and integrate the derivative in u along the segment. The constant term vanishes, leaving a matrix times v . For the slow scalar graph the correction contains $D_w N_+ - Dh_s(n)D_w N_0$ integrated along that segment. Thus its bound requires control of Dh_s as well as of the nonlinear field derivatives; a simple unstable eigenvalue by itself is not sufficient.

Proposition G.26 (Rapid Lyapunov growth and slow scalar growth). *Under [Assumption G.25](#), there are $c, C, \mu_r, \mu_s > 0$ such that, until exit from the corresponding full certified chart,*

$$ce^{\mu_r t/a^3} |v(0)| \leq |v(t)| \leq Ce^{Ct/a^3} |v(0)|, \quad e^{\mu_s t/a} |\omega(0)| \leq |\omega(t)| \leq e^{2\gamma t/a} |\omega(0)|, \quad (\text{G.112})$$

where the scalar estimate is for histories in $\mathcal{S}_a$ and one can take $\mu_s = \gamma/2$ after reducing a_0 .

Proof. The matrix $W = \int_0^\infty e^{-rJ_u^*} e^{-rJ_u} dr$ is positive definite and satisfies $J_u^* W + W J_u = I$. For N_a sufficiently close to J_u ,

$$\frac{d}{ds}(v^* W v) = v^*(N_a^* W + W N_a)v \geq \frac{1}{2}|v|^2 \geq (2\|W\|)^{-1} v^* W v.$$

Integrate and use the extremal eigenvalues of W to obtain the lower rapid bound. A uniform upper bound on N_a gives the upper estimate by Gronwall. For the slow scalar factor, $\gamma/2 \leq \lambda_+(a) + r_a \leq 2\gamma$ at small a ; its exact integral solution gives the other two inequalities. Converting each time variable separately yields [\(G.112\)](#). $\square$

Theorem G.27 (Two necessary normal precision scales for local residence). *On the separate invariant normal-factor chart of [Assumption G.25](#), suppose the same original finite action satisfies the stated graph and factorization certificate. This chart is not identified with the rapid spectral packet of [Theorem G.16](#). There are nested locally invariant constrained graphs*

$$\mathcal{D}_a \subset \mathcal{S}_a \subset \mathcal{P}_a, \quad \dim \mathcal{D}_a = 311, \quad \dim \mathcal{S}_a = 312, \quad \dim \mathcal{P}_a = 323.$$

For a prescribed physical time $T > 0$, local residence in the certified charts requires the rapid physical normal displacement to satisfy

$$\delta_r(0) \leq Ca \exp(-\mu_r T/a^3). \quad (\text{G.113})$$

Inside $\mathcal{S}_a$, the additional slow physical normal must satisfy

$$\delta_s(0) \leq Ca \exp(-\mu_s T/a). \quad (\text{G.114})$$

Consequently a tube whose every exact history remains in these charts through T cannot have algebraic thickness in either indicated normal sector. The constants are fixed-cutoff constants. These are residence obstructions, not statements of curvature blowup or nonexistence after chart exit.

Proof of [Theorem G.27](#). Residence through T bounds the terminal rapid normal by its fixed chart radius. The lower growth bound gives $|v(0)| \leq C \exp(-\mu_r T/a^3)$. Multiply by the physical normal scale a to obtain [\(G.113\)](#). Within $\mathcal{S}_a$ the same argument for ω gives [\(G.114\)](#). A smaller available slow radius can only decrease that bound. Apply the rapid inequality to the rapid radius of any putative resident tube, and apply the slow inequality to its intersection with $\mathcal{S}_a$. Since $ae^{-cT/a^k} = o(a^m)$ for every fixed m and $k > 0$, neither radius can be bounded below by a fixed algebraic power of a . The dimension counts subtract eleven and then one normal coordinate from 323; the eleven rapidly contracting directions are retained. $\square$

For example, if $|v(0)| \geq ca^m$, the residence time is at most $(a^3/\mu_r)[m \log(1/a) + C_m]$, as long as the initial point lies in the chart and the history exists until its exit. The analogous slow estimate is $(a/\mu_s)[m \log(1/a) + C_m]$. In an anchored control with $|v(0)| \asymp a^3|e|$ the rapid expression instead contains $m+3$ when $|e| \geq ca^m$. A trajectory may leave the full chart through a tangential boundary sooner; none of these estimates asserts growth after that exit. Exact zero normal coordinates remain zero while local invariance applies, so the theorem does not exclude a longer trajectory on the selected graph.

H Slow exact-action compatibility and obstruction theory

This appendix proves the slow compatibility and obstruction assertions of [Theorem 6.6](#). It uses the amplitude-graded original-action chart and source projections of [Appendix G](#), at fixed spatial cutoff. The first step identifies necessary bounded-rate limits, the second measures the initial acceleration defect, and the final steps reduce the quadratic slope condition. Necessary algebraic compatibility is not identified with nonlinear positive-time propagation.

H.1 State-dependent principal terms on the slow physical scale

Set $t = a\tau$, $s = \tau/a^2$, $w = (x, az)$ and $\xi = (x, z)$. The unbalanced coordinate is then $v = a\xi$. On bounded ξ -sets the physical chart gives

$$X - X_a^0 = O(a^2), \quad \lambda_A - \lambda_{a,A}^0 = O(a^2), \quad \lambda_W - \lambda_{a,W}^0 = O(a^2). \quad (\text{H.1})$$

In these coordinates the exact slow system is

$$\begin{aligned} x_\tau &= b_s + \Sigma_0(\xi) + a\Sigma_1(a, \xi), \\ z_\tau &= a^{-2}\Phi_0(\xi) + a^{-1}\Phi_1(\xi) + b_w + \Phi_2(a, \xi), \end{aligned} \quad (\text{H.2})$$

Lemma H.1 (Exact anisotropic slow expansion). *The graded analytic remainders in [\(G.93\)](#) give exactly [\(H.2\)](#), where the remainders Σ_1, Φ_2 are uniformly bounded on bounded ξ -sets and*

$$\begin{aligned} \Phi_0(x, z) &= C_{\text{red}}x + Bz, \\ \Phi_1(\xi) &= D_a D_v R_f(0, 0)[\xi] + \tfrac{1}{2} D_v^2 R_f(0, 0)[\xi, \xi], \\ \Sigma_0(\xi) &= \tfrac{1}{2} D_a^2 D_v R_s(0, 0)[\xi] + \tfrac{1}{2} D_a D_v^2 R_s(0, 0)[\xi, \xi]. \end{aligned} \quad (\text{H.3})$$

In particular $\Sigma_0(0) = \Phi_1(0) = 0$ and both maps have degree at most two.

Proof. Since $v' = a^3\xi_\tau$, substitution in the unbalanced equations gives $x_\tau = b_s + a^{-3}R_s(a, a\xi)$ and $z_\tau = a^{-2}(C_{\text{red}}x + Bz) + b_w + a^{-3}R_f(a, a\xi)$. The analytic order bounds force $R_s(a, a\xi) = a^3\Sigma_0(\xi) + a^4\Sigma_1(a, \xi)$ and $R_f(a, a\xi) = a^2\Phi_1(\xi) + a^3\Phi_2(a, \xi)$. Taylor expansion shows that exactly the displayed mixed derivatives survive in the leading coefficients; the constant remainders are $O(a^4)$. Analyticity on the original chart gives uniform boundedness on any fixed ξ -ball for sufficiently small a . This proves the expansion before any trajectory estimate is used. $\square$

The rank assertions needed for the next step are additional finite tests on the same coefficient matrices. Let $\mathcal{H}$ be the three-dimensional constant-shift space and $\mathcal{Z}_g$ a two-dimensional weak-zero space. Choose a direct decomposition

$$\mathbb{R}^{29} = \text{Ran } B \oplus \mathcal{Z}_g \oplus \mathcal{H}, \quad \text{rank } B = 24, \quad (\text{H.4})$$

with p_g, p_H its two indicated projections, annihilating $\text{Ran } B$. Assume $p_H C_{\text{red}} = 0$ and $p_g C_{\text{red}}$ is onto $\mathcal{Z}_g$. In applications the decomposition and covector identifications must use the retained positive source pairing, not an unjustified orthogonal projection for an indefinite matrix.

Proposition H.2 (Rank and cokernel of the leading slow constraint). *Under [\(H.4\)](#) and the stated coupling tests,*

$$\text{rank } \Phi_0 = 26, \quad \dim \ker \Phi_0 = 297, \quad \text{coker } \Phi_0 \cong \mathcal{H}. \quad (\text{H.5})$$

Proof. The image of B contributes 24 directions. Modulo that image, C_{red} contributes exactly the two directions in $\mathcal{Z}_g$ and none in $\mathcal{H}$. Therefore $\text{rank}[C_{\text{red}} B] = 26$. Rank-nullity in the 323-dimensional domain gives 297, and the quotient of the 29-dimensional codomain is the remaining three-dimensional space. $\square$

Define $\mathbf{m}(\xi) = p_H \Phi_1(\xi)$ and $\mathbf{g}(\xi) = p_g C_{\text{red}}(b_s + \Sigma_0(\xi))$. Their ordered pair is

$$\mathfrak{C}_{\text{sl}}(\xi) = (p_g C_{\text{red}}[b_s + \Sigma_0(\xi)], p_H \Phi_1(\xi)) \in \mathbb{R}^2 \oplus \mathbb{R}^3. \quad (\text{H.6})$$

This is a finite polynomial map extracted from the unchanged action, not a replacement constraint added to its finite dynamics.

Theorem H.3 (Necessary slow compatibility and the bounded-rate readout). *Let $a_j \downarrow 0$ and suppose original exact histories exist on $0 \leq t \leq Ta_j$ in the indicated chart, with*

$$\sup_j \sup_{0 \leq \tau \leq T} (|\xi_{a_j}(\tau)| + |\partial_\tau \xi_{a_j}(\tau)|) < \infty. \quad (\text{H.7})$$

Every uniformly convergent subsequence has a Lipschitz limit ξ_0 with

$$\Phi_0(\xi_0) = 0, \quad (x_0)_\tau = b_s + \Sigma_0(\xi_0), \quad \mathfrak{C}_{\text{sl}}(\xi_0) = 0. \quad (\text{H.8})$$

The differential identity holds almost everywhere. The physical histories satisfy $X_t, \lambda_t, X_{tt} = O(a_j)$ and the corresponding metric and link curvature is $O(a_j)$ on their full intervals, subject to the readout conditions in [Assumption G.17](#).

Proof. The bound gives equicontinuity, hence uniform subsequential compactness. Multiply the weak equation by a^2 :

$$\Phi_0(\xi_a) + a\Phi_1(\xi_a) = a^2[z_{a,\tau} - b_w - \Phi_2(a, \xi_a)]. \quad (\text{H.9})$$

Its right-hand side is $O(a^2)$, so $\Phi_0(\xi_0) = 0$. The regular x equation passes in integral form to the second identity. Applying p_H to (H.9), using $p_H \Phi_0 = 0$ and dividing by a , gives $\mathbf{m}(\xi_a) = O(a)$ and hence $\mathbf{m}(\xi_0) = 0$. Differentiate $C_{\text{red}}x_0 + Bz_0 = 0$ almost everywhere and apply p_g . The B term vanishes, leaving $\mathbf{g}(\xi_0) = 0$ after substitution of the regular x drift. Continuity extends the polynomial identities to all times.

In the physical chart, $p_W = az$ and $t = a\tau$. The bounded slow rates and (G.95) imply $X_t, \lambda_t = O(a)$. The exact canonical equation with bounded derivatives then gives $X_{tt} = O(a)$. Apply the original readout estimate as in [Theorem G.21](#). These bounds rely on (H.7); the polynomial equalities alone do not provide them. $\square$

Proposition H.4 (First correction and initial slow tangency). *For $\xi_0 \in \ker \Phi_0$, the equation $\Phi_0\xi_1 + \Phi_1(\xi_0) = 0$ is solvable exactly when $\mathbf{m}(\xi_0) = 0$. If R_0 is a right inverse of Φ_0 on its range, one solution is $\xi_1 = -R_0\Phi_1(\xi_0)$, modulo $\ker \Phi_0$. Moreover, if a limiting path in [Theorem H.3](#) is C^1 at zero with $\xi_0(0) = 0$ and $\xi_0'(0) = b = (b_s, b_w)$, then*

$$D\mathfrak{C}_{\text{sl}}(0)[b] = (p_g C_{\text{red}} D\Sigma_0(0)[b], p_H D\Phi_1(0)[b]) = 0. \quad (\text{H.10})$$

Proof. The range of Φ_0 is the kernel of p_H by [Proposition H.2](#). Thus the first-correction equation is solvable if and only if $p_H \Phi_1(\xi_0) = 0$, and a right inverse gives the stated solution. Taylor expansion then gives $\Phi_0(\xi_0 + a\xi_1) + a\Phi_1(\xi_0 + a\xi_1) = O(a^2)$. For the final assertion differentiate $\mathfrak{C}_{\text{sl}}(\xi_0(\tau)) = 0$ at zero. The actual derivative of the limiting path is part of the hypothesis; convergence of derivatives at one initial point alone is not substituted for it. $\square$

The transport identity gives $b \in \ker \Phi_0$ and $\mathfrak{C}_{\text{sl}}(0) = 0$. It does not evaluate $D\mathfrak{C}_{\text{sl}}(0)[b]$. Along the particular affine ansatz $\xi = \tau b$, the three mean conditions reduce to $D\mathfrak{m}(0)[b] = 0$ and $D^2\mathfrak{m}(0)[b, b] = 0$, since $\mathfrak{m}$ is quadratic and vanishes at zero. These six scalar contractions test that ansatz; they are not imposed on every curved slow path. A vanishing first tangency vector likewise does not prove invariance of the nonlinear five-equation zero set.

H.2 Initial acceleration defect and first Lyapunov–Schmidt reduction

For the initial-cut audit it is useful to retain the regular and singular source terms separately. Rewrite (H.2) as

$$x_\tau = f(\xi) + as(a, \xi, u), \quad (H.11)$$

$$z_\tau = a^{-2}\Phi_0(\xi) + a^{-1}g(\xi) + h(a, \xi, u), \quad \xi = (x, z), \quad (H.12)$$

where

$$f = b_s + \Sigma_0, \quad g = \Phi_1, \quad h = b_w + \widehat{\Phi}_2, \quad (H.13)$$

and $s = \widehat{\Sigma}_1$ denotes the corresponding regular order- a source. The companion coordinate u contains all remaining finite initializer/source variables. Define

$$\mathfrak{c}_g(\xi) = p_g C_{\text{red}} f(\xi), \quad \mathfrak{c}_H(\xi) = p_H g(\xi). \quad (H.14)$$

The leading bounded-rate differential–algebraic system is therefore

$$\Phi_0(\xi) = 0, \quad x_\tau = f(\xi), \quad \mathfrak{c}_g(\xi) = 0, \quad \mathfrak{c}_H(\xi) = 0. \quad (H.15)$$

This is a necessary slow limit of exact histories; it is not imposed as an additional finite equation.

Write $s_0 = s(0, 0, u_0)$ and $h_0 = h(0, 0, u_0)$. Suppose an exact family, defined on an amplitude-dependent interval if necessary, satisfies

$$\xi_a(0) = a\kappa + O(a^2), \quad \partial_\tau \xi_a(0) = b + o(1), \quad b = (b_s, b_w), \quad u_a(0) \rightarrow u_0, \quad p_g C_{\text{red}} b_s = 0. \quad (H.16)$$

No uniform second-time-derivative bound is assumed. The regular equation gives

$$\partial_\tau x_a(0) = b_s + a\{Df(0)\kappa + s_0\} + o(a). \quad (H.17)$$

Define

$$\mathcal{L}_0 \kappa = \begin{pmatrix} p_g C_{\text{red}} Df(0)\kappa \\ p_H Dg(0)\kappa \end{pmatrix}, \quad (H.18)$$

$$f_0 = \begin{pmatrix} p_g \{C_{\text{red}} s_0 + Dg(0)b\} \\ p_H (h_0 - b_w) \end{pmatrix}. \quad (H.19)$$

Theorem H.5 (Initial acceleration-defect identity). *Under (H.16), $\kappa \in \mathcal{K}_0$ and*

$$(\mathcal{L}_0 \kappa + f_0)_H = 0. \quad (H.20)$$

If the exact histories and the composed source $h(a, \xi_a, u_a)$ are twice, respectively once, differentiable at the initial cut, then the following limit exists and

$$(\mathcal{L}_0 \kappa + f_0)_g = \lim_{a \downarrow 0} a p_g \left\{ z_{a, \tau\tau}(0) - \frac{d}{d\tau} h(a, \xi_a, u_a) \Big|_{\tau=0} \right\}. \quad (H.21)$$

Hence the complete five-row equation $\mathcal{L}_0 \kappa = -f_0$ holds if and only if the two-row acceleration defect in (H.21) vanishes.

Proof. Multiplying (H.12) by a at $\tau = 0$ gives

$$a\partial_\tau z_a(0) = \Phi_0\kappa + O(a) + g(\xi_a(0)) + ah(a, \xi_a(0), u_a(0)).$$

All terms except $\Phi_0\kappa$ tend to zero, so $\Phi_0\kappa = 0$. Applying p_H to (H.12) and using $p_H\Phi_0 = 0$ gives in the initial limit

$$p_H b_w = p_H Dg(0)\kappa + p_H h_0,$$

which is (H.20).

At positive amplitude the exact identity

$$\Phi_0\xi_a + ag(\xi_a) = a^2\{\partial_\tau z_a - h(a, \xi_a, u_a)\} \quad (\text{H.22})$$

holds. Differentiate it at $\tau = 0$ and apply p_g :

$$p_g C_{\text{red}} \partial_\tau x_a(0) + ap_g Dg(\xi_a(0)) \partial_\tau \xi_a(0) = a^2 p_g \left\{ z_{a,\tau\tau}(0) - \frac{dh}{d\tau} \Big|_0 \right\}. \quad (\text{H.23})$$

Subtract $p_g C_{\text{red}} b_s = 0$, divide by a , and use (H.17) and (H.16). The left-hand side converges to

$$p_g \{C_{\text{red}} Df(0)\kappa + C_{\text{red}} s_0 + Dg(0)b\},$$

which proves (H.21). $\square$

Corollary H.6 (Sufficient acceleration certificate). *If the projected difference in (H.21) is $o(a^{-1})$, then $\mathcal{L}_0\kappa = -f_0$. In particular, uniform bounds for $z_{a,\tau\tau}(0)$ and $\frac{d}{d\tau}h(a, \xi_a, u_a)|_0$ are sufficient.*

The qualification is not removable by analyticity of the initial record alone.

Proposition H.7 (Analytic preparation does not imply the two-row cancellation). *The autonomous polynomial Lagrangian*

$$L(\alpha, z, \lambda; \alpha', z', \lambda') = \frac{1}{2}\alpha^2(z')^2 - \frac{1}{2}(z - \alpha)^2 + \lambda\alpha' \quad (\text{H.24})$$

has exact initial records analytic in $a > 0$, with $\alpha(0) = a$ and bounded analytic initial velocities, for which the analogue of the upper homological row is nonzero. The exact solutions nevertheless have uniformly bounded first derivatives on every fixed finite interval.

Proof. The Euler equations are

$$\alpha' = 0, \quad (\alpha^2 z')' = \alpha - z, \quad \lambda' = \alpha(z')^2 + (z - \alpha).$$

Set $\alpha = a$, $x = a^2 z'$ and $x(0) = z(0) = \lambda(0) = 0$. Then

$$x' = -z + a, \quad z' = a^{-2}x, \quad x = a^2 \sin(\tau/a), \quad z = a\{1 - \cos(\tau/a)\}. \quad (\text{H.25})$$

The multiplier equation is solved by integration and has derivative bounded by $2a$. Thus all Euler equations hold and the first derivatives remain bounded, but $az''(0) = 1$. The right-hand side of (H.21) is therefore nonzero in the corresponding one-row reduction. The example uses one unchanged polynomial action for every a ; amplitude labels the initial record rather than changing the action. $\square$

The action itself supplies the required acceleration test. On the normalized multiplier slice let

$$\mathcal{B}(Q) + \mathcal{M}(Q)r(Q) = 0, \quad Q = (X, \lambda), \quad V = (F^{\text{can}}, r). \quad (\text{H.26})$$

Proposition H.8 (Original-action multiplier acceleration). *Where $\mathcal{M}$ is invertible on the normalized slice,*

$$\mathcal{M}\dot{r} = -D_Q\mathcal{B}[V] - D_Q\mathcal{M}[V]r. \quad (\text{H.27})$$

Thus the two-row quantity in (H.21) is computable from a directional derivative of the unchanged finite action together with the actual amplitude-graded decoder; it is not obtained by differentiating an independently prescribed target rate.

Proof. Differentiate (H.26) along the exact physical solution and solve for $\dot{r}$. $\square$

H.3 Finite-jet evaluation and preparation dependence of the acceleration defect

All coefficients in this subsection are evaluated at the initial cut, at fixed spatial regulator and in the fixed mean-lapse normalization. The symbol $[a^j]$ denotes a Taylor coefficient, including its factorial normalization. It does not denote a finite difference of sampled values. Write $r_a = \lambda_t(0)$ and set

$$\mathcal{S}_a = D_Q\mathcal{B}(Q_a)[V_a] + D_Q\mathcal{M}(Q_a)[V_a]r_a, \quad V_a = (F^{\text{can}}(Q_a), r_a). \quad (\text{H.28})$$

This expression uses the actual initialized rate and the derivative of the original action. A prescribed rate at one cut is not differentiated as though it were an identity along the history.

Proposition H.9 (Exact elimination of the active acceleration). *Suppose the normalized multiplier mass on the analytic initial family has*

$$\mathcal{M}_a = \begin{pmatrix} a^2 A(a) & a^3 L(a) \\ a^3 L(a)^\top & a^4 C(a) \end{pmatrix}, \quad D(a) = C(a) - L(a)^\top A(a)^{-1} L(a), \quad (\text{H.29})$$

where $A(0)$ and $D(0)$ are invertible. For $\mathcal{S}_a = (\mathcal{S}_A, \mathcal{S}_W)$ put

$$\chi(a) = \mathcal{S}_W - aL^\top A^{-1} \mathcal{S}_A, \quad v(a) = D(a)^{-1} \chi(a). \quad (\text{H.30})$$

Then the physical accelerations satisfy exactly

$$\lambda_{tt,W} = -a^{-4}v(a), \quad \lambda_{tt,A} = -a^{-2}A^{-1}\mathcal{S}_A + a^{-3}A^{-1}Lv(a). \quad (\text{H.31})$$

Proof. The acceleration equation is $\mathcal{M}_a \lambda_{tt} = -\mathcal{S}_a$. Its active row gives $\lambda_{tt,A} = -a^{-2}A^{-1}\mathcal{S}_A - aA^{-1}L\lambda_{tt,W}$. Substitution into the weak row gives $a^4 D\lambda_{tt,W} = -\chi$; back-substitution proves the other formula. $\square$

In the balanced chart $\lambda_W = \lambda_W^0(a) + a^2 z$ and $t = a\tau$, with chart centre independent of time, one has $z_{\tau\tau} = \lambda_{tt,W}$. This decoder identity is part of the coefficient test; a different time-dependent chart must retain its chain-rule terms. Define

$$\theta(a) = ap_g \frac{d}{d\tau} h(a, \xi_a, u_a) \Big|_0. \quad (\text{H.32})$$

Theorem H.10 (Coefficient-complete acceleration certificate). *Assume the analytic mass and source hypotheses of Proposition H.9. Write $D(a) = \sum_{j \geq 0} a^j D_j$, $\chi(a) = \sum_{j \geq 0} a^j \chi_j$. The four coefficients $v_0, \dots, v_3$ are obtained using only the same invertible matrix D_0 :*

$$D_0 v_j = \chi_j - \sum_{i=1}^j D_i v_{j-i}, \quad 0 \leq j \leq 3. \quad (\text{H.33})$$

If $\theta(a) = \sum_{k=-3}^0 a^k \theta_k + O(a)$, the two-row acceleration defect has a finite limit precisely when

$$p_g v_j + \theta_{j-3} = 0 \quad (0 \leq j < 3). \quad (\text{H.34})$$

In that case its value is

$$\Delta_g = -p_g v_3 - \theta_0. \quad (\text{H.35})$$

For a family satisfying [Theorem H.5](#), this value also equals the entirely initial-data expression

$$\Delta_g = p_g \{C_{\text{red}} Df(0) \kappa + C_{\text{red}} s_0 + Dg(0) b\}. \quad (\text{H.36})$$

Thus the action-side coefficient calculation and the initialized slow-source calculation must give the same two-vector. When $\chi_0 = \chi_1 = \chi_2 = 0$ and $\theta(a) = o(1)$, this simplifies to $\Delta_g = -p_g D_0^{-1} \chi_3$. Neither simplification is inferred merely from an $O(a)$ initial rate.

Proof. Compare coefficients in $Dv = \chi$. The exact projected expression is $-a^{-3} p_g v(a) - \theta(a)$, by [\(H.31\)](#). Its three possible negative powers vanish exactly under [\(H.34\)](#); its constant coefficient is [\(H.35\)](#). The initial-data formula is the acceleration-defect identity. If $\chi_j = 0$ for $j < 3$, the recursion gives $v_j = 0$ for $j < 3$ and $v_3 = D_0^{-1} \chi_3$. $\square$

If h is uniformly C^1 and both $\partial_\tau \xi_a(0)$ and $\partial_\tau u_a(0)$ are bounded, the chain rule gives $\theta(a) = O(a)$. If those companion-velocity bounds are not available, the corresponding θ_k must be included. Lower coefficients can vanish after projection without χ itself being divisible by a^3 . The recurrence, rather than a presumed leading order, covers this case. Inverses of $A(a)$ needed for D_j and χ_j are obtained similarly: for $U(a) = A(a)^{-1}$, $A_0 U_0 = I$ and $A_0 U_j = -\sum_{i=1}^j A_i U_{j-i}$. Thus only finitely many action jets and solves with A_0, D_0 enter the certificate.

Proposition H.11 (Dependence on preparation jets). *Consider two exact initialized families in the same slow-source chart. If they have the same $\kappa, b, s_0, Df(0), Dg(0), C_{\text{red}}, p_g$, then they have the same Δ_g . For a differentiable family of initial preparations with the last four maps fixed,*

$$D_d \Delta_g[\dot{d}] = p_g \{C_{\text{red}} Df(0) D_d \kappa[\dot{d}] + C_{\text{red}} D_d s_0[\dot{d}] + Dg(0) D_d b[\dot{d}]\}. \quad (\text{H.37})$$

Consequently a higher target jet preserving this full determining tuple cannot remove a nonzero defect. A nonzero derivative of the acceleration source alone is not a certificate for the derivative of the complete defect.

Proof. Apply [\(H.36\)](#) to the two families and then differentiate that formula. If the source maps themselves vary, their product-rule terms must also be retained. The formula concerns exact initialized records, so implicit changes of those records are included. $\square$

For example, the same polynomial Lagrangian in [Proposition H.7](#) admits the exact solutions

$$x = a^2 \sin(\tau/a) + a^4 d \cos(\tau/a), \quad z = a\{1 - \cos(\tau/a)\} + a^3 d \sin(\tau/a).$$

Their initial weak rate changes by $a^2 d$, while κ , the limiting first rate and the source coefficients are unchanged. Nevertheless $az''(0) = 1$ for every fixed d . This example tests the logical distinction; it does not assign a coefficient to the gravitational event.

H.4 Defect-aware Lyapunov–Schmidt reduction

It remains useful to retain the algebraic reduction for arbitrary forcing. Let $J = \{0, \dots, 25\} \setminus \{2, 24\}$ and $F = \{2, 24\}$. The certified principal block K_{JJ} is invertible, so

$$Z_g = \begin{pmatrix} -K_{JJ}^{-1}K_{JF} \\ I_2 \end{pmatrix} \quad (\text{H.38})$$

spans $\mathcal{Z}_g$. Let K_{Ag} be the active/gradient first-Poisson block and $A_{78} \succ 0$ the certified active mass, and define

$$G_A = K_{Ag}Z_g, \quad Q_g = G_A^\top A_{78}^{-1} G_A. \quad (\text{H.39})$$

The verified $LDL^\top$ pivots obey

$$d_1 > 1.2143 \times 10^7, \quad d_2 > 8.00 \times 10^5, \quad (\text{H.40})$$

so Q_g is positive definite. Therefore the induced upper two-row vertical map $\mathcal{A}_g : \mathcal{Z}_g \rightarrow \mathcal{Z}_g$ is an isomorphism; for $z \in \mathcal{Z}_g$,

$$\langle z, \mathcal{A}_g z \rangle = -\langle K_{Ag}z, A_{78}^{-1} K_{Ag}z \rangle. \quad (\text{H.41})$$

Choose a complement $\mathcal{X}_h$ of the two vertical directions in $\mathcal{K}_0$, with inclusion E_x , and write the complete block map as

$$\mathcal{L}_0(J_g z + E_x x) = \begin{pmatrix} \mathcal{A}_g z + \mathcal{B}_x x \\ \mathcal{C}_g z + \mathcal{D}_x x \end{pmatrix}, \quad \mathcal{H}_x = \mathcal{D}_x - \mathcal{C}_g \mathcal{A}_g^{-1} \mathcal{B}_x. \quad (\text{H.42})$$

No assumption that the upper block vanishes on an arbitrary complement is made.

Theorem H.12 (Lyapunov–Schmidt reduction of the initial gate). *For arbitrary $f = (f_g, f_H) \in \mathcal{Z}_g \oplus \mathcal{H}$, $\mathcal{L}_0 \kappa = -f$ is solvable if and only if*

$$f_H - \mathcal{C}_g \mathcal{A}_g^{-1} f_g \in \text{Ran } \mathcal{H}_x, \quad (\text{H.43})$$

and

$$\text{rank } \mathcal{L}_0 = 2 + \text{rank } \mathcal{H}_x. \quad (\text{H.44})$$

Thus full five-row surjectivity is equivalent to $\text{rank } \mathcal{H}_x = 3$. If the actual initialized coefficient is $\kappa_* = J_g z_* + E_x x_*$ and $\mathcal{L}_0 \kappa_* + f_0 = (\Delta_g, 0)$, then

$$\mathcal{H}_x x_* + f_{0,H} - \mathcal{C}_g \mathcal{A}_g^{-1} f_{0,g} = -\mathcal{C}_g \mathcal{A}_g^{-1} \Delta_g. \quad (\text{H.45})$$

There exists some coefficient $\tilde{\kappa}$ with $\mathcal{L}_0 \tilde{\kappa} = -f_0$ if and only if $\mathcal{C}_g \mathcal{A}_g^{-1} \Delta_g \in \text{Ran } \mathcal{H}_x$. This algebraic existence is distinct from the assertion that the given initializer has zero defect.

Proof. For arbitrary x , solve the upper rows by $z = -\mathcal{A}_g^{-1}(\mathcal{B}_x x + f_g)$. Substitution gives $\mathcal{H}_x x + f_H - \mathcal{C}_g \mathcal{A}_g^{-1} f_g = 0$ and the rank formula follows by invertible block operations. For the actual initializer, the upper equation instead gives $z_* = -\mathcal{A}_g^{-1}(\mathcal{B}_x x_* + f_{0,g}) + \mathcal{A}_g^{-1} \Delta_g$. Its original lower rows vanish, so substitution proves (H.45). Finally, $\mathcal{L}_0(\tilde{\kappa} - \kappa_*) = -(\Delta_g, 0)$ has a solution exactly when $\mathcal{C}_g \mathcal{A}_g^{-1} \Delta_g$ lies in $\text{Ran } \mathcal{H}_x$, by the same block elimination. $\square$

Proof of Theorems H.5 and H.12. The initial identity and its qualification are Theorem H.5 and Corollary H.6; (H.27) is Proposition H.8. Positivity of (H.39) gives invertibility of $\mathcal{A}_g$, and Theorem H.12 gives the finite block reduction. $\square$

H.5 Quadratic slow-slope gate

We next eliminate the two generalized-zero equations before differentiating in slow time. Let $J_g : \mathcal{Z}_g \hookrightarrow \mathbb{R}^{29}$ and $J_H : \mathcal{H} \hookrightarrow \mathbb{R}^{29}$ be the weak insertions. When composed with a map on $\mathcal{K}_0$, these symbols mean the insertions $(0, J_g \cdot)$ and $(0, J_H \cdot)$. On the certified prepared event the literal source identities are

$$\begin{aligned} \mathbf{c}_g(0) = \mathbf{c}_H(0) = 0, \quad Df(0)J_H = 0, \\ D\mathbf{c}_g(0)J_H = D\mathbf{c}_H(0)J_H = 0, \quad \mathcal{A}_g = D\mathbf{c}_g(0)J_g \in GL_2(\mathbb{R}). \end{aligned} \quad (\text{H.46})$$

The harmonic identities in (H.46) are proved below from the finite constant-shift Ward structure; they are pointwise identities at the prepared event and are not extrapolated to an arbitrary neighbouring rate.

Let

$$\mathcal{X} = \ker(p_g C_{\text{red}}) \subset \mathbb{R}^{294}, \quad \dim \mathcal{X} = 292, \quad (\text{H.47})$$

choose an isomorphism $X : \mathbb{R}^{292} \rightarrow \mathcal{X}$ and inverse L_X , and let $B^\#$ denote the inverse of B on $\text{Ran } B$, extended by zero on $\ker B$. Put

$$\mathbf{E}r = (Xr, -B^\# C_{\text{red}} Xr), \quad \Xi_0(r, z_g, y) = \mathbf{E}r + J_g z_g + J_H y. \quad (\text{H.48})$$

Lemma H.13 (Exact two-row elimination). *The map Ξ_0 is a linear isomorphism onto $\mathcal{K}_0$. There is a unique local analytic map $\gamma(r, y) \in \mathcal{Z}_g$, with $\gamma(0, 0) = 0$, such that*

$$\Xi(r, y) = \mathbf{E}r + J_g \gamma(r, y) + J_H y, \quad \mathbf{c}_g(\Xi(r, y)) = 0. \quad (\text{H.49})$$

Moreover $\gamma_y(0, 0) = 0$.

Proof. If $\Phi_0(x, z) = 0$, applying p_g gives $x \in \mathcal{X}$. Conversely, for $x \in \mathcal{X}$, $C_{\text{red}}x$ lies in $\text{Ran } B$ because both its p_g and p_H projections vanish. Every solution of $Bz = -C_{\text{red}}x$ is $-B^\# C_{\text{red}}x$ plus one unique element of $\ker B = \mathcal{Z}_g \oplus \mathcal{H}$, proving the linear statement. The derivative of $\mathbf{c}_g(\Xi_0)$ in z_g at zero is $\mathcal{A}_g$, so the analytic implicit-function theorem gives (H.49). Differentiating in y and using (H.46) gives $\gamma_y(0, 0) = 0$. $\square$

Define

$$\mathcal{F}(r, y) = L_X f(\Xi(r, y)), \quad \mathcal{H}(r, y) = \mathbf{c}_H(\Xi(r, y)). \quad (\text{H.50})$$

Theorem H.14 (Reduced three-constraint leading system). *A local absolutely continuous path in the coordinate neighbourhood solves (H.15) if and only if it is $\xi = \Xi(r, y)$ with*

$$r_\tau = \mathcal{F}(r, y), \quad \mathcal{H}(r, y) = 0. \quad (\text{H.51})$$

At zero,

$$\mathcal{H}(0, 0) = 0, \quad \mathcal{F}_y(0, 0) = \mathcal{H}_y(0, 0) = 0. \quad (\text{H.52})$$

Proof. The coordinate map enforces $\Phi_0 = 0$ and the implicit graph enforces $\mathbf{c}_g = 0$; the remaining algebraic equation is $\mathbf{c}_H = 0$. The regular equation has $x = Xr$, so it is equivalent to $r_\tau = L_X f$. The identities at zero follow from (H.46) and $\gamma_y(0, 0) = 0$. $\square$

Set

$$v = \mathcal{F}(0, 0), \quad u_0 = \frac{1}{2} \mathcal{F}_r(0, 0)v, \quad \rho_0 = \mathcal{H}_r(0, 0)v, \quad (\text{H.53})$$

and define

$$\mathcal{Q}(c) = \mathcal{H}_r \mathcal{F}_r v + \mathcal{H}_{rr}[v, v] + 2\mathcal{H}_{ry}[v, c] + \mathcal{H}_{yy}[c, c]. \quad (\text{H.54})$$

Proposition H.15 (Slope coefficients from the original compatibility rows). *Use the physical harmonic row convention of (H.50), so that $\mathcal{Q}_{\text{can}} = \mathcal{Q}$ in these coordinates. Define*

$$\widehat{E} = E - J_g \mathcal{A}_g^{-1} D\mathbf{c}_g(0)E, \quad b_* = \widehat{E}v, \quad b(c) = b_* + J_H c, \quad a_* = \widehat{E}L_X Df(0)b_*. \quad (\text{H.55})$$

Then the complete constant coefficient and the slope polynomial are

$$\begin{aligned} \mathcal{Q}(c) &= D\mathbf{c}_H(0)a_* + D^2\mathbf{c}_H(0)[b(c), b(c)] \\ &\quad - \mathcal{C}_g \mathcal{A}_g^{-1} D^2\mathbf{c}_g(0)[b(c), b(c)]. \end{aligned} \quad (\text{H.56})$$

In another fixed invertible harmonic row convention, the entire formula, including $q_{0,\text{can}}$ and $\mathcal{C}_g$, is transformed together.

Proof. Implicit differentiation of $\mathbf{c}_g(\Xi) = 0$ gives $\Xi_r(0) = \widehat{E}$, $\Xi_y(0) = J_H$, and $\mathcal{A}_g \gamma_{\alpha\beta} = -D^2\mathbf{c}_g(0)[\Xi_\alpha, \Xi_\beta]$. The chain rule for $\mathcal{H} = \mathbf{c}_H \circ \Xi$ therefore gives the last two terms. Since $\mathcal{F}_r = L_X Df(0)\widehat{E}$, the regular acceleration contribution is $D\mathbf{c}_H(0)a_*$. Insert these identities into (H.54). $\square$

Theorem H.16 (Necessary quadratic slope gate). *Let (r, y) be a Lipschitz solution of (H.51) on $[0, T]$ with $r(0) = y(0) = 0$. Then $\rho_0 = 0$ and*

$$r(\tau) = \tau v + \tau^2 u_0 + O(\tau^3), \quad \mathcal{Q}(y(\tau)/\tau) = O(\tau). \quad (\text{H.57})$$

Every cluster point of $y(\tau)/\tau$ is a real zero of $\mathcal{Q}$. If the admissible slope ball contains no real zero, no such Lipschitz trajectory exists.

Proof. Lipschitz continuity gives $r, y = O(\tau)$. The integral regular equation first yields $r = \tau v + O(\tau^2)$. Taylor expansion of $\mathcal{H}(r, y) = 0$, using $\mathcal{H}_y(0, 0) = 0$, gives $\rho_0 = 0$. Since $\mathcal{F}_y(0, 0) = 0$,

$$r_\tau = v + \mathcal{F}_r r + O(\tau^2) = v + \tau \mathcal{F}_r v + O(\tau^2),$$

so integration gives the displayed expansion for r . With $k(\tau) = y(\tau)/\tau$, the second-order Taylor expansion of $\mathcal{H}$ gives

$$0 = \frac{\tau^2}{2} \mathcal{Q}(k(\tau)) + O(\tau^3),$$

which proves the estimate and the cluster-point statement. $\square$

H.6 Source-specific rank-two factorization of the slope gate

We now use the constant-shift structure of the same finite Hamiltonian. For a multiplier test ν write

$$P_\nu = \partial_\lambda H_h[\nu], \quad \mathcal{B}_\nu = D_X P_\nu F^{\text{can}}, \quad \mathcal{M} = H_{\lambda\lambda}, \quad (\text{H.58})$$

and let $H_c : \mathbb{R}^3 \rightarrow \mathbb{R}^{108}$ insert the literal constant shifts. On the prepared first-field chart the finite coefficient identities used below are

$$F^{\text{can}}(0, e) = 0, \quad \mathcal{M}(X, \lambda) = O(\|X\|^2), \quad (\text{H.59})$$

For $c \in \text{Ran } H_c$, the coefficient branch used below retains

$$\begin{aligned} P_{c,1} &= 0, & P_{c,2}(U, p) &= \langle p, \delta_c U \rangle, \\ \mathcal{M}_2(X, e)c &= 0 \quad \text{for every } X, & [\delta_c, \delta_d] &= 0, & \mathbb{K}_1(Y)c &= 0. \end{aligned} \quad (\text{H.60})$$

Here subscripts denote homogeneous degree in X at $\lambda = e$ and $\mathcal{M}_2(X, e) = \frac{1}{2}D_X^2\mathcal{M}(0, e)[X, X]$. The quadratic mass-column identity is required on the coefficient space, not just at one value of X . It is an explicit finite-action identity of this branch; neither a weak-rank count nor $\mathcal{M} = O(\|X\|^2)$ alone implies it. The joint constraint initializer is analytic in the first field and first multiplier tilt and has

$$I(a, Y, \ell) = aY + a^2X_2(Y) + O(a^3), \quad (\text{H.61})$$

with X_2 independent of ℓ . Its harmonic drift has the exact graded form

$$H_c^\top \mathcal{B}(I(a, Y, \ell), e + a\ell) = a^3\{t(Y) + \mathcal{R}_Y\ell\} + a^4\mathcal{U}(a, Y, \ell), \quad \mathcal{R}_Y H_c = 0, \quad (\text{H.62})$$

with analytic $\mathcal{U}$. These are coefficient identities of the finite action on the declared chart, not continuum symmetry assumptions.

Lemma H.17 (Constant-shift derivative orders). *Along $X_a = aY + O(a^2)$ and $\lambda_a = e + O(a)$,*

$$\|D_X \mathcal{B}_c\| = O(a), \quad \|D_\lambda \mathcal{B}_c\| = O(a^2), \quad D_\lambda \mathcal{B}_c[H_c d] = O(a^3) \quad (\text{H.63})$$

for literal constant shifts c, d , uniformly on compact sets of the finite parameters.

Proof. The vanishing linear constant-shift constraint coefficient gives $D_X P_c = O(a)$, while $F^{\text{can}} = O(a)$ and their required derivatives are bounded. Differentiation of $\mathcal{B}_c$ gives the first estimate. The identity $\mathcal{M}_2(X, e)c = 0$ for every X , symmetry of the mass, and analyticity give $D_X \mathcal{M}(X_a, \lambda_a)[c, \nu] = O(a^2)$: the cubic remainder at e contributes $O(a^2)$, and changing λ by $O(a)$ changes the quadratic derivative by $O(a^2)$. For the multiplier derivative use

$$D_\lambda \mathcal{B}_c[\nu] = D_X \mathcal{M}[c, \nu]F^{\text{can}} + \{P_c, P_\nu\}.$$

The first term is $O(a^3)$. The linear Poisson coefficient vanishes by the first-Poisson constant-shift identity, giving $O(a^2)$. When $\nu = H_c d$, the quadratic bracket is the bracket of two commuting phase translations and vanishes, leaving $O(a^3)$. $\square$

Fix the actual initial rate $ar_*(a) = O(a)$ and put

$$\mathcal{F}_a(Q) = \mathcal{B}(Q) + \mathcal{M}(Q)ar_*(a). \quad (\text{H.64})$$

A pure harmonic second-record direction $J_H c$ is realized by an exact-constrained static tangent satisfying

$$\delta X = O(a^5), \quad \delta \lambda = a^2 H_c c + O(a^3). \quad (\text{H.65})$$

Let $q_4(\xi; J_H c)$ denote the a^4 coefficient of the literal constant-shift row of the directional residual at a fixed-first-field second-record base ξ .

Theorem H.18 (Base-uniform harmonic quartic annihilation). *For every sufficiently small second-record base with the prepared first field and first tilt fixed,*

$$q_4(\xi; J_H c) = 0 \quad (c \in \mathbb{R}^3). \quad (\text{H.66})$$

Consequently its derivative with respect to any admissible second-record base direction also vanishes.

Proof. At the initialized base, differentiate (H.62) with respect to the first multiplier tilt. The physical multiplier tangent is $a^2 H_c c$ and the associated canonical tangent starts at order a^4 ; since $\mathcal{R}_Y H_c = 0$, the a^4 constant-row coefficient vanishes. The higher canonical and multiplier parts of the tangent do not change that coefficient by Lemma H.17 and (H.59).

At a nearby second-record base the canonical contribution is $O(a)O(a^5) = O(a^6)$, the leading harmonic multiplier contribution is $O(a^3)O(a^2) = O(a^5)$, and every nonharmonic $O(a^3)$ multiplier correction contributes only $O(a^5)$. Finally $D_Q \mathcal{M}[\delta Q] ar_*(a) = O(a^5)$. Thus no a^4 coefficient is created. Differentiating the resulting identity with respect to the base gives the final assertion. $\square$

This coefficient audit also gives the harmonic identities in (H.46): the pure harmonic weak direction has no regular first derivative and no first slow-compatibility column at the prepared event.

Let

$$\mathfrak{g}_2(u, w) := D^2 \mathfrak{c}_g(0)[u, w] = p_g C_{\text{red}} D^2 \Sigma_0(0)[u, w] \in \mathcal{Z}_g, \quad (\text{H.67})$$

let b_* be the zero-harmonic initial tangent after the two-row vertical correction, and put

$$\mathcal{C} = \mathcal{C}_g \mathcal{A}_g^{-1}, \quad \mathcal{R}(c) = 2\mathfrak{g}_2(b_*, J_H c) + \mathfrak{g}_2(J_H c, J_H c). \quad (\text{H.68})$$

Let $\mathcal{Q}_{\text{can}}$ denote the physical three-row normalization of (H.54) and $q_{0,\text{can}} = \mathcal{Q}_{\text{can}}(0)$.

Theorem H.19 (Canonical rank-two factorization). *On the record-preserving native-lift branch,*

$$\mathcal{Q}_{\text{can}}(c) = q_{0,\text{can}} - \mathcal{C} \mathcal{R}(c). \quad (\text{H.69})$$

Consequently

$$D\mathcal{Q}_{\text{can}}(c) = -\mathcal{C} D\mathcal{R}(c), \quad \text{rank } D\mathcal{Q}_{\text{can}}(c) \leq \text{rank } \mathcal{C}_g \leq 2. \quad (\text{H.70})$$

Proof. Before applying the Ward identity, the linear and quadratic harmonic-slope coefficients split into a lower regular-row Hessian contribution and a quartic constant-row contribution. Theorem H.18 removes the latter whenever one argument is a pure harmonic weak direction. The surviving coefficients are

$$L_{\text{can}} e_i = -2\mathcal{C}_g \mathcal{A}_g^{-1} \mathfrak{g}_2(b_*, J_H e_i),$$

and

$$B_{\text{can}}[e_i, e_j] = -\mathcal{C}_g \mathcal{A}_g^{-1} \mathfrak{g}_2(J_H e_i, J_H e_j).$$

Reassembling the quadratic polynomial gives (H.69); differentiation gives (H.70). $\square$

Corollary H.20 (Cokernel obstruction). *For every $\lambda \in \ker \mathcal{C}_g^\top$,*

$$\lambda^\top \mathcal{Q}_{\text{can}}(c) = \lambda^\top q_{0,\text{can}} \quad \text{for every } c. \quad (\text{H.71})$$

Hence

$$q_{0,\text{can}} \notin \text{Ran } \mathcal{C}_g \implies \mathcal{Q}_{\text{can}}(c) \neq 0 \quad \text{for every } c. \quad (\text{H.72})$$

Proof. The image of $\mathcal{C}$ equals the image of $\mathcal{C}_g$. Pair (H.69) with a left-null covector. $\square$

Assume now

$$\text{rank } \mathcal{C}_g = 2, \quad q_{0,\text{can}} \in \text{Ran } \mathcal{C}_g. \quad (\text{H.73})$$

Then $\mathcal{C}$ is an isomorphism from $\mathcal{Z}_g$ onto its range; let $y_0 \in \mathcal{Z}_g$ be defined by $\mathcal{C} y_0 = q_{0,\text{can}}$.

Theorem H.21 (Two-equation physical slope reduction). *Under (H.73),*

$$\mathcal{Q}_{\text{can}}(c) = 0 \iff \mathcal{R}(c) = y_0. \quad (\text{H.74})$$

At every root where $D\mathcal{R}$ has rank two, the zero set is locally a one-dimensional real-analytic submanifold of $\mathbb{R}^3$.

Proof. Equation (H.69) becomes $\mathcal{Q}_{\text{can}} = \mathcal{C}(y_0 - \mathcal{R})$. Injectivity of $\mathcal{C}$ proves the equivalence, and the regular-level-set theorem gives the local dimension. $\square$

For a rigorous numerical certificate one should work on a transverse 2-plane, rather than invert the singular 3×3 slope Jacobian. Choose an affine splitting

$$c = c_* + Ey + sn, \quad E : \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad [E \ n] \in GL_3(\mathbb{R}), \quad (\text{H.75})$$

and put

$$F_s(y) = \mathcal{R}(c_* + Ey + sn) - y_0. \quad (\text{H.76})$$

Proposition H.22 (Two-dimensional contraction certificate). *Fix $s = s_0$, a centre $\bar{y} \in \mathbb{R}^2$, an invertible $M \in \mathbb{R}^{2 \times 2}$ and $r > 0$. Write*

$$F_{s_0}(\bar{y} + q) = F_{s_0}(\bar{y}) + Jq + \mathbb{B}[q, q] \quad (\text{H.77})$$

with symmetric bilinear $\mathbb{B}$, and set

$$\eta = \|MF_{s_0}(\bar{y})\|, \quad \theta = \|I - MJ\|, \quad \beta = 2\|M\mathbb{B}\|, \quad q_* = \theta + \beta r. \quad (\text{H.78})$$

If

$$q_* < 1, \quad \eta + q_*r < r, \quad (\text{H.79})$$

then F_{s_0} has a unique zero in the closed radius- r ball about $\bar{y}$, lying in its interior. If the resulting 2×2 derivative remains invertible on a product neighbourhood in (s, y) , these slice roots continue to a local analytic root curve.

Proof. The map $T(q) = q - MF_{s_0}(\bar{y} + q)$ has derivative norm at most $q_* < 1$ on the ball and satisfies $\|T(0)\| + q_*r < r$. Banach's fixed-point theorem gives the unique zero; the parameterized implicit-function theorem gives the final statement. $\square$

The only quadratic source data required after the range test are

$$\mathfrak{g}_2(b_*, J_H e_i), \quad 1 \leq i \leq 3, \quad \mathfrak{g}_2(J_H e_i, J_H e_j), \quad 1 \leq i \leq j \leq 3, \quad (\text{H.80})$$

namely eighteen scalar entries in the two-dimensional output. Since $\mathfrak{c}_g$ is quadratic on this source class, they can be obtained by exact polarization; for example

$$\mathfrak{g}_2(u, w) = \mathfrak{c}_g(u + w) - \mathfrak{c}_g(u) - \mathfrak{c}_g(w) + \mathfrak{c}_g(0) \quad (\text{H.81})$$

when the arguments are expressed in the normalized coefficient chart. No quartic constant-row packet is required for these harmonic slope directions.

H.7 Finite range certificates and a quantitative slope obstruction

The range condition can be tested before any quadratic slope coefficient is computed. All inner products in this subsection are fixed positive coefficient inner products, not the indefinite Cartan form. Choose an orthonormal basis of the three harmonic output rows and write $C_* := C_g \in \mathbb{R}^{3 \times 2}$ and $q_* := q_{0,\text{can}}$. These must be coefficients at the same prepared record, in the same row convention as the slope factorization.

Theorem H.23 (All-rank range test and sharp affine obstruction). *Put $e_* = (I - C_* C_*^\dagger)q_*$. Then*

$$q_* \in \text{Ran } C_* \iff \text{rank}(C_* \ q_*) = \text{rank } C_* \iff e_* = 0. \quad (\text{H.82})$$

For every harmonic slope c ,

$$\|\mathcal{Q}_{\text{can}}(c)\| \geq \|e_*\|. \quad (\text{H.83})$$

This is the exact distance from zero to the affine space $q_* - \text{Ran } C_*$; it need not be the minimum over the quadratic image of $\mathcal{R}$. If $C_* = (c_1 \ c_2)$ has rank two, set $n = c_1 \times c_2$. Then

$$\|e_*\| = \frac{|n^\top q_*|}{\|n\|} = \frac{|\det(c_1, c_2, q_*)|}{\|c_1 \times c_2\|}. \quad (\text{H.84})$$

For rank one, a nonzero column u gives membership exactly when $u \times q_* = 0$; for rank zero it is exactly $q_* = 0$.

Proof. $C_* C_*^\dagger$ is the orthogonal projection onto $\text{Ran } C_*$. Projection of the slope factorization onto its orthogonal complement therefore gives e_* , independently of c . Orthogonal projection is contractive and the nearest point in an affine subspace proves the distance assertion. In rank two its orthogonal complement is spanned by n ; the other cases are the corresponding line and zero-space tests. $\square$

This bound is unaffected by a change of the two-dimensional domain basis. Under a change of harmonic row coordinates the physical inner product must be transported as well. Membership is invariant without a norm choice; a numerical Euclidean distance is not invariant under an arbitrary unaccompanied row rescaling.

Certification with enclosing errors. Suppose $\|c_i - \hat{c}_i\| \leq \epsilon_i$ and $\|q_* - \hat{q}\| \leq \epsilon_q$. Let $\hat{n} = \hat{c}_1 \times \hat{c}_2$ and put

$$\begin{aligned} \epsilon_n &= \epsilon_1 \|\hat{c}_2\| + \|\hat{c}_1\| \epsilon_2 + \epsilon_1 \epsilon_2, \\ \epsilon_\Omega &= \epsilon_n \|\hat{q}\| + (\|\hat{n}\| + \epsilon_n) \epsilon_q. \end{aligned} \quad (\text{H.85})$$

The bilinear cross-product estimate and the triangle inequality give $\|n - \hat{n}\| \leq \epsilon_n$ and $|n^\top q_* - \hat{n}^\top \hat{q}| \leq \epsilon_\Omega$. In particular, if $|\hat{n}^\top \hat{q}| > \epsilon_\Omega$, rank two and nonmembership are both certified, and every slope satisfies

$$\|\mathcal{Q}_{\text{can}}(c)\| \geq \frac{|\hat{n}^\top \hat{q}| - \epsilon_\Omega}{\|\hat{n}\| + \epsilon_n} > 0. \quad (\text{H.86})$$

An interval containing zero proves neither membership nor rank loss. Exact membership instead requires an exact dependence identity, or a validated root argument when the prepared event is allowed to vary.

Evaluation before constructing a kernel basis. A lower-order coefficient realization may supply an invertible harmonic mass D_c and a map $R_c : \mathbb{R}^{26} \rightarrow \mathbb{R}^3$ with the exact incidence

$$C_* = -D_c^{-1} R_c Z_g, \quad d = D_c q_*, \quad (\text{H.87})$$

where Z_g is the kernel basis in (H.38). This incidence must be checked in the physical source normalization; it is not inferred from the rank of K_g . Membership is then exactly $d \in R_c(\ker K_g)$. The minus sign is immaterial for membership but is retained when solving for a coefficient.

For an even skew matrix K , use the convention $\text{Pf} \begin{pmatrix} 0 & k \\ -k & 0 \end{pmatrix} = k$ and define its polynomial Pfaffian adjugate $P(K)$ by $P(K)_{ij} = (-1)^{i+j} \text{Pf}(K_{\widehat{ij}})$ for $i < j$ with one-based indices, skew completion, and $\text{Pf}(0_{0 \times 0}) = 1$. Then $KP(K) = \text{Pf}(K)I$. Consider

$$\mathbb{B}(K, R, d) = \begin{pmatrix} K & R^\top & 0 \\ -R & 0_3 & d \\ 0 & -d^\top & 0 \end{pmatrix}. \quad (\text{H.88})$$

Proposition H.24 (One bordered Pfaffian for the range obstruction). *Let $K \in \mathbb{R}^{2m \times 2m}$ have rank $2m - 2$ and set $S = RP(K)R^\top$, $n_P = (S_{23}, -S_{13}, S_{12})^\top$. Then*

$$\text{Pf } \mathbb{B}(K, R, d) = n_P^\top d. \quad (\text{H.89})$$

One has $n_P \neq 0$ exactly when $\text{rank}(R|_{\ker K}) = 2$; on that branch the displayed Pfaffian vanishes exactly when $d \in R(\ker K)$. At membership the border has rank $2m + 2$, and away from membership it is invertible. Thus for $K = K_g$ and (H.87) this is a 30×30 test of the same range obstruction, not a different slow-time compatibility condition.

Proof. The image of $P(K)$ is contained in $\ker K$. A nonzero principal $(2m - 2)$ -Pfaffian exists, so $P(K)$ is nonzero and has rank two. For any ordered kernel basis (z_1, z_2) it is therefore a nonzero scalar multiple of $z_1 z_2^\top - z_2 z_1^\top$. Its pushforward is nonzero exactly when Rz_1, Rz_2 are independent, and n_P is then a nonzero multiple of their cross product. Expanding the border Pfaffian in its final index, and then in the two remaining added indices, gives $d_1 S_{23} - d_2 S_{13} + d_3 S_{12}$ with the stated convention.

To check the rank at membership, split off an invertible skew $(2m - 2)$ -block of K by congruence. The remaining six-dimensional skew matrix couples the two kernel coordinates to the three harmonic coordinates by a rank-two map, so it has rank at least four. Membership makes its Pfaffian zero, hence its even rank is at most four. Restoring the invertible block gives rank $2m + 2$. Away from membership the full Pfaffian is nonzero. $\square$

The smaller pivot implementation is sufficient: solve $K_{JJ}Z_J = -K_{JF}$, set $Z_g = \begin{pmatrix} Z_J \\ I_2 \end{pmatrix}$, and test the triple product of the two columns of $R_c Z_g$ with d . Its zero set agrees with that of the full border. Exact skew elimination or a rigorous interval implementation may be used; floating-point Pfaffian algorithms alone do not certify a zero [72].

The quadratic calculation after membership. On the rank-two membership branch, put $\varphi(c) = \mathbf{c}_g(b_* + J_H c)$. Provided the displayed arguments lie in the coefficient chart, the quadratic source identities imply

$$\mathcal{R}(c) = 2\{\varphi(c) - \varphi(0)\}. \quad (\text{H.90})$$

Indeed $D\mathbf{c}_g(0)J_H = 0$ and the Hessian of $\mathbf{c}_g$ is $\mathbf{g}_2$. This identity does not require a real slope root. For a nonzero sufficiently small δ , the ten values at 0 , $\pm\delta e_i$, and $\delta(e_i + e_j)$ for $i < j$ determine the entire quadratic φ : if $\varphi(c) = \varphi(0) + \sum_i \ell_i c_i + \sum_i b_{ii} c_i^2 + 2 \sum_{i < j} b_{ij} c_i c_j$, then

$$\begin{aligned}\ell_i &= \frac{\varphi(\delta e_i) - \varphi(-\delta e_i)}{2\delta}, \\ b_{ii} &= \frac{\varphi(\delta e_i) + \varphi(-\delta e_i) - 2\varphi(0)}{2\delta^2}, \\ b_{ij} &= \frac{\varphi(\delta(e_i + e_j)) - \varphi(\delta e_i) - \varphi(\delta e_j) + \varphi(0)}{2\delta^2}.\end{aligned}\tag{H.91}$$

These are exact polynomial identities, not asymptotic differencing formulas. The root test remains [Proposition H.22](#). A root curve of this necessary polynomial condition is not, by itself, a trajectory of the nonlinear slow differential–algebraic system.

Remark H.25 (Scope of a finite coefficient verdict). The acceleration recurrence and the range tests use actual coefficient data at one identified event. The rank and spectral inequalities in [Section G.9](#) do not specify the entries of χ_j , the differentiated-source terms, C_* or q_* . No value or sign of either event-specific test is asserted here without those entries and their exact or enclosing arithmetic. In particular, a verified algebraic identity for arbitrary input matrices is distinct from a verified numerical outcome for the gravitational event. Neither test alone supplies positive-time propagation or spatial-refinement uniformity.

H.8 Scope of exact-action realization and the remaining propagation problem

There are now four distinct exact-action statements. The original finite response identities retain their all-cutoff scope wherever their own rank hypotheses hold. The boundary realization on the amplitude-graded chart yields actual constrained short histories, with readout accuracy measured against that same history. The exact initialized family pays the three mean rows at the first finite-amplitude cut, while the remaining two rows are governed by the acceleration certificate above; their feedback is retained in the reduced mean rows. The coefficient-complete acceleration recurrence and all-rank range test are finite evaluation procedures, while arbitrary forcing admits the defect-aware Lyapunov–Schmidt reduction. Bounded-rate positive-time limits additionally obey the quadratic slope gate and its source-specific rank-two factorization. The normal factorizations and slow expansion identify separate obstructions to forward residence and longer bounded-rate evolution. Their fixed-cutoff hypotheses cannot be interchanged with the cofinal hypotheses of the Einstein handoff.

In particular, the $a^{-1}\Phi_1(\xi)$ contribution in slow time must be retained in the principal compatibility problem. A frozen spectrum cannot remove it: the elementary system $x_\tau = 1$, $h_\tau = a^{-1}x^2$ has no growing linear mean term at the origin but gives $h_\tau = \tau^2/a$ from zero initial data. This example establishes only the logical need for the nonlinear mean test, not a value or sign for a coefficient of the gravitational action.

A direct exact-action Einstein limit would require a common family of full finite coefficient certificates, a regular solution and propagation of the nonlinear slow compatibility system, differentiated source and readout control, and a noncollapsed physical interval independent of spatial cutoff. A physical forward selection mechanism must additionally account for the rapid and slow precision requirements without using future endpoint data. None of these conclusions follows from homogeneous blocking, a successful static determinant test, or an improved numerical quadrature alone. Conversely, local residence failure is not a no-go theorem for other exact histories, other gauges with controlled comparisons, or singular restoration of continuum gauge symmetry.

I Homogeneous and Gowdy sectors; explicit geometric regulators

This appendix proves the sector-specific statements of Section 7. The homogeneous part retains independent lapse variation and the complete trial normalizer; the Gowdy part retains its smooth-slab and finite-offset assumptions. The final part collects the flat and de Sitter spatial and exact-link estimates. These use the earlier ADM map and classified action but have their own evolution and access conditions.

I.1 Homogeneous dynamics and action matching

This appendix proves the homogeneous dynamical statements of Section 7. Arbitrary positive homogeneous renewal variables are first varied with the full lapse constraint retained; the flat/de Sitter profiles are solved only after the stationary equations have been obtained. A second layer constructs a finite positive trial law whose measured probabilities reconstruct the corresponding finite action off shell.

I.1.1 Homogeneous ADM map and reduced vacuum action

On the root-symmetric branch $k_{\alpha,h} = \kappa/(8h^2)$ and $m_h = \varrho h^3$. Since $\sum_{\alpha \in \Phi_{A_3}} \alpha \alpha^\top = 8I$, one has $\mathcal{B} = \kappa I$ and $\beta = 0$. Substitution in (3.8) gives (7.2).

For completeness, let

$$ds^2 = -N(t)^2 dt^2 + a(t)^2 \delta_{ij} dx^i dx^j.$$

The nonzero Christoffel symbols are

$$\Gamma_{00}^0 = \dot{N}/N, \quad \Gamma_{ij}^0 = a\dot{a} \delta_{ij}/N^2, \quad \Gamma_{0j}^i = \dot{a} \delta_j^i/a.$$

Direct contraction gives

$$R_{00} = -3 \left(\ddot{a}/a - \dot{a}\dot{N}/(aN) \right), \quad (\text{I.1})$$

$$R_{ij} = \left(a\ddot{a}/N^2 + 2\dot{a}^2/N^2 - a\dot{a}\dot{N}/N^3 \right) \delta_{ij}, \quad (\text{I.2})$$

$$R = 6 \left(\ddot{a}/(aN^2) + \dot{a}^2/(a^2N^2) - \dot{a}\dot{N}/(aN^3) \right). \quad (\text{I.3})$$

Thus

$$Na^3 R = 6 \frac{d}{dt} \left(\frac{a^2 \dot{a}}{N} \right) - \frac{6a\dot{a}^2}{N}. \quad (\text{I.4})$$

With the orientation/sign convention used here, the bare torsion-free Einstein–Hilbert representative on $[t_0, t_1] \times \mathbb{T}^3$ is therefore

$$\frac{S_{\text{EH}}}{\chi V_0} = 6 \left[\frac{a^2 \dot{a}}{N} \right]_{t_0}^{t_1} + \int_{t_0}^{t_1} \left[-\frac{6a\dot{a}^2}{N} - 2\Lambda Na^3 \right] dt. \quad (\text{I.5})$$

The Gibbons–Hawking–York Dirichlet boundary term [61, 62] contributes

$$\frac{S_{\text{GHY}}}{\chi V_0} = -6 \left[\frac{a^2 \dot{a}}{N} \right]_{t_0}^{t_1}, \quad (\text{I.6})$$

so $S_{\text{EH}} + S_{\text{GHY}} = \chi V_0 S[a, N]$ with $S[a, N]$ exactly (7.3). The corresponding Dirichlet problem fixes the induced spatial metric at the two endpoint slices, $\delta a(t_0) = \delta a(t_1) = 0$, while the lapse remains

independently variable in the interior. The finite action (7.9) is a midpoint variational discretization of this already boundary-completed first-order functional; hence fixing q_0, q_M is precisely the reduced discrete counterpart of the same induced-metric boundary condition. Substituting $a = \varrho^{1/3}$ and $N = \varrho^{1/3}\sqrt{\kappa}$ gives (7.4).

Detailed proof of Theorem 7.1. Varying the lapse and scale factor independently gives

$$E_N = 2a^3(3\mathcal{H}^2 - \Lambda), \quad E_a = 6Na^2(2D_\tau\mathcal{H} + 3\mathcal{H}^2 - \Lambda). \quad (\text{I.7})$$

Because $a = e^{v/3}$ and $N = e^{u/2+v/3}$,

$$E_u = \frac{N}{2}E_N, \quad E_v = \frac{a}{3}E_a + \frac{N}{3}E_N,$$

which proves (7.5). The Jacobian of $(u, v) \mapsto (\log a, \log N)$ is $-1/6 \neq 0$, so this change of variables loses no variation. The reconstructed Einstein residual is

$$G_{00} + \Lambda g_{00} = N^2(3\mathcal{H}^2 - \Lambda), \quad G_{ij} + \Lambda g_{ij} = -a^2(2D_\tau\mathcal{H} + 3\mathcal{H}^2 - \Lambda)\delta_{ij}, \quad (\text{I.8})$$

with $G_{0i} = 0$. Hence the two renewal Euler equations are equivalent to every Einstein component within this ansatz. $\square$

Proof of Corollary 7.2. For $\Lambda > 0$, the lapse constraint gives $\mathcal{H}^2 = H_0^2$ with $H_0 = \sqrt{\Lambda/3}$. Continuity on a connected interval fixes one sign σ , while $D_\tau\mathcal{H} = 0$ makes it constant. Integrating $D_\tau \log a = \sigma H_0$ gives (7.7); the renewal variables then follow from (7.2). The cases $\Lambda = 0$ and $\Lambda < 0$ follow from the same squared constraint. $\square$

Remark I.1 (Why the lapse is varied before gauge fixing). If $q = a^{3/2}$ and one sets $N = 1$ before variation, the remaining equation is $q'' = \omega^2 q$ with $\omega = \sqrt{3\Lambda/4}$. Its general solution $q = Ae^{\omega t} + Be^{-\omega t}$ satisfies the omitted lapse constraint $(q')^2 = \omega^2 q^2$ only when $AB = 0$. Thus a positive cosh-type solution is a spurious critical point of the prematurely gauge-fixed action.

I.1.2 Finite lapse-varied action and stationary recurrence

This is the midpoint discretization of the already Gibbons–Hawking–York-completed reduced functional (7.3), not of the bare second-derivative Einstein–Hilbert expression. The Dirichlet endpoint data are therefore q_0, q_M fixed (equivalently the induced endpoint three-metrics are fixed), while every interior q_j and every lapse-weighted interval s_j are varied independently. Consequently

$$\delta S_d = \sum_{j=1}^{M-1} E_{q_j} \delta q_j + \sum_{j=0}^{M-1} E_{s_j} \delta s_j, \quad \delta q_0 = \delta q_M = 0, \quad (\text{I.9})$$

with no omitted endpoint acceleration term. If one discretized the uncompleted Einstein–Hilbert expression instead, a matching discrete gravitational boundary term would have to be retained explicitly. Through

$$\varrho_j = q_j^2, \quad N_{j+1/2} = s_j / \Delta t_j, \quad \kappa_{j+1/2} = \frac{(s_j / \Delta t_j)^2}{q_j^{A/3}}, \quad (\text{I.10})$$

this is a smooth finite-dimensional action in positive renewal coordinates; no exponential history appears in its definition.

Put $q_j > 0$, $s_j > 0$, $\bar{q}_j = (q_j + q_{j+1})/2$, and define (7.9). Because this discretizes the boundary-completed first-order action, the discrete Dirichlet problem fixes q_0, q_M and varies $(q_1, \dots, q_{M-1}, s_0, \dots, s_{M-1})$ independently. A direct summation-by-parts calculation gives exactly (I.9); the endpoint momenta multiply δq_0 and δq_M and therefore vanish under the fixed induced endpoint geometry. No condition is imposed on $\sum_j s_j$, since that would modify the lapse variation and hence the Hamiltonian constraint. The change of variables (I.10), together with $q_j = \sqrt{\bar{\varrho}_j}$, is a smooth bijection on the positive finite domain. Hence critical points in (q, s) and in the corresponding positive renewal coordinates are equivalent.

Proof of the Euler equations and stationary classification in Theorem 7.3. Differentiating an edge term with respect to s_j gives

$$A_0 \frac{(q_{j+1} - q_j)^2}{s_j^2} - B_0 \bar{q}_j^2,$$

which is (7.10). At an interior node the right derivative of the preceding edge and left derivative of the following edge give (7.11).

For $\Lambda > 0$, the lapse equation says $v_j = \sigma_j \omega \bar{q}_j$ with $\sigma_j \in \{\pm 1\}$. Solving for the terminal amplitude gives

$$q_{j+1} = q_j \frac{1 + \sigma_j \omega s_j / 2}{1 - \sigma_j \omega s_j / 2}.$$

Positivity requires $\omega s_j / 2 < 1$. Substitution in the interior momentum matching condition shows $2A_0 \omega q_j (\sigma_j - \sigma_{j-1}) = 0$, so every positive stationary path has one common sign. Conversely a common-sign recurrence satisfies every finite Euler equation. For $\Lambda = 0$ the lapse equation forces constant q ; for $\Lambda < 0$ it has no positive real solution.

To prove (7.13), set $x_j = \omega s_j / 2$. For $0 \leq x \leq b$,

$$0 \leq 2 \operatorname{arctanh} x - 2x = 2 \int_0^x \frac{t^2}{1 - t^2} dt \leq \frac{2x^3}{3(1 - b^2)}.$$

Taking logarithms of the recurrence and summing gives

$$|\log(q_j/q_0) - \sigma \omega \tau_j| \leq \frac{\omega^3}{12(1 - b^2)} \sum_i s_i^3 \leq \frac{\omega^3 T}{12(1 - b^2)} h_t^2.$$

The power relations between q, a, ϱ, κ then give the stated convergence of renewal variables. $\square$

For later residual estimates it is useful to define

$$R_j^\sigma = q_{j+1} - q_j - \sigma \omega s_j \bar{q}_j. \quad (\text{I.11})$$

A direct expansion gives the exact factorization

$$S_d = -A_0 \sum_j \frac{(R_j^\sigma)^2}{s_j} - A_0 \sigma \omega (q_M^2 - q_0^2). \quad (\text{I.12})$$

Thus at fixed endpoints the finite stationary branch is exactly the zero set of the nonnegative residual energy $\mathcal{E} = A_0 \sum_j (R_j^\sigma)^2 / s_j$.

Lemma I.2 (Full finite variation controlled by the residual energy). *For endpoint-vanishing nodal variations w_j and lapse variations $\delta s_j = s_j n_j$, put*

$$\mathcal{B}_s(w, n; q) = \sum_j s_j \left(\frac{w_{j+1} - w_j}{s_j} - \sigma \omega \bar{w}_j - \sigma \omega \bar{q}_j n_j \right)^2. \quad (\text{I.13})$$

Then

$$|\delta S_d[w, sn]| \leq 2\sqrt{A_0 \mathcal{E}} \sqrt{\mathcal{B}_s(w, n; q)} + \mathcal{E} \|n\|_\infty. \quad (\text{I.14})$$

If q_j^* is the exact common-sign stationary recurrence with the same q_0, s_j , and $\omega \max s_j/2 \leq b < 1$, then on a fixed positive amplitude range

$$\max_j |q_j - q_j^*| \leq C\sqrt{\mathcal{E}}. \quad (\text{I.15})$$

Proof. Differentiate (I.12); weighted Cauchy–Schwarz gives (I.14). For the path estimate, write the forced recurrence as $q_{j+1} = r_j q_j + R_j/d_j$ with $d_j = 1 - \sigma \omega s_j/2$ and $r_j = (1 + \sigma \omega s_j/2)/d_j$. Iteration, $d_j \geq 1 - b$, and $\sum_j |R_j| \leq \sqrt{T \sum_j R_j^2/s_j}$ give the result. $\square$

I.1.3 Measured quadratic costs and exact off-shell action matching

We now prove that the finite action can be read from an actual normalized trial law rather than appended as an unrelated deterministic writer. Let $E_s(x, y)$ be (7.14).

Proposition I.3 (Three-cost identification and rank defect). *Fix $s > 0$ and distinct $u, v > 0$. Within the declared quadratic class, $E_s(u, u)$, $E_s(u, v)$, and $E_s(v, u)$ determine*

$$B = \frac{E_s(u, u)}{su^2}, \quad (\text{I.16})$$

$$C = \frac{E_s(u, v) - E_s(v, u)}{2(v^2 - u^2)}, \quad (\text{I.17})$$

$$A = \frac{s}{2(v - u)^2} \left[E_s(u, v) + E_s(v, u) - 2Bs \left(\frac{u + v}{2} \right)^2 \right]. \quad (\text{I.18})$$

Moreover, with $\mathfrak{D} = AB - C^2$, $c = -C/A$, and $\nu = \mathfrak{D}/A$,

$$E_s(x, y) = \frac{A}{s} \left[y - x - cs \frac{x + y}{2} \right]^2 + \nu s \left(\frac{x + y}{2} \right)^2. \quad (\text{I.19})$$

Thus $\mathfrak{D} = 0$ is exactly the rank-one compatibility condition.

Proof. The diagonal measurement determines B . Reversal exchanges x and y , leaving the first two terms invariant and changing the sign of the C term, which determines C ; the reversal sum then determines A . Completing the square gives (I.19). $\square$

Proof of Theorem 7.4. From (7.17), $Z_s Q_s = p_s a_s$. Taking logarithms and using $a_s = e^{-E_s/\varepsilon}$ proves (7.18). Summing over one path gives

$$\varepsilon \sum_j [\log(Q_j/p_j) + \log Z_j] = - \sum_j E_{s_j}(q_j, q_{j+1}).$$

The C term telescopes to $C(q_M^2 - q_0^2)$, which is exactly cancelled by the endpoint term in (7.19); this proves the off-shell identity (7.20). Differentiating that identity on the smooth positive parametric domain proves equality of all interior and lapse first variations.

For the mismatch statement, minimizing $y \mapsto E_s(x, y)$ gives, for small s , a ratio with logarithmic drift $c = -C/A$, so the limiting scale factor has Hubble rate $2c/3$. Hence its geometric cosmological parameter is $\Lambda_{\text{geom}} = 4C^2/(3A^2)$. Dividing the bulk action by A/A_0 shows that its parameter is $\Lambda_{\text{act}} = 4B/(3A)$. Therefore

$$G[g] + \Lambda_{\text{act}}g = (\Lambda_{\text{act}} - \Lambda_{\text{geom}})g = \frac{4(AB - C^2)}{3A^2}g,$$

which is (7.21). $\square$

Remark I.4 (Why the acceptance normalizer is physical). Retaining only p_s and Q_s would not determine the action: replacing $E_s(x, y)$ by $E_s(x, y) + c_s(x)$ leaves Q_s unchanged but multiplies $Z_s(x)$ by $e^{-c_s(x)/\varepsilon}$. Such a source-dependent term can change the lapse variation. The total acceptance frequency is therefore a load-bearing part of the complete physical trial table.

I.1.4 An explicit finite stochastic cylinder

We give the construction underlying Theorem 7.5. Set $s = T/M$ and, for large M ,

$$d = 1 - \sigma\omega s/2, \quad r = \frac{1 + \sigma\omega s/2}{1 - \sigma\omega s/2}, \quad \delta_M = M^{-3}, \quad \varepsilon_M = \frac{M^{-5}}{1 + \log M}. \quad (\text{I.20})$$

Starting from $F_0 = \{q_0\}$, define support intervals recursively by

$$L_0 = q_0/2, \quad U_0 = 3q_0/2, \quad L_{j+1} = rL_j - \delta_M, \quad U_{j+1} = rU_j + \delta_M,$$

and let F_j be finite grids of mesh at most δ_M in these intervals. Each source proposes uniformly from F_{j+1} and accepts with

$$a_j(x, y) = \exp \left[-\frac{A_0 d^2 (y - rx)^2}{s\varepsilon_M} \right]. \quad (\text{I.21})$$

Because $d(y - rx) = y - x - \sigma\omega s(x + y)/2$, this is exactly the rank-one cost (7.22).

Lemma I.5 (History-level entropy–energy estimate). *At stage j , let a finite reference row p_j and nonnegative cost E_j define $Q_j = p_j e^{-E_j/\varepsilon}/Z_j$. Let $\mathbb{Q}$ be the product of these accepted rows and $\mathbb{P} \ll \mathbb{Q}$ any complete path law with the same initial value, possibly with memory. If each relevant row contains a proposal of cost at most e_j and every proposal weight is at least $p_{*,j}$, then*

$$\mathbb{E}_{\mathbb{P}} \sum_j E_j \leq \sum_j e_j + \varepsilon \left[\sum_j \log(1/p_{*,j}) + D_{\text{KL}}(\mathbb{P} \parallel \mathbb{Q}) \right]. \quad (\text{I.22})$$

Proof. For a fixed history prefix, direct substitution gives

$$\mathbb{E}_{\mathbb{P}} E_j = \varepsilon D_{\text{KL}}(P_j \parallel Q_j) - \varepsilon D_{\text{KL}}(P_j \parallel p_j) - \varepsilon \log Z_j.$$

The candidate bound gives $Z_j \geq p_{*,j} e^{-e_j/\varepsilon}$. Integrating over prefixes, summing, and using the chain rule for path relative entropy proves (I.22). $\square$

Proof of Theorem 7.5. The rational stationary multiplier and its inverse are uniformly bounded on $0 \leq j \leq M$ when $\omega s/2 \leq b < 1$. Iterating the support recursion shows that all F_j lie in one fixed interval $0 < q_- \leq q \leq q_+ < \infty$ and have cardinality $O(M^3)$. For every source x , the exact target rx lies inside the next support and has a grid point within δ_M . The corresponding cost is at most $C\delta_M^2/s = CM^{-5}$. Hence the proposal floor is $p_* \geq cM^{-3}$ and the total acceptance probability has a positive explicit lower bound.

Apply Lemma I.5 to the ideal accepted product law $\mathbb{Q}_M$ itself. With the scales (I.20),

$$\mathbb{E}_{\mathbb{Q}_M} \mathcal{E}_M \leq CM^{-4}. \quad (\text{I.23})$$

The deterministic estimates (I.14) and (I.15), Jensen's inequality, and the second-order stationary-profile estimate then give the two bounds in (7.23) and an $O(M^{-2})$ expected first variation against every fixed smooth scale/lapse test.

To make every cutoff history genuinely finite, permit at most

$$R_M = \lceil 7\underline{Z}_M^{-1} \log M \rceil$$

independent proposals at each stage, where $\underline{Z}_M$ is the explicit row-wise acceptance lower bound. A union bound gives total failure probability at most M^{-6} . Conditional on complete success, the path-law correction has relative entropy at most $-\log(1 - M^{-6})$, which changes (I.23) only by a summable term. The expected residual energies are summable in M , so Tonelli and the first Borel–Cantelli lemma give $\mathcal{E}_M \rightarrow 0$ almost surely under any common coupling. The deterministic path estimates then imply uniform and strong H^1 convergence to $q_0 e^{\sigma\omega\tau}$.

Finally, (7.24) is the positive ADM inverse on the proper-time nodal branch. The A_3 root identity gives exactly $\mathcal{B}_j = \kappa_j I$, and ordinary Taylor expansion of the symmetric root generator yields the spatial continuum coefficient $\frac{1}{2}q_*^{-4/3}\Delta$. Together with the homogeneous Einstein identity, this proves the vacuum limit with $\Lambda = 4\omega^2/3$. $\square$

Remark I.6 (Exact scope of the operational construction). The quadratic duration law, the rank-one condition $\mathfrak{D} = 0$, the nonzero action normalization, and the value of Λ are not deduced from the bare predictive quotient. They are finite operational conditions that can be measured and can fail. The construction proves that this matched class is nonempty and robust under vanishing measured action/likelihood errors; it does not prove generic microscopic attraction to that class.

I.1.5 Exact conservative blocking and the relative defect

Put $u = A/s + Bs/4$, $v = A/s - Bs/4$, $\mu = \sqrt{AB}$ and $\theta = 2 \operatorname{artanh}((s/2)\sqrt{B/A})$. Then $u^2 - v^2 = \mu^2$, $u = \mu \coth \theta$ and $v = \mu \operatorname{csch} \theta$, giving (7.26).

Lemma I.7 (Two-edge Schur identity). *Let $E_i(x, y) = p_i x^2 - 2q_i xy + r_i y^2$ with $p_i, q_i, r_i > 0$ and $d_i = p_i r_i - q_i^2 \geq 0$. Then the minimizer of $E_1(x, z) + E_2(z, y)$ is positive on positive endpoints and the blocked quadratic has*

$$p' = p_1 - \frac{q_1^2}{r_1 + p_2}, \quad q' = \frac{q_1 q_2}{r_1 + p_2}, \quad r' = r_2 - \frac{q_2^2}{r_1 + p_2}, \quad (\text{I.24})$$

with

$$p'r' - (q')^2 = \frac{r_2 d_1 + p_1 d_2}{r_1 + p_2}. \quad (\text{I.25})$$

Proof. Complete the square in z . The minimizer is $z_* = (q_1 x + q_2 y)/(r_1 + p_2) > 0$. The three coefficients in (I.24) are the remaining quadratic coefficients; direct expansion gives (I.25). $\square$

Proof of Theorem 7.6. For $\mathcal{E}_{\mu,\theta_i,C}$ put $u_i = \mu \coth \theta_i$ and $v_i = \mu \operatorname{csch} \theta_i$. The two intermediate Cz^2 terms cancel. Applying Lemma 1.7 and $u_i^2 - v_i^2 = \mu^2$ gives

$$u' = \frac{u_1 u_2 + \mu^2}{u_1 + u_2}, \quad v' = \frac{v_1 v_2}{u_1 + u_2}, \quad C' = C. \quad (\text{I.26})$$

The hyperbolic addition formulas identify these with $\mu \coth(\theta_1 + \theta_2)$ and $\mu \operatorname{csch}(\theta_1 + \theta_2)$, proving (7.28). Converting the two-identical-edge result back to midpoint coefficients gives (7.29). Hence AB , C , $\mathfrak{D}$, and $\Delta_{\text{rel}} = \mathfrak{D}/(AB)$ are invariant.

For a change of amplitude and time units, $x = r\tilde{x}$ and $s = \ell\tilde{s}$, followed by common action rescaling by $d > 0$, one has $\tilde{A} = r^2 A/(d\ell)$, $\tilde{B} = r^2 \ell B/d$, and $\tilde{C} = r^2 C/d$; therefore $C^2/(AB)$ is unchanged.

For the finite positive statement, let $F \subset (0, \infty)$ be finite and write the subprobability kernel as

$$L_C(x, y) = e^{Cx^2/\varepsilon} L_0(x, y) e^{-Cy^2/\varepsilon}, \quad L_0(x, y) = L_0(y, x) > 0. \quad (\text{I.27})$$

Then $L_C^n = D_C L_0^n D_C^{-1}$ and hence, for $x \neq y$,

$$\frac{\varepsilon}{2(x^2 - y^2)} \log \frac{L_C^n(x, y)}{L_C^n(y, x)} = C. \quad (\text{I.28})$$

Thus finite blocking preserves the measured endpoint twist exactly. If the intermediate positive amplitudes are integrated over a grid of covering error d and cardinality m , completion of the positive interior quadratic form around its exact bridge minimizer gives

$$0 \leq E_{F,n}^\varepsilon(x, y) - \mathcal{E}_{\mu,n\theta,C}(x, y) \leq 4u(n-1)d^2 + \varepsilon(n-1) \log m. \quad (\text{I.29})$$

At fixed total duration one may take $d_n = n^{-2}$, $m_n = O(n^2)$, and $\varepsilon_n = n^{-3}/(1 + \log n)$, making the right side $O(n^{-2})$. The three-probe coefficient reconstruction is continuous on a compact calibrated range with $AB > 0$, so every initial $\Delta_{\text{rel}} > 0$ remains bounded away from zero for sufficiently fine finite blocking. This proves the operational statement. $\square$

I.1.6 The cross-scale optimizer test

Matching is an exact cross-scale consistency condition. The absence of an attracting flow does not leave the matched surface without an intrinsic test. The same blocking operation characterizes it by asking whether local minimization commutes with the elimination of an intermediate amplitude.

Proposition I.8 (Optimizer composition characterizes action-law matching). *Let $m_\theta(x)$ minimize $y \mapsto \mathcal{E}_{\mu,\theta,C}(x, y)$ over $y > 0$ for $x > 0$, and use $\rho = C/\mu$ and Δ_{rel} from (7.25). Then*

$$m_\theta(x) = r_\theta x, \quad r_\theta = (\cosh \theta + \rho \sinh \theta)^{-1}, \quad (\text{I.30})$$

and, for every $\theta, \phi > 0$,

$$r_{\theta+\phi}^{-1} - (r_\theta r_\phi)^{-1} = \Delta_{\text{rel}} \sinh \theta \sinh \phi. \quad (\text{I.31})$$

Thus $AB = C^2$ is equivalent to $m_{\theta+\phi} = m_\phi \circ m_\theta$ for every positive pair, and already to that equality for one positive pair at one $x > 0$. On this branch $r_\theta = e^{-\rho\theta}$ with $\rho = \pm 1$.

The proof is in Appendix I.1. Equation (I.31) gives a second finite diagnostic of the relative defect, independent of its reconstruction from three cost entries. On compact calibrated ranges away from zero denominators it is stable under small measurement errors. It concerns the minimizing maps of the reconstructed cost; identifying these with observed low-noise updates is a further test of the accepted law. It is a characterization of matching, not a mechanism which makes an arbitrary law approach it.

Proof of Proposition I.8. The coefficient of y^2 in $\mathcal{E}_{\mu,\theta,C}(x,y)$ is $\mu \coth \theta + C > 0$, since $|C| \leq \mu$ and $\theta > 0$. Its unique minimizer is therefore

$$y = \frac{\mu \operatorname{csch} \theta}{\mu \coth \theta + C} x = \frac{x}{\cosh \theta + \rho \sinh \theta} > 0.$$

Set $d_\theta = \cosh \theta + \rho \sinh \theta$. Hyperbolic addition gives

$$d_{\theta+\phi} - d_\theta d_\phi = (1 - \rho^2) \sinh \theta \sinh \phi,$$

which proves (I.31). All denominators and both hyperbolic sines are positive. Equality of the maps for even one positive pair and one nonzero amplitude therefore forces $1 - \rho^2 = 0$, equivalently $AB = C^2$. Conversely, $\rho = \pm 1$ gives $d_\theta = e^{\rho\theta}$ and exact composition for every pair. For $\Delta_{\text{rel}} > 0$ the formula also gives $r_{\theta+\phi} < r_\theta r_\phi$.

Solving the same identity for the defect gives

$$\Delta_{\text{rel}} = \frac{r_{\theta+\phi}^{-1} - (r_\theta r_\phi)^{-1}}{\sinh \theta \sinh \phi}. \quad (\text{I.32})$$

On a compact range where the observed ratios and $\sinh \theta \sinh \phi$ are bounded away from zero, the right-hand side has bounded first derivatives in its measured arguments. It is consequently locally Lipschitz there. This is stability of a finite diagnostic, not transverse attraction under the conservative blocking operation. $\square$

I.2 The nonlinear two-polarization regulator

This appendix proves Theorem 7.7. The purpose is not to establish unrestricted four-dimensional global regularity, but to show that the finite renewal evolution, constraint, readout, and independent action tests can be carried through a genuinely nonlinear spatially inhomogeneous vacuum sector. We use the unpolarized Gowdy T^3 class in areal coordinates [64]; the lapse and shift are nevertheless retained as independent variables until after the action variation.

I.2.1 First-order frame system and constraints

On $[t_0, t_1] \times \mathbb{T}^3$, $t_0 > 0$, consider

$$g = t^{-1/2} e^{\lambda/2} (-dt^2 + d\theta^2) + t [e^P (dx + Q dy)^2 + e^{-P} dy^2]. \quad (\text{I.33})$$

With $a = P_t$, $b = P_\theta$, $c = e^P Q_t$, $d = e^P Q_\theta$ and

$$e = \frac{1}{2}(a^2 + b^2 + c^2 + d^2), \quad f = ab + cd, \quad (\text{I.34})$$

the vacuum equations take the first-order form

$$\begin{aligned} a_t &= b_\theta - a/t + c^2 - d^2, & b_t &= a_\theta, \\ c_t &= d_\theta - c/t - ac + bd, & d_t &= c_\theta + ad - bc, \\ P_t &= a, & Q_t &= e^{-P}c, & \lambda_t &= 2te, \end{aligned} \quad (\text{I.35})$$

with

$$P_\theta = b, \quad e^P Q_\theta = d, \quad \lambda_\theta = 2tf. \quad (\text{I.36})$$

A direct differentiation using (I.35) gives

$$e_t - f_\theta = -\frac{a^2 + c^2}{t}, \quad f_t - e_\theta = -\frac{f}{t}. \quad (\text{I.37})$$

Consequently $\partial_t(\lambda_\theta - 2tf) = 0$ whenever the displayed equations hold, and $\partial_t \int_{\mathbb{T}^1} tf \, d\theta = 0$. Thus the momentum constraint and the periodic zero-mean condition for λ propagate.

I.2.2 Independent lapse and shift variation before areal gauge

Let $(R, \lambda, P, Q, N, \beta)$ be independent fields with $R, N > 0$, put $v_u = \dot{u} - \beta u'$, and define

$$\begin{aligned} \mathcal{L}_G &= -\frac{v_R(v_\lambda - 4\beta')}{2N} + \frac{R}{2N}(v_P^2 + e^{2P}v_Q^2) \\ &\quad + N \left[-2R'' + \frac{R'\lambda'}{2} - \frac{R}{2}\{(P')^2 + e^{2P}(Q')^2\} \right]. \end{aligned} \quad (\text{I.38})$$

The canonical momenta are

$$p_R = -\frac{v_\lambda - 4\beta'}{2N}, \quad p_\lambda = -\frac{v_R}{2N}, \quad \pi = \frac{Rv_P}{N}, \quad \varpi = \frac{Re^{2P}v_Q}{N}. \quad (\text{I.39})$$

Lapse and shift variation give respectively

$$\mathcal{C} = -2p_R p_\lambda + \frac{\pi^2 + e^{-2P}\varpi^2}{2R} + \frac{R}{2}\{(P')^2 + e^{2P}(Q')^2\} + 2R'' - \frac{1}{2}R'\lambda', \quad (\text{I.40})$$

$$\mathcal{D} = p_R R' + p_\lambda \lambda' + \pi P' + \varpi Q' - 4p'_\lambda. \quad (\text{I.41})$$

On a solution of (I.35), set

$$R = t, \quad N = 1, \quad \beta = 0, \quad p_\lambda = -\frac{1}{2}, \quad p_R = -te, \quad \pi = ta, \quad \varpi = te^P c. \quad (\text{I.42})$$

Substitution of (I.36) shows $\mathcal{C} = 0$ and $\mathcal{D} = 0$; the remaining area equation follows from (I.37). Hence the areal-gauge solution is the restriction of a critical point of the unreduced symmetry-class action rather than a solution obtained after deleting the lapse and shift equations.

I.2.3 Local splitting and exact staggered momentum

Set

$$U = (u_1, u_2, u_3, u_4) = \sqrt{t}(a, b, c, d), \quad \mathcal{J}(U) = u_1 u_2 + u_3 u_4, \quad n(U) = |U|^2. \quad (\text{I.43})$$

The local source flow, with the evolution parameter denoted by s , is

$$\begin{aligned}\frac{du_1}{ds} &= -\frac{u_1}{2t} + \frac{u_3^2 - u_4^2}{\sqrt{t}}, & \frac{du_2}{ds} &= \frac{u_2}{2t}, \\ \frac{du_3}{ds} &= -\frac{u_3}{2t} + \frac{-u_1u_3 + u_2u_4}{\sqrt{t}}, & \frac{du_4}{ds} &= \frac{u_4}{2t} + \frac{u_1u_4 - u_2u_3}{\sqrt{t}}, \\ \frac{dP}{ds} &= \frac{u_1}{\sqrt{t}}, & \frac{dQ}{ds} &= e^{-P} \frac{u_3}{\sqrt{t}}, & \frac{dt}{ds} &= 1,\end{aligned}\tag{I.44}$$

with $d\lambda/ds = 0$. Let S_σ be its near-identity implicit-midpoint map over source duration σ .

On a periodic grid of spacing ℓ , let $S_\pm$ be the two shifts, $D_+ = (S_+ - I)/\ell$, and $A_+ = (I + S_+)/2$. Put

$$w^+ = \frac{1}{\sqrt{2}}(u_1 + u_2, u_3 + u_4), \quad w^- = \frac{1}{\sqrt{2}}(u_1 - u_2, u_3 - u_4), \quad g^\pm = |w^\pm|^2.\tag{I.45}$$

At frozen physical time define the transport map by

$$(w^+)^+ = S_+ w^+, \quad (w^-)^+ = S_- w^-, \quad \lambda^+ = \lambda + \frac{\ell}{2}[(I + S_+)g^+ + (I + S_-)g^-],\tag{I.46}$$

leaving P, Q fixed.

Theorem I.9 (Exact staggered momentum under symmetric splitting). *With synchronized full step $h = \ell$, the source-half/transport/source-half composition preserves exactly*

$$\mathcal{K}_\ell = D_+ \lambda - 2A_+ \mathcal{J}(U).\tag{I.47}$$

The update is nearest neighbour, its background increment is a nonnegative sum of squared characteristic amplitudes, and it is second-order consistent on bounded smooth reference charts with $t \geq t_0 > 0$.

Proof. For the source flow, direct substitution in (I.44) gives $\nabla \mathcal{J}(U) \cdot \dot{U} = 0$. The difference of a quadratic function at two endpoints equals its derivative at the arithmetic midpoint paired with the increment, so implicit midpoint preserves $\mathcal{J}$ exactly while leaving λ fixed. For the transport step,

$$\Delta \mathcal{J} = \frac{1}{2}[(S_+ - I)g^+ - (S_- - I)g^-], \quad D_+ \Delta \lambda = 2A_+ \Delta \mathcal{J},\tag{I.48}$$

which proves exact preservation without invoking an assumed discrete product rule. The λ increment in (I.46) is a positive sum of squared norms. Characteristic translation is exact for the transport fields, while the background increment is a trapezoidal approximation to the integral of the translated squares with local error $O(\ell^3)$. Implicit midpoint has the same local order for the smooth source, and symmetric composition therefore gives second-order consistency. $\square$

I.2.4 Residual, constraint, metric, and action budgets

Let $X_j = (U_j, P_j, Q_j, \lambda_j)$ satisfy

$$X_{j+1} = \Phi_h(t_j, X_j) + r_j, \quad h = \ell,\tag{I.49}$$

and define

$$\mathcal{E}_h = \sum_j \frac{\|r_j\|_{2,\ell}^2}{2h}, \quad \mathcal{F}_h = \ell^{-(2r+1)} \mathcal{E}_h.\tag{I.50}$$

Here S_ℓ denotes grid sampling.

Proposition I.10 (Residual and constraint control). *On a fixed smooth reference chart, with initial C_ℓ^r error ϵ_0 and either a smallness bootstrap or an enforced offset envelope,*

$$\max_j \|X_j - \mathbf{S}_\ell X_*(t_j)\|_{r,\infty,\ell} \leq C(\epsilon_0 + h^2 + \sqrt{\mathcal{F}_h}). \quad (\text{I.51})$$

Writing $\widehat{U}_j$ for the deterministic predicted frame, the exact constraint increment is

$$\begin{aligned} \mathcal{K}_\ell(X_{j+1}) - \mathcal{K}_\ell(X_j) &= D_+ r_j^\lambda \\ &\quad - 2\mathbf{A}_+[\nabla \mathcal{J}(\widehat{U}_j) \cdot r_j^U + \mathcal{J}(r_j^U)]. \end{aligned} \quad (\text{I.52})$$

For $r \geq 1$, its accumulated norm is bounded by the initial residual plus $C\sqrt{\mathcal{F}_h}$.

Proof. The deterministic split map has smooth-chart Lipschitz constant $1 + Ch$ and consistency error Ch^3 . Discrete Gronwall therefore bounds the global error by $C(\epsilon_0 + h^2 + \sum_j \|r_j\|_{r,\infty,\ell})$. The one-dimensional inverse grid estimate and Cauchy–Schwarz bound the last sum by $C\sqrt{\mathcal{F}_h}$, proving (I.51). Expanding the quadratic $\mathcal{J}$ about $\widehat{U}_j$ and using exact preservation (I.47) gives (I.52); the same inverse estimate controls its accumulated norm. $\square$

For the pathwise curvature statement specialize the finite offset alphabet so that every accepted offset obeys

$$\|\epsilon_j\|_\infty \leq c_0 h^4, \quad (\text{I.53})$$

and take initial C_h^1 error $O(h^2)$. The retained source implementation may add a solver error $O(h^5)$ in C_h^1 . Since $\|D_h v\|_\infty \leq 2h^{-1}\|v\|_\infty$, the offset has C_h^1 size $O(h^3)$. The one-step Lipschitz factor $1 + Lh$, second-order consistency, and discrete Gronwall then give, pathwise,

$$\max_j \|X_j - \mathbf{S}_h X_*(t_j)\|_{1,\infty,h} \leq Ch^2. \quad (\text{I.54})$$

For $f = P, Q, \lambda$, subtraction of the reference step and division by h also gives an $O(h^2)$ bound for the nodal time differences of the errors.

Use the unchanged derivative records

$$\begin{aligned} u &= \left(u_1/\sqrt{t}, e^{-P} u_3/\sqrt{t}, |U|^2 \right), \\ v &= \left(u_2/\sqrt{t}, e^{-P} u_4/\sqrt{t}, 2\mathcal{J}(U) \right), \quad w = D_0 u, \end{aligned} \quad (\text{I.55})$$

where $D_0 = (S_+ - S_-)/(2h)$. These are respectively the time, space, and mixed jet records for (P, Q, λ) . On each spacetime square let $\mathcal{H}_h(f, u, v, w)$ be the tensor-product cubic Hermite interpolant using

$$\begin{aligned} H_0(q) &= 2q^3 - 3q^2 + 1, & H_1(q) &= -2q^3 + 3q^2, \\ G_0(q) &= q^3 - 2q^2 + q, & G_1(q) &= q^3 - q^2. \end{aligned}$$

The identities $H_0 + H_1 = 1$, $H'_0 + H'_1 = 0$, and $H''_0 + H''_1 = 0$ imply the stability estimate

$$\|F\|_\infty + \|DF\|_\infty \leq C\eta, \quad \|D^2 F\|_\infty \leq C\eta/h, \quad (\text{I.56})$$

whenever the two nodal jet banks and the first nodal differences of their value records differ by at most η . Shared corner jets make the interpolant C^1 across cell interfaces, so its weak second derivative is the cellwise second derivative and has no interface measure.

Theorem I.11 (Pathwise full-curvature control on the Gowdy alphabet). *Assume the smooth reference has bounded derivatives through the order required by the Hermite remainder, (I.53), and (I.54). For the shared Hermite reconstruction of $f = (P, Q, \lambda)$,*

$$\|f_h - f_*\|_{W^{1,\infty}} \leq Ch^2, \quad \|D^2 f_h - D^2 f_*\|_{L^\infty} \leq Ch. \quad (\text{I.57})$$

For the reconstructed metric (I.33),

$$\|g_h - g_*\|_{W^{1,\infty}} \leq Ch^2, \\ \|D^2 g_h - D^2 g_*\|_{L^\infty} + \|\text{Riem}(g_h) - \text{Riem}(g_*)\|_{L^\infty} \leq Ch. \quad (\text{I.58})$$

Hence on every compact positive subslab

$$\sup_h \|\text{Riem}(g_h)\|_{L^2} < \infty, \quad \|\text{Ric}(g_h)\|_{L^\infty} \leq Ch. \quad (\text{I.59})$$

All estimates are pathwise and the curvature limit is the Levi-Civita curvature of g_* . No all-time curvature norm is assumed.

Proof. The right sides of (I.55) are smooth functions on the standing chart. Equation (I.54), the nodal time difference estimate, and centered-difference consistency therefore make every corner value/first/mixed jet error $O(h^2)$. Apply (I.56); the exact-reference cubic Hermite remainder is $O(h^{4-k})$ through derivative order $k = 2$. This proves (I.57).

On $t \geq t_0 > 0$ the Gowdy metric map is smooth on the bounded chart and has determinant $-te^\lambda$, so its inverse stays uniformly bounded. The metric estimates follow by differentiating this smooth composition twice. In coordinates, $\Gamma = g^{-1} * Dg$, hence $\|\Gamma_h - \Gamma_*\|_\infty \leq Ch^2$ and $\|D\Gamma_h - D\Gamma_*\|_\infty \leq Ch$. Because g_h is C^1 across interfaces there is no distributional surface term. Substitution in $\text{Riem} = D\Gamma + \Gamma * \Gamma$ gives (I.58). The reference is vacuum, so contraction with the uniformly bounded metric and inverse gives the Ricci estimate; finite volume gives the L^2 curvature certificate. $\square$

A finite positive realization is obtained by giving each predicted local state a finite offset list containing zero and contained in (I.53), a strictly positive reference row, and acceptance probability $\exp[-E_j/\varepsilon]$ for a nonnegative deviation cost. If $p_{*,j}$ is the reference mass of the zero offset, the exact Gibbs identity gives

$$\mathbb{E} \sum_j E_j \leq \varepsilon \left[\mathcal{K}_{\text{path}} + \sum_j \log(1/p_{*,j}) \right], \quad (\text{I.60})$$

where $\mathcal{K}_{\text{path}}$ is the chain-rule relative entropy of the observed history law against the declared accepted law. Product reference masses count all local trials; solver residuals, finite retry failures, and iteration errors remain separate budgets. The positive selection cost is not identified with the signed ADM action. Since the curvature estimates above are deterministic and uniform over the permitted alphabet, every successful history inherits (I.59).

I.2.5 Measured proposal access and finite execution

Measured access to the curvature-controlled histories. A finite alphabet with admissible offsets establishes realizability, but does not by itself give a uniform probability of proposing those offsets. The following result separates this operational issue from the deterministic curvature estimate.

Proposition I.12 (Finite proposal-access certificate). *At stage j , let p_j be the measured finite proposal row and let $\mathcal{G}_j$ contain only proposals satisfying the offset and solver bounds of [Theorem 7.7](#). Suppose $p_j(\mathcal{G}_j) \geq \mu_j > 0$ and $0 \leq E_j \leq c_j$ on this set, uniformly over the supported source histories. Accept with the actual rule $\mathbf{1}_{\mathcal{G}_j} \exp(-E_j/\varepsilon_j)$. Then*

$$\mu_j e^{-c_j/\varepsilon_j} \leq Z_j \leq p_j(\mathcal{G}_j), \quad D_{\text{KL}}(Q_j \| p_j) \geq -\log p_j(\mathcal{G}_j). \quad (\text{I.61})$$

With a genuine reset to the same source between retries, caps R_j give

$$\Pr(\text{some stage fails}) \leq \sum_j \exp[-R_j \mu_j e^{-c_j/\varepsilon_j}]. \quad (\text{I.62})$$

Every successful history inherits the pathwise curvature and independent action-test estimates of [Theorem 7.7](#). If the displayed failure bounds are summable along a countable refinement, these conclusions hold on every sufficiently fine successful refinement almost surely.

The proof in [Appendix I.2](#) also gives a sufficient proposal floor from finite launch-share and completed-writer records. A missing safe proposal mass is a genuine obstruction which changing temperature cannot remove. These are bounds on trial counts; an imposed physical deadline must also accommodate execution of the capped trials. The selection rule and its normalizer remain physical records, and its positive deviation cost is not identified with the signed gravitational action.

The deterministic second-jet estimate is uniform over the stated offset alphabet. It therefore survives changes of the actual proposal weights, provided the acceptance rule retains that support and successful proposals really occur. At a fixed stage and source history write $\mu = p(\mathcal{G})$ for the measured mass of the admissible set, and use

$$a(u) = \mathbf{1}_{\mathcal{G}}(u) e^{-E(u)/\varepsilon}, \quad Z = \sum_u p(u) a(u), \quad Q(u) = p(u) a(u) / Z.$$

This is a declared selection rule, not an unrecorded replacement of the original microscopic transition.

Proof of [Proposition I.12](#). Since $0 \leq E \leq c$ on $\mathcal{G}$, $\mu e^{-c/\varepsilon} \leq Z \leq \mu$. For $\mu > 0$, let $p_{\mathcal{G}} = p/\mu$ on $\mathcal{G}$. Every $Q \ll p$ supported on $\mathcal{G}$ obeys the exact identity

$$D_{\text{KL}}(Q \| p) = D_{\text{KL}}(Q \| p_{\mathcal{G}}) - \log \mu \geq -\log \mu.$$

In particular $\mu = 0$ precludes such an accepted row altogether. No choice of temperature can correct this support failure.

On the accepted support the complete trial table still reconstructs its cost: $E(u) = -\varepsilon \log(ZQ(u)/p(u))$. Averaging also gives $\mathbb{E}_Q E = -\varepsilon \log Z - \varepsilon D_{\text{KL}}(Q \| p)$. These identities concern the recorded positive deviation cost; the independently varied signed action is unchanged.

With independent reset retries conditional on the same source, the probability that R_j attempts fail is $(1 - Z_j)^{R_j} \leq e^{-R_j Z_j}$. The same bound follows for history-dependent retries when the displayed lower success bound holds conditionally after every failed attempt. Taking the supported lower bound for each stage and using a union bound proves [\(I.62\)](#). For J stages and total failure target $0 < \delta < 1$, one sufficient finite choice is

$$R_j = \left\lceil \mu_j^{-1} e^{c_j/\varepsilon_j} \log(J/\delta) \right\rceil. \quad (\text{I.63})$$

Conditioning the entire path on success may reweight its intermediate source states. No equality with the unconditioned product of accepted rows is needed: every successful path has the same deterministic admissible support, and hence the same curvature and action bounds. Summability of the complete failure probabilities and the first Borel–Cantelli lemma give the final assertion, without independence between refinements. $\square$

Lemma I.13 (A finite execution certificate for the proposal floor). *At a retained source snapshot let a launch race have finitely many bins. Conditional on no earlier launch, let a_ℓ be the probability of any launch and f_ℓ the probability of a predictor-marked launch in bin ℓ . Set $S_0 = 1$, $S_\ell = \prod_{i \leq \ell} (1 - a_i)$. Suppose*

$$f_\ell \geq \rho_* a_\ell, \quad -\log S_J \geq A_* > 0, \quad 0 < \rho_* \leq 1. \quad (\text{I.64})$$

If, conditional on every predictor launch, the actually executed writer returns a certified member of $\mathcal{G}$ before the attempt deadline with probability at least $q_ > 0$, then*

$$p(\mathcal{G}) \geq q_* \rho_* (1 - e^{-A_*}). \quad (\text{I.65})$$

Proof. The disjoint first-launch events give predictor launch probability $\sum_{\ell=1}^J S_{\ell-1} f_\ell$. Since $S_{\ell-1} a_\ell = S_{\ell-1} - S_\ell$, this is at least $\rho_*(1 - S_J) \geq \rho_*(1 - e^{-A_*})$. On each such event the conditional completed-safe-output probability is at least q_* . Multiplication of conditional probabilities and summation prove the claim. Launch, interruption, completion and output certificates belong to the actual same-history record; a possible but unexecuted numerical predictor supplies none of these probabilities. $\square$

The execution probability and deadline in this lemma are physical inputs, not consequences of a fast geometric root clock. For example, one root clock competing with D clocks of equal rate has share $1/(D+1)$, regardless of their common rate scale. Moreover, a cap on the number of retries bounds their physical duration only when each attempt has a certified duration bound. With attempt durations at most d_j , the allowed stage duration must accommodate $R_j d_j$. Thus the proposition removes the need to assume a separate continuum proposal profile on a class with the displayed finite execution budgets, without asserting a uniform computation or sampling cost for every microscopic law.

For the general residual/action estimate below put

$$\kappa_h := \epsilon_0 + h^2 + \sqrt{\mathcal{F}_h}. \quad (\text{I.66})$$

Finally discretize (I.38) by midpoint time sampling and centered spatial differences, keep N and β independent, and replace $-2NR''$ by $2N'R'$ under the periodic boundary convention. Evaluation at the recorded areal-gauge fields gives, for every fixed smooth compactly supported independent test $(\delta R, \delta \lambda, \delta P, \delta Q, \delta N, \delta \beta)$,

$$|\delta S_h - \delta S_*| \leq C \kappa_h. \quad (\text{I.67})$$

The estimate follows from the same first-jet comparison used above, and the variation is taken before the areal-gauge restriction. Hence $\mathbb{E}\mathcal{F}_h = O(h^4)$ and $\epsilon_0 = O(h^2)$ yield second-order mean metric and action-test errors. Exact preservation of $\mathcal{K}_\ell$ is one constraint statement and is not substituted for the independent lapse/shift action tests.

Proof of Theorem 7.7. Item (G1) is Theorem I.9. Item (G2) is Proposition I.10. The pathwise C_h^1 estimate and Theorem I.11 give item (G3). The unreduced variational calculation (I.38)–(I.42), the action test (I.67), and the Gibbs identity (I.60) give item (G4) while retaining the lapse and shift variations before gauge restriction. $\square$

I.3 Flat and de Sitter regulator limits

This section proves the spatial and Cartan-continuum parts of Theorem I.14. The temporal profiles are no longer posited here: they are the stationary renewal solutions derived in Appendix I.1. The

present appendix verifies that the associated A_3 spatial generators have the required normalization and compact screens, and that the reconstructed coframes and exact link transports converge to the corresponding flat and de Sitter Cartan geometries.

Stationary geometric profiles. For $\Lambda = 0$, choose the constant stationary branch with $a_0 = 1$. Then

$$\mathcal{B}_h = I_3, \quad \varrho_h = 1, \quad g_h = I_3, \quad N_h = 1, \quad \beta_h = 0, \quad (\text{I.68})$$

and the graph generators converge to $\frac{1}{2}\Delta_{\mathbb{T}^3}$. The Lorentzian limit is

$$(\mathbb{R} \times \mathbb{T}^3, -dt^2 + dx_1^2 + dx_2^2 + dx_3^2). \quad (\text{I.69})$$

Its curvature and vacuum Einstein tensor vanish.

For $\Lambda > 0$, choose the expanding stationary branch of [Corollary 7.2](#) with $a_0 = 1$, $t_0 = 0$, and $H = H_0 = \sqrt{\Lambda/3}$. Then

$$\mathcal{B}_h(t) = e^{-2Ht} I_3, \quad \varrho_h(t) = e^{3Ht}, \quad g_h(t) = e^{2Ht} I_3, \quad N_h = 1, \quad \beta_h = 0. \quad (\text{I.70})$$

The continuum coframe and connection are

$$\vartheta^0 = dt, \quad \vartheta^a = e^{Ht} dx^a, \quad \omega^a{}_0 = H\vartheta^a, \quad \omega^a{}_b = 0. \quad (\text{I.71})$$

Cartan's equations give zero torsion and constant sectional curvature H^2 , so

$$g_H = -dt^2 + e^{2Ht}(dx_1^2 + dx_2^2 + dx_3^2), \quad G_{\mu\nu} + 3H^2(g_H)_{\mu\nu} = 0. \quad (\text{I.72})$$

The spatial quotient is globally hyperbolic and is the compact spatial quotient of the flat de Sitter slicing.

Theorem I.14 (Stationary flat and de Sitter renewal regulators). *For the operational predictive carrier and geometric branch of [Theorems 2.2](#) and [3.6](#), the stationary homogeneous families above have the following properties.*

- (i) *their rate and density profiles solve the lapse-varied homogeneous renewal Euler system of [Theorem 7.1](#) and arise as the continuum limits of the finite stationary system in [Theorem 7.3](#);*
- (ii) *each spatial cutoff defines a normalized finite A_3 renewal law whose root moments and speed densities reconstruct the displayed ADM variables without a primitive metric;*
- (iii) *the flat sequence converges to [\(I.69\)](#), with $\Lambda = 0$, while the curved sequence converges to [\(I.72\)](#), with $\Lambda = 3H^2$;*
- (iv) *on every compact time slab the cut, Poincaré, low-energy counting, and compact-screen margins are uniform in h ;*
- (v) *the flat curvature vanishes identically, while on each compact de Sitter slab the reconstructed curvature and observer/transport coefficients are uniformly bounded; the initial-energy and integrated propagation/source budgets of [Theorem 5.2](#) therefore pass directly, and in fact the curvature is uniformly bounded in L^∞ ;*
- (vi) *along $h_n = 2^{-n^2}$ with time mesh h_n and screen $[-n, n]$, exact parallel transport of the reconstructed Levi-Civita connection gives metric-unitary links, while the normalized torsion, plaquette-curvature, and Palatini first-variation defects are $O_T(h_n)$ and summable.*

In addition, [Theorem 7.5](#) supplies a nontrivial finite positive accept/reject realization whose measured action and accepted dynamics converge to the same expanding stationary branch.

The detailed homogeneous variational and operational proofs are in [Appendix I.1](#); the spatial generator, compact-screen, Cartan-curvature, and exact-link estimates are in [Appendix I.3](#). The result is not an attractor theorem: it does not select Λ , the time orientation, or the quadratic operational class from arbitrary microscopic renewal data.

I.3.1 The A_3 identity and periodic regulators

Let

$$W_0 = \left\{ x \in \mathbb{R}^4 : \sum_{i=1}^4 x_i = 0 \right\}, \quad \Lambda_{A_3} = W_0 \cap \mathbb{Z}^4, \quad (\text{I.73})$$

and let

$$\Phi_{A_3} = \{e_i - e_j : 1 \leq i \neq j \leq 4\} \quad (\text{I.74})$$

be the twelve oriented roots. The scalar product on W_0 is the restriction of the standard Euclidean scalar product on $\mathbb{R}^4$.

Lemma I.15 (A_3 second-moment identity). *On W_0 ,*

$$\sum_{\alpha \in \Phi_{A_3}} \alpha \alpha^\top = 8I_{W_0}. \quad (\text{I.75})$$

Equivalently, for every $v \in W_0$,

$$\sum_{i \neq j} (v_i - v_j)^2 = 8\|v\|^2. \quad (\text{I.76})$$

Proof. For $v \in W_0$, expand

$$\sum_{i \neq j} (v_i - v_j)^2 = \sum_{i \neq j} (v_i^2 + v_j^2 - 2v_i v_j).$$

Each v_i^2 occurs six times in the first two sums, while

$$\sum_{i \neq j} v_i v_j = \left(\sum_i v_i \right)^2 - \sum_i v_i^2 = - \sum_i v_i^2.$$

Hence the displayed expression is $6 \sum_i v_i^2 + 2 \sum_i v_i^2 = 8\|v\|^2$. Polarization gives [\(I.75\)](#). □

For an integer $d \geq 2$, put $h = d^{-1}$ and

$$V_h = \Lambda_{A_3} / d\Lambda_{A_3}. \quad (\text{I.77})$$

We regard hx , $x \in V_h$, as a periodic mesh of the flat torus $\mathbb{T}_{A_3}^3 = W_0 / \Lambda_{A_3}$. An oriented root α connects x to $x + \alpha$ in V_h . Let $m_h > 0$ be the mass of each vertex and let $k_{\alpha,h} > 0$ be the jump rate along the oriented root α . The Markov generator on functions $f : V_h \rightarrow \mathbb{C}$ is

$$(L_h f)(x) = \sum_{\alpha \in \Phi_{A_3}} k_{\alpha,h} [f(x + \alpha) - f(x)]. \quad (\text{I.78})$$

Its positive Dirichlet form on $L^2(V_h, m_h)$ is

$$\mathcal{E}_h(f) = -\langle f, L_h f \rangle_{L^2(m_h)} = \frac{m_h}{2} \sum_{x \in V_h} \sum_{\alpha \in \Phi_{A_3}} k_{\alpha, h} |f(x + \alpha) - f(x)|^2, \quad (\text{I.79})$$

whenever the rates are reversal symmetric.

Lemma I.16 (Fourier symbol and flat generator limit). *Suppose*

$$k_{\alpha, h} = \frac{1}{8h^2} \quad (\alpha \in \Phi_{A_3}). \quad (\text{I.80})$$

For a dual-lattice mode $e_k(x) = e^{2\pi i h \langle k, x \rangle}$, $-L_h$ has eigenvalue

$$\lambda_h(k) = \frac{1}{4h^2} \sum_{a=1}^6 [1 - \cos(2\pi h \langle r_a, k \rangle)], \quad (\text{I.81})$$

where $r_1, \dots, r_6$ is one orientation of the six root pairs. For every fixed k ,

$$\lambda_h(k) \longrightarrow 2\pi^2 \|k\|^2, \quad (\text{I.82})$$

and, under the natural piecewise-constant embeddings,

$$L_h \longrightarrow \frac{1}{2} \Delta_{\mathbb{T}_{A_3}^3} \quad (\text{I.83})$$

in strong resolvent sense and strongly at the semigroup level.

Proof. Pair the roots α and $-\alpha$ in (I.78). This gives (I.81). Taylor expansion yields

$$\lambda_h(k) = \frac{\pi^2}{2} \sum_{a=1}^6 \langle r_a, k \rangle^2 + O_k(h^2).$$

Because the sum over all twelve roots is twice the sum over the six unoriented roots, Lemma I.15 gives $\sum_{a=1}^6 r_a r_a^\top = 4I$, and therefore the leading term is $2\pi^2 \|k\|^2$.

The trigonometric symbol is nonnegative and converges pointwise on the common Fourier core. On each fixed finite Fourier band the convergence is uniform. The corresponding quadratic forms converge on trigonometric polynomials and have the common lower bound zero. Their closure gives Mosco convergence to the form of $\frac{1}{2} \Delta$; the resolvent and semigroup conclusions follow from the Mosco-resolvent equivalence. Equivalently, one may apply dominated convergence to the Fourier multipliers and then use density. $\square$

Lemma I.17 (Uniform spatial screens for the periodic A_3 graph). *Let the vertex mass be $m_h = a^3 h^3$ and the symmetric root rate be $k_{\alpha, h} = a^{-2}/(8h^2)$, where a ranges in a compact interval $0 < a_- \leq a \leq a_+ < \infty$. Then there are constants $c_P, C_W, c_{\text{cut}} > 0$, depending only on a_-, a_+ and the fixed torus, such that for all sufficiently small h :*

- (i) *the first nonzero eigenvalue of $-L_h$ is at least c_P ;*
- (ii) *the low-energy counting function satisfies*

$$N_h(R) := \#\{\lambda \in \text{spec}(-L_h) : \lambda \leq R\} \leq C_W(1 + R^{3/2}); \quad (\text{I.84})$$

- (iii) with edge conductance $c_{e,h} = m_h k_{\alpha,h} = ah/8$, every vertex set A with $0 < \mu_h(A) \leq \mu_h(V_h)/2$ obeys

$$h c_h(\partial A) \geq c_{\text{cut}} \mu_h(A)^{2/3}. \quad (\text{I.85})$$

Consequently the spectral projections $P_{h,R} = \mathbf{1}_{[0,R]}(-L_h)$ have uniformly bounded rank and

$$\|(I - P_{h,R})f\|_{L^2(m_h)}^2 \leq R^{-1} \mathcal{E}_h(f). \quad (\text{I.86})$$

Thus they form a uniformly exhausted compact-screen family.

Proof. The scalar a^{-2} multiplies the flat symbol (I.81), and the compact interval for a changes all spectral comparisons only by fixed constants. For $|2\pi h \langle r_a, k \rangle| \leq \pi$, the inequalities $2s^2/\pi^2 \leq 1 - \cos s \leq s^2/2$, together with the root-frame identity, give

$$c_1 \|k\|^2 \leq \lambda_h(k) \leq c_2 \|k\|^2 \quad (\text{I.87})$$

for modes in a fundamental dual cell, with constants independent of h . The first nonzero dual vector has fixed positive length, proving (i). Counting dual-lattice points in a three-dimensional ball gives (ii).

For (iii), retain only the three basis-root directions $\pm e_1, \pm e_2, \pm e_3$ from (3.6). Their Cayley subgraph is a periodic three-dimensional grid and each retained edge has conductance $ah/8 \geq a_-h/8$. The discrete edge-isoperimetric inequality on the d^3 periodic grid follows by slicing A along the three coordinate directions and applying the Loomis–Whitney inequality to the three coordinate projections; in physical units it reads $h^2 \# \partial A \geq c_0 (h^3 \# A)^{2/3}$. Since $\mu_h(A) = a^3 h^3 \# A$ and $c_{e,h} = ah/8$, it follows that $h c_h(\partial A) \geq [c_0/(8a_+)] \mu_h(A)^{2/3}$. Adding the other A_3 edges only increases this boundary capacity, proving (I.85). Finally, $-L_h \succeq R(I - P_{h,R})$ gives (I.86) by the spectral theorem. $\square$

I.3.2 Flat renewal vacuum

Construction I.18 (Flat periodic renewal regulator). At mesh $h = d^{-1}$, take the persistent spatial state $Z_n \in V_h$, one point in every nongeometric fibre, the twelve root branches (I.80), vertex mass

$$m_h = h^3, \quad (\text{I.88})$$

the identity root router, the identity slow row, and zero nonvacuum reward. The event alphabet consists of the root-race outcomes together with the protected orientation and one-point spectator fibres. Its conditional law is normalized directly, so the predictive carrier and scheduler records are literal parts of the operational regulator.

Theorem I.19 (Flat ADM and vacuum limit). *The regulator of Construction I.18 satisfies*

$$\mathcal{B}_h = I_3, \quad \varrho_h = 1, \quad g_h = I_3, \quad N_h = 1, \quad \beta_h = 0. \quad (\text{I.89})$$

Every undirected root edge has conductance $h/8$. The spatial generators converge to $\frac{1}{2} \Delta_{\mathbb{T}_{A_3}^3}$, and the Lorentzian continuum is

$$(M_0, g_0) = (\mathbb{R} \times \mathbb{T}_{A_3}^3, -dt^2 + dx_1^2 + dx_2^2 + dx_3^2). \quad (\text{I.90})$$

Its Levi–Civita connection and curvature vanish, so $G(g_0) = 0$. The finite torsion, curvature, and vacuum Palatini first-variation residuals vanish identically.

Proof. The root identity and rates give

$$\mathcal{B}_h = h^2 \sum_{\alpha \in \Phi_{A_3}} \frac{1}{8h^2} \alpha \alpha^\top = I_3.$$

Reversal symmetry gives $\beta_h = 0$. The mass density per physical volume is one, so $\varrho_h = 1$. Substitution in the ADM inversion theorem [Theorem B.11](#) gives (I.89). Multiplication of one directed rate by one vertex mass gives $h/8$.

Generator convergence is [Lemma I.16](#). The coframe $\vartheta^0 = dt$, $\vartheta^a = dx^a$ is constant, and the exact identity router gives the zero spin connection. Hence $T^A = d\vartheta^A + \omega^A_B \wedge \vartheta^B = 0$ and $\Omega^A_B = d\omega^A_B + \omega^A_C \wedge \omega^C_B = 0$. Therefore $\text{Ric} = 0$, $R = 0$, and $G = 0$. Any face-based finite Cartan and Palatini writer constructed from these exact constant data is identically zero, as is its first variation. Normalization and predictive persistence follow directly from the operational event law and [Theorem A.9](#). $\square$

I.3.3 Homogeneous de Sitter renewal vacuum

Let $\Lambda > 0$, $H = \sqrt{\Lambda/3}$, and choose the expanding proper-time stationary branch of [Corollary 7.2](#) with $a_0 = 1$ and $t_0 = 0$. On a compact time slab the resulting renewal rates and masses are

$$k_{\alpha,h}(t) = \frac{e^{-2Ht}}{8h^2}, \quad m_h(t) = e^{3Ht}h^3, \quad (\text{I.91})$$

with physical speed density $\varrho_h(t) = e^{3Ht}$. These profiles are stationary outputs of the homogeneous renewal action, not assumptions made in this appendix.

Theorem I.20 (Reconstructed de Sitter ADM packet). *For every h and t ,*

$$\mathcal{B}_h(t) = e^{-2Ht}I_3, \quad \varrho_h(t) = e^{3Ht}, \quad g_h(t) = e^{2Ht}I_3, \quad N_h(t) = 1, \quad \beta_h(t) = 0. \quad (\text{I.92})$$

Each undirected edge has conductance

$$c_{e,h}(t) = \frac{e^{Ht}}{8}h. \quad (\text{I.93})$$

On every compact time slab, the cut, Poincaré, counting, and compact-screen constants are uniform in h and t .

Proof. By [Lemma I.15](#),

$$\mathcal{B}_h(t) = h^2 \frac{e^{-2Ht}}{8h^2} \sum_{\alpha \in \Phi_{A_3}} \alpha \alpha^\top = e^{-2Ht}I_3.$$

The rates are reversal symmetric, so $\beta_h = 0$. Formula (3.8) gives

$$g_h = e^{2Ht}I_3, \quad N_h = (e^{3Ht})^{1/3}(e^{-6Ht})^{1/6} = 1.$$

The conductance is the product of vertex mass and directed rate: $e^{3Ht}h^3e^{-2Ht}/(8h^2) = e^{Ht}h/8$. On $[-T, T]$, the scalar $a(t) = e^{Ht}$ is bounded above and below by positive constants, so [Lemma I.17](#) applies uniformly. $\square$

Let internal indices $A, B \in \{0, 1, 2, 3\}$ be raised and lowered with $\eta = \text{diag}(-1, 1, 1, 1)$. Define

$$\vartheta^0 = dt, \quad \vartheta^a = e^{Ht}dx^a, \quad \omega^a_0 = H\vartheta^a, \quad \omega^a_b = 0, \quad (\text{I.94})$$

with $\omega_{AB} = -\omega_{BA}$.

Theorem I.21 (Cartan curvature and the vacuum Einstein equation). *The connection (I.94) is the torsion-free Levi-Civita spin connection of*

$$g_H = -dt^2 + e^{2Ht} \sum_{a=1}^3 (dx^a)^2. \quad (\text{I.95})$$

Its curvature two-form is

$$\Omega^A{}_B = H^2 \vartheta^A \wedge \vartheta_B. \quad (\text{I.96})$$

Consequently

$$\text{Ric}_{\mu\nu} = 3H^2 g_{\mu\nu}, \quad R = 12H^2, \quad G_{\mu\nu} + 3H^2 g_{\mu\nu} = 0. \quad (\text{I.97})$$

The slices $t = \text{const}$ are compact Cauchy hypersurfaces, so the spatial quotient is globally hyperbolic.

Proof. Since $d\vartheta^a = H\vartheta^0 \wedge \vartheta^a$,

$$T^a = d\vartheta^a + \omega^a{}_0 \wedge \vartheta^0 = H\vartheta^0 \wedge \vartheta^a + H\vartheta^a \wedge \vartheta^0 = 0,$$

and $T^0 = 0$. The coframe is nondegenerate, so this metric-compatible connection is Levi-Civita.

Cartan's second equation gives

$$\Omega^a{}_0 = d(H\vartheta^a) = H^2 \vartheta^0 \wedge \vartheta^a, \quad \Omega^a{}_b = \omega^a{}_0 \wedge \omega^0{}_b = H^2 \vartheta^a \wedge \vartheta^b,$$

which is (I.96) in invariant notation. Hence $R_{ABCD} = H^2(\eta_{AC}\eta_{BD} - \eta_{AD}\eta_{BC})$, so contraction gives $\text{Ric}_{AB} = 3H^2\eta_{AB}$ and $R = 12H^2$. Therefore $G = \text{Ric} - \frac{1}{2}Rg = -3H^2g$, proving (I.97).

The function t has timelike gradient and every inextendible causal curve meets each compact slice $\{t\} \times \mathbb{T}_{A_3}^3$ exactly once; this is the standard global-hyperbolicity criterion for a product metric with positive scale factor. $\square$

I.3.4 Exact links and summable finite Cartan–Palatini defects

Let K_h be the spacetime cell complex obtained by combining the spatial root mesh with a time mesh of size h . For an oriented edge e , let

$$U_{e,h} = \mathcal{P} \exp \left(- \int_e \omega \right) \quad (\text{I.98})$$

be exact parallel transport of the reconstructed continuum connection along that edge. Then $U_{e,h}^\top \eta U_{e,h} = \eta$ exactly. For an oriented face p with area $|p|$, define the normalized curvature defect

$$\mathfrak{r}_{p,h} = |p|^{-1} \log U_{\partial p} + \Omega(x_p), \quad (\text{I.99})$$

where x_p is the face midpoint and the logarithm is taken on the principal chart for sufficiently small h . Define the normalized torsion defect by the covariant developed-edge closure

$$\mathfrak{t}_{p,h} = |p|^{-1} \sum_{e \subset \partial p} U_{x_p \leftarrow s(e),h} \int_e \vartheta. \quad (\text{I.100})$$

Finally, let $\delta S_h(e_h, U_h)$ be the midpoint quadrature of the Palatini–cosmological first variation and let $\delta S(e, \omega)$ be its continuum value.

Theorem I.22 (First-order consistency of the exact-link regulator). *For every compact time slab $[-T, T]$ there is $C_T < \infty$, independent of h , such that*

$$\sup_p (\|\mathbf{r}_{p,h}\| + \|\mathbf{t}_{p,h}\|) + \sup_{\|v_h\|_{C^1} \leq 1} |\delta S_h[v_h] - \delta S[v_h]| \leq C_T h. \quad (\text{I.101})$$

For the flat regulator the left-hand side is zero. For the de Sitter regulator, take

$$h_n = 2^{-n^2}, \quad T_n = n. \quad (\text{I.102})$$

Then every normalized defect is summable along the diagonal exhaustion:

$$\sum_{n=1}^{\infty} C_{T_n} h_n < \infty. \quad (\text{I.103})$$

Proof. On a compact slab, the coframe, connection, curvature, and the finitely many derivatives needed below are bounded. The nonabelian Stokes expansion for the exact link gives

$$\log U_{\partial p} = -|p|\Omega(x_p) + O_T(h^3),$$

because $|p| = O(h^2)$ and the first variation of the curvature across the face is $O_T(h)$. Division by $|p|$ gives the curvature part of (I.101). The covariant sum of developed edge coframes is the integral of the torsion over the face plus an $O_T(h^3)$ quadrature remainder. Since the continuum torsion vanishes, division by area gives the torsion bound.

The Palatini–cosmological first variation is a finite sum of products of coframe, curvature, and test-field coefficients. Midpoint quadrature on each cell has local error $O_T(h^5)$ for a four-dimensional cell of volume $O(h^4)$; summing $O(h^{-4})$ cells on a fixed slab gives $O_T(h)$. Equivalent estimates follow directly from Taylor expansion of the finite wedge and holonomy writers. In the flat case all coefficients are constant and the exact links are identities, so the defects vanish.

For de Sitter, derivatives of order bounded independently of n grow at most exponentially on $[-n, n]$: $C_{T_n} \leq Ce^{cn}$ for fixed constants C, c . Hence $C_{T_n} h_n \leq C \exp(cn - (\log 2)n^2)$, whose series converges. $\square$

I.3.5 Proof of Theorem I.14

Proof of Theorem I.14. The predictive carrier at every cutoff is reconstructed by Theorem A.9. The homogeneous temporal profiles and their finite stationary origin are Theorems 7.1 and 7.3 and Corollary 7.2; the nonempty operational matching family is Theorem 7.5. The spatial ADM reconstruction is Theorems I.19 and I.20. The strong-resolvent and semigroup spatial limits follow from Lemma I.16; the scalar time-dependent factor is uniformly controlled on compact slabs. Uniform cut, Poincaré, counting, and compact-screen bounds are Lemma I.17. The flat Einstein equation follows from the zero connection in Theorem I.19. The de Sitter Cartan and Einstein calculation is Theorem I.21. Exact metric-unitary links and the $O_T(h)$, diagonally summable consistency estimates are Theorem I.22. These statements establish the six numbered clauses and the operational realization asserted in the main theorem. $\square$

Remark I.23 (Meaning of constructive vacuum emergence). The temporal rate profiles used here are not chosen in order to reproduce a target metric: on the homogeneous branch they solve the lapse-varied renewal Euler system of Appendix I.1, and a nonempty finite accept/reject class reconstructs the same reduced action from its measured trial probabilities. The present appendix

verifies the downstream spatial and Cartan geometry of those stationary profiles. The result still does not claim that an arbitrary renewal table selects the quadratic operational class, the rank-one compatibility condition, the value of Λ , or the expanding sign, nor that de Sitter is a universal attractor under coarse graining.

References

- [1] T. Regge, General relativity without coordinates, *Il Nuovo Cimento* **19** (1961), 558–571.
- [2] D. P. Rideout and R. D. Sorkin, Classical sequential growth dynamics for causal sets, *Physical Review D* **61** (2000), 024002.
- [3] J. Myrheim, Statistical geometry, CERN preprint CERN-TH-2538 (1978).
- [4] D. A. Meyer, *The Dimension of Causal Sets*, Ph.D. thesis, Massachusetts Institute of Technology, 1989.
- [5] D. M. T. Benincasa and F. Dowker, Scalar curvature of a causal set, *Physical Review Letters* **104** (2010), 181301.
- [6] L. Bombelli, J. Henson, and R. D. Sorkin, Discreteness without symmetry breaking: a theorem, *Modern Physics Letters A* **24** (2009), 2579–2587.
- [7] J. Ambjørn, J. Jurkiewicz, and R. Loll, Emergence of a 4D world from causal quantum gravity, *Physical Review Letters* **93** (2004), 131301.
- [8] J. Ambjørn, J. Jurkiewicz, and R. Loll, The spectral dimension of the universe is scale dependent, *Physical Review Letters* **95** (2005), 171301.
- [9] A. Perez, The spin-foam approach to quantum gravity, *Living Reviews in Relativity* **16** (2013), 3.
- [10] E. Hawkins, F. Markopoulou, and H. Sahlmann, Evolution in quantum causal histories, *Classical and Quantum Gravity* **20** (2003), 3839–3854.
- [11] L. Hardy, Probability theories with dynamic causal structure: a new framework for quantum gravity, arXiv:gr-qc/0509120 (2005).
- [12] O. Oreshkov, F. Costa, and Č. Brukner, Quantum correlations with no causal order, *Nature Communications* **3** (2012), 1092.
- [13] H. Jaeger, Observable operator models for discrete stochastic time series, *Neural Computation* **12** (2000), 1371–1398.
- [14] M. L. Littman, R. S. Sutton, and S. Singh, Predictive representations of state, in *Advances in Neural Information Processing Systems 14*, MIT Press, Cambridge, MA, 2002, pp. 1555–1561.
- [15] H.-P. Breuer and B. Vacchini, Quantum semi-Markov processes, *Physical Review Letters* **101** (2008), 140402.
- [16] B. Vacchini, Quantum renewal processes, *Scientific Reports* **10** (2020), 5592.

- [17] S. Gielen, D. Oriti, and L. Sindoni, Cosmology from group field theory formalism for quantum gravity, *Physical Review Letters* **111** (2013), 031301.
- [18] S. Gielen and L. Sindoni, Quantum cosmology from group field theory condensates: a review, *SIGMA* **12** (2016), 082.
- [19] N. Lashkari, M. B. McDermott, and M. Van Raamsdonk, Gravitational dynamics from entanglement “thermodynamics”, *Journal of High Energy Physics* **2014** (2014), no. 4, 195.
- [20] T. Jacobson, Entanglement equilibrium and the Einstein equation, *Physical Review Letters* **116** (2016), 201101.
- [21] C. Cao, S. M. Carroll, and S. Michalakis, Space from Hilbert space: recovering geometry from bulk entanglement, *Physical Review D* **95** (2017), 024031.
- [22] S. Holst, Barbero’s Hamiltonian derived from a generalized Hilbert–Palatini action, *Physical Review D* **53** (1996), 5966–5969.
- [23] F. Besnard and N. Bizi, On the definition of spacetimes in noncommutative geometry: Part I, *Journal of Geometry and Physics* **123** (2018), 292–309.
- [24] T. J. R. Hughes, T. Kato, and J. E. Marsden, Well-posed quasi-linear second-order hyperbolic systems with applications to nonlinear elastodynamics and general relativity, *Archive for Rational Mechanics and Analysis* **63** (1977), 273–294.
- [25] H. Lindblad and I. Rodnianski, The global stability of Minkowski space-time in harmonic gauge, *Annals of Mathematics* **171** (2010), 1401–1477.
- [26] K. G. Wilson, Confinement of quarks, *Physical Review D* **10** (1974), 2445–2459.
- [27] C. J. Isham and N. Linden, Continuous histories and the history group in generalized quantum theory, *Journal of Mathematical Physics* **36** (1995), 5392–5408.
- [28] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, Space-time as a causal set, *Physical Review Letters* **59** (1987), 521–524.
- [29] S. Surya, The causal set approach to quantum gravity, *Living Reviews in Relativity* **22** (2019), 5.
- [30] T. Konopka, F. Markopoulou, and S. Severini, Quantum graphity: a model of emergent locality, *Physical Review D* **77** (2008), 104029.
- [31] J. Ambjørn, J. Jurkiewicz, and R. Loll, Quantum gravity, or the art of building spacetime, in *Approaches to Quantum Gravity*, D. Oriti, ed., Cambridge University Press, Cambridge, 2009, pp. 341–359.
- [32] D. Oriti, ed., *Approaches to Quantum Gravity: Toward a New Understanding of Space, Time and Matter*, Cambridge University Press, Cambridge, 2009.
- [33] A. Connes, *Noncommutative Geometry*, Academic Press, San Diego, 1994.
- [34] T. Jacobson, Thermodynamics of spacetime: the Einstein equation of state, *Physical Review Letters* **75** (1995), 1260–1263.

- [35] E. Verlinde, On the origin of gravity and the laws of Newton, *Journal of High Energy Physics* **2011** (2011), no. 4, 029.
- [36] M. Van Raamsdonk, Building up spacetime with quantum entanglement, *General Relativity and Gravitation* **42** (2010), 2323–2329.
- [37] C. R. Shalizi and J. P. Crutchfield, Computational mechanics: pattern and prediction, structure and simplicity, *Journal of Statistical Physics* **104** (2001), 817–879.
- [38] N. F. Travers and J. P. Crutchfield, Equivalence of history and generator ϵ -machines, *Physical Review E* **89** (2014), 062107.
- [39] S. E. Marzen and J. P. Crutchfield, Structure and randomness of continuous-time, discrete-event processes, *Journal of Statistical Physics* **169** (2017), 303–315.
- [40] A. M. Jurgens and J. P. Crutchfield, Divergent predictive states: the statistical complexity dimension of stationary, ergodic hidden Markov processes, *Chaos* **31** (2021), 083114.
- [41] W. Feller, *An Introduction to Probability Theory and Its Applications, Vol. II*, Wiley, New York, 1971.
- [42] A. M. Garsia and J. Lamperti, A discrete renewal theorem with infinite mean, *Commentarii Mathematici Helvetici* **37** (1962), 221–234.
- [43] K. B. Erickson, Strong renewal theorems with infinite mean, *Transactions of the American Mathematical Society* **151** (1970), 263–291.
- [44] F. Caravenna and R. Doney, Local large deviations and the strong renewal theorem, *Electronic Journal of Probability* **24** (2019), Paper No. 72, 1–48.
- [45] W. F. Stinespring, Positive functions on C^* -algebras, *Proceedings of the American Mathematical Society* **6** (1955), 211–216.
- [46] M.-D. Choi, Completely positive linear maps on complex matrices, *Linear Algebra and its Applications* **10** (1975), 285–290.
- [47] A. Strohmaier, On noncommutative and pseudo-Riemannian geometry, *Journal of Geometry and Physics* **56** (2006), 175–195.
- [48] N. Franco, Temporal Lorentzian spectral triples, *Reviews in Mathematical Physics* **26** (2014), 1430007.
- [49] N. Franco and M. Eckstein, An algebraic formulation of causality for noncommutative geometry, *Classical and Quantum Gravity* **30** (2013), 135007.
- [50] K. van den Dungen, M. Paschke, and A. Rennie, Pseudo-Riemannian spectral triples and the harmonic oscillator, *Journal of Geometry and Physics* **73** (2013), 37–55.
- [51] K. van den Dungen, Krein spectral triples and the fermionic action, *Mathematical Physics, Analysis and Geometry* **19** (2016), 4.
- [52] A. Devastato, S. Farnsworth, F. Lizzi, and P. Martinetti, Lorentz signature and twisted spectral triples, *Journal of High Energy Physics* **2018** (2018), no. 3, 089.

- [53] A. Devastato, M. Filaci, P. Martinetti, and D. Singh, Actions for twisted spectral triple and the transition from the Euclidean to the Lorentzian, *International Journal of Geometric Methods in Modern Physics* **17** (2020), suppl. 1, 2030001.
- [54] G. Nieuviarts, Emergence of time from a twisted spectral triple in almost-commutative geometry, *Journal of Noncommutative Geometry* (2026), to appear; arXiv:2512.15450.
- [55] G. Nieuviarts, Emergence of Lorentz symmetry from an almost-commutative twisted spectral triple, *International Journal of Geometric Methods in Modern Physics* (2026), to appear; arXiv:2502.18105.
- [56] L. M. Bregman, The relaxation method of finding the common point of convex sets and its application to the solution of problems in convex programming, *USSR Computational Mathematics and Mathematical Physics* **7** (1967), 200–217.
- [57] F. R. K. Chung, *Spectral Graph Theory*, CBMS Regional Conference Series in Mathematics, vol. 92, American Mathematical Society, Providence, RI, 1997.
- [58] L. Saloff-Coste, *Aspects of Sobolev-Type Inequalities*, London Mathematical Society Lecture Note Series, vol. 289, Cambridge University Press, Cambridge, 2002.
- [59] A. Anderson, Y. Choquet-Bruhat, and J. W. York, Jr., Einstein–Bianchi hyperbolic system for general relativity, arXiv:gr-qc/9710041, version 2 (1998).
- [60] S. Klainerman, I. Rodnianski, and J. Szeftel, The bounded L^2 curvature conjecture, *Inventiones Mathematicae* **202** (2015), 91–216.
- [61] J. W. York, Jr., Role of conformal three-geometry in the dynamics of gravitation, *Physical Review Letters* **28** (1972), 1082–1085.
- [62] G. W. Gibbons and S. W. Hawking, Action integrals and partition functions in quantum gravity, *Physical Review D* **15** (1977), 2752–2756.
- [63] J. E. Marsden and M. West, Discrete mechanics and variational integrators, *Acta Numerica* **10** (2001), 357–514.
- [64] V. Moncrief, Global properties of Gowdy spacetimes with $T^3 \times \mathbb{R}$ topology, *Annals of Physics* **132** (1981), 87–107.
- [65] G. Chiribella, G. M. D’Ariano, and P. Perinotti, Theoretical framework for quantum networks, *Physical Review A* **80** (2009), 022339.
- [66] F. A. Pollock, C. Rodríguez-Rosario, T. Frauenheim, M. Paternostro, and K. Modi, Non-Markovian quantum processes: Complete framework and efficient characterization, *Physical Review A* **97** (2018), 012127.
- [67] J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Encyclopedia of Mathematics and its Applications, vol. 137, Cambridge University Press, Cambridge, 2010.
- [68] A. Corichi, I. Rubalcava-García, and T. Vukašinac, Actions, topological terms and boundaries in first order gravity: A review, arXiv:1604.07764 (2016).
- [69] J. Simon, Compact sets in the space $L^p(0, T; B)$, *Annali di Matematica Pura ed Applicata* **146** (1986), 65–96.

- [70] R. M. M. Mattheij, Decoupling and stability of algorithms for boundary value problems, *SIAM Review* **27** (1985), 1–44.
- [71] P. W. Bates, K. Lu, and C. Zeng, Invariant foliations near normally hyperbolic invariant manifolds for semiflows, *Transactions of the American Mathematical Society* **352** (2000), 4641–4676.
- [72] M. Wimmer, Algorithm 923: Efficient numerical computation of the Pfaffian for dense and banded skew-symmetric matrices, *ACM Transactions on Mathematical Software* **38** (2012), no. 4, Article 30, 1–17.
- [73] A. Bisio, G. Chiribella, G. M. D’Ariano, and P. Perinotti, Minimal computational-space implementation of multi-round quantum protocols, *Physical Review A* **83** (2011), 022325.
- [74] L. Lindblom, M. A. Scheel, L. E. Kidder, R. Owen, and O. Rinne, A new generalized harmonic evolution system, *Classical and Quantum Gravity* **23** (2006), S447–S462.